

IRAN DEFENCE PRODUCTS

**MINISTRY OF DEFENCE
ISLAMIC Rep. of IRAN**

Talking with Dear Readers

Mindex center as the publisher of this book , intend to introduce part of Iran`s Defence potential in both production and engineering.

Since being the fourth issue of “Iran Defence Products”, as anyone who has worked in this area knows, it takes a considerable amount of time to collect such an intelligent document.

Due to much of this information provided in this book comes from vast related industries,Mindex center has made all efforts to obtain accurate material by research & acknowledgment.

However,we are still hoping to soon fill in the probable existing gaps. In the meantime, readers could send their ideas /suggestions by email and also inform us of your book`s serial number.

Any update to the content of this book will appear on line as it occurs , see: www.mindexcenter.ir

The content of this book is only for the purpose of providing information, though the publisher will not be liable for any damages,direct or indirect arising from any misuse of the information contained.

At the end, we should express our utmost appreciation to all persons in different ranks who have helped us to complete this selected issue.

S/N:

Issue 2013 - 2016



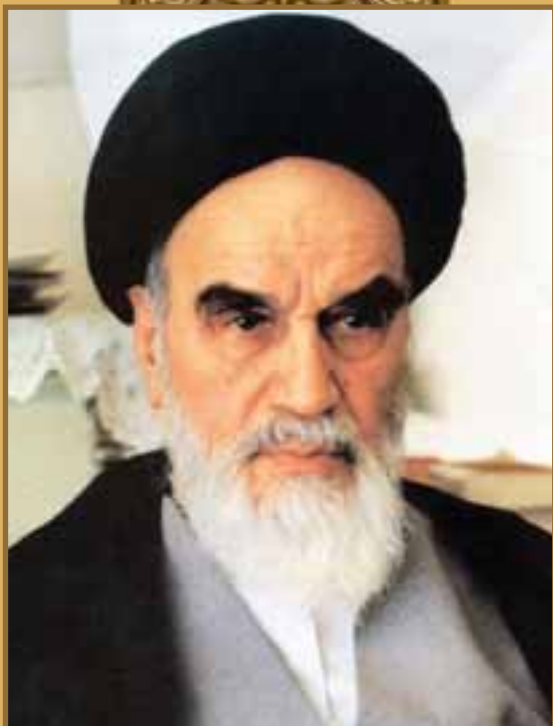
IN THE NAME OF ALLAH
The most Compassionate
The most Merciful



With Compliment

Signature

Date



**AYATOLLAH KHOMEINI
LATE LEADER AND FOUNDER
OF ISLAMIC REP. OF IRAN**



**AYATOLLAH SEYED ALI KHAMENEI
SUPREME LEADER OF THE
ISLAMIC REP. OF IRAN
AND SUPREME COMMANDER
OF ALL FORCES**

I.R.IRAN Defence Ministers Since Feb. 1979 (After Islamic Revolution)

	Ahmad Madani	1979 (2 months)		Akbar Torkan	1989-1993
	Mostafa Chamran	1979-1981 Martyred in Iraq War		Mohammad Forozandeh	1993-1997
	Javad Fakori	1981 (2 months) Martyred in Iraq war		Ali Shamkhani	1997-2005
	Seyyed Mosa Namjo	1981 Martyred in Iraq War		Mostafa Mohammad Nadjar	2005-2009
	Mohammad Salimi	1981-1985		Ahmad Vahidi	2009-2013
	Mohsen Rafighdoost *	1981-1988 Minister of Sepah		Hossain Dehghan	2013-on wards
	Mohammad Hossain Jalali	1985-1989	* Months after 1979 Iranian revolution, a para-military Revolutionary Guards Corps established. At the beginning of the eight years of Iraqi imposed war against Islamic Republic, RGC logistically commissioned to support the Corps inside the country and in the war fronts. Then became capable to form and resamble to a geniune army in all respects. In 1980, logistic branch of RGC formed into a ministry alongside of Ministry of Defense (1981 till 1988) that was inserted into the Ministry of Defense in 1989 almost as soon as the aforesaid war ended in 1988. The Ministry now Known as Ministry of Defense and Armed Forces Logistics.		

Exporter 's Comment

In summer 1989, a M.o.D international relation department was instructed to form a centre for ordnance export activities. In a very short time several highly commercially specialist officers were recruited to take up the job which soon followed by few foreign contracts.

The Centre soon received defence minister's special blessing to introduce the defensive products more to the potential customers and expand marketing activity globally. Export Centre as a result pioneered the unprecedented commercial sales of military equipment and hardware made in Iran, the move in which was welcomed by all military producers and organizations in the Islamic Republic.

After 24 years of credible reputation, this M.o.D's exporting entity has arrived to the point that can proudly announce undeniable unique role in formation of military relation and export of arms and equipment to as many as 35 countries in four continents.

In 1992, other M.o.D's producing organizations were encouraged to enter into the marketing arena and export their own military products independently. The supreme Export Council chaired by the honourable minister himself was then established to lead the export promotion and legislate the required laws on arms sales, supervise the activities and their conformity to the prevailing UN resolutions and arms control policies. The council's initial members were the Defence Minister, Vice Defence Minister, Deputy Defence Minister for Logistics, Chief of Export Centre and the Chairman of the Defence Industries Organization. Today the number of the council's permanent member has increased into ten. The council formed a technical committee chaired by the Deputy Defence Minister for Logistics who is responsible for all military exports and permissions.

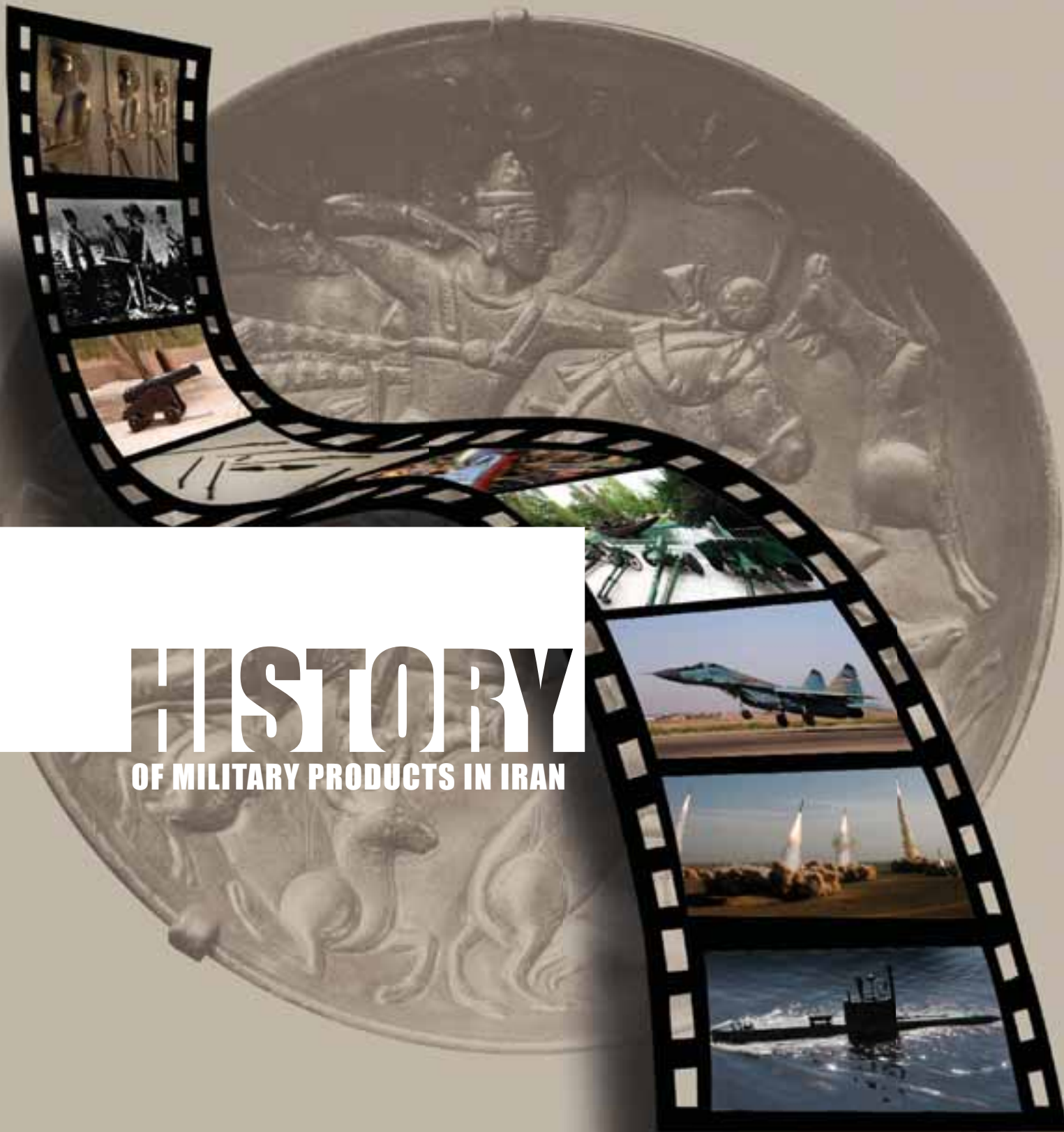
The Iranian Ministry of Defence, like all other producers and military exporters, shortly considered the importance of publicity and showing the global audience its achievements. Therefore next to advertising in prime international military magazines, it participates regularly and periodically in a few of the main international military exhibitions.

One of the advertising mean was to publish and provide enough technical literature so to satisfy the recipients. Though taking the above into consideration, in 2002 the idea of publishing a workable, technical and comprehensive defence product book came into light, a hectic, important but worthy task. Literally Export Centre managed to publish successfully the first edition not sooner than the year 2005, then the 2006/07 and the 2008/09 editions. presently the new 2013-2016 issue contains a variety of Iranian manufactured items with several new chapters, one consists of 12 sections, under the title of "transfer of technology" and the other provides good information on the history of arms manufacturing in Iran. We have also introduced in this collection, a military manufactured coding system known as product Identification Code (PIC), help to recognize the products by referring to the codes. The technical tolerances are constantly reviewed, needless to say, the data necessarily subject to change without notice. To ease the reader's searching through the pages, the two newly inserted sections, being: "Alphabetic Index" and "Acronyms", can come handy.

Readers who require more detailed information or have specific suggestions, are most welcome to contact us on our correspondence shown in this book.

Thank you

550 BC ...



HISTORY

OF MILITARY PRODUCTS IN IRAN

... Today

The history of arms production in Iran (Persia) begins as early as 550 BC., in ancient Persia and by the Cyrus the Great, the founder of the Achaemenid Empire, well-recognized for his achievements in human rights, conquering four corners of the world, also been known for his innovation in building projects. Cyrus the Great took all steps to form a standing and well-trained army with defensive instruments; long spears, shields, helmets, breast-plates, swords, javelins and of course bow and arrows with which conquered vast territories.

In the later era in the 16th century, too many threads and greed to conquer the empire attracted pertinacious monarchs and rulers of the world.

Although the militarism never paved its way into the daily lives of the nation, however this motherland fostered men & women to become warriors and braver-ies against all foreign insurgents throughout the history and defended her territories and fate. Ancient Persian armies have always been in the position to make their necessary armaments and defensive particles.

In 1590, during the reign of King Abbas I ,Isfahan was the center of the government and the ruling system . Some industries developed as well as moulding method of making cannons and a single barrel rifle and gunsmithing .The skilled men & group of blacksmiths expanded the relative activities amid lack of extrater-ritorial assistance.

For over hundred years the Persians learned more and gathered more knowledge how to handle gun-powder and the gun itself .The so-called industry developed during Afsharia (1703) to build boats in the city of Bushehr and in the Zandiya Dynasty (1780) in which the city of Shiraz became another center to make can-nons & flint guns .

In the early eighteenth century (1807- 1852) Mirza Taghi Farahani , known as Amir Kabir (chief minister to Naser al-din shah of Qajar reign) had been consid-ered as a modernizer of his time . After 1840 he was put in charge of army's provisions and through divine efforts during his authority before 1852 , he uplift the Persian arsenal with the help of Austrians in Isfahan and Tehran. He eagerly collected information of military value to the nation and established special school for military education and scholarship. In 1888 new place in the center of Tehran and next to the building of ministry of defense was chosen to accommodate the Ghorkhane (the military manufacturing establish-ment) and took first steps to purchase three ships to



Persian Soldiers at Persepolis Monument



First Machine Gun in Iran



Some Ancient Rifles



A Type of Flint Rifle

form a very young navy force. Seven years after the 1st. World War, and up to the year 1925, Ghorkhane was closed several times due to lack of skill, equipment and well – trained personnel. In 1925 , starts a-fifty four-year period of father (Reza) and the son (Mohammad Reza) Pahlavi dynasty that ended by the 1979 Islamic Revolution .

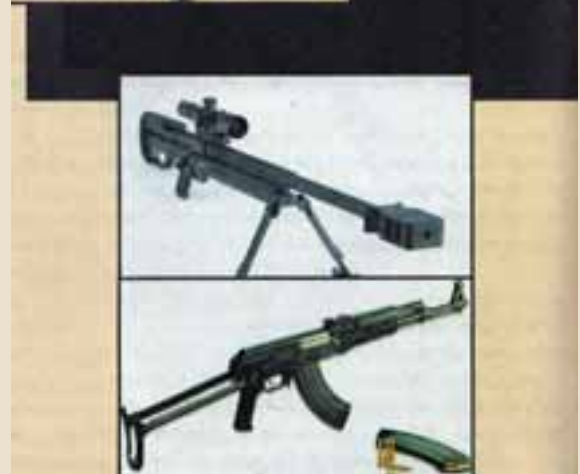
During the two decades ruling of Reza Shah, great significance were given to the creation of a classic strong army. An army that can serve the king in frontiers, national security and against political opponents .To fulfill the king's ambitions, recruitment of all skilled workers in steel mills and metal workshops was the first step towards establishment of a repairing center , followed by commissioning of 18 persons to Europe to be trained in this profession. Gradually during 1922-1929 by bringing tools, equipment and samples from Europe, the factory was able to manufacture barrels, rifles and thousands rounds of bullets per day and produce necessary gun-powder in chemical factory .Thanks to the Germans , the Iranian ammunition manufacturing plant and the armament manufacturing plant were constructed under separate arrangements between 1935 to 1940 .Very soon contracts were signed with German manufacturers to transfer the technology to produce 75mm , 205 mm , 105 mm ammunitions. The brass plant though was built before 1940.

In 1941 and under the command of the pahlavi II reign period , western governments increased their influence in Iran and controlled most of the Iranian oil and other natural resources . To guard the so-called western interests, country needed a strong army , something that required most of the revenue to be paid on weapons and equipments in the land that most of the people lived in extreme poverty. The Shah (king) established himself as an indispensable ally of the west, thus major arms industries and unprecedented defensive war material and machines were purchased and imported into the country , together with vast number of US and European military and technical advisers & missionaries. During the first decade, the Military Industries Organization (MIO) was formed and indisputable development of manufacturing of arms & ammunition , chemicals , air & sea industries came in to light .

Heckler & Koch manufacturer exported to Iran the G.3 rifles technology and the MP5 series, 9mm pistols and MG3 machines Guns. During the process, the MIO also acquired the capability to produce gun- powder, dynamite, acids & nitrate explosives. Another leading ammu-



Zanboorak (Light Cannon)



nition manufacturer, Fritz Werner sold the main semi-mechanical ammunition maker that changed the whole industry and trained and brought knowledge & skill into the lines. 9×19mm, 7.62 ×51.... , and many other different calibers and ammunition parts & spares.

MIO in its renovation programs had to be self – sufficient and bring the chain factories into efficiency . The Organization soon came into the conclusion that with its qualified staff, it should move towards self-reliance and cover other sectors of the defensive products. In the sixties, vehicles, communication, battery producing and electronic parts and devices were the areas that the Organization was almost capable to handle in all respects. In another sector, Iran Aviation Industries originally was constructed by Textron and intended to produce Bell 214 series, initially 350 helicopters plus transfer of technology. This project later was expanded enormously and US International Bell co. trained these personnel that before 1977 well-trained staff were sufficiently capable to logistically support the fleet in the national army.

As the Pahlavi's era ended in 1979, the Iranian post revolution inherited immense amount of land , air and naval war-machine , and due to the west's hatred towards creation of a truly Islamic state in the Middle East , flamed the old disputes into eight-year –long war with the eastern neighbor. With systematic refusal of social and commercial relations as from the beginning of the revolution followed by a great degree of worsening sanctions, literally paved the way towards a bright future for the creative, active & capable nation to stand-up against all international & technological challenges .

On the military might, the country necessarily had to be self-reliance by overriding all difficulties and restrictions. The nation with momentous ideals beside a revolution with propositions and message for the world to pursue, soon recognized its threats and opportunities.

The Iranian military industries entered into a circumstance that had been dealt with prudence. Acquiring Eastern technology together with the existing Western know-how, increased the capability of the army much in a greater level among the countries in the region. What the dear readers would find in this book is just a tip of an iceberg. This illustrated collection would give you somehow a thorough introduction to the current products and equipments and its various types. Achievements in great extent shall take the lead in no time.

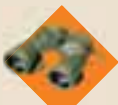


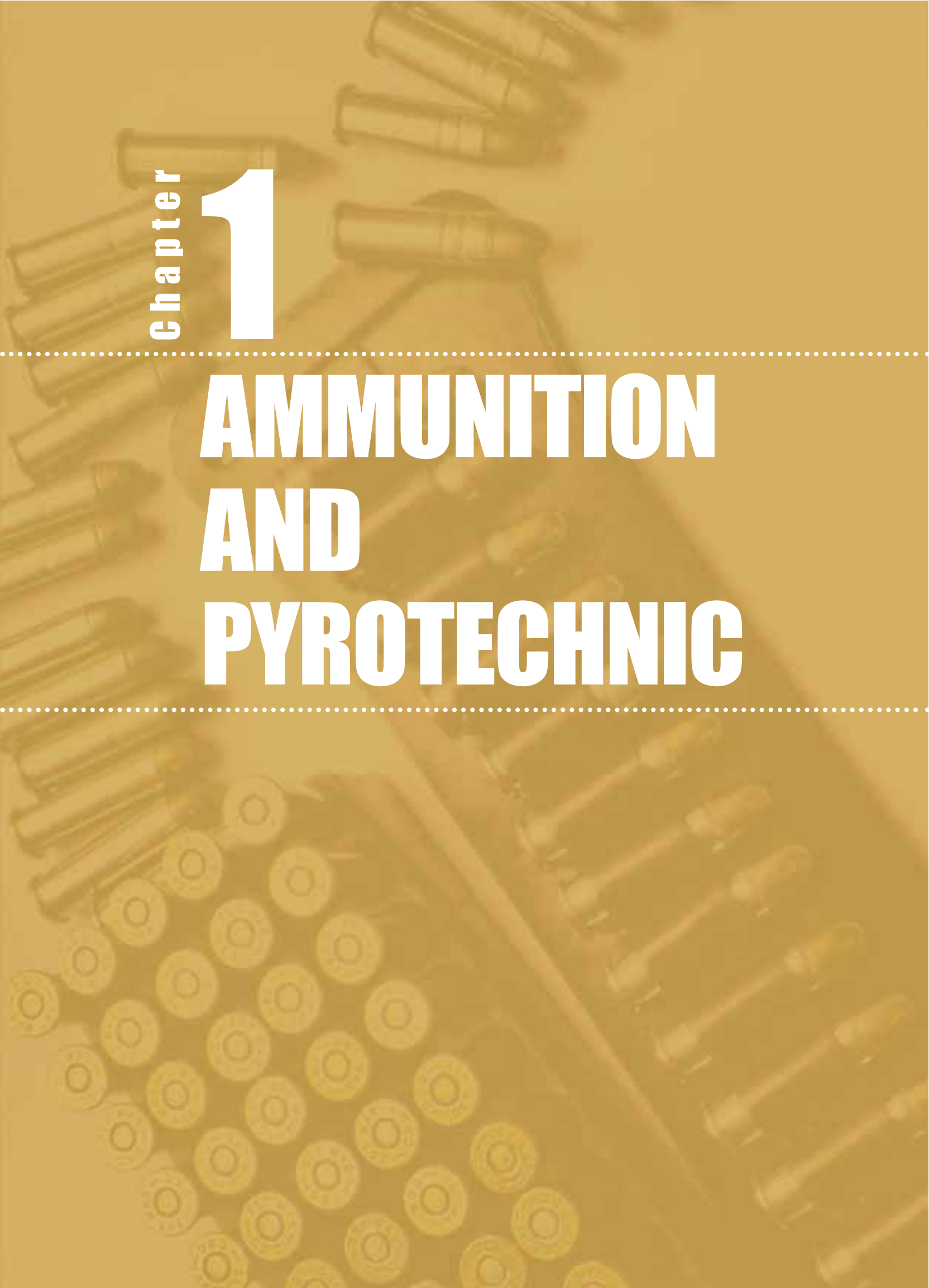
Acronym Index

A / C	Aircraft	Canister	An anti - personnel round that consists of a large number of balls's slugs,pellets or flechettes inside a sheet metal casing or canister
A /G	Air to Ground	Cartridge	A Complete self - contained round of ammunition / A complete - contained round
A/ A	Air to Air	CCL	Command and Control Interface
AAA	Anti Aircraft Artillery	CEP	Circular Error Probability
AAAV	Advanced Amphibious Assault Vehicle	CG	Centre of Gravity
AAG	Anti Aircraft Gun	CH	Channel
AAM	Air to Air Missile	Chaff	Thin silvers of radar- reflective material to protect friendly aircraft
ABS	Anti - Skid Braking System	CKD	Component Knocked down
ACV	Armoured Combat Vehicle	CMOS	Complementary Metal Oxide Semi-conductor
ADF	Automatic Direction Finding	Comint	Communication Intelligence
AFCS	Automatic Fire Control System	CRT	Cathode Ray Tube
AFV	Armoured Fighting Vehicle	DF	Direction Finding
AGL	Assault Grenade Launcher	Drill or dummy	An ammunition that is completely inert and provided for training use
AIFV	Armoured Infantry Fighting Vehicle	E - O	Electro - Optical
AIV	Armoured Infantry Vehicle	ECCM	Electronic Counter Counter Measures
AM	Amplitude Modulation	ECM	Electronic Counter Measures
AMC	Armoured Mortar Carrier	EFP	Explosivly Formed Penetrator
AMS	Armoured Mortar System	EHF	Extra High Frequency
AP	Armour - Piercing	ELF	Extremly Low Frequency
APC	Armor Personnel Carrier	Elint	Electronic Inteligence
APFSDS	Armor Piercing Fin Stabilised Discarding Sabot	ERA	Explosive Reactive Armour /Extended Range Ammunition
APGM	Autonomous Precision-Guided Munition	ERFB	Extended Range Full-Bore
API	Armour Piercing Incendiary Tracer	ERFB-BB	Extended Range Full- Bore - Base Bleed
APIT	Armour Piercing Incendiary Tracer	ESM	Electronic Support (or Surveillance) Measures
APU	Auxiliary Power Unit	EW	Electronic Warfare
APV	Armoured Patrol Vehicle	FCS	Fire Control System
ARTY	Artillery	FM	Frequency Modulation
ARV	Armored Recovery Vehicle	FMJ	Full Metal Jacket
AT	Anti Tank	FoV	Field of View
ATC	Air Traffic Control	GCS	Ground Control Station (or System)
ATG	Anti Tank Gun	GPS	Global Positioning System
ATGW	Anti Tank Guided Weapon	HAS	High Safey Ammunition
ATM	Anti Tank Mine	HC	Hollow charge
AVIO-Avionics	Aviation Electronics	HE	High Explosive
B.B.	Base Bleed	HE - Frag	High Explosive Fragmentation
Ball	Universal Military Cartridges	HEAT	High Explosive Anti Tank
Baton round	A round made from rubber to decrease lethal in riot control	HEAT	High Explose Anti Tank
Berdan Priming	A Metallic Primer the anvil of which is formed integrally with The cartridge case in the bottom of the Primer Pocket	HEAT-FS	High Explosive Anti Tank - Fin stabilised
BL	Blank / A cartrides has no projectile	HEAT-T	High Eplosive Anti Tank-Tracer
C2	Command and Control	HEAT-TP-T	High Explosive Anti Tank - Target Practice - Tracer
C3	Command, Control and Communication	HEI	High Explosive Incendiary
C3I	Command, Control and Communication and Intelligence	HERA	High Explosive Rocket - Assisted
C4	Command, Control, Communication and Computers	HESH	High Explosive Squash Head
Cal.	Diameter of gun's bore excluding the depth of the rifling grooves		

HF	High Frequency	RPG	Rocket Propelled Grenade
HOW	Howitzer	rpm	Rotation or Round per minute
HP	Hollow Point / Horse Power	RPV	Remotely Piloted Vehicle(same as UAV)
HPBT	Hollow Point Boat Tail	RWR	Radar Warning Receiver
HTTB	High Technology Test Bed (US)	RXD	Research and Developmet
HUD	Head Up Display	S / L	Sea Level
IAFV	Infantry Armour Fighting Vehicle	SAM	Surface to Air Missile
IFV	Infantry Fighting Vehicle	SAR	Search and Rescue
II	Image Intensification	Satcom	Satellite Communication
ILL	Illuminating	Self-Des / Self-Destructing	A feature of fuzes that function after a predetermined of time
Incendiary bullet	A bullet with small ammunition of flammable material	SF	Special Force
JATO	jet - Assisted Take - Off	SP	Self Propelled
Kg	Kilogram	SPG	Self Propelled Gun
LAW	Light Anti - tank Weapon	SPH	Self Propelled Howitzer
LCD	Liquid Crystal Display	Sqn	Squadron
LED	Light Emitting Diod	SRTA	Short Range Training Ammunition
IFF	Identification Friend or Foe	SSM	Surface to Surface Missile
IFR	Instrument Flight Rules	TBO	Time Between Overhauls
Impact Fuze	Any fuze that is designed to function when it strikes the target	TCS	Tactical Control System
IMU	Inertial Measurement Unit	TFT	Thin Film Transistor
INS	Inertial Navigation System	TI	Thermal Imaging
Log	Lodgestics	TIS	Thermal Imaging System
LoS	Line of Sight	TNT	Tri Nitro Toluene
IR	Infra Red	TOW	Tube - launched Optically tracked Wire - guided
LRF	Laser Range Finder	Tracer	Tactical Reconnaissance Armored Combat Equipment Requirement
MBT	Main Battle Tank	Transceiver	Radio Transmitter / Receiver
MG	Machine Gun	Transponder	Radio transmitter triggered automatically by a particular received signal
MIL - STD	Military Standard	TV	Television
MLRS	Multiple Launch Rocket System	UV	Ultra Violet
MRL	Multiple Rocket Launcher	UAV	Unmanned Aerial Vehicle (same as RPV)
MTBF	Meantime Between Failures	UHF	Ultra High Frequency
MV	Muzzle Velocity	VCR	Video Cassette Recorder
NATO	North Atlantic Treaty Organization	VFR	Visual Flight Rules
NBC	Nuclear Biological and Chemical	VHF	Very High Frequency
NDT	Non Destructive Testing	VLF	Very Low Frequency
nm	Nautical Miles	VP	Variable Pitch (propeller)
OEO	Optical and Electro Optical	WAS	Wide Area Search
OPV	Offshore Patrol Vessel	WC	Wadcutter bullet
PCM	Pulse Code Modulation	WP	White Phosphorous
PIBD	Point Initiating Base Detonating	WP- T	White Phosphorous - Tracer
RADAR	Radio Detection And Ranging	XMIT	Transmit
RAM	Random Access Memory		
RAP	Rocket Assisted Projectile		
RCS	Radar Cross- Section		
RDX	Cyclotrimethylene-Trinitramine, a widely used as secondary high explosive		
RF	Radio Frequency		

»»» CONTENTS

	1	AMMUNITION AND PYROTECHNIC	1
	2	PROPELLANT AND HIGH EXPLOSIVE	49
	3	ARMAMENT	79
	4	MILITARY VEHICLES	115
	5	ROCKET & MISSILE	139
	6	COMMUNICATION AND RADAR	171
	7	OPTICS	217
	8	N.B.C PROTECTIVE PRODUCTS	235
	9	AVIATION	251
	10	MARINE	277
	11	MISCELLANEOUS	299
	12	TRANSFER OF TECHNOLOGY	335



chapter

1

AMMUNITION AND PYROTECHNIC



PREVIEW



Ammunition Industries is one of the largest industries of Iran Defense Industries Organization which is in charge of producing a wide range of ammunition from small arms up to large caliber artillery ammunition covering both western and eastern standards.

Having more than seven decades of experience in manufacturing military products, it has met the requirement of domestic and international markets in supplying any type of conventional ammunition as well as all components there to.

This industry composing of 8 subsidiaries located in different cities of Iran is utilizing modern technology, sophisticated machineries and equipment as well as specialized personnel to manufacture qualified products in both military and non military field.

This has led industries to achieve various certificates in ISO and other international standards.

Ammunition products include:

SMALL ARMS ammunition from 7.62 mm to 12.7 mm in different types of **Ball**, **AP** and Tracer as well as shotgun and rifle cartridges in various gauges and caliber.

MEDIUM CALIBER ammunition consisting of 23 mm, 30 mm, and 35 mm each in different types of **HE**, **HEI**, **HEIT** and **AP** for aircraft and anti aircraft gun. The 30mm is also in two categories; one for BMP2 APCs and the other for NAVAL purposes.

MORTAR BOMBS in various caliber of 60mm, 81mm and 120mm in different types of **HE**, **SMOKE**, **ILLUMINATING** and **RA**(Rocket Assisted for 120mm). These Mortar Bombs are also available with LONG RANGES.

TANK & ANTI-TANK ammunition from 105 mm to 125mm and in different types of **HE** and **APFSDS**.

FIELD ARTILLERY ammunition from 105mm to 203mm and in different types of **HE**, **SMOKE**, **ILLUMINATING** and **BASE BLEED**.





CHAPTER 1



ITEM	TABLE OF CONTENT	PAGE
1	LAW ENFORCEMENT CARTRIDGES	4
2	7.62 MILITARY CARTRIDGES	5
3	0.38 S&W CARTRIDGES	5
4	12.7×108 MM CARTRIDGES	6
5	ANTI RIOT CARTRIDGES	7
6	SHOTGUN & SPORTING CARTRIDGES	7
7	20×102 MM (HEI) - AIRCRAFT AMMUNITION	8
8	20×128 MM (TP-T) ANTI – AIRCRAFT	8
9	23×152MM - ANTI AIRCRAFT AMMUNITION	9
10	30 MM NAVAL - ANTI AIRCRAFT AMMUNITION	10
11	30×165 MM - ANTI AIRCRAFT AMMUNITION	11
12	30MM AIRCRAFT AMMUNITION	12
13	35MM ANTI AIRCRAFT AMMUNITION	13
14	40MM NAVAL AMMUNITION	14
15	60 MM MORTAR BOMB	15
16	81 MM MORTAR BOMB	16
17	81 MM MORTAR BOMB ILLUMINATING – M301A2	17
18	120 MM MORTAR BOMB (HE, SMOKE & ILLUMINATING)	18
19	120 MM MORTAR BOMB - ROCKET ASSISTED HE	19
20	120 MM MORTAR BOMB (HE/LR)	20
21	76/62 MM NAVAL AMMUNITION	21
22	100 MM ANTI – AIRCRAFT AMMUNITION	21
23	105 MM TANK AMMUNITION	22
24	106 MM HE & HEAT AMMUNITION	23
25	120 & 122 MM TANK AMMUNITION	24
26	125 MM HE & HEAT	25
27	125 MM APFSDS – T (BM 22 & 42)	26
28	105 MM HOW - HIGH EXPLOSIVE & ILLUMINATING	27
29	105MM HOW (SMOKE, WP & HC)	28
30	122 MM HOW - HIGH EXPLOSIVE	29
31	122 MM HOW - SMOKE & HC	30
32	122 MM HOW - ILLUMINATING	31
33	130 MM HE & BB	32
34	130 MM SMOKE & ILLUMINATING	33
35	155 MM HOW (HE & ERFB – BB)	34
36	155 MM SMOKE & ILLUMINATING	35
37	203 MM (8IN) HOW - HE, M106	36
38	40 MM HE & TRAINING GRENADE	37
39	HAND GRENADES	38
40	RIFLE GRENADE	39
41	MORTAR BOMB FUZES	40
42	AMMUNITION FUZES	41
43	AMMUNITION FUZES	42
44	EITF FUZE	43
45	F61 FUZE	43
46	PRIMERS & DETONATORS	44
47	PLAIN & SHOCK TUBE DETONATORS	45
48	SMOKE POT & GRENADE	46
49	IRRITANT & COLORED SMOKE GRENADE	47



LAW ENFORCEMENT CARTRIDGES



PIC: 000199

DESCRIPTION

The cartridges required for the Law Enforcement purposes are other products of Iran which include 9mm, 0.38", 0.32", 0.40" and 0.45" calibres as well as 7.65 Browning.

These cartridges are intended for use with Pistol, Revolver, and submachine-gun and are vastly used by Police and Law Enforcement forces all over the world.



SPECIFICATIONS

TYPE	Ballistic Data		Technical Data				
	Muzzle Velocity (m/s)	Max. Pressure (kp/cm ²)	Bullet Weight (g)	Round Length (mm)	Case Length (mm)	Propellant Powder	Primer
9 × 18 Makarov:	318	2,000	6.08	25	18	Single Base	Boxer
9 × 19 Luger Ball:	385	2,000	8.03	29.7	19	Single Base	Boxer
9 × 19 Luger Blank:	-	200	-	29	29	Single Base	Boxer
0.45 Auto:	300	1,400	14.9	31.88	22.88	Double Base	Boxer
0.40 S. & W.:	320	2,400	12.3	27.56	21.59	Double Base	Boxer

PACKING

TYPE	Rounds per Carton	Cartons per PVC Vaccumed Bag	PVC Bags per Box	Box Weight (kg)	Box per Pallet
9 × 18, 19:	50	10	2	16	50
0.45, 0.40:	50	10	2	25	50



7.62 MM MILITARY CARTRIDGES



PIC: 000101



SPECIFICATIONS

TYPE	7.62x39	7.62x39	7.62x39	7.62x51	7.62x51	7.62x51	7.62x54	7.62x54
Model:	Tracer	FMJ(Ball)	AP	Tracer	FMJ (Ball)	AP	FMJ (Ball)	AP
Type of Core:	Lead – Tracing Material	Lead	Steel	Lead – Tracing Material	Lead	Steel	Lead	Steel
Type of Gun:	KL	KL	KL	G3 – MG3	G3	G3	Grinov (PKM)	Grinov (PKM)
Muzzle Velocity (m/s):	700 - 730	695 - 725	720 - 760	820 - 860	823 - 853	820 - 860	770 - 800	780 - 820
Max. Gas Pressure (kgf/cm ²):	P ≤ 3100	P ≤ 3100	P ≤ 3100	P ≤ 3448	P ≤ 3448	P ≤ 3448	P ≤ 3400	P ≤ 3400
Accuracy at 100 m (cm):	D ≤ 12.7	a+b ≤ 18.5	a+ b ≤ 15	a+b ≤ 20	a+b ≤ 20	a+b ≤ 20	a+b ≤ 18	a+b ≤ 20
Tracing in 800 m from Muzzle:	≤ 85%	-	-	≥ 80%	-	-	-	-
Penetration in Steel Plate (st 37):	-	-	≥ 80% (Thickness 15 mm at 15m)	-	-	≥ 80% (Thickness 20 mm at 15m)	-	≥ 80% (Thickness 25 mm at 15m)
Quantity per Lot:	400,000	400,000	400,000	500,000	500,000	500,000	200,000	200,000
Packing:	Standard	Standard	Standard	Standard	Standard	Standard	Standard	Standard

0.38 S & W CARTRIDGES

SPECIFICATIONS

TYPE	FLAT NOSE	FMJ
Core:	Lead	Lead
Gun:	0.38 S & W	0.38 S & W
Muzzle Velocity (m/s):	228 - 258	265 - 295
Max. Gas Pressure (kgf/cm ²):	P ≤ 1200	P ≤ 1200
Accuracy at 25 m (cm):	D ≤ 12	D ≤ 9
Quantity per Lot:	30,000	30,000
Packing:	Standard	Standard





12.7×108 MM CARTRIDGES



PIC: 000102



SPECIFICATIONS

TYPE	TRACER	AP	BALL
Core Lead:	Lead – Tracing Material	Steel – Lead	Steel – Lead
Gun:	DSHK	DSHK	DSHK
Muzzle Velocity (m/s):	810 - 840	810 - 840	810 - 840
Max. Gas Pressure (kgf/cm ²):	P ≤ 3300	P ≤ 3300	P ≤ 3300
Accuracy at 100 m (cm):	a+b ≤ 25	a+b ≤ 25	a+b ≤ 20
Tracing in 800 m from Muzzle:	≥80%	-	-
Quantity per Lot:	100,000	100,000	100,000
Packing:	Standard	Standard	Standard





ANTI RIOT CARTRIDGES



PIC: 000103

SPECIFICATIONS

TYPE	GAUGE 12 (RUBBER)
Wads:	Plastic
Case Length (mm):	70
Base Height (mm):	12.9
Shot Weight (gr):	5.6
Shot Number:	14
Muzzle Velocity (m/s):	370 - 410
Max. Gas Pressure (bar):	P < 650
Shot Material:	Rubber
Packing (pcs/box):	25



SHOTGUN & SPORTING CARTRIDGES

PIC: 000104



SPECIFICATIONS

■ SHOTGUN CARTRIDGE GAUGE 12

TYPE	STAR	FLAT	HIGH PRESSURE	HIGH CHARGE	TRAP	SKEET
Wads:	Plastic	Plastic	Plastic	Plastic	Plastic	Plastic
Case Length (mm):	70	70	70	70	70	70
Base Height (mm):	12.9	12.9	12.9	12.9	12.9	12.9
Shotshells (gr):	34	32	37	37	24	24
Shotshell Numbers:	00-8	00-8	5-6	5-6	7	7
Muzzle Velocity (m/s):	385	385	370	370	385 - 415	385 - 415
Max. Gas Pressure (bar):	P ≤ 650	P ≤ 650	P ≤ 845	P ≤ 745	P ≤ 650	P ≤ 650
Packig (pcs/box):	25	25	25	25	25	25

■ Above cartridge is also available in gauge 16 and flat type with shotshells weight of 28 gr.



20 × 102 MM (HEI) AIRCRAFT AMMUNITION



PIC: 000105

DESCRIPTION

This ammunition can be fired by **Vulcan** guns mounted on helicopter.
The rimless bottle necked case is of steel and electrically primed.

SPECIFICATIONS

TYPE	HEI (20×102mm)
Length of Shell (mm):	168
Weight of Shell (gr):	266
Type of Primer:	Electric
Muzzle Velocity (m/s):	1030
Armament:	VULCAN



20 × 128 MM (TP-T) ANTI – AIRCRAFT

PID: 000106

DESCRIPTION

This ammunition is produced for training purposes and is fired by 20 mm anti – aircraft cannons.

The shell body is same as **HEIT** type with front section empty and rear section filled with tracer.

SPECIFICATIONS

TYPE	20 ×128 MM(TP-T)
Length of Shell (mm):	203.5
Weight of Shell (gr):	340
Length of Projectile (mm):	92.3
Weight of Projectile (gr):	125
Type of Primer:	Percussion
Muzzle Velocity (m/s):	1050
Armament:	OERLIKON





23×152MM ANTI AIRCRAFT AMMUNITION



PIC: 000107

DESCRIPTION

» 23×152 MM (APIT)

This round forms part of 23mm series fired by **ZU-23** air defense gun.

The projectile is a solid steel body with steel ballistic cap and tracer assembly in the base. The tracer burns for at least 5 seconds.



APIT

» 23×152 MM (HEI)

This round is another type of 23 mm Anti Aircraft series fired By **ZU-23** air defense gun.

The projectile consists of a steel shell filled with explosive material and ignited by a **PD MG-25** nose fuze and the round is equipped with self destruction with duration of 5-11 seconds.



HEI

» 23×152 MM (HEIT)

This round is the same as **HEI** type being fired by **ZU-23** air defense gun but with tracer assembly burning for at least 5 seconds.



HEIT

SPECIFICATIONS

TYPE	APIT	HEI	HEIT
Tracer Time (s):	Min. 5	-	Min. 5
Self Destruction Duration(s):	-	5-11	5-11
Length of Shell (mm):	235	237	237
Weight of Shell(gr):	450	450	450
Length of Projectile (mm):	99	108.8	108.8
Weight of Projectile (gr):	190	180	185
Type of Primer:	Percussion	Percussion	Percussion
Muzzle Velocity (m/s):	960	960	960
Armament:	ZU-23	ZU-23	ZU-23





30 MM NAVAL ANTI AIRCRAFT AMMUNITION



PIC: 000108

DESCRIPTION

» 30×210 MM (HEI)

This ammunition is produced to be used with 30 mm **AK630** naval gun intended for the short range defense of warships or as the main armament of small vessels.

The **HEI** type of this ammunition is equipped with self destruction device with duration of 12 to 28 seconds and primed electrically.



» 30×210 MM (TRACER)

This ammunition is produced to be used with 30mm type **69** or **AK630** naval gun intended for tracing purposes.

The tracer of this ammunition burns for at least 6 seconds.



» 30×210 MM (TP)

The Target Practice type of 30 mm anti aircraft naval ammunition (**TP**) is produced to be used with the same gun as **HEI** and Tracer types and intended for training purposes.

The round ballistically matches the **HEI** type.



SPECIFICATIONS

TYPE	HEI	TRACER	TP
Self Destruction Duration (s):	12-28	-	-
Length of Shell (mm):	307	307	307
Weight of Shell (gr):	1085	1085	1085
Length of Projectile (mm):	128.7	120	128.75
Weight of Projectile (gr):	356	356	356
Type of Primer:	Electric	Electric	Electric
Muzzle Velocity (m/s):	1035	1035	1035
Armament:	AK630	AK630	AK630





30×165 MM ANTI AIRCRAFT AMMUNITION



PIC: 000109

DESCRIPTION

» 30 × 165 MM (HEIT)

This is the **HEIT** type of 30mm Anti Aircraft Ammunition which is fired by dual feed cannon **2A42** and is the same as **HEI** type but with tracer assembly burning for at least 6 seconds.

» 30 × 165 MM (HEI)

This is the **HEI** type of 30mm Anti Aircraft Ammunition which is fired by dual feed cannon **2A42** fitted to **BMP-2** and intended to be used against lightly armored targets as well as low flying aerial targets such as helicopters. It is equipped with a **PD** fuze having a self-destructed device functioning between 6 and 14 seconds.

» 30 × 165 MM (APT)

This is the **APT** type of 30mm Anti Aircraft Ammunition which is fired by dual feed cannon **2A42** fitted to **BMP-2** and intended to be used against lightly armored targets as well as low flying aerial targets such as helicopters.



SPECIFICATIONS

TYPE	HEIT	HEI	APT
Tracer Time (s):	Min. 6	-	Min. 3.5
Self Destruction Duration (s):	6 - 14	6 -14	-
Length of Shell (mm):	292	291	292
Weight of Shell (gr):	835	845	860
Length of Projectile (mm):	147.5	151	148.5
Weight of Projectile (gr):	387	394	405
Type of Primer:	Percussion	Percussion	Percussion
Muzzle Velocity (m/s):	960	960	960
Armament:	2A42 - 2A38 - 2A72	2A42 - 2A38 - 2A72	2A42 - 2A38 - 2A72





30MM AIRCRAFT AMMUNITION



PIC: 000110

DESCRIPTION

» 30×165 MM(HEI)

This round is one of 30mm Aircraft cannon ammunition range being fired by **GSH-301** aircraft gun mounting on **MIG-29** fighter. It has a nose-fuzed (**PD**) streamlined steel-walled projectile containing explosive payload fixed with brass case primed electrically.

HEI



» 30×165 MM (TP)

This round is the Target Practice type of 30mm Aircraft cannon ammunition range being fired by **GSH-301** mounted on fighter **MIG-29** aircraft and intended to be used for Training Purposes.

TP



» 30×155 MM (TP)

This round is just for Training Purposes and fired by air to ground gun type 1 being mounted on aircraft **F7** fighter.

TP



SPECIFICATIONS

TYPE	HEI (30×165mm)	TP (30×165mm)	TP (30×155mm)
Length of Shell (mm):	283	283	268
Weight of Shell (gr):	835	835	840
Length of Projectile (mm):	143.5	143.5	187.5
Weight of Projectile (gr):	385	385	97
Type of Primer:	Electric	Electric	Percussion
Muzzle Velocity (m/s):	870	870	790
Armament:	GSH-301, GSG30K, GSH-6-30, GSH-30	GSH-301, GSG30K, GSH-6-30, GSH-30	30mm gun





35MM ANTI AIRCRAFT AMMUNITION



PIC: 000111

DESCRIPTION

The **35mm** Anti Aircraft ammunition is produced to operate against enemy's air attacks and to defend the air territory of cities and important military and industrial places.

It is fired from **Oerlikon** series cannon and provided in **3** types as **HEI**, **HEIT** and **TPT**.

These types are all ballistically the same but different in purpose of use.

The target practice type (**TPT**) of 35mm Anti Aircraft is produced to be used for training purposes.
The tracer duration of this type is 7 seconds.



SPECIFICATIONS

TYPE	HEI	HEIT	TPT
Round Length (mm):	387	387	385
Round Weight (g):	1545	1565	1580
Projectile Weight (g):	558	550	480
Case Length (mm):	227	227	227
Max. Chamber Pressure (bar):	3900	3900	3600
Muzzle Velocity (m/s):	1175	1175	1175
Self Destruction Time (sec):	6-16	6-16	-
Max. Range (m):	11000	11000	11000
Trace Duration (sec):	-	5	7



PACKING

Round in Wooden Box:	25
Wooden Box Weight (kg):	58
Wooden Box Dim. (mm):	520 x 420 x 370
Wooden Box per Pallet:	12
Pallet Weight (kg):	710



40MM NAVAL AMMUNITION



PIC: 000112

DESCRIPTION

The **40 mm L/70** Prefragmented High Explosive is a fixed round anti aircraft ammunition with a brass cartridge case crimped rigidly to its projectile. The projectile has a shell body manufactured from special high-fragmentation steel and a copper drive band encircles the projectile just above the cartridge case/projectile junction.

The Target Practice Purposes (**TPT**) is simulate the ballistic properties of service projectile for training in marksman ship. The projectile, which is loaded with an inert material, is fuze with either a dummy or an inert **PD** fuze.

The **HE-T** is High Explosive including Tracer.



SPECIFICATIONS

TYPE	PFHE	TPT	HET
Total Weight of Round (g):	2450	2500	2450
Total Weight of Projectile (g):	895	-	895
Weight of Propellant (g):	480	480	480
Weight of Explosive (g):	116	-	114
Type of Explosive:	Hegzal	NC1066	Hegzal
Type of Primer:	M1	M39sx	M1
Length of Round (mm):	534	534	Max. 535
Length of Projectile (mm):	365	-	1025
Muzzle Velocity (m/s):	1025	1005	FZ104M12
Type of Fuze:	PF-HE40	Dummy	-
Type of Primer:	M 39 SX	M39sx	-
Safety Time Fuze (sec):	8.5	M39sx	-
Firing Rate (round /min):	300	300	-
Max. Range (km):	11	11	-





60 MM MORTAR BOMB



PIC: 000113

DESCRIPTION

The high flexibility and easy handling on one hand and the accuracy of operation on the other, have made the Mortars to be considered as the most worldwide suitable weapons in infantry fighting particularly in hilly or jungle terrains fields.

The **60 mm Mortar** shell is the smallest mortar bomb produced by Iran Defence Industries and characterized by the special process of production and high quality material for the body, resulting in more effective blast and fragments and higher lethality for the ammunition.

The **smoke** type of this mortar bomb intended to be used for concealing the combatants and materials from enemy observation or for signaling purposes. They use White Phosphorus or Red Phosphorus as the smoke generating agent.

Long range of 60 mm mortar bomb is improvement type of that. The improvement is not only in the range but also in their performance with more volume of explosive material. The new types have more explosive power and as a result causing much better fragments than the previous types, resulting in a better performance. All these improvements are made without any changes to the mortar launchers.



SPECIFICATIONS

TYPE	HIGH EXPLOSIVE	SMOKE	HIGH EXPLOSIVE (LONG RANGE)
Lenght, Fused (mm):	265	242	305
Weight, Fused (kg):	1.270	1.801	1.635
Caliber (mm):	60	60	60
Color:	Olive green	Green	Olive green
Filler Material (gr):	170 TNT	150-200 WP	210 TNT
Charge:	1red + 2 white	M8 & M9 powder type II	1red + 3 white
Body:	Forged steel	Forget steel	Forged steel
Tail:	Extruded Aluminum	Extruded Aluminum	Extruded Aluminum
Fuze:	AZ111-A2	AZ111-A2	AZ111-A2
Primer:	M35	M35	M35
Muzzle Velocity (m/s):	190	158	230
Max. Chamber Pressure (bar):	350	350	450
Max. Range (m):	2300	1812	3200
Armament:	60 mm mortars	All 60 mm smoothbore mortars	All 60 mm smoothbore mortars

PACKING

Quantity per Box:	10
Box Weight (kg):	22.5
Box Dim. (mm):	445 × 330 × 220
Box per Pallet:	45, (30)
Pallet Dim. (mm):	1380 × 1065 × 1300





81 MM MORTAR BOMB



PIC: 000118

DESCRIPTION

The high flexibility and easy handling on one hand and the accuracy of operation on the other, have made the mortars to be considered as the most worldwide suitable weapons in infantry fighting particularly in hilly or jungle terrains fields.

Special process of production and high quality material for the body, resulting in more effective blast and fragments and higher lethality for the ammunition.

81 mm Smoke mortar bomb is one of the smoke generating ammunition intended to be used for concealing the combatants and materials from enemy observation or for signaling purposes. They use White Phosphorus or Red Phosphorus as the smoke generating agent.

The improvement of this type of mortar shell achieved **Long Range** one. The improvement is not only in the range but also in their performance with more volume of explosive material. The new types have more explosive power and respectively much better fragments than the previous types, resulting in a better performance. All these improvements are made without any changes to the mortar launchers.



SPECIFICATIONS

TYPE	HIGH EXPLOSIVE	SMOKE	HIGH EXPLOSIVE (L.R)
Length, Fused (mm):	370	324	522
Weight, Fused (kg):	3.990	3.100	4525
Caliber (mm):	81	81	81
Color:	Olive green/grey	Green	Olive green/grey
Filler Material (g):	540 TNT	725 Wp	900 TNT
Charge:	6 white	M8 & M9 Powder type II	6 pcs white
Body:	Forged steel	Forged steel	Forged steel
Tail:	Extruded Aluminum	Extruded Aluminum	Extruded Aluminum
Fuze:	AZ111-A2	AZ111-A2	AZ111-A2
Primer:	M35	M71A2	M 35
Muzzle Velocity (m/s):	295	264	320
Max. Chamber Pressure (bar):	790	674	790
Max. Range (m):	5300	5250	6350
Armament:	All 81 mm smoothbore mortars	All 81 mm smoothbore mortars	All 81 mm smoothbore mortars

PACKING

TYPE	HE & SMOKE	LONG RANGE (HE)
Quantity per Box:	4	4
Box Weight (kg):	24	28
Box Dim.(mm):	480 × 255 × 265	675 × 255 × 280
Box per Pallet:	32	32
Pallet Dim. (mm):	970 × 1080 × 1200	1250 × 1080 × 1200





81 MM MORTAR BOMB ILLUMINATING – M301A2



PIC: 000119

DESCRIPTION

This Mortar Shell is intended for illuminating a desired point or area under the conditions of poor visibility.

The complete round consists of a time fuze, thin-walled steel body-tube assembly, parachute assembly steel tail cone, fin assembly containing propellant increment charges, ignition cartridge and percussion primer.

The illuminant assembly consists of a first-fire charge and an illuminant contained in a box board case. The parachute assembly is attached to the illuminant assembly casing by a steel suspension wire.



SPECIFICATIONS

TYPE	ILLUMINATING, M301A2
Purpose of Use:	Illuminating in Night Missions
Length, Fuzed (mm):	571
Weight, Fuzed (kg):	4.5
Body:	Forged Steel
Illuminating Material (g):	621
Illuminating Intensity (cd):	Min. 400,000
Burning Time (sec.min):	60
Rate of Descent (m/s):	5-6
Charge (s):	M8 & M9 Powders type II
Charge Weight (g):	52
Fuze:	M84A1E1 or Equivalent
Primer:	M34
Muzzle Velocity (m/s):	181
Max. Range (m):	2,150
Max. Chamber Pressure (kp/cm ²):	674
Armament:	81 mm Mortar



PACKING

Quantity per Box (rds):	3
Box Weight (kg):	24
Box Dimensions (mm):	700 × 340 × 118
Box per Pallet:	42
Pallet Weight (kg):	1035
Pallet Dimensions (mm):	1430 × 1060 × 1300



120 MM MORTAR BOMB HE, SMOKE & ILLUMINATING



PIC: 000120

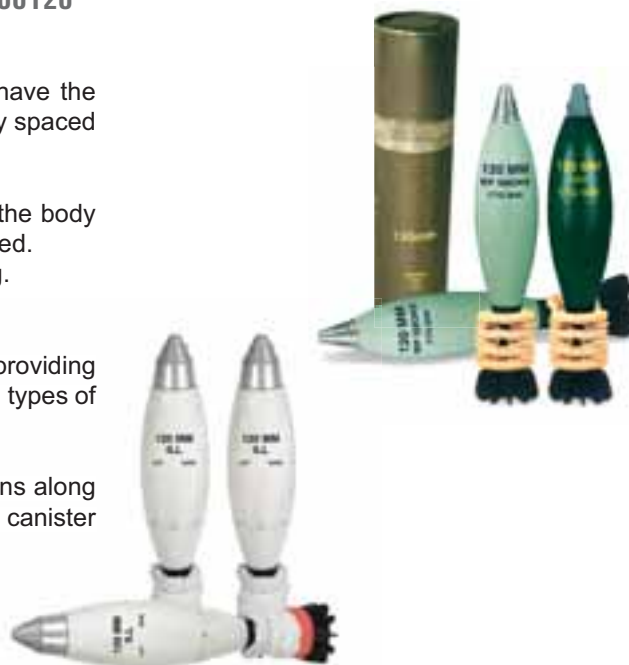
DESCRIPTION

Both **HE** and **SMOKE** types of **120 mm** Mortar Bombs have the symmetrical steel body, with five gas check rings irregularly spaced around the waist.

The nose is threaded for an impact fuze (**AZ111A2**) and the body ends in a screwed spigot to which a steel tail unit is attached. The only difference between two types is the internal filling.

The **120 mm illuminating** Mortar Shell is intended for providing intense illumination for identification and engagement of all types of stationary and moving objects.

The complete round consists of a time fuze, which functions along the trajectory at the pre-set time enabling illuminant canister supported by parachute to eject and deploy in the air.



SPECIFICATIONS

TYPE	HIGH EXPLOSIVE	SMOKE	ILLUMINATING
Length, Fused (mm):	580	581	591
Weight, Fused (kg):	12.60	13	12.10
Caliber (mm):	120	120	120
Color:	Olive green	Green	White
Filler Material (g):	2300 TNT	1.8 - 2.1 WP	-
Charge (s):	1red+8 white	M8 & M9 Powder	M8 & M9 Powder
Body:	Forged steel	Forged steel C60	-
Tail:	Extruded Aluminum	Extruded Aluminum	-
Fuze:	AZ111-A2	AZ111-A2	M84A1E1 or Equivalent
Primer:	M34	M134B1	M134B1
Muzzle Velocity (m/s):	310	301	232
Max. Chamber Pressure (bar):	900	900	900
Max. Range (m):	6,600	6,000	3,850
Armament:	All 120 mm smoothbore mortars	All 120 mm smoothbore mortars	All 120 mm smoothbore mortars

PACKING

TYPE	HE & SMOKE	ILLUMINATING
Quantity per Box:	2	2
Box Weight (kg):	35	35
Box Dim. (mm):	670 × 355 × 210	720 × 330 × 180
Box per Pallet:	30(15)	30
Pallet Dim. (mm):	1050 × 1370 × 1200	1020 × 1480 × 1140





120 MM MORTAR BOMB ROCKET ASSISTED - HE



PIC: 000121

DESCRIPTION

This is an enhanced range of **Mortar Bomb** to be used against light material and personnel, functioning with both fragmentation and blast effect.

The complete round consists of a symmetrical forged C60 steel projectile body, internal solid-fuel rocket motor and **HE** filling. There are 6 gas check rings spaced around the waist.

The rear taper is considerably thicker than normal and the tail fins are close to the body. The central cartridge container extends behind the fins and the horse-shoe shape increments are fitted around it.

The filler is 2.15 kg of **Composition B** and the Bomb is drop loaded and fired in the usual manner. Once the Bomb is fired and enroute to its target, the rocket motor initiates during the final few seconds of the flight, adding approximately 4 km range over conventional bombs.

SPECIFICATIONS

TYPE	120 MM HE ROCKET ASSISTED
Length, Fused (mm):	740
Weight, Fused (kg):	16.85
Caliber (mm):	120
Color:	Olive Green
Filler Material (g):	2,100 Composition B
Charger (s):	1 red + 8 White
Body:	Forged Steel
Tail:	Extruded Aluminum
Fuze:	AZ111-A2
Primer:	M34
Muzzle Velocity (m/s):	265
Max. Chamber Pressure (bar):	1,000
Max. Range (m):	9,800
Armament:	All 120 mm Smoothbore Mortars

PACKING

Quantity per Box (rds):	2
Box Weight (kg):	50
Box Dimensions (mm):	1080 × 330 × 200
Box per Pallet:	15
Pallet Weight (kg):	765
Pallet Dimensions (mm):	1100 × 1000 × 1200





120 MM MORTAR BOMB /HE/LR



PIC: 000122

DESCRIPTION

Iran Defence Industries has improved its type of Mortar Shells to achieve **Long Range** for 120 mm.

The improvement is not only in the range but also in their performance with more volume of explosive material.

The new types have more explosion power and as a result, much better fragments than the previous types, resulting in a better performance.

All these improvements are made without any changes to the mortar launchers.



SPECIFICATIONS

TYPE	120MM HE (LONG RANGE)
Length, Fused (mm):	665
Weight Fused (g):	13,650
Caliber (mm):	120
Color:	Olive Green
Filler Material (g):	2,100 TNT
Charge (s):	1 red + 8 White
Body:	Forged Steel
Tail:	Extruded Aluminum
Fuze:	AZ111-A2
Primer:	M34
Muzzle Velocity (m/s):	320
Max. Chamber pressure (bar):	900
Max. Range (m):	7,200
Armament:	All 120 mm Smoothbore Mortars

PACKING

TYPE	120 MM HE-LR
Quantity per Box:	2
Box Weight (kg):	50
Box Dim. (mm):	1080 × 330 × 200
Box per Pallet:	15
Pallet Dim. (mm):	1100 × 1000 × 1200





76/62 MM NAVAL AMMUNITION



PIC: 000114

DESCRIPTION

76/62 mm naval ammunition is a high explosive round intended for firing against a wide range of targets either surface or air ones. By application of a point detonating (PD) fuze, it is used against surface targets and by applying a radio proximity fuze, it is used against air targets.

SPECIFICATIONS

TYPE	76/62 MM
Max. Range (surface)(m):	16,000
Effective Range (vertical)(m):	7,000
Muzzle Velocity (m/s):	900
Chamber Pressure (bar):	3,500
Type of Nose Fuze:	Proximity Fuze / PD
Weight of Explosive (gr):	455
Type of Explosive:	Comp. A3
Armament:	76/62 OTOMELARA



100 MM ANTI AIRCRAFT AMMUNITION

PID: 000115

DESCRIPTION

100 mm Anti-aircraft ammunition manufactured to counter aerial threats posed by fighters, helicopters. This ammunition is equipped with electronic time fuze enabling it to explode and terminate the target from a close distance even if it fails to hit it directly. The anti-aircraft gun is a system that can create a missile shield against enemies aircrafts in medium and low altitudes.

SPECIFICATIONS

TYPE	100 MM
Total Length, Fuzed (mm):	1,110
Total Projectile Weight (gr):	30,360
Shell Dia. (mm):	99.8
Type and Weight of Filler Material (gr):	1,840 TNT
Propellant Charge Weight (gr):	5,600
Fuze:	WC 100
Shell Material:	C60
Max. Muzzle Velocity (m/s):	900
Max. Chamber Pressure (mp):	300
Max. Range (vertical) (m):	11,000 to 12,000
Max. Range (m):	21,000
Effective Range (m):	18,000





105 MM TANK AMMUNITION



PIC: 000123

DESCRIPTION

» 105MM HEAT (M456A1)

This is a fixed round high explosive anti tank ammunition intended for use against armored targets. The ammunition consists of a steel body projectile and a brass case containing the propellant charges. The body contains a funnel shaped copper liner, a **PIBD** fuze at the bottom, a nose cap retaining piezoelectric element and is loaded with composition B. When the projectile is detonated on impact by fuze functioning, the cone collapses, creating a high velocity shock wave and a jet of metal particles which penetrate the target.

» 105MM HE-TK (M60)

The product is a High Explosive ammunition with the projectile secured to the brass cartridge case and is fired with **M60** series tank guns. Having a muzzle velocity of 864 m/s the projectile reaches a max. range of 17,000 m in the field being effective against enemy's personnel and materials by producing both fragmentation and blasts.

» 105MM APFSDS-T

Anti tank 105 mm **APFSDS – T** ammo is one of the **T – 55** or **M60** Tank ammunitions intended to be used against tanks and armored vehicles. This ammo is equipped with a long rod **KE** penetration. High muzzle velocity and high density of its penetrator, create great amount of impact kinetic energy to penetrate in **RHA** armours.

SPECIFICATIONS

TYPE	M456A1 (HEAT)	M60 (HE –TK)	APFSDS-T
Total Length, Fuzed (mm):	997	1,040	-
Total Weight, Fuzed (kg):	22	27	~19
Filler Material (kg):	1.1 Composition B	1.65 TNT	-
Propellant Charge:	M30 seven perforated type, 1-5.18 kgs	Triple base M26 powder - 4 bags 1 kg	5.3
Cartridge Case Weight (kg):	W/O Chargers 6.1 With Chargers 11.18	-	-
Fuze:	PIBD M509A1 or M509A2	PDM 557 or PDM 572	-
Primer:	Electric M83/percussion	Electric M63	-
Muzzle Velocity (m/s):	1174	864	-
Max. Range (m):	7,000	17,000	Effective 2,000
Max. Chamber Pressure (kp/cm ²)	3,300	3,450	4,110
Penetration:	356 mm RHA plate at 90°	-	460 mm (in RHA target)
Armament:	Tank M60A1 Gun 105 mm	M60A1 Tank Gun	M60 or T55



HEAT



HE-TK



APFSDS-T

PACKING

TYPE	M456A2 (HEAT)	M60 (HE-TK)
Quantity per Box (rds):	2	2
Box Weight (kg):	65	76
Box Dimensions (mm):	1120 × 375 × 240	1200 × 380 × 220
Box per Pallet:	12	12
Pallet Weight (kg):	795	925
Pallet Dimensions (mm):	1220 × 1110 × 1050	1200 × 1140 × 960





106 MM HE & HEAT AMMUNITION



PIC: 000124

DESCRIPTION

This product is provided in two types as **HE** and **HEAT**. The **HE** type is applied against enemy's personnel while the **HEAT** type against tanks and armored vehicles. Both are fired with Recoilless Rifles **M40** or **M40A1**.

The base of the projectile contains the **PIBD M509A1** fuze and the fin assembly. The cartridge case has a steel perforated body and contains the propellant charge and fitted with a percussion primer at the base.



SPECIFICATIONS

TYPE	HE	HEAT
Total Length Fuzed (mm):	999	999
Total Weight, Fuzed (kg):	16.35	16.25
Filler Material:	700 g A4	1,200 gr Comp. B
Cartridge Case Length (mm):	609.6	609.6
Propellant Charge (kg):	M26 type 1 seven perforated 3.8	M26 type 1 Seven Perforated 3.8
Cartridge Case Weight (kg):	4.3 (W/O Charges) 7.8 (With Charges)	4.3 (W/O Charges) 7.8 (With Charges)
Fuze:	PDM121A	PIBD M509A1 or M509A2
Primer:	M57	M57
Muzzle Velocity (m/s):	530	503
Max. Range (m):	6,000	6,000
Max. Chamber pressure (kp/cm ²):	700	702
Penetration:	-	152 mm RHA Plate at 60°
Armament:	M40 or M40A1 Recoilless Rifle	M40 or M40A1 Recoilless Rifle

PACKING

TYPE	HE & HEAT
Quantity per Box (rds):	2
Box Weight (kg):	52
Box Dim. (mm):	1125 × 340 × 225
Box per Pallet:	15
Pallet Weight (kg):	795





120 & 122 MM TANK AMMUNITION

PIC: 000125

DESCRIPTION

» 120 MM HESH

This product is a general purpose round with a good anti-armour performance. It consists of a thin-walled projectile which collapses when striking a target allowing the explosive to be spread over the surface of the target before the base fuze function.

The resultant detonation creates shockwaves which force off a large high velocity scab of material from the inner surface of the targeted armour.

» 122 MM HEAT HOW

In order to have ammunition with **HEAT** property but capable of firing from Howitzer Guns, Iran Defence Industry has provided its **HOW HEAT** types in 122 mm caliber. They can be fired in different types for **D30** , **T-83** and **T-54** Guns or similar.

PIC: 000126



SPECIFICATIONS

TYPE	120 MM HESH	122 MM T83	122 MM T54
Weight of Complete Round (kg):	17.1	27	25
Length of Complete Round (mm):	510	700	590
Length of Cartridge Case (mm):	-	394	284
Max. Range (m):	2530	8,000	4,000
Max. Chamber Pressure (kp/cm ²):	-	2,600	2,350
Muzzle Velocity (m/s):	670	618	450
Penetration:	-	220/60°	220/60°
Fuze:	L56	V-15 PIBD	V-15 PIBD
Primer :	-	MC 22/30	MC 22/30
Propellant Charge:	NQIS 27-09	2.7 kg NC + CBI	2kg NC + CBI
Armament :	L30A1 (CHARM) Tank Gun	T83 or Similar	T54 or Similar

PACKING

TYPE	HESH
Quantity per Box:	2
Box Weight (kg):	64
Box Dim. (mm):	600 × 400 × 200
Box per Pallet:	15
Pallet Dim. (mm):	610 × 410 × 220





125 MM HE & HEAT



PIC: 000127

DESCRIPTION

» 125 MM HE (FS,30F22)

The product is fin stabilized and separate loading ammunition which is fired with **2A46M** main gun fitted to **T-72 MBT** and **T-80 UD**. It is automatically loaded into the breech followed by a semi – combustible propellant case.

Having a muzzle velocity of 850 m/s the projectile reaches a max. range of 9,200 m in the field being effective against enemy's personnel and materials by producing both fragmentation and blasts.

» 125MM HEAT (FS,3BK18M)

This is a separate loading ammunition which is intended for use against armoured target and fired with **2A46M** main gun fitted to **T-72 MBT**. It is automatically loaded into the breech followed by a semi – combustible cartridge case.

The projectile has a steel body encircled by two driving bands and its front, carries a standoff spike inside which is a piezoelectric contact sensor connected by an internal wire to a base – mounted fuze. This ammunition can penetrate 550 mm of Rolled Homogenous Armor (**RHA**) of normal angle or 273 mm at an angle of 60 deg. to the normal.



SPECIFICATIONS

TYPE	HE,FS,30F22	HE, ANTI TANK, FS,3BK18M
Total Length, Fuzed (mm):	677	677
Total Weight, Fuzed (kg):	23	19.34
Filler Material (kg):	3.4 TNT	1.637 Composition A4
Propellant Charge (kg):	* 52 - 5	* 52 - 5
Cartridge Case Weight (kg):	3.4 (W/O Charges) 10 (With Charges)	408
Fuze:	V429E	B-15
Primer:	RYB – 7	RYB – 7
Muzzle Velocity (m/s):	850	905
Max. Range (m):	9,200	4,000
Max. Chamber Pressure (kp/cm ²):	3,500	3,000
Armament:	Tank T-72 & T-80 Gun	Tank T-72 Gun

PACKING

TYPE	15 MM HE	125 MM HEAT
Quantity per Box (rds):	1	1
Box Weight (kg):	52	52
Box Dim. (mm):	800 × 480 × 280	823 × 544 × 295
Box P/Pallet:	8	8
Pallet Weight (kg):	430	460





125 MM APFSDS – T (BM 22 & 42)



PIC: 000128

DESCRIPTION

These products are advanced types of **125 mm APFSDS-(BM22 & 42)** using long rod penetrator.

The high muzzle velocity and high density of penetrator produce a great amount of kinetic impact energy for penetration in **RHA**, Multi layer targets and advanced new armors.

The **BM22 & 42** ammunitions fired by **T-72** tanks and intended to be used against tanks and armored vehicles.



SPECIFICATIONS

TYPE	BM 22	BM 42
Caliber (mm):	125	125
Projectile Length (mm):	560	572
Projectile Weight (kg):	6.5	7.05
Effective Range (m):	2,000	2,000
Penetration Depth (mm):	Min. 360 (in RHA target)	Min. 420 (in RHA target)
Total Weight (including steel packing) (kg):	~ 30	~ 30
Max. Chamber Pressure (kgf/cm ²):	5,100	5,100

PACKING

TYPE	BM 22 & 42
Quantity Box:	1
Box Weight (kg):	30
Box Dim. (mm):	1208 × 200 × 200
Box P/Pallet:	20
Pallet Weight (kg):	530





105 MM HOW HIGH EXPLOSIVE & ILLUMINATING



PIC: 000129

DESCRIPTION

105 mm howitzer ammunition series are semi- fixed rounds with the projectile being a free fit in the cartridge case to allow free access to the propellant increments inside the case for adjustment.

The projectile is of a high-tensile steel forging with a streamlined ogive a boat tailed base and a welded base cover and a single copper driving band.



SPECIFICATIONS

TYPE	HIGH EXPLOSIVE, M1	ILLUMINATING, M314A2E1
Projectile Length Fuzed (mm):	495	471
Projectile Weight Fuzed (kg):	14	14.85
Cartridge Case Length (mm):	370.8	371.8
Propellant Charge (kg):	M67-1.35	M67-1.35
Cartridge Case Weight (kg):	2.68 (W/O charges) / 4.03 (With Charges)	2.68 (W/O charges) / 4.03 (With Charges)
Filler Material (kg):	2.5 TNT	789 gr Illuminant in Pellets
Illuminating Instensity (cd):	-	700,000
Burning Time (sec):	-	60
Rate of Descent (m/s):	-	5 - 6
Primer:	M28A2, M39B2 or M28C1	M28A2, M39B2 or M28C1
Fuze:	PDM 557 or PDM572	PDM 557 or PDM572
Muzzle Velocity (m/s):	494	494
Max. Range (m):	11,500	11,500
Max. Chamber Perssure (kp/cm ²):	2,285	2,285
Armament:	102 mm HOW Gun M2A2	102 mm HOW Gun M2A2

PACKING

TYPE	HIGH EXPLOSIVE & ILLUMINATING
Quantity per Box (rds):	2
Box Weight (kg):	52
Box Dimensions (mm):	920 × 310 × 210
Box per Pallet:	15
Pallet Weight (kg):	795
Pallet Dimensions (mm):	940 × 920 × 1150



105MM HOW-SMOKE, WP & HC



PIC: 000130

DESCRIPTION

The **105 mm smoke** cartridges are intended for screening and spotting purposes and are provided in two types depending in their filler materials and the process of smoke disposal.

One type uses **WP** to screen and spot with a limited incendiary effect and the other type uses **HC** or Colored Smoke material for screening spotting and signalling purposes.

In the first type, **WP** is dispersed in the air, creating a dense white smoke cloud and in the other type which is fitted with a mechanical time fuze, the canisters containing smoke materials are ejected from the base and create the smoke cloud.



SPECIFICATIONS

TYPE	SMOKE, HC	SMOKE WP
Projectile Length Fuzed (mm):	790	790
Projectile Weight Fuzed (kg):	15.65	15.65
Caltridge Case Length (mm):	371.8	371.8
Propellant Charge (kg):	M67 - 1.28	M 67-1.35
Cartridge Case Weight (kg):	2.68 (W/O charges) – 4.03 (With Charges)	2.68 (W/O charges) 4.04 (With Charges)
Filler Material (kg):	2.73 HC	1.57 WP
Primer:	M28A2, M28B2 or M28C1	M28A2, M28B2 or M28C1
Fuze:	M564 or M500A1	M564 or M500A1
Muzzle Velocity (m/s):	494	494
Max. Range (m):	11,500	11,500
Max. Chamber Pressure (kp/cm ²):	2,285	2,285
Armament:	105mm How Gun M2A2	105mm How Gun M2A2

PACKING

TYPE	105 MM HOW - SMOKE, HC, WP
Quantity per Box:	2
Box Weight (kg):	52
Box Dimensions (mm):	920 × 310 × 210
Box per Pallet:	15
Pallet Weight (kg):	795
Pallet Dimensions (mm):	940 × 920 × 1150



122 MM HOW - HIGH EXPLOSIVE



PIC: 000131

DESCRIPTION

122 mm Howitzer ammunition series are separate loading munitions widely used all over the world since world war II as reliable artilleries in military services.

High Explosive **OF-462** intended for use with **D-30** Howitzer Gun and High Explosive Hollow Base (**HEHB**) to be fired with **PL83** Howitzer Gun as well as their relevant smoke and illuminating types are the various available products.

The shell bodies are forged high-fragmentation steel with a relatively short streamlined ogive, a pronounced bourrelet, a distinct boat tail and a single copper driving band. The cartridge cases are brass made inside which are the propellant charges which vary in quantity for full and reduced charges.



SPECIFICATIONS

TYPE	FRAG-HE, OF-462	HEHB
Projectile Length Fuzed (mm):	560	653.5
Projectile Weight Fuzed (kg):	21.760	21.760
Cartridge Case Length (mm):	446.4	394 ± 0.1
Propellate Charge (kg):	Full – 3.8 / Reduced – 2.4	2.70 kgs 7-perforated + CBI
Cartridge Case Weight (kg):	5.15 (W/O Charges)	5.15 (W/O Charges)
Filler Material (kg):	3.55 TNT	3.20 TNT
Primer:	MC22/30	MC22/30
Fuze:	Fuze PDM 557 or PDM 572	Fuze PDM 557
Muzzle Velocity (m/s):	Full chrg:690 / Reduced Chrg. : 565	618
Max. Range (m):	Full chrg:16,800 / Reduced Chrg. : 13,430	15.600
Max. Chamber Pressure (bar):	Full chrg:2,900 / Reduced Chrg. : 2,500	2,600
Armament:	122 mm HOW D30	122 mm HOW PL83 & D30

PACKING

TYPE	FRAG-HE, OF-462, HEHB
Quantity per Box (rds):	1
Box Weight (kg):	39
Box Dimensions (mm):	720 × 360 × 220
Box per Pallet:	15
Pallet Weight (kg):	600
Pallet Dimensions (mm):	1100 × 1200 × 750



122 MM HOW - SMOKE & HC



PIC: 000132

DESCRIPTION

The **122 mm smoke** ammunition is supplied in two types, depending on their filler materials and the process of smoke disposal.

One type uses **WP** to screen and spot with a limited incendiary effect and the other type uses **HC** or Colored Smoke material for screening spotting and signaling purposes.

The ammunition is also separate loading with the projectile and cartridge case loaded separately so that where appropriate the propellant charges can be altered to suit the fire mission.



SPECIFICATIONS

TYPE	WP, D-462	HC, (Base Ejection)
Projectile Length Fuzed (mm):	571.8	571.8
Projectile Weight Fuzed (kg):	21.100 - 21.610	22.05
Cartridge Case Length (mm):	446.4	446.4
Propellate Charge (kg):	Full: 3.8 / Reduced: 2.4	Full: 3.8 / Reduced: 2.4
Cartridge Case Weight (kg):	5.15 (W/O Charges)	5.15 (W/O Charges)
Filler Material (kg):	2.20 WP	3.60 HC
Primer:	MC22/30	MC22/30
Fuze:	Fuze PDM 557	MTSQ – M564
Muzzle Velocity (m/s):	Full Chrg:690 / Reduced Chrg. : 565	Full Chrg:690 / Reduced Chrg : 565
Max. Range (m):	Full Chrg:16800 / Reduced Chrg. : 13430	Full Chrg:16800 / Reduced Chrg. : 13430
Max. Chamber Pressure (bar):	Full Chrg:2900 / Reduced Chrg. : 2500	Full Chrg:2900 / Reduced Chrg. : 2500
Armament:	122 mm HOW D30	122 mm HOW D30

PACKING

TYPE	122 MM HOW - SMOKE & HC
Quantity per Box:	1
Box Weight (kg):	40
Box Dimensions (mm):	640 × 330 × 220
Box per Paller:	15
Pallet Weight (kg):	620
Pallet Dimensions (mm):	1100 × 1500 × 750



122 MM HOW - ILLUMINATING



PIC: 000133

DESCRIPTION

This ammunition is intended for signaling or illuminating a designated area under conditions of poor visibility.

The projectile is hollow and contains a payload consisting of a parachute and illuminant assembly.

The illuminant is ignited and the parachute and illuminant assembly are ejected from the projectile by an expelling charge adjacent to a time fuze.

The parachute and burning illuminant assembly slowly descend, lighting the target area.



SPECIFICATIONS

TYPE	122 MM HOW - ILLUMINATING
Projectile Length, Fuzed (mm):	517
Projectile Weight, Fuzed (kg):	21.20
Cartridge Case Length (mm):	446.4 - 0.8
Propellant Charge (kg):	FULL:3.800 / Reduced: 2.40
Expelling Charge (g):	Black powder: 50
Cartridge Case Weight (kg):	5.15 (W/O Charges)
Illuminating Material (g):	750 in Pallets
Illuminating Instensity (cd):	700,000
Burning Time (sec.min):	50
Rate of Descent (m/s):	7
Fuze:	MTSQ - M565
Primer:	MC 22/30
Muzzle Velocity (m/s):	Full Chrg.: 690 / Reduced Chrg.: 565
Max. Range (m):	Full Chrg.: 16,800 / Reduced Chrg.: 13,430
Max. Chamber Pressure (kp/cm ²):	Full Chrg.: 2,900 / Reduced Chrg.: 2,500
Armament:	122 mm HOW D30

PACKING

TYPE	122 MM HOW - ILLUMINATING
Quantity per Box:	1
Box Weight (kg):	40
Box Dimensions (mm):	640 × 330 × 220
Box per Pallet:	15
Pallet Weight (kg):	620
Pallet Dimensions (mm):	1100 × 1500 × 750



130 MM HE & BB



PIC: 000134

DESCRIPTION

» 130 MM HE (0F482M)

130 mm gun ammunition are separate loading series of munitions provided in various types as high explosive, extended range, smoke and illuminating. The extended range has base bleed unit which brings an enhanced range of 36,000 m for ammunition.

» 130MM HE (BB)

This is an improved type of 130 mm **HE** ammunition which has a different aerodynamic shape to the normal one and is equipped with a gas generating system at its base called as **Base Bleed** unit. The generated gas replaces the vacuum created at the rear side of the shell and therefore results reduction of the drag force and consequently a remarkable increasing in the range which is around 37,800 at 1,000 m above sea level for this ammunition.

» 130 MM HE (HERA)

This is a new developed type of 130 mm HE-BB which in addition to the **Base Bleed** system, a rocket motor has been added to it and resulted its range to be increased up to 42,000 m.



HE (0F482M)



HE (BB)



HE (HERA)

SPECIFICATIONS

TYPE	FRAG-HE (0F482M)	HE (BB)	HE (HERA)
Projectile Length Fuzed (mm):	662.8	722 ± 2	800
Projectile Weight Fuzed (kg):	33.30	32	32
Cartridge case length (mm):	845.3	-	-
Propellant Charge (kg):	Full: 13.5 / Reduced: 6.75	Full: 13.5	Full: 13.5
Cartridge Case Weight (kg):	11.32 (W/O Charges)	-	-
Filler Material:	3.64 TNT	TNT	TNT or Comp.B
Primer:	MC22/30	MC22/30	MC22/30
Fuze:	PDM 572	B429	B429
Muzzle Velocity (m/s):	Full Chrg.: 930 Reduced Chrg.: 705	970 ± 2%	970 ± 2%
Max. Range (m):	Full Chrg.: 27,500 Reduced Chrg.: 19,130	37,800	42,000
Max. Chamber Pressure (bar):	Full Chrg.: 3,500 Reduced Chrg.: 2,800	-	-
Armament:	130 mm (M-46, Type 59-1) Gun	T59-1 or M46	T59-1 or M46

PACKING

TYPE	HE (0F482M)	HE (BB)	HE (HERA)
Quantity per Box:	1	1	1
Box Weight (kg):	75	68	75
Box Dimensions (mm):	980 × 410 × 230	1200 × 300 × 200	980 × 410 × 230
Box per Pallet:	12	12	12
Pallet Weight (kg):	920	830	920
Pallet Dimensions (mm):	1000 × 1300 × 1000	1000 × 1300 × 1090	1000 × 1300 × 1000



130 MM SMOKE & ILLUMINATING



PIC: 000135

DESCRIPTION

» 130 MM SMOKE (WP,DTS-1)

This product is also separate loading ammunition allowing the variable charges (full and reduced), contained in a brass case to vary to suit any particular fire mission and as the other smoke ammunition, it is used to provide screening or marking smoke .

The shell contains White Phosphorus as smoking agent which upon contacting the air , ignites and produces a thick cloud of white smoke.

» 130 MM ILLUMINATING (SP – 46)

130 mm illumination is another type of 130 mm ammunition having illuminating assembly and parachute to be used for signaling or illuminating a target area in military operation under conditions of poor visibility. Illuminating intensity is 500,000 cd and min. burning time is 60 sec.



SPECIFICATIONS

TYPE	SMOKE & (WP, DTS - 1)	ILLUMINATING (SP-46)
Projectile Length Fuzed (mm):	628.8 - 2.5	661 ± 5
Projectile Weight Fuzed (kg):	32.97 - 33.30	29.7 ± 0.3
Cartridge Case Length (mm):	845.3 - 1	-
Propellate Charge (kg):	Full Chrg.: 13.500 / Reduced: 6.75	-
Cartridge Case Weight (kg):	11.32 (W/O Charges)	-
Filler Material (kg):	3.23 WP	2.6 Illuminant
Primer:	MC22 / 30	MC22 / 30
Fuze:	PDM 572 / V-429 E	ET fuze – DRS 12
Muzzle Velocity (m/s):	Full Chrg.: 930 / Reduced Chrg.: 705	930 ± 15
Max. Range (m):	Full Chrg.: 27,500 / Reduced Chrg.: 19,130	27,000
Max. Chamber Pressure (kp/cm ²):	Full: 3,200 / Reduced: 2,700	-
Armament:	130 mm (M-46, Type 59-1) Gun	130 mm M46

PACKING

TYPE	SMOKE (WP,DTS-1)	ILLUMINATING (SP-46)
Quantity per Box:	1	1
Box Weight (kg):	75	Approx. 65
Box Dimensions (mm):	980 × 410 × 230	1000 × 450 × 256
Box per Pallet:	12	15
Pallet Weight (kg):	920	Approx. 1065
Pallet Dimensions (mm):	100 × 1300 × 1000	-



155 MM HOW (HE & ERFB – BB)



PIC: 000136

DESCRIPTION

155 mm HOW ammunition series are supplies in **HE, M107**, Extended Range with Base Bleed unit as well as smoke and illuminating types.

The extended range one is equipped with base bleed unit resulting to an enhanced range of 32 km for the ammunition fired with the same armament.



SPECIFICATIONS

TYPE	HIGH EXPLOSIVE, M107	HE-ERFB/BB
Projectile Length Fuzed (mm):	702	871
Projectile Weight Fuzed (kg):	43	41.40 - 43.40
Weight Zone Group:	4	5
Filler Material (kg):	6.5 - 6.9 TNT	8.2 - 8.6 TNT
Propellate Charge (kg):	White bag M4A2 - 6 / M119A2 9.200	Zone 10 / 15.80 - 16.30
Fuze:	PDM 557, PDM 572 or PDM 739	PDM 557 or PDM 739
Primer:	M82 or MK2A4	M82 or MK2A4
Muzzle Velocity (m/s):	564	914
Max. Range (m):	14,600	32,000
Max. Chamber Pressure (bar):	2,790	3,500
Armament:	155 Howitzer M1A2	GHN 45 A1

PACKING

TYPE	HE, M107	FUZES	PROPELLANT CHARGES
Quantity per Box:	-	20	4
Box Weight (kg):	-	30	50
Box Dimensions (mm):	-	320 × 400 × 180	730 × 425 × 440
Box per Pallet:	-	30	12
Quantity per Pallet:	8	600	48
Pallet Weight (kg):	355	920	620
Pallet Dimensions (mm):	700 × 360 × 800	1150 × 1000 × 1200	1460 × 1275 × 900



155 MM SMOKE & ILLUMINATING



PIC: 000137

DESCRIPTION

» 155 MM SMOKE (WP)

155 mm smoke is used to provide screening or marking smoke. It has the same general appearance as the 155 mm high-explosive (HE) M107 and has the same design features. The shell contains White Phosphorus and is normally fitted with a nose mounted Point Detonating (PD) fuze.

On impact, the fuze ignites the internal buster charge, which then ruptures the shell casing to release WP filling. As it contacts the air the WP ignites to product thick clouds of white smoke.

» 155 MM ILLUMINATING

This is a separate-loading ammunition of 155 mm series using a hollow forget steel body with a steel base plug press fitted to the projectile and held in place by shear pins and twist pins.

The shell interior contains a primary expulsion charge, a canister assembly and a drag parachute. The canister assembly contains a secondary expulsion charge, a delay holder, an illumination composition and the main parachute.



SPECIFICATIONS

TYPE	SMOKE (WP)	ILLUMINATING
Projectile Length Fuzed (mm):	700	700
Projectile Weight Fuzed (kg):	44.60	41.7
Cartridge Case Length (mm):	4	4
Propellate Charge (kg):	White Bag M4A: 2-6 / Green Bag M3A1: 2.5	White Bag M4A: 2-6 / Green Bag M3A1: 2.5
Filler Material (kg):	7 wp	2.6 Illuminating
Primer:	MK2A4	MK2A4 or M82
Fuze:	PDM 557/ PDM 572	M565MT or M577MTSQ
Muzzle Velocity (m/s):	564	576
Max. Range (m):	15,000	14,000
Min. Smoke / Burning Time (sec):	100	100
Max. Chamber Pressure (kp/cm ²):	2,790	2,790
Armament:	155mm HOW M1A2	155 mm HOW M1A2 - IH

PACKING

TYPE	SMOKE (WP)	ILLUMINATING
Quantity per Box (rds):	8	8
Box Weight (kg):	375	350
Box Dimensions (mm):	700 × 340 × 800	700 × 340 × 800



203 MM (8IN) HOW HE, M106



PIC: 000138

DESCRIPTION

This is the largest and heaviest artillery ammunition which is intended for use with **M2 or M2A1** Howitzer Cannon to provide heavy, long range artillery fire against personnel and installations.

The projectile consists of a hollow steel forging with a streamlined ogive, a boat tail base and a copper driving band encircling the shell body near the base and protected by a grommet during storage, transport and handling. The fuze cavity is occupied by a threaded lifting plug during transit and handling.

Before firing, the plug is replaced by any of standard point detonating fuzes such as **PDM557**, **PDM 572** or **DPM739**.



SPECIFICATIONS

TYPE	HIGH EXPLOSIVE, M106
Projectile Length Fuzed (mm):	895
Projectile Weight Fuzed (kg):	90
Filler Material (kg):	16 TNT
Propellant Charge (kg):	12.725
Fuze:	PDM 572, PDM 739 or M557
Primer:	MK2A4
Muzzle Velocity (m/s):	595
Max. Range (m):	17,000
Max. Chamber Pressure (kp/cm ²):	3,000
Armament:	M2, M2A1

PACKING

TYPE	HE, M106	FUZES	PROPELLANT CHARGES
Quantity per Box:	-	20	4
Box Weight (kg):	-	30	50
Box Dimensions (mm):	-	320 × 400 × 180	730 × 425 × 440
Box per Pallet:	-	30	12
Quantity per Pallet:	6	600	48
Pallet Weight (kg):	550	920	620
Pallet Dimensions (mm):	630 × 420 × 900	1150 × 1000 × 1200	1460 × 1275 × 900



40 MM HE & TRAINING GRENADE



PIC: 000139

DESCRIPTION

The **40 mm** Cartridge is a high velocity **HE** Grenade which is fired from automatic grenade launcher **M19** either manual or automatic against enemy 's personnel. In comparison to the low velocity grenades, this cartridge is much stronger and has a much thicker base into which the forward part of the high pressure chamber is formed. The larger quantity of its propellant results in higher final chamber pressure and thus greater velocity and range.

The **M385** Cartridge is just for training purpose and as the same as **M383** type can be fired from automatic grenade launcher **M19** or manual one.



M385 (TRAINING)



M383 (HE)



SPECIFICATIONS

TYPE	HE	TRAINING
Caliber (mm):	40	40
Total Length (mm):	112	112
Total Weight (gr):	≤ 340	≤ 340
Average Muzzle Velocity (m/sec):	≥ 240	≥ 240
Max. Range(m):	~2,200	~2,200
Effective Range (m):	~1500	-

PACKING

TYPE	HE & TRAINING
Quantity:	60 linked Rounds in Wooden Box



HAND GRENADES



PIC: 000140

DESCRIPTION

» FRAGMENTATION

This product is a high quality grenade having both blast and fragmentation effects and designed for use against personnel or material in a close combat.

It achieves incapacitation of injury within a range of 8 m of the burst point, as the result of high velocity dispersion of steel shots. There are two types in use right now, one having steel shots in its body and the other without any shots. In the latter, the body which is made of cast iron will act as fragments after blast.



Fragmentation
(With Cast Iron)



Fragmentation
(With Steel Shots)



» PRACTICE (SOUND & SMOKE)

The **Practice Sound** Hand Grenade is a little bit different in shape and color to the fragmentation one and by only changing the fuze, the body can be used for 1000 times. That is why the body and the fuze should be proportionally selected. Maximum sound decible is 100 ± 10 db.

The **Practice Smoke** hand Grenade has the same shape and size of fragmentation type but in blue color body and without any steel shots or explosives and the smoke material is **HC**.



SOUND



SMOKE

SPECIFICATIONS

TYPE	FRAGMENTATION (With Steel Shots)	FRAGMENTATION (With Cast Iron)	SOUND	HC (SMOKE)
Length (mm):	100	114.5	88 ± 2	116 ± 2
Dia. Of the Body:	62	57	60 ± 2	60 ± 1
Total Weight (gr):	480	620	470 ± 20	410 ± 10
Filling Material (gr):	PETN, 45	PETN, 38	-	-
Delay (sec):	3.5 - 6.5	3.5 - 6.5	1.5 - 4	3.5 - 4.5
Lethal Radius (m):	8	8	-	-
Steel Shots:	300g dia. 2.05 - 3mm	300g dia. 2.05 - 3mm	-	-

PACKING

TYPE	SOUND
Quantity per Box:	20
Box Weight (kg):	17
Box per Pallet:	30
Pallet Weight (kg):	550
Pallet Dimensions (mm):	1150 × 100 × 720





RIFLE GRENADE



PIC: 000141

DESCRIPTION

This product is a blast and fragmentation **BTU** rifle grenade with the ability to engage and defeat, light structures, material target and personnel.

The grenade may be launched from any standard military **5.56** or **7.62** mm rifle using standard ball ammunition.



SPECIFICATIONS

TYPE	RIFLE GRENADE
Caliber of the Grenade (mm):	35
Height (mm):	290 ± 7
Weight (g):	470 ± 10
Filler Material:	50 g Comp A4
Fuze:	PDF61
Max. Firing Range (m/s):	400 ± 50
Operation Temp. (°C):	-33 to +50
Warehousing (years):	15

PACKING

TYPE	RIFLE GRENADE
Quantity per Box:	10
Box Weight (kg):	17.5
Box Dimensions (mm):	510 × 410 × 240
Box per Pallet:	30
Pallet Weight (kg):	520
Pallet Dimensions (mm):	1200 × 1170 × 1030





MORTAR BOMB FUZES



PIC: 000142

DESCRIPTION

There are three types of electronic & mechanical fuzes which are used by different model of mortar bombs.



SPECIFICATIONS

TYPE		PD FUZE (AZ111A2)	ELECTRONIC TIME FUZE (ETM-86)	TIME FUZE (M84A1E1)
Compatibility:	Weapon (mm)	60, 81, 120 Mortar	81 & 120 Mortars	81, 120 Mortar
	Ammunition	HE	HE, Illumination, Cargo	Illumination
Operation Mode:		Super quick & delay	-	Time, Adjustable
Arming Data:	First Safety	Pull wire	Manual	Pull Wire
	Second Safety	40 m	99.9 sec.	Set back: 500 g non arm / 1000 g arm
Safety:	Set Back (g)	600	600	7070
Function Temperature (°C):		-40 to +60	-33 to +60	-
Physical Data:	Weight (g)	201 ± 10	365 ± 1	826
	Length (mm)	87.8	121.5 ± 1	97.8
	Cross Section (mm)	49	49 ± 0.15	76.2
	Intrusion	27.7	27.7	16 mm dp
	Thread	1.5" - 12 UNF-1A	1.5" - 12UNF - 2A	8.4 - 18 UNS - 1A
Max. Velocity (m/s):		-	301	250



PACKING

TYPE	AZ111A2	ETM-86	M84A1E1
Quantity(pcs):	100	As per Request	26
Dimensions (cm):	62 X 33 X 27	-	47 X 42 X 16.5



AMMUNITION FUZES



PIC: 000143



SPECIFICATIONS

TYPE		PD (B429 & B429E)	PIBD (M509 A1)	PIBD (V-15)
Compatibility:	Weapon	85,122,125,130, 152,155 mm	90 mm Gun M1 M2 ,105 Tank Gun, 106 mm Rifle M40A6	Tank-125
	Ammunition	HE , WP	Cartridge 90 mm HEAT-TM431 M348 105 HEAT-T m456 A2. 106mm HEAT-T m344 A1	125mm HEAT-T
Operation mode:		Super Quick & Delay	Super Quick	-
Arming Data:	First Safety	-	2500 g non- arm, 4300 g arm Set Back 1.5 to 3 ms	-
	Second Safety Acceleration	min. : 2716 g max : 17043 g	Setback Electrical Switch, Approx 1800 g (only in M456 A2)	2.5 m
Safety:	Set Back	-	40,000 g	-
Physical Data:	Weight	438 g	141.0 g	710 ± 10 g
	Length	105.71 mm	49.8 mm	114 mm
	Cross Section	Ø 40 mm max	34.3 mm	42 mm
	Intrusion	46.8 mm	49.8mm dp, 32.0mm dia	-
	Thread	M36.18X10 Thds/in	-	M42X1.5 LH
Max. Velocity:		930 m/s	1173 m/s	905 m/s





AMMUNITION FUZES



PIC: 000144



SPECIFICATIONS

TYPE	PD (M572)		PROXIMITY
Compatibility:	Weapon	122mmD30,105, 130,175,203 mm	76/62 mm Naval Ammunition
	Ammunition	105,122,130, 203 mm HE, WP	-
Operation Mode:	Super Quick & Delay		Proximity
Arming Data:	First Safety	16 rps. non-arm, 33 rps arm	-
	Second Safety	Delay: 40 m	
Safety:	Set Back	21,000 g	Mechanical Distance 70m
Physical Data:	Weight	1,043 g	930 g
	Length	152.1 mm	203 mm
	Cross Section	61.3 mm	62 mm
	Intrusion	56.1 mm dp 44.0 mm dio	-
	Thread	2-12 UNS-1A	M58 x 1.5 - 6 g
Max. Velocity:	914 m/s		Rotation 24,600 rpm





EITF FUZE



PIC: 000145

SPECIFICATIONS

TYPE		100MM INDUCTION TIME FUZE	
Compatibility:		Ammunition 100 mm Anti Aircraft	
Arming Data:		2000 rpm	
Operation Mode:		ET fuze	
Setting Data:	Setting Mode	Manual Induction (by setting unit)	
	Total Setting Time	Less than 1 sec	
	Max. Setting Time	31 sec	
	Min. Setting Time	3 sec	
	Accuracy	0.1 Sec	
Power Supply :		Reserve Battery	
Max. Velocity:		930 m/sec	
Function Temperature:		-40°C to + 60°C	
Physical Data:	Weight	685 g	
	Length	135 mm	
	Cross Section	63.5 mm	
Safety:		400XCaliber	



F61 FUZE

PIC: 000146

SPECIFICATIONS

TYPE		PD (F61)
Compatibility:	Weapon	G3, M16
	Ammunition	BTU M262A1, Rifle Grenade
Operation Mode:		Super Quick
Arming Data:	First Safety	Pull Wire
	Second Safety	12 m
Safety:		Set Back 600 g
Physical Data:	Weight	114 ± 5g
	Length	59 mm
	Cross Section	35 mm
	Intrusion	7.4
	Thread	M32X16 g
Max. Velocity:		100 m/s
Packing:		As per Request
Function Temperature:		- 40°C to + 60°C





PRIMERS & DETONATORS



PIC: 000147

DESCRIPTION

All primers and detonators ranging from small arms up to artilleries are produced in Iran Defence Industries.

The primers for small arms consist of small capsules in either Berdan or Boxer form and when the question is large calibers then they will be accompanied with primer seat and igniting tube to be a complete primer.

The detonators are required for completing the explosive chain in fuzes and other parts.



SPECIFICATIONS

PRIMER No.	TYPE	APPLICATION	DETONATOR	TYPE	APPLICATION
5608	Berdan	7.62 mm	M24	Stab	Fuze M557, M572
8001	Berdan	12.7mm & 0.50	M14	Flame	Fuze M557, M572
8216	Boxer	105 mm	DM 1019	Stab	Fuze AZ 111A2
5337	Boxer	Mortars	A30T	Flame	23 mm (fuze)
7206	Boxer	Shogun cartridge	M31	Stab	Mine AT
5625	Berdan	AA35 mm	M50	Stab	Mine AT
7522	Boxer	5.56 mm	-	-	-
7022	Boxer	9 mm & 0.38	-	-	-
No. 1	Stab	23 mm (fuze)	-	-	-
No. 2	Stab	23 mm (fuze)	-	-	-
No. 3	Percussion	23 mm(propellant charge)	-	-	-
M82	Percussion	Artillerires	-	-	-
MK2A4	Percussion	Artillerires	-	-	-
M28	Percussion	Artillerires	-	-	-



PLAIN & SHOCK TUBE DETONATORS

PIC: 000148

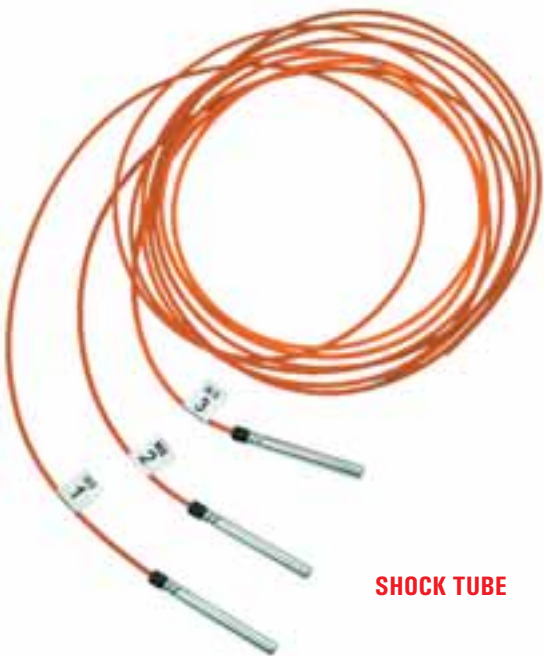
DESCRIPTION

Plain detonator no.8 is a non electric detonator being used for blasting operations close to high pressure power transmtion lines, except coal mines.

SHOCK TUBE detonators (Instantaneous & millisecond delay) are various blasting operations except coal mines. Insensitivity of this detonator to electricity & its water proof properties have enabled it to be usable in even bad weather conditions.



PLAIN



SHOCK TUBE



SPECIFICATIONS

TYPE	PLAIN (NO. 8)	SHOCK TUBE
Shell Material:	Aluminium	Aluminium
Shell Length of Instantaneous Type (mm):	42	52
Shell Length of Delay Type (mm):	-	62 ~ 82
Tube Material:	-	P.V.C
Outside Diameter of the Tube (mm):	6.9	3
Inside Diameter of the Tube (mm):	-	1.2
Shock Wave Velocity (m/s):	-	2,000
Interval Time (In Delay Type) (Millisecond):	-	25
Period Number (In Delay Type):	-	1 - 10





SMOKE POT & GRENADE



PIC: 000149

DESCRIPTION

» SMOKE POT (HC)

This is another pyrotechnic product which is intended for screening movements, live stationary and moving targets in wide areas as well as various objects and floaters in different tactical situations.

The dense smoke of this product disturbs the visibility of the enemy's **Helicopter or Aircraft** igniting the smoke pot can be done either manually or electrically.

» 81 MM SMOKE GRENADE (HC-T72)

This is a smoke generator of 81 mm caliber which is intended to be used for screening purposes fired by special launchers mounted on **T72** tank.



SMOKE POT



SMOKE GRENADE

SPECIFICATIONS

TYPE	81 MM SMOKE GRENADE	SMOKE POT
Hight (mm):	215 ± 3	247.5
Diameter (mm):	80.2	213
Weight (kg):	2.80	15.5
Smoke Material (g):	726 g HC	HC
Burning Time:	90 sec ± 30	~20 min.
Capsule Delay:	7-12 sec	-
Smoke Screen:	10-30m × 3-10m at 2-5 m/s Wind Speed	-
Primer:	Electrical 3 KB/30M	Electrical
Max. Range (m):	350	-
Armament:	81 mm launcher of T-72 or T-80 UD	-



PACKING

TYPE	81 MM GRENADE	POT
Quantity per Box (pcs):	8	1
Box Weight (kg):	33	21
Box Dimensions (mm):	470 × 350 × 270	350 × 290 × 320



IRRITANT & COLORED SMOKE GRENADE



PIC: 000150

DESCRIPTION

IRRITANT CARTRIDGES & HAND GRENADE

Riot control irritant products are provided in two types **CS Hand Grenade** and **Tear Gas Cartridges**. They are used for establishing a safe distance law between enforcement officers and crowds of demonstrators.

The delay in ignition in hand grenade ensures that the thrower will not be deflected by the emission and the rate of emission is such that there is almost no possibility of the grenade being retrieved and thrown back.

The cartridge type can be fired with either pistol or rifle having 38 mm of caliber and can disperse a large crowd and prevent re – grouping. This product is an effective, simple-to-operate CS tear gas generator which creates a large dense cloud of CS smoke for crowd control and dispersal.



COLORED SMOKE HAND GRENADE (M18)

These are **M18** colored smoke hand grenades in various colors and are used to emit smoke for signalling purposes as ground to air or ground to ground, marking of landing places for parachuters and to ascertain wind direction.

These are provided in smoke colors **YELLOW, BLUE, GREEN, ORANGE, RED, VIOLETE & GREY**.



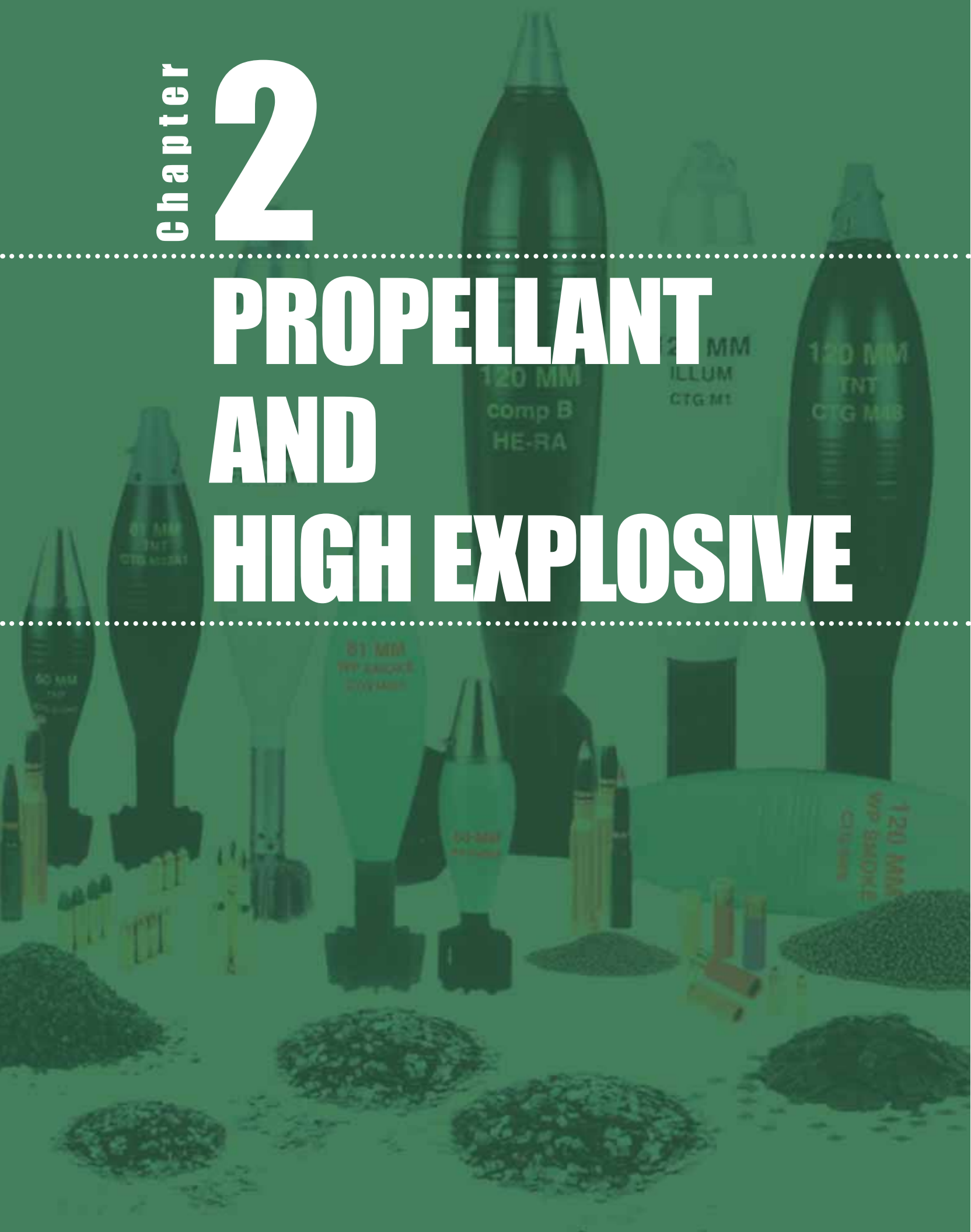
SPECIFICATIONS

TYPE	MK2	CS HAND GRENADE	M18
Caliber (mm):	38	-	-
Length (mm):	122.5	145	-
Diameter (mm) :	-	60	60
Weight (gr):	210 ± 5	355 g	Max. 450
Fuze Type:	-	Impact	Igniting Type
Chemical:	CS Mixture	CS Mixture	Colored
Smoke Emission (sec):	Min. 20	-	Min. 50
Delay Time (sec):	-	0.7-2	0.7-2
Burning Time:	-	Min. 20 sec	-
Max. Range (Rifle)(m):	150	-	-
Max. Rang (Pistol)(m):	75	-	-



chapter 2

PROPELLANT AND HIGH EXPLOSIVE





Chemical Industries Group (**CIG**) is a large-size technical and developed branch of Iran Defense Industries which is affiliated to the Ministry of Defense. CIG was established in 1939 and since then produced military and civil products by making use of a very long experience and taking advantages of the latest advanced technologies and high scientific efficiency for the production of powders, propellant charges and mines explosive materials within a range of more than 200 products.

All products are manufactured as per international skills with the best quality in modern factories, presenting after sales services.

CIG has been awarded international certifications such as **ISO 9001**, **ISO 14001**, **ISO 10015** and Military Standard which have enabled it to compete successfully in worldwide competitions. CIG is proud of making a safe and reliable image, for its products worldwide, while keeping its undertakings with customers at home and abroad.

The range of products and materials are as follows:

- Single, Double and Triple Base propellants.
- TNT, DNT, HMX, RDX, Composition (A, B, C4).
- Hexal, Torpex, Cyclotol, PETN, Octal, Ocfol.
- Detonating Cord.
- Emulite (Emulsion Explosive).
- Cellulose Nitrates.
- Increment Charges.
- Ammonium Perchlorate.
- Black Powder.





CHAPTER 2



ITEM	TABLE OF CONTENT	PAGE
1	SINGLE BASE PROPELLANT	52
2	7.62 × 51 NATO	52
3	9 × 19 MM - ONE PERFORATED	53
4	12.7 X 99 MM - ONE & SEVEN PERFORATED	54
5	40 MM L60 & 76.62 MM NAVAL - SEVEN PERFORATED	55
6	105 MM HOWITZER - ONE AND SEVEN PERFORATED	56
7	122 MM HOWITZER - REDUCED & FULL CHARGE PROPELLANT	57
8	125 MM - ONE AND SEVEN PERFORATED	58
9	152 MM - REDUCED CHARGE	59
10	130 MM - REDUCED CHARGE	60
11	155 MM - HOWITZER	61
12	203 MM HOWITZER - SINGLE BASE PROPELLANT	62
13	DOUBLE BASE PROPELLANT	63
14	5.56 × 45 MM (WC 844) AND 6.68 MM (WIN 270)	64
15	7.62 × 39 MM & 7.62 × 51 MM	64
16	7.62 × 54 MM & 9 MM (WC 230)	65
17	12.7 × 108 MM & 20 × 139 MM	65
18	130 & 152 MM FULL CHARGE	66
19	100 MM ANTI AIRCRAFT	67
20	BALLISTITE	67
21	60/81/120 MM SPLIT INCREMENT CHARGE	68
22	60/81/120 MM IGNITION CARTRIDGE	69
23	START IGNITION CHARGE RPG7	69
24	TRIPLE BASE PROPELLANT - NAVAL 4.5 INCH MK8 & 105 MM (HEAT)	70
25	155 MM B.B - TRIPLE BASE PROPELLANT	71
26	HIGH EXPLOSIVE	72
27	COMPOSITION A	72
28	COMPOSITION B - HEXOTOL- HEXOLITE	73
29	COMPOSITION C4	73
30	TNT	74
31	HEXOGEN (RDX)	75
32	PETN	75
33	BLACK POWDER	76
34	DYNAMITE	77
35	CARTRIDGE EMULSION EXPLOSIVE	77



SINGLE BASE PROPELLANT

DESCRIPTION

Single base propellant mainly consist of Nitrocellulose & Stabilizers as well as other additives. solvent can be produced with different Geo Material shape such as (needles, cylindrical and single, seven perforated and tubular).

These types of propellant can be manufactured with **NC** and mixture of Ether and Alcohol, which converts into Gelatinized paste.

In addition , **NC** powder passes little other material as additive such as Diphenylamine as stabilizer.

D.N.T & D.B.P act as plasticizer, salt of Potassium Sulfate act as flame reducer and camphor, Centralite act retarding the combustion rate.

For some types of Single Base propellant same as Hunting powder, Potassium Nitrate also is added. Due to dissolving in water , the powder becomes poroused.

7.62 x 51 NATO

PIC: 000201

DESCRIPTION

Single base one perforated propellant **7.62 x 51 Nato** is containing Nitrocellulose (**NC**) and Diphenylamine (**DPA**) and potassium sulfate. This propellant is used for 7.62 x 51 Nato rifle and machine gun.

SPECIFICATIONS

TYPE	ONE PERFORATED
Length (mm):	1.1 - 2.2
Diameter (mm):	0.8 - 0.9
Hole Diameter (mm):	0.25
Web Thickness (mm):	-
Total Volatiles (%):	1.2
External Moisture (%):	≤1.2
Residual Solvent (%):	-
Stability at 134.5°C (min.):	60



BALLISTIC SPECIFICATIONS

TYPE	ONE PERFORATED
Total Charge (gr):	3
Velocity (m/s):	837 ± 7
Max. Pressure (kg/cm²):	3,200



9 X 19 MM - ONE PERFORATED

PIC: 000202

DESCRIPTION

Single base one perforated propellant cylindrical grains, 9x19 mm containing Nitrocellulose (**NC**) and Diphenylamine (**DPA**) and potassium nitrate and ethyle centralite.

This propellant is used for **9x19mm parabelum** and **pistol 0.45**.



SPECIFICATIONS

TYPE	ONE PERFORATED
Length (mm):	1.1 - 1.7
Diameter (mm):	0.4 - 0.6
Hole Diameter (mm):	0.1 - 0.2
Web Thickness (mm):	-
Total Volatiles (%):	1.7 max
External Moisture (%):	≤1.2
Residual Solvent (%):	≤0.5
Stability at 134.5°C (min.):	≥60



BALLISTIC SPECIFICATIONS

TYPE	ONE PERFORATED
Total Charge (gr):	≤0.37
Velocity (m/s):	385 ± 15
Max. Pressure (kg/cm ²):	≤2,500



2

12.7 X 99 MM & 12.7 x 108 MM ONE & SEVEN PERFORATED

PIC: 000203

DESCRIPTION

Single base one perforated propellant grains, 12.7x108 mm containing nitrocellulose (**NC**), diphenylamine (**DPA**), dibutyle phthalate(**DBP**) and added dinitrotoluene(**DNT**), for 12.7 x 99 mm.

The 12.7 x 99 mm and 12.7 x 108 mm are used for **Rifle & DSHKA** machine gun.



SPECIFICATIONS

TYPE	ONE PERFORATED	SEVEN PERFORATED
Length (mm):	2.2	3.2 ± 0.1
Diameter (mm):	0.9	3 ± 0.1
Hole Diameter (mm):	0.2	0.35 ± 0.05
Web Thickness (mm):	-	-
Total Volatiles (%):	1.2	1.5
External Moisture (%):	≤0.7	≤1.2
Residual Solvent (%):	≤0.5	≤0.7
Stability at 134.5°C (min.):	≥60	≥60

BALLISTIC SPECIFICATIONS

TYPE	ONE PERFORATED	SEVEN PERFORATED
Total Charge (gr):	7 - 7.3	16 - 18
Velocity (m/s):	550 - 560	820 ± 10
Max. Pressure (kg/cm ²):	≤ 2,500	≤ 3,300



40 MM L60 & 76.62 MM NAVAL SEVEN PERFORATED

PIC: 000204

DESCRIPTION

Single base seven perforated propellant cylindrical grains, 40mm L60 containing nitrocellulose (**NC**) and diphenylamine (**DPA**) and dinitrotoluene (**DNT**) and dibutyle phthalate (**DBP**).

These propellants are used for both **40 mm L60 BOFORS** and **OTOMELARA** naval gun.



SPECIFICATIONS

TYPE	40MM L60	76.62 MM NAVAL
Length (mm):	5 ± 0.1	14.2 ± 0.1
Diameter (mm):	2.6 ± 0.1	6.12 ± 0.02
Hole Diameter (mm):	0.18	0.47 ± 0.01
Web Thickness (mm):	0.5 ± 0.1	1.26 ± 0.05
Total Volatiles (%):	Max. 2	Max. 1
External Moisture (%):	≤ 1.2	≤ 1.2
Residual Solvent (%):	≤ 0.5	≤ 0.5
Stability at 134.5°C (min.):	≥ 60	≥ 60



BALLISTIC SPECIFICATIONS

TYPE	40 MM L60	76.62 MM NAVAL
Total Charge (gr):	≤ 380	$\sim 2,400$
Velocity (m/s):	860 ± 10	930 ± 10
Max. Pressure (kg/cm ²):	$\leq 2,900$	$\leq 3,600$



2

105 MM HOWITZER ONE AND SEVEN PERFORATED

PIC: 000205

DESCRIPTION

Single base one & seven perforated propellant cylindrical grains, 105 mm containing Nitrocellulose (NC) and diphenylamine (DPA) and dinitrotoluene (DNT) and Dibutyle phthalate (DBP) and Potassium Sulfate.

These propellants (one & seven perforated) are used for **105 mm Howitzer**.



SPECIFICATIONS

TYPE	ONE PERFORATED	SEVEN PERFORATED
Length (mm):	6 ± 1	4 ± 1
Diameter (mm):	1.2 ± 0.1	4 ± 0.2
Hole Diameter (mm):	0.4 ± 0.05	0.6 ± 0.05
Web Thickness (mm):	0.4 ± 0.05	0.56 ± 0.05
Total Volatiles (%):	Max. 1.6	Max. 1.8
External Moisture (%):	0.4 - 0.8	0.4 - 0.6
Residual Solvent (%):	≤ 0.7	≤ 1
Stability at 134.5°C (min.):	≥ 40	≥ 40

BALLISTIC SPECIFICATIONS

TYPE	ONE PERFORATED	SEVEN PERFORATED
Total Charge (gr):	280	960
Velocity (m/s):	472 ± 2	470 ± 2
Max. Pressure (kg/cm ²):	≤ 2,450	≤ 2,450



122 MM HOWITZER REDUCED&FULL CHARGE PROPELLANT

PIC: 000206

DESCRIPTION

Single base seven perforated propellant cylindrical grains, category 9/7 is a kind of single base propellant containing Nitrocellulose (NC) and Diphenylamine (DPA).

This propellant plus increments of one perforated granular propellant type 4/1, and type 3/1 as flash reducer which is used as reduced charge for **122 mm gun**.

This propellant plus two increments, (a) one perforated tubular propellant type 12/1-27, (b) granular propellant type 3/1 as flash reducer which is used as full charge for **122 mm gun**.



Reduced charge propellant

Full charge propellant

SPECIFICATIONS

TYPE	REDUCED CHARGE			FULL CHARGE		
FORM	9/7	4/1	3/1	12/7	12/1-27	3/1
Length (mm):	11.5 - 12.6	5.5 - 7.5	1.6 - 2.1	13.5 - 15.6	258 - 270	1.6 - 2.1
Hole Diameter (mm):	0.4 - 0.6	0.25 - 0.35	0.1 - 0.25	0.50 - 0.70	2.4 - 2.8	0.1 - 0.25
Web Thickness (mm):	0.95 - 1.10	0.4 - 0.8	0.28 - 0.36	1.15 - 1.25	1.10 - 1.25	0.28 - 0.36
Total Volatiles (%):	≤ 4.5	≤ 3.8	≤ 3.7	≤ 4.5	≤ 4.5	≤ 3.7
External Moisture (%):	1.0 - 1.8	1.0 - 1.8	1.0 - 1.8	1.0 - 1.8	1.0 - 1.8	1.0 - 1.8
Residual Solvent (%):	≥ 0.9	≥ 0.6	≥ 0.5	≥ 1.0	≥ 1.2	≥ 0.5
Stability at 134.5°C (min.):	≥ 40	≥ 40	≥ 40	≥ 40	≥ 40	≥ 40
Weight of Charge (gr):	1,885	600	20	≈ 3,600	200	75
Mean Muzzle Velocity (m/s):	565	-	-	690	-	-
Average Max. Pressure (kg/cm ²):	2,500	-	-	≤ 2,500	-	-



2

125 MM ONE AND SEVEN PERFORATED

PIC: 000207

DESCRIPTION

» 125MM HEAT & HE

Single base one & seven perforated propellant cylindrical grains, 125 mm containing Nitrocellulose (NC) and Diphenylamine (DPA).

These propellants are used for **125mm (HEAT and HE)** gun fitted on **T-72 & T-80**.

» 125MM APFSDS

Single base one perforated tubular propellant (15/1-240, 15/1-352) containing Nitrocellulose (NC) and Diphenylamine (DPA) and seven perforated 12/7 VA.

This propellant is used for **125mm anti-tank** ammunition.



SPECIFICATIONS

TYPE	125 MM (HE & HEAT)		125 MM APFSDS		
FORM	ONE PERFORATED	SEVEN PERFORATED	SEVEN PERFORATED 12/7 VA	TUBULAR 15.1 - 240	TUBULAR 15.1 - 352
Length (mm):	2.6 ± 1	14.2 ± 0.2	13.5 - 15.6	235 - 240	344 - 352
Diameter (mm):	0.7 - 0.9	6.3 - 6.5	-	-	-
Hole Diameter (mm):	0.25 ± 0.05	0.35 ± 0.02	0.50 - 0.70	1.50 - 1.60	1.85 - 2.0
Web Thickness (mm):	-	1.34 ± 0.04	1.15 - 1.25	1.45 - 1.55	1.45 - 1.55
Total Volatiles (%):	0.6 ± 0.2	≤ 2	≤ 4.5	≤ 3.5	≤ 3.5
External Moisture (%):	≤ 1.2	0.6 - 0.8	1 - 1.8	1 - 1.6	1 - 1.6
Residual Solvent (%):	≤ 0.7	0.6	≥ 1	≥ 1.3	≥ 1.3
Stability at 134.5°C (min.):	≥ 40	≥ 60	60	≥ 60	≥ 60

BALLISTIC SPECIFICATIONS

TYPE	125 MM (HE)		125 MM APFSDS
FORM	One Perforated	Seven Perforated	One Perforated
Total Charge (gr):	45	5,200 ± 200	7,800
Velocity (m/s):	-	850 ± 15	1,700
Max. Pressure (kg/cm ²):	-	≤ 3,500	≤ 5,200



152 MM REDUCED CHARGE

PIC: 000208

DESCRIPTION

Single base seven perforated propellant cylindrical grains, category 9/7 is a kind of single base propellant containing Nitrocellulose (**NC**) and Diphenylamine (**DPA**).

This propellant plus increments of one perforated granular propellant type 4/1 and tubular type 8/1 which is used as reduced charge for **152 mm** gun type **D-20**.



PROPELLANT AND HIGH EXPLOSIVE

SPECIFICATIONS

TYPE	152 MM REDUCED CHARGE		
	9/7	4/1	8/1
FORM			
Length (mm):	11.5 - 12.6	5.5 - 7.5	395 - 410
Hole Diameter (mm):	0.4 - 0.6	0.25 - 0.35	2.6 - 3.0
Web Thickness (mm):	0.95 - 1.10	0.4 - 0.8	0.70 - 0.85
Total Volatiles (%):	≤ 4.5	≤ 4.0	≤ 3.5
External Moisture (%):	1.0 - 1.8	1.0 - 1.8	1.0 - 1.8
Residual Solvent (%):	≥ 0.9	≥ 0.6	≥ 0.9
Stability at 134.5°C (min.):	≥ 40	≥ 40	≥ 40
Weight of Charge (gr):	2,650	1,175	100
Mean Muzzle Velocity (m/s):	511 ± 5	-	-
Average Max. Pressure (kg/cm ²):	2,100	-	-



130 MM - REDUCED CHARGE

PIC: 000209

DESCRIPTION

Single based one & seven perforated propellant cylindrical grains, category 9/7 containing Nitrocellulose (**NC**) and diphenylamine (**DPA**).

This propellant plus increment of one perforated tubular propellant type **12/1-72** is used as reduced charge for **130mm Gun** type **M-46** and type **59**.



SPECIFICATIONS

TYPE	9/7	12/1-72
Length (mm):	11.5 - 12.6	690 - 720
Hole Diameter (mm):	0.4 - 0.6	2.4 - 2.8
Web Thickness (mm):	0.95 - 1.10	1.1 - 1.25
Total Volatiles (%):	≤ 4.5	≤ 4.5
External Moisture (%):	1.0 - 1.8	1.0 - 1.8
Residual Solvent (%):	≥ 0.9	≥ 1.2
Stability at 134.5°C (min.):	≥ 40	≥ 40

BALLISTIC SPECIFICATIONS

TYPE	No.2	No.3	No.4
Charge Weight for propellant type 9/7 (kg):	5.7	4.45	3.20
Charge Weight for propellant type 12/1-72 (kg):	0.7	0.7	0.7
Initial velocity (m/s):	705	617 - 623	525
Initial velocity probable error (m/s):	≤ 1.5	-	≤ 1.5
Average Maximum Bore Pressure (Mpa):	≤ 264.8	-	-



155 MM HOWITZER

PIC: 000210

DESCRIPTION

Single base one & seven perforated propellant cylindrical grains, 155 mm Howitzer containing Nitrocellulose(**NC**) and diphenylamine (**DPA**) and dinitrotoluene (**DNT**) and dibutyl Phthalate(**DBP**) and Potassium Sulfate.

One perforated is used in **zone 1-5** and seven perforated in **zone 3-7** propellant charge.



SPECIFICATIONS

TYPE	ONE PERFORATED	SEVEN PERFORATED
Length (mm):	7.2	9 ± 1
Diameter (mm):	1.2 - 1.4	4.8 ± 0.2
Hole Diameter (mm):	0.4 ± 0.05	0.9 ± 0.05
Web Thickness (mm):	-	-
Total Volatiles (%):	1.2 ± 0.2	1.2 ± 0.2
External Moisture (%):	0.6 ± 0.2	0.6 ± 0.2
Residual Solvent (%):	≤ 0.8	≤ 1.2
Stability at 134.5°C (min.):	≥ 40	≥ 40

BALLISTIC SPECIFICATIONS

TYPE	ONE PERFORATED	SEVEN PERFORATED
Total Charge (gr):	2,500	6,200
Velocity (m/s):	375 ± 5	564 ± 5
Max. Pressure (kg/cm ²):	1,265 - 1,968	2,250 - 2,672

PROPELLANT AND HIGH EXPLOSIVE



2

203 MM HOWITZER SINGLE BASE PROPELLANT

PIC: 000211

DESCRIPTION

Single base one & seven perforated propellant cylindrical grains, **203 mm** containing Nitrocellulose (**NC**) and Diphenylamine (**DPA**) and Potassium nitrate and Dibutyl Phthalate (**DBP**).

This propellant is used in **zone 5-7** for **203 mm Howitzer**.



SPECIFICATIONS

TYPE	SEVEN PERFORATED
Length (mm):	14 ± 0.3
Diameter (mm):	6.3 ± 0.2
Hole Diamete (mm):	0.8 - 1
Web Thickness (mm):	0.94
Total Volatiles (%):	1.3 max
External Moisture (%):	≤ 0.7+0.2
Residual Solvent (%):	≤ 0.5
Stability at 134.5°C (min.):	≥ 60

BALLISTIC SPECIFICATIONS

TYPE	SEVEN PERFORATED
Total Charge (gr):	12,800 - 13,000
Velocity (m/s):	595 ± 8
Max. Pressure (kg/cm ²):	2,400 - 2,600



DOUBLE BASE PROPELLANT



DESCRIPTION

Double base powder consist of Nitrocellulose (Smokeless) as a base and Nitroglycerin as a plasticizer. Production process for double base powder normally is as solventless process. The advantage of this

process is the safe production of the so-called wet paste, which is prepared underwater by mixing water-wet Nitrocellulose, Nitroglycerine and other water insoluble ingredients. The resulting Wet paste is fairst

manufactured and dewatered. Incorporation of water soluble ingredients takes place by Kneading. Subsequently the dough is taken to the rolling operations, extruded, dried, annealed and further finished.

» BALL PROPELLANT

Ball powder is manufactured in the form of small spheres and it is containing no visible foreign matter.

Smokeless powder is black ball grains. They are used for loading bullet of rifle and machingun with calibre as fallow:

Their bulk density is about 0.9-1.025

gr/cm³.

Ball powder is intended to function by burning, so it must be protected against accidental exposure to flame, sparks or high tempratures.

For these reasons it is desirable that storage enclosures be made of insularing materials to protect the

powder from external heat sources.

One ball powder begins to burn it will normally continue to burn and generate gas pressure untill it is consumed.



2

5.56 X 45 MM (WC 844) AND 6.68 MM (WIN 270)

PIC: 000212

DESCRIPTION

» PERFORMANCE ADVANTAGE:

- Superior chemical stability
- Charging uniformity
- Increased propellant charge
- Superior ballistic stability
- Improves barrel life

SPECIFICATIONS

TYPE	5.56x45	6.68(Win270)
Bulk Density (gr/lit):	≥ 930	945 - 1025
Average Grains Diameter "AGD"(mm):	0.550 ± .05	0.55 ± 0.05
Stability at 120°C (min.):	Min. 60	Min. 60
Heat of Combustion (cal/gr):	Max. 1025	950 ± 50



BALLISTIC SPECIFICATIONS

TYPE	5.56x45	6.68 (Win270)
Total Charge (gr):	1.5 - 1.7	2.7 - 3
Velocity (m/s):	965 ± 6	876 ± 5
Max. Pressure (kg/cm ²):	≤ 3,870	≤ 3,300



7.62 X 39 MM & 7.62 X 51 MM

PIC: 000213

DESCRIPTION

» PERFORMANCE ADVANTAGE:

- Superior chemical stability
- Charging uniformity
- Increased propellant charge
- Superior ballistic stability
- Improves barrel life

SPECIFICATIONS

TYPE	7.62 x 39	7.62 x 51(WC846)
Bulk Density (gr/lit):	≥ 910	≥ 910
Average Grains Diameter "AGD"(mm):	0.550 ± .05	0.650 ± 0.05
Stability at 120°C (min.):	Min. 40	Min. 40
Heat of Combustion (cal/gr):	≤ 1,025	≤ 950



BALLISTIC SPECIFICATIONS

TYPE	7.62x39	7.62x51(WC846)
Total Charge (gr):	1.6 - 1.8	2.5 - 3
Velocity (m/s):	716 ± 10	838 ± 5
Max. Pressure (kg/cm ²):	3,180	3,660





7.62×54 MM & 9 MM (WC 230)

PIC: 000214

DESCRIPTION

» PERFORMANCE ADVANTAGES :

- Superior chemical stability
- Charging uniformity
- Increased propellant charge
- Superior ballistic stability
- Improved barrel life

SPECIFICATIONS

TYPE	7.62x54	9 (WC230)
Bulk Density (gr/lit):	≥900	850 ± 10
Average Grains Diameter "AGD"(mm):	0.65 ± 0.05	0.38 ± 0.05
Stability at 120°C (min.):	Min. 40	Min. 40
Heat of Combustion (cal/gr):	≤950	≤950



PROPELLANT AND HIGH EXPLOSIVE

BALLISTIC SPECIFICATIONS

TYPE	7.62x54	9 (WC230)
Total Charge (gr):	2.8 - 3.2	0.23 - 0.25
Velocity (m/s):	785 ± 7	380 ± 10
Max. Pressure (kg/cm²):	3,180	2,100



12.7 X 108 MM & 20 X 139 MM

PIC: 000215

DESCRIPTION

» PERFORMANCE ADVANTAGES :

- Superior chemical stability
- Charging uniformity
- Increased propellant charge
- Superior ballistic stability
- Improved barrel life

SPECIFICATIONS

TYPE	12.7x108	20x139 (WC 870)
Bulk Density (gr/lit):	900 - 1,000	900
Average Grains Diameter "AGD"(mm):	0.85 ± 0.1	0.85 ± 0.1
Stability at 120°C (min.):	60	60
Heat of Combustion (cal/gr):	950 ± 50	1000 ± 20

BALLISTIC SPECIFICATIONS

TYPE	12.7x108	20x139 (WC 230)
Total Charge (gr):	16 - 18	Appr 37 - 40
Velocity (m/s):	820 ± 10	1,030 ± 8
Max. Pressure (kg/cm²):	3,200	3,867





2

130 & 152 MM FULL CHARGE

PIC: 000216

DESCRIPTION

» 130 MM FULL CHARGE

One perforated tubular propellant, category 23/1-37 is a kind of double base propellant mainly containing Nitrocellulose(NC) and Nitroglycerine(NG).

This propellant is used as full charge for **130mm Cannon type M-46 and type 59.**

» 152 MM FULL CHARGE

One perforated tubular propellant, is a kind of double base propellant mainly containing Nitrocellulose (NC) and Nitroglycerine (NG).

This propellant plus auxiliary charge is used as full charge for **152 mm D-20 HOWITZER.**



SPECIFICATIONS

TYPE	130 MM	152 MM
Length (mm):	362 - 370	282 - 290
Hole Diameter (mm):	2.3 - 2.6	1.3 - 1.6
Web Thickness (mm):	2.2 - 2.35	1.65 - 1.75
External Diameter (mm):	6.7 - 7.3	4.6 - 5.0
Water Content (%):	≤ 0.7	≤ 0.7
Heat of Explosion (cal/g):	765 ± 10	765 ± 10
Stability at 120°C (min.):	≥ 40	≥ 40
Weight of Charge (gr):	12,900	8,150
Weight of Auxiliary Charge (gr):	-	2,750
Mean Muzzle Velocity (m/s):	930	655
Average Chamber Pressure (kg/cm ²):	≤ 3,150	≤ 2,350



100 MM ANTI AIRCRAFT

PIC: 000217

DESCRIPTION

One perforated tubular propellant, category 18/1-26 is a kind of double base propellant mainly containing Nitrocellulose (**NC**) and Nitroglycerine (**NG**). This propellant is used as full charge for **100 mm Cannon** type **KS-19**.

SPECIFICATIONS

TYPE	100 MM
Lenght (mm):	252 - 260
Hole Diameter (mm):	2.0 - 2.2
Web Thickness (mm):	1.70 - 1.85
External Diameter (mm):	5.4 - 5.9
Water Content (%):	≤ 0.7
Heat of Explosion (cal/g):	765 ± 10
Stability at 120°C (min.):	≥ 40
Weight of Charge (gr):	5,500
Mean Muzzle Velocity (m/s):	900
Average Max. Pressure (kg/cm ²):	≤ 3,000



PROPELLANT AND HIGH EXPLOSIVE

BALLISTITE

DESCRIPTION

Ballistite is a double base propellant used as a rocket propellant. it is composed of two explosive substances Nitrocellulose and Nitroglycerine, blended together with centralite, wich acts as a stabilizer.

It burns with a considerable amount of flash and smoke, and generates a great volume of gas.

Ballistite burns progressively, but at a rate dependent upon the composition and physical characteristics of the powder grain, the temprature of the powder grain before ignition and the pressure during reaction. it is produced in various shapes to fit the rocket motor hosing.

Due to high burning tempratures this powder cause erosin of rifled barrels and therefor they are very suitable for use from smooth-bore barrels. Primarily for mortars **60,81** and **120** mm for ignition cartridges and increment charges.

The ballistites are used for artillery ammunition also particularly with short barrels.





2

60 / 81 / 120MM SPLIT INCREMENT CHARGE

PIC: 000218

DESCRIPTION

This increment charge is fired by all kind of 60/81 and 120 mm medium pressure mortars. It is a kind of double base propellant containing NC, NG, plasticizer and stabilizer.

» PACKING SPLIT RING:

It is charged in Viscose rayon ring shape bag, and each set is packed inside **PE** bags.

» PACKING FLAKE:

It is charged in horse shoe celluloid and each set is packed inside **PE** bags.



SPECIFICATIONS

TYPE	60 MM		81 MM		120 MM	
FORM	SPLIT RING	FLAKE	SPLIT RING	FLAKE	SPLIT RING	FLAKE
Dimension (mm):	-	1.5 x 1.5	-	2.5 x 2.5	-	5 x 5
Inner Diameter (mm):	24.3 ± 0.5	-	32.2 ± 0.5	-	55 ± 0.5	-
Outer Diameter (mm):	50 ± 1	-	60.5 ± 0.5	-	105 ± 1	-
Web Thickness (mm):	0.08	0.1 - 0.2	0.08	0.1 - 0.2	0.08	0.08
Moisture (%):	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5
Heat of Combustion (cal/gr):	1,250 ± 20	1,250 ± 20	1,250 ± 20	1,250 ± 20	1,250 ± 20	1,250 ± 20
Stability at 120°C (min.):	40	40	40	40	40	40

BALLISTIC SPECIFICATIONS

TYPE	60 MM		81 MM	120 MM
FORM	SPLIT RING	FLAKE		
Total Charge (gr):	18 (ch 3)	18 (ch 3)	100 (ch 6)	360 (ch 9)
Velocity (m/s):	198 ± 3	198 ± 3	286 ± 3	310 ± 4
Max. Pressure (kg/cm ²):	≤ 350	≤ 350	790	900





60 / 81 / 120 MM IGNITION CARTRIDGE

PIC: 000219

DESCRIPTION

They are used for initial ignition that fires increment charges. It is a kind of double base propellant containing NC, NG, plasticizer and stabilizer.

SPECIFICATIONS

TYPE	60	81	120
Form	Flake	Flake	Flake
Dimension (mm):	1.5 x 1.5	2 x 2	2 x 2
Web Thickness (mm):	0.1 - 0.2	0.1 - 0.2	0.1 - 0.2
Moisture (%):	< 0.5	< 0.5	< 0.5
Heat of Combustion (cal/gr):	1,300 ± 20	1,300 ± 20	1,300 ± 20
Stability at 120°C (min.):	40	40	40

BALLISTIC SPECIFICATIONS

TYPE	60	81	120
Total Charge (gr):	3	9	30
Velocity (m/s):	89 ± 2	68 ± 2	80 ± 2



PROPELLANT AND HIGH EXPLOSIVE

START IGNITION CHARGE RPG7

PIC: 000220

DESCRIPTION

This charge is fired by 40mm anti-tank rocket projectiles. It is a kind of double base propellant containing NC, NG, plasticizer and stabilizer..

SPECIFICATIONS

TYPE	RPG7 CHARGE
Form:	Strip
Dimension (mm):	225 ± 1.5
Web Thickness (mm):	0.45 ± 0.04
Moisture (%):	< 0.8
Heat of Combustion (cal/gr):	1,240 ± 10
Stability at 120°C (min.):	40
Charge (gr):	130 ± 2
Max. Velocity (m/s):	
At - 30°C	112 ± 2
At - 20°C	120 ± 2
At - 60°C	125 ± 2
Min. Pressure (kg/cm²): at-30°C	550
Max. Pressure (kg/cm²): at-20°C	900
Max. Pressure (kg/cm²): at-60°C	1,100





TRIPLE BASE PROPELLANT NAVAL 4.5 INCH MK8 & 105 MM (HEAT)

PIC: 000221

DESCRIPTION

Seven perforated cylindrical propellant is a kind of triple base powder mainly containing Nitrocellulose (NC), Nitroglycerine (NG) and Nitroguanidine (NGu).

This propellant is used for **4.5 inch Navy gun MK 8.**

Seven perforated cylindrical propellant type **M30-15/7** is a kind of triple base propellant mainly containing Nitrocellulose (NC), Nitroglycerine (NG) and Nitroguanidine (NGu).

This propellant is used for anti tank ammunition (**HEAT**) in **105 mm gun.**

SPECIFICATIONS

TYPE	NAVAL MK8
Length (mm):	11 - 12
Hole Diameter (mm):	0.58 - 0.64
Web Thickness (mm):	0.75 - 0.9
External Diameter (mm):	4.8 - 5.2
Stability at 120(min.):	≥ 40
Heat of Explosion (cal/gr):	730 - 760
Weight of Charge (gr):	~ 7,000
Mean Muzzle Velocity (m/s):	870
Average Chamber Pressure (kgf/cm ²):	3,000



SPECIFICATIONS

TYPE	105 MM (HEAT)
Average Length (mm):	~ 16.5
Average Hole Diameter (mm):	~ 0.5
Average Web Thickness (mm):	~ 1.5
Average External Diameter (mm):	~ 7.5
Length-Diameter Ratio (L/D):	2.10 - 2.50
Outer Dia.-Hole Dia. ratio (D/d):	10 - 15
Stability at 120°C (min.):	≥ 40
Heat of Explosion (cal/g):	953 - 1,001
Weight of Charge (gr):	~ 5,100
Mean Muzzle Velocity (m/s):	1173.5
Average Max. Pressure (MPa):	≤ 436.40





2

155 MM B.B TRIPLE BASE PROPELLANT

PIC: 000222

DESCRIPTION

Nineteen perforated cylindrical propellant type **M30A1**, is a kind of triple base powder mainly containing Nitrocellulose (NC), Nitroglycerine (NG) and Nitroguanidine (NGu).

This triple-base propellant is used in zone 10 for Gun **155mm GC-45 HOWITZER**



PROPELLANT AND HIGH EXPLOSIVE

SPECIFICATIONS

TYPE	M30-A1
Average Length (mm):	16.5 - 17.5
Average Hole Diameter (mm):	0.30 - 0.38
Average Inner Web Thickness (mm):	2.20 - 2.50
Average Outer Web Thickness (mm):	2.50 - 2.90
Average External Diameter (mm):	16.0 - 16.6
Heat of Explosion (cal/g):	940 - 980
Weight of Charge (gr):	~ 16,000
Mean Muzzle Velocity (m/s):	897
Mean Chamber Pressure (MPa):	≤ 348





HIGH EXPLOSIVE

2



DESCRIPTION

Military high explosives are safe to handle, have a long shelf life, their energy density is high and the propagation of the explosion (detonation reaction) is very fast. The energetic compounds are combined with polymers and other substances to match the properties as required. To initiate a detonation, a shock wave is needed.

This is accomplished with a small quantity of a primary explosive.

Modern, extremely insensitive formulations have been developed in most countries to eliminate accidents with ammunition.

Apart from insensitive energetic compounds, new inert or energetic binder systems will be introduced to improve the vulnerability. The search for new energetic materials with highest performance characteristics is a never-ending task.

These products include insensitive explosives, plastic explosives, plastic explosives and castable composition based on **PETN**, **RDX**, **HMX** and aluminized composition.

In the previous pages we explained some of formulations and we can also produce some other combinations, such as desensitized **PETN** (pentastit), **CH-6** and etc.

COMPOSITION A

DESCRIPTION

Composition A is wax-coated granular form of RDX and a phlagmetizing wax. Five different types of composition A have been developed and designated **A-1**, **A-2**, **A-3**, **A-4**, and **A-5**.

Composition A is used as bursting charges in **Navy 2.5** and **5-Inch** rockets, as well as in land mines, shaped charge for destruction and generally in pressed charges boosters, missile **warhead & RPG** in ammunition primer.

We produce the below compounds of composition A.

PIC: 000223



SPECIFICATIONS

TYPE	COMPOSITION A3	COMPOSITION A4	COMPOSITION (RDX.Stearic Acid & Graphite)	COMPOSITION (RDX.Wax & Graphite)
Appearance	Yellow to brown - yellow granulate free from dust and mechanical impurities	Yellow to brown - yellow free from dust and mechanical impurities	Gray - black powder free from mechanical impurities	
RDX (%) :	91 ± 0.7	97 ± 0.5	Min. 98	94 ± 1
WAX (%) :	9 ± 0.7	3 ± 0.5	-	5 ± 1
Stearic Acid (%) :	-	-	1.6 - 1.8	-
Graphite (%) :	-	-	0.2 - 0.4	1 ± 0.3
Volatile Matter (%) :	-	-	Max. 0.05	-



COMPOSITION B HEXOTOL-HEXOLITE

PIC: 000224

DESCRIPTION

Composition B is **RDX** mixed with moldable amounts of TNT. Sometimes other agents are added to make it more stable. Composition B is used primarily in army rockets and land mines, as well as in other projectiles, Airplane bombs, Torpedoes, Anti-tank weapon, molding charges such as missile warhead and munition's charges.



SPECIFICATIONS

TYPE	COMPOSITION B	COMPOSITION B4
Appearance	Yellow to brown - yellow granulate free from dust and mechanical impurities	Yellow to brown - yellow granulate free from dust and mechanical impurities
RDX (%) :	59.5 ± 2	60 ± 2
TNT (%) :	39.5 ± 2.3	39.5 ± 2
WAX (%) :	1 ± 0.3	-
Moisture (%) :	Max. 0.2	-
Acidity as CH₃COOH (%) :	Max. 0.02	-
Calcium Silicate (%) :	-	0.5 ± 0.15

COMPOSITION C4

PIC: 000225

DESCRIPTION

This is a military plastic explosive, consisting of cyclonite (RDX) and plasticizer, which itself may or may not be explosive.

It is used for explosive welding of metals, casting metals, sweeping mine fields, reactive armors, and hollow charge of destruction.



SPECIFICATIONS

TYPE	COMPOSITION C4
Appearance	Pasty material
RDX (%) :	90.5 ± 0.7
Polyisobutylene Binder PIB+Oil+DOA(%) :	9.5 ± 0.7
Plasticity (%) :	Max. 0.018





TNT

PIC: 000226

DESCRIPTION

Military grade TNT is widely used as a bursting charge for high explosive shell, aircraft bombs, grenades, naval mines, torpedoes, etc. Either alone or mixed in given proportions with other explosives such as hexogen to form hexotol (composition B, RDX), hexogen and aluminium to form hexotonal (torpex), PETN to form pentolite, ammonium nitrate to form amatol, etc.

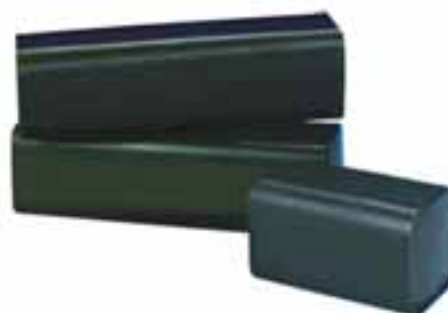
It is also used in boosters, detonators, demolition charges, land mines and other blasting material and even as a constituent of some kinds of smokeless powders.

SPECIFICATIONS

TYPE	TNT
Appearance:	Flake of light yellow to yellow orange colour
Solidification Point (°C):	Min. 80.20
Moisture (%):	Max. 0.1
Acidity as H₂SO₄ (%):	Max. 0.005
Insoluble Material (%):	Max. 0.05
Ash Content (%):	Max. 0.05
Alkalinity:	None



» TNT BLOCKS



SPECIFICATIONS

TYPE	200 G FLAKE 80.2	500 G FLAKE 80.2
Without Case (mm):	45 x 45 x 70	45 x 45 x 167
With Case (mm):	49 x 49 x 89	49 x 49 x 178
Dimensions of Detonator Housing (mm):	Dia 7.5 Length 60	Dia 7.5 Length 70
Insensitivity Figure:	F	F
Detonator Type:	No.8	No.8
Detonation Rate(m/s):	6,900	6,900
Power (kj/kg):	4,310	4,310
Storage Temp (°C):	0 to 35	0 to 35





HEXOGEN (RDX)

PIC: 000227

DESCRIPTION

Hexogen (RDX) is a white crystalline powder. Production process for hexogen among the manufactures in the world is different. Our process is according to **Batchman method**.

Phlegmatized and pressed RDX is used as a highly brisant material for the manufacture of boosters and hollow charges.

Non phlegmatized RDX in combination with TNT is also used as a parable mixture for hollow charges and brisant explosive charges, (composition B), mixture of cyclonite with aluminum powder are used as torpedo charges (hexotonal, torpex, trialent).

Sometime this material may be used as an additive in the manufacture of smokeless powders in manufacturing explosive charges which are required to have a certain mechanical strength or rubber elastic toughness, cyclonite is incorporated into curable plastic materials such as polyurethanes, polybutadiene or polysulfide and is poured into molds (plastic explosive).



PROPELLANT AND HIGH EXPLOSIVE

PETN

PIC: 000228

DESCRIPTION

PETN is one of the most powerful and most brisant explosives, its stability is satisfactory and its sensitivity is not extreme.

It is used in high-efficiency blasting-cap fillings and detonation cords. If phlegmatized with a small amount of wax and pressed, it may be used to produce boosters and fillings for smaller caliber projectiles. **PETN** can be incorporate into gelatinous, industrial explosives(e.g. for seismic prospecting).

It is also used in fuse, explosive cap of primers, pentolite, inductive charges, molding charges, primary demolition of naval mines, small bullets, grenades, mixture with Nitrocellulose or synthetic rubbers for production of plastic explosives.



SPECIFICATIONS

TYPE	PETN
Appearance:	White crystalline
Melting Point (°C):	Min. 140
Insoluble in Acetone (%):	Max. 0.1
Acidity as HNO ₃ (%):	Max. 0.01
Alkalinity (%):	Max. 0.01
Vacuum Stability at 120°C (ml):	Max. 5





BLACK POWDER

PIC: 000229

DESCRIPTION

Black Powder is a mixture of potassium nitrate, sulphur and charcoal processed to different grain sizes. It is used for ignition charge, primer charge, propelling charge and blasting charges as explosives.

It is commonly used for fire works and crackers. To grain size it is classified as seven grades according i.e. No. 1, 2, 3, 4, 5, 6 and 7

● APPEARANCE:

Glossy, uniform grains with greyish black to black color.



SPECIFICATIONS

CLASS NO.	GRAIN DIAMETER (MM)
1	2 - 4
2	0.5 - 1.2
3	0.3 Max
4	0.5 - 1.0
5	0.4 - 1.0
6	0.6 - 1.0
7	0.2 - 0.5

NOTE: It is produced in different sizes by applying the granulation change based on the user's request.

SPECIFICATIONS

TYPE	BLACK POWDER
Potassium Nitrate (%):	75 ± 1.0
Sulfur (%):	10 ± 1.0
Charcoal (%):	15 ± 1.0
Moisture (%):	1.0 Max
Specific Gravity (gr/cm ³):	1.26 - 1.8
Bulk Density (gr/cm ³):	0.9 Max



DYNAMITE

PIC: 000230

DESCRIPTION

This Dynamite is as safe as salt-sheated coal mine explosive used for blasting operations in coal mines with high level of marsh gas and coal dust.

PACKING

Shell of cartridge: Waxed paper
Middle packing: Polyethylene film bag
Net weight of package: 20 Kg

SHELF LIFE

4 months

SPECIFICATIONS

CARTRIDGE DIAMETER (MM)	CARTRIDGE LENGTH (MM)	CARTRIDGE WEIGHT (GR.)	DENSITY (GR./CM ³)	TRAUZL TEST (ML)	BRISANCE (MM)	DETONATION TRANSMISSION (CM)
G2	30	195	1.2 - 1.3	>150	>8	>5
G3	30	195	1.2 - 1.3	>150	>8	>5



CARTRIDGE EMULSION EXPLOSIVE

PIC: 000231

DESCRIPTION

Cartridge emulsion explosive is more reliable and safer than gelatine dynamite and has excellent water resistance.

PACKING

PE cartridge 27mm, 30mm, 35mm, 45mm, 50mm, 65mm.

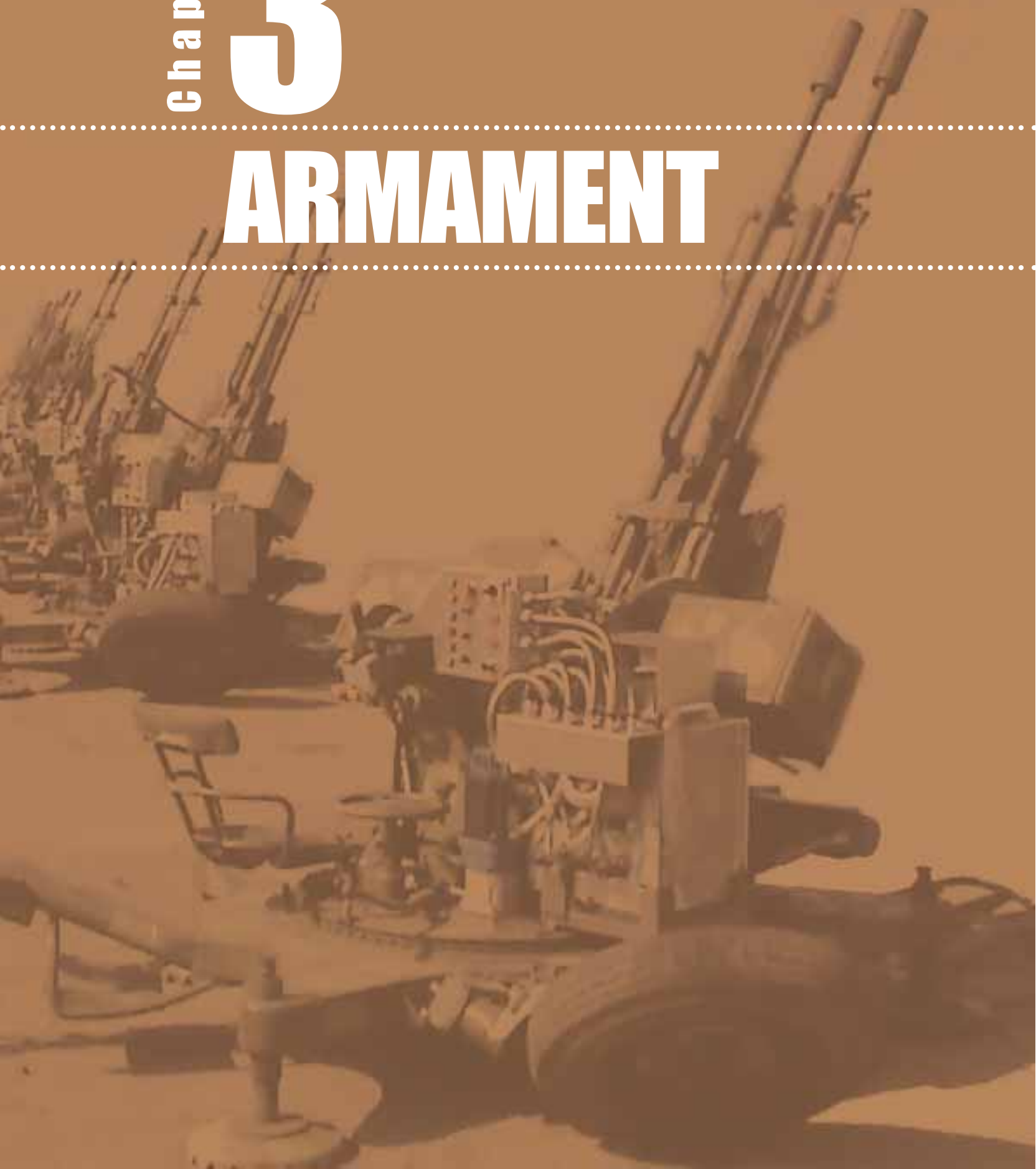
SPECIFICATIONS

TYPE	CARTRIDGE EMULSION EXPLOSIVE
Density (gr/cm ³):	1.15 - 1.30
Oxygen Balance (gr/100gr):	- 4
Velocity of Detonation (Diameter=30mm) (m/s):	> 5,000
Theoretical Weight Energy (Cal/g):	1,300
Theoretical Bulk Energy (Cal/cm ³):	1,560
Gas Volume (lit/kg):	700
Relative Strength:	1
Cap Sensitivity:	Cap no. 8
Impact Sensitivity (J):	> 50
Friction Sensitivity (N):	> 353
Shock Sensitivity (air gap sensitivity) (cm):	> 2
Brisance (mm):	16



chapter 3

ARMAMENT





PREVIEW

The previous record of fire arms production in M.o.D had begun centuries ago but mass production began on 1934 in an old factory of weapon manufacturing (Gorkhane). Since then new factory were established in 1937 for producing guns and machine guns and had been developed and completed through the years, and in 1966 the production line for G3 guns and machine guns was established and by the end of 1977 a considerable amount of various mentioned weapons and related equipments were produced.

These factories have been considerably developed for production and related activities, and with utilization of modern technology, it has been able to present its military products with excellent quality and increase the variety.

In part of medium and large caliber gun, Armament Group as one of the most noticeable part of Iran Defence Industries is able to manufacture different type of standard mortars, rocket launchers and variety of propelled, self-propelled Howitzers and many kind of barrels.

The Armament Industries with the skillful experts, competent staff and valued experience during the last two decades in designing, manufacturing, rebuilding and upgrading weapons has the honour to fulfil customer's enquiry.

Receiving international awards and certificates on high quality and performances such as ISO 9001_2008, ISO 17025_2005, ISO 10002_2004 & OHSAS 18001_2007, ISO 14001_2004 and military standards have enabled Iran to enter the worldwide competition.





CHAPTER 3



ITEM	TABLE OF CONTENT	PAGE
1	ZOAF-PC9 PISTOL	82
2	T9K-LIGHT SUBMACHINE GUN	83
3	T9 & T9S AUTOMATIC WEAPON	84
4	AK-47 (KL-7.62MM) ASSAULT RIFLE	85
5	G3 RIFLE AUTOMATIC WEAPON	86
6	MGA3 MACHINE GUN	87
7	SVD SNIPER RIFLE	88
8	PKM (TYPE T-80) MACHINE GUN	89
9	DOSKA 12.7MM MACHINE GUN	90
10	12.7MM SNIPER RIFLE	90
11	38MM MULTI PURPOSE GUN AND PISTOL	91
12	LP 26.5 & LP 26.5 W.T	91
13	40MM GRENADE LAUNCHER	92
14	40MM AUTOMATIC GRENADE LAUNCHER	92
15	RPG-7 ANTI TANK ROCKET LAUNCHER	93
16	RPG7 TRAINING AID	93
17	23 MM SINGLE BARREL ANTI-AIRCRAFT MACHINE GUN	94
18	23 MM TWIN BARREL ANTI-AIRCRAFT MACHINE GUN	95
19	MESBAH - AIR DEFENCE SYSTEM	96
20	MESBAH 1 - AIR DEFENCE SYSTEM	97
21	SAMAVAT - 35 MM AIR DEFENCE SYSTEM	98
22	60MM MORTAR LAUNCHER STANDARD & LONG RANGE	99
23	60MM COMMANDO MORTAR LAUNCHER	99
24	81MM STANDARD MORTAR LAUNCHER	100
25	81MM SHORT BARREL MORTAR LAUNCHER	100
26	120 MM MORTAR LAUNCHER	101
27	73 MM SPG-9 - RECOILLESS ANTI TANK GUN	102
28	106 MM - RECOILLESS ANTI TANK GUN	103
29	107 MM SINGLE & DOUBLE BARREL ROCKET LAUNCHER	104
30	107 MM MULTI ROCKET LAUNCHER	105
31	107 MM NAVAL 11 & 19 BARREL ROCKET LAUNCHER	106
32	122 MM ROCKET LAUNCHER	107
33	122 MM & 155 MM HOWITZER	108
34	40 MM MARINE GUN	109
35	100 MM ANTI AIRCRAFT GUN	109
36	FADJR-27-76MM NAVAL GUN	110
37	240 MM NAVAL ROCKET LAUNCHER	111
38	CANNON BARRELS	112
39	INFANTRY MOBILE UNIT	113
40	WIRELESS FIRE CONTROL	113



ZOAF - PC9 PISTOL

PIC: 000232

DESCRIPTION

The **PC9** combat pistol has been designed with advanced technology specifically for military and disciplinary forces. This pistol works with short recoil and has double action trigger, automatic pin safety lock, decocking lever with external slide catch.

This pistol is automatically loaded by magazine and after firing of final shot the slide is stopped.

ADVANTAGES OF PC9 PISTOL:

- Minimal reaction time
- Double action trigger
- Simple operation
- High accuracy
- Low weight



SPECIFICATIONS

TYPE	PC9 PISTOL
Calibre (mm):	9
Length, Overall (mm):	196
Height, Overall (mm):	139
Barrel Length (mm):	112
Number of Grooves:	6 Right Hand
Weight of Weapon without Magazine (gr):	770
Weight of Empty Magazine (gr):	95
Capacity of Magazine (rds):	15
Muzzle Energy (J):	500
Number in a Case (pcs):	1
Total weight Related Attachment (kg):	1.62





T9K-LIGHT SUBMACHINE GUN

PIC: 000233

DESCRIPTION

The light submachine gun **T9K**, cal. 9 mm is an automatic weapon and can be used for either single or bursts shots selectively from all position.

The weapon is loaded directly with pressure of powder gas.

This weapon is specially used in personal protection guards.



T9K

SPECIFICATIONS

TYPE	T9K
Calibre (mm):	9
Number of Rifling Grooves:	6 Right Hand
Weight of Gun without Magazine (kg):	2.03
Weight of Empty Magazine (gr):	175
Length of Weapon (mm):	325
Length of Barrel (mm):	116
Rate of Fire (rounds/min):	900
Muzzle Velocity (m/sec):	375
Muzzle Energy (J):	570
Number in a Case (pcs):	10
Total Weight of the Case (kg):	38



T9K



3

T9 & T9S - AUTOMATIC WEAPON

PIC: 000234

DESCRIPTION

The submachine gun **T9, cal. 9 mm** is an automatic weapon and can be used for either single or bursts shots selectively from all position. The weapon is loaded directly with pressure of powder gas and its barrel cooled with air.

Cartridges are fed from 30 round box type magazine. In order to carry the weapon easily and better usage, it is usually equipped with retractable butt stock, if required it can be replaced with fixed butt stock.



T9



T9-S



T9

SPECIFICATIONS

TYPE	T9	T9S
Caliber (mm):	9	9
Number of Rifling Grooves:	6 Right Hand	6 Right Hand
Weight of Weapon without Magazine (kg):	2.93	2.63
Weight of Gun with Full Magazine (kg):	3.46	3.16
Length . Open Stock (mm):	670	690
Length . Closed Stock (mm):	500	---
Weight Magazine Full (gr):	492	492
Length of Barrel (mm):	225	225
Max. Effective Range (M):	200	200
Muzzle Velocity (m/sec):	400	400
Muzzle Energy (J):	650	650
Rate of Fire (rounds/min):	800	800
Capacity of Magazine (rounds):	30	30
Weight of Empty Magazine (kg):	0.16	0.16
Number in a Case (pcs):	5	5
Total Weight Related Attachment (kgs):	27	26





AK-47 (KL-7.62MM) ASSAULT RIFLE

PIC: 000235

DESCRIPTION

The assault rifle type **AK-47 (KL-7.62 mm)** is a gas-operated rifle and capable both of semi-automatic and automatic fire, and intended for taking out hostile individuals and groups, also can be used in close combat by bayonet and butt stock.

It has become the weapon of many army forces specially suited for the guerrilla warfare in every weather conditions.

The rifle is supplied in various types, such as; fixed butt stock (**KLS**), under folding butt stock (**KLF**) & side folding butt stock (**KLT**).

Also it should be mentioned that this rifle has a multipurpose bayonet.



KLF



KLT



KLS

SPECIFICATIONS

TYPE	AK-47
Calibre (mm):	7.62 x 39
Rate of Fire Theoretical (rds/min):	600
Rate of Fire Practical (rds/min):	90 - 100
Effective Range (m):	300
Final Range(m):	2000
Weight of Empty Magazine (gr):	350
Weight of Cartridge (gr):	16.2
Length of Barrel (mm):	415
Feed:	30 round magazine
Weight of Rifle (with empty magazine)	
Model KLS (kg):	3.800
Model KLF (kg):	3.850
Model KLT (kg):	3.600
Length of Rifle with (extended butt-folded butt)	
Model KLS (mm):	890
Model KLF (mm):	895 - 655
Model KLT (mm):	895 - 690



3

G3 RIFLE - AUTOMATIC WEAPON

PIC: 000236

DESCRIPTION

The **G3 Rifle** is an automatic weapon which can fire single shots or bursts from all firing positions. The **Rifle** is a recoil operated weapon incorporating a delayed roller locked bolt system. Ammunition is fed from a 20-rounds magazine or can be inserted manually. Telescopic sight or infrared scopes can be employed without modifying the rifle. Rifle grenades can be launched with the rifle. This requires a **7.62 × 51 mm** propellant cartridge. A blank attachment permits 7.62 × 51 blank ammunition to be fired.

The **Rifle G3** is available in **two** models:

1. Fixed Butt stock G3 assault Rifle (**G3-A3**)
2. Retractable Butt stock G3 Rifle (**G3-A4**)

SPECIFICATIONS

TYPE	G3-A3 RIFLE
Caliber (mm):	7.62 x 51
Rate of Fire (rds/min.):	500-600
Muzzle Energy (kpm):	290- 300
Effective Range (m):	400
Maximum Range (m):	3750
Number of Grooves:	4 Right Hand
Length of Weapon (mm):	1020
Length of Barrel (mm):	450
Muzzle Velocity (m/sec):	780 - 800
Weight of Weapon without Magazine (kg):	4.3
Weight of Empty Magazine (gr):	284
Weight of Cartridge (gr):	24
Sight Radius (mm):	572



G3-A4

G3-A3

SPECIFICATIONS

TYPE	G3-A4 RIFLE
Length of Weapon: Butt Stock Retracted(mm):	840
Butt Stock Extended (mm):	1020
Weight of Weapon without Magazine (kg):	4.6
Weight of Empty Magazine (gr):	284





MGA3 - MACHINE GUN

PIC: 000237

DESCRIPTION

The **MGA3** is an open fully automatic weapon for sustained firing or firing in bursts. It can be utilized by all services, army, navy, air force for combating point and area targets.

The weapon may be used as "standard machine gun" for different purposes, when used with the bipod it is designated as a "light machine gun" (LMG). When used with the Tripod mount, it is designated as a "heavy machine gun" (HMG).

Special mountings are available for installation on vehicles, armored vehicles and aircrafts.



SPECIFICATIONS

TYPE	MGA3 - MACHINE GUN
Calibre (mm):	7.62
Rate of Fire (rounds/min.):	1000 - 1300
Muzzle Velocity (m/sec):	820
Muzzle Energy (kpm):	300
Effective Range (m):	800
Maximum Range (m):	4000
Number of Grooves (right hand):	4
Weight of Weapon without Bipod (kg):	10.5
Weight of Bipod (kg):	1
Weight of Tripod (kg):	15.800
Weight of Pod (kg):	30.5
Weight of Barrel (kg):	1.800
Length of Weapon (mm):	1225
Length of Barrel (mm):	565
Sight Radius (m):	200-1600





SVD - SNIPER RIFLE

PIC: 000238

DESCRIPTION

The **SVD** is a well-designed, light-weight and well-built precision rifle, generally used in search and destroy missions from a long distance.

The rifle is a semi – automatic, gas-operated and air – cooled arm with 10 round magazine, can be used with various optical and night - vision sights.



SPECIFICATIONS

TYPE	SVD - SNIPER RIFLE
Calibre (mm):	7.62 x 54 R
Muzzle Velocity (m/sec):	800 - 830
Rate of Fire (rds/min.):	30 - 40
Effective Range with Mechanical Sight (m):	1200
Effective Range (optical sight) (m):	1300
Weight with Optical Sight and Butt Check Rest (kg):	3.8
Weight without Optical Sight and Butt Check Rest (kg):	4.5
Magazine Capacity (rd):	10
Ammunition (mm):	7.62 x 54 (special coating)
Length of Rifle (m):	1.22
Length of Barrel (mm):	620
Sight Radius (mm):	585





PKM (TYPE T-80) MACHINE GUN

PIC: 000239

DESCRIPTION

The **PKM** is a gas-operated and light machine gun which is intended for use for regular and guerilla warfares.

The reasons for extensive usages of this rifle in most armed forces are; simple operation, light weight , high accuracy and efficiency.

This machine gun is an infantry support gun and normally fired from it's bipod mount, however it can be installed in tripod and is used in the classic role of a heavy machine gun to provide long-range area fire and cover final protective lines at range of 1000 meters.



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SPECIFICATIONS

TYPE	PKM T-80
Calibre (mm):	7.62 x 54
Muzzle Velocity (m/s):	820 - 850
Rate of Fire (rds/min):	700 - 800
Effective Range (m):	1500
Final Range (m):	3800
Weight (kg):	7.900
Ammunition (mm):	7.62 x 54 (special coating)
Length (mm):	1192
Length of Barrel (mm):	605
Sight Radius (mm):	663
Feed Device:	Belt cartridge
Weight of Weapon with Tripod (kg):	12.700
Capacity of Magazine (pcs):	100 & 200.





DOSHKA - 12.7MM MACHINE GUN

PIC: 000240

DESCRIPTION

12.7x108 mm heavy machine gun, **DOSHKA** is an automatic heavy machine gun which is armed by indirect pressure of powder gas. It has two barrels, main & spare one, which shall be cooled with ambient temperature or water, if necessary. This kind of gun is left-fed with 70 rounds ammunition box. Primary mission of the gun is intended for shooting against ground targets such as HMG parapets, individuals, moving vehicles. etc ... & the secondary, against air targets such as gliders, low - height helicopters and similar.

SPECIFICATIONS

TYPE	DOSHKA
Calibre (mm):	12.7
Muzzle Velocity (m/s):	800
Rate of Fire (rds/min):	540 - 600
Rifling:	8 grooves RH
Total Weight with Tripod (kg):	95.35
Weight of Cradle-rest (kg):	16.45
Weight of the Gun (kg):	33
Weight of Magazine with Unloaded Cartridge Belt (kg):	3.4
Weight of Magazine Loaded with 70 Cartridge (kg):	13.6
Total length (mm):	2030



12.7 MM SNIPER RIFLE

PIC: 000241

DESCRIPTION

The **AM-50** (12.7 mm anti-material) is a single-shot, manual bolt-action, multi purpose, long range sniper rifle for infantry snipers to annihilate individual important targets, light vehicles and pillboxes. This rifle is fitted with a highly efficient muzzle brake and recoil is sufficiently reduced to permit fine firing.

SPECIFICATIONS

TYPE	SNIPER RIFLE
Calibre (mm):	12.7 x 99
Muzzle Velocity (m/s):	850
Rate of Fire (r/m):	3-5
Effective Range (m):	1200
Barrel Length (mm):	913
Weight (kg):	12.2 ± 0.2
Overall Length (mm):	1480
Trigger:	Two stages by pulling force 1.8 - 2.4 kg
Safety:	Two stages press button (Fire - Safe)





3

38MM - MULTI PURPOSE GUN AND PISTOL

PIC: 000242

DESCRIPTION

Signal rifle are applied to shoot gas, illumination, igniting, and smoke rounds. **TPT** is a single-shot launcher rifle, there is no rifling inside the barrel and arming mechanism is break action. Making smoke for war attacks, preventing enemy attacks, illuminating the operation zones at night, signalling, igniting fuel sources & oil wells, direction determination of wind blowing, round recording for artillery shootings, riot controlling, are among the prominent usages of **TPT**.

SPECIFICATIONS

TYPE	MULTI PURPOSE GUN & PISTOL
Calibre (mm):	38
Weight (gr):	2130
Length (mm):	680
Height (mm):	180
Range (m):	150
Kind of Launcher Rounds:	Different type of gas, illumination, igniting & smoke, caliber 38 mm rounds.
Feed Mechanism:	Break action, single shot
Rate of Fire:	5 Rounds per minute, practical



TPT



KPT



A R M A M E N T

LP 26.5 & LP 26.5 W.T

PIC: 000243

DESCRIPTION

These two models of signal pistol are used for specifying of situation of infantry forces and are armed with 26.5 mm illuminating cartridge. The triggered model (LP 26.5) is loaded by folding of barrel and shoots through barrel.

Trigger less model(LP 26.5 W.T) is loaded by bottom and shoots with a stroke on pistol bottom.

SPECIFICATIONS

TYPE	LP 26.5	LP 26.5 W.T
Calibre (mm):	26.5	26.5
weight of Weapon (gr):	900	450
Range (m):	100	100
Length of Weapon (mm):	210	185
Length of Barrel (mm):	150	150



LP 26.5

LP 26.5 W.T





3

40MM GRENADE LAUNCHER

PIC: 000244

DESCRIPTION

The 40 mm grenade launcher is a light weight weapon which can launch one grenade in every turn of arming. Its arming system is hand drawing and it can launch a good variety of low-velocity 40 mm projectiles. Having a 90 degree aiming device, it is capable of being attached on **M16**, **G3** and **AKM** rifles and similar types of weapons. Since its firing mechanism is fully independent, it is possible to use it as a stand – alone weapon or as a pistol.



SPECIFICATIONS

TYPE	GRENADE LAUNCHER
Calibre (mm):	40
Muzzle Velocity (m/s):	71
Max. Range (m):	400
Effective Range (m):	350
Max. Accurate Range (m):	150
Grenade Weight (kg):	1.36
Bullet Weight (gr):	277
Fire Rate (r/m):	5 - 7



40MM AUTOMATIC GRENADE LAUNCHER

PIC: 000245

DESCRIPTION

- Anti tank
- Anti personnel
- Anti equipment
- Installable on all military vehicles, firing boats, warships and logistic helicopters
- Suitable to use in land, air and marine forces
- High rate of firing

SPECIFICATIONS

TYPE	AUTOMATIC GRENADE LAUNCHER
Caliber (mm):	40 x 53
Weight of Weapon without Magazine (kg):	36
Weight of Tripod, Magazine with Ammunition (kg):	44
Capacity of Magazine (grenades):	30
Max. Range (m):	2200
Muzzle Velocity (m/s):	240
Rate of Firing (r/m):	250 - 300
Dimensions (mm):	1120 x 136 x 206





RPG-7 ANTI TANK ROCKET LAUNCHER

PIC: 000246

DESCRIPTION

40 mm anti tank rocket launcher (**RPG-7**) is an effective weapon against tanks, moving targets including armored vehicles, fortifications and artilleries.

This rocket launcher with high precision, light weight and with ability to use under all atmospheric conditions is proved to be a most applicable weapon.

SPECIFICATIONS

TYPE	RPG-7	COMMANDO VERSION
Caliber (mm):	40	40
Effective Range (m):	330	330
Maximum Range (m):	1000	1000
Penetration Depth into Armored Surface (mm):	110	---
Rate of Fire (r/m):	2 upto 4	6
Length of the Rocket Launcher (mm):	950	738
Weight of the Rocket Launcher with the Sight (kg):	6.2	5.28
Weight of the Rocket (projectile) (kg):	2.2	---



COMMANDO VERSION

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RPG-7 TRAINING AID

PIC: 000247

DESCRIPTION

ADVANTAGES OF TRAINING BY SUB - RIFLE

- Lower cost (saving real ammunition)
- Higher safety
- Similar results as if real ammunition is used
- Utilization of smaller fire-field
- No wear-out of barrel of PRG7

SPECIFICATIONS

TYPE	RPG7 SUB-RIFLE
Caliber (mm):	7.62
Maximum Range (m):	400
Length (mm):	910
Diameter (mm):	85
Effective Range (m):	200
Weigth (kg):	3.8

The Sectional cut for Training aid is also provided.





3

23 MM SINGLE BARREL ANTI-AIRCRAFT MACHINE GUN

PIC: 000248

DESCRIPTION

23 MM SINGLE BARREL anti-aircraft machine gun, standing type is a suitable weapon against targets at a distance of 2500m and height of 1500 m.

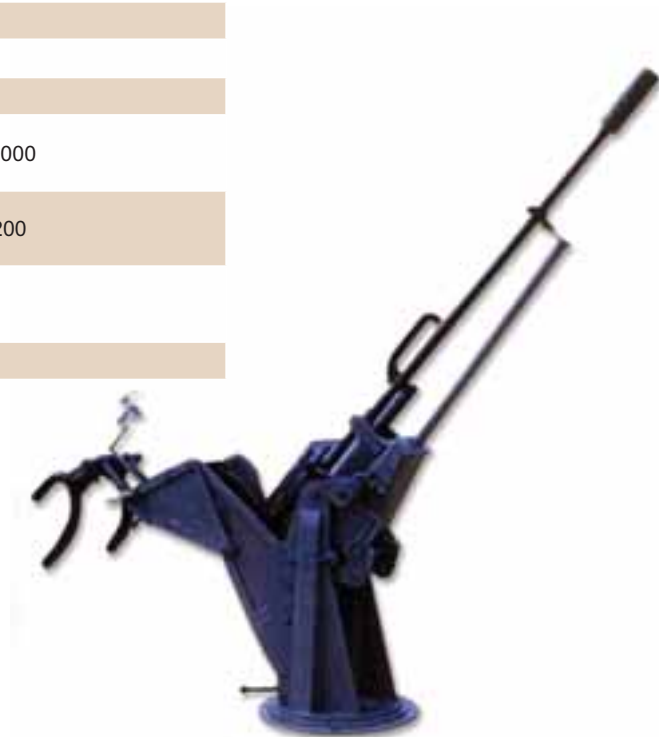
This type of gun has been designed and manufactured for mounting on light & fast boat and is used by coast-guard. Despite strength and high fire power, it has a very delicate design.

Also, this gun is made of anti-corrosive marine steel. The weapon is produced in two versions of sitting and standing types with/without support.



SPECIFICATIONS

TYPE	SINGLE-BARREL AA MACHINE GUN
Caliber of Barrel (mm):	23
Length of Barrel with Fire Shield (mm):	2,010
Length of Barrel without Fire Shield (mm):	1,880
Length of Machine-Gun with Fire Shield (mm):	2,780
Width of Machine-Gun with Cartridge Box (mm):	2,425
Max. Vertical Distance (m):	1,500
Max. Horizontal Distance (m):	2,000
Max. Inclined Distance (m):	2,500
Tempo of Shooting/with Both Machine Guns (Shots per Min.):	800 - 1000
Firing Rate/with Both Machine Guns (Shots per Min.):	Up to 200
Weight of Cartridge Box with a Cartridge Belt Furnished with 50 Cartridges (kg):	35.5
Without Cartridge Box Weight of Machine (kg):	222





23 MM TWIN BARREL ANTI-AIRCRAFT MACHINE GUN

PIC: 000249

DESCRIPTION

23 MM TWIN BARREL anti-aircraft machine gun is a powerful means for shooting at airborne targets to a distance of 2500 m and a height of 1500 m. It is designed as an anti-aircraft defence by airborne landing forces.

The installations constructive characteristic allow it to be used for shooting at ground light-armored target and fire nests to a distance of 2000 m.

Further more the installation can be used for destroying of concentrated living force, in open places or behind light concealments of a field type.

For firing at airborne and ground targets are using cartridges with common incendiary tracer /CIT/ and armour piercing incendiary tracer /APIT/. The feeding of submachine gun is done with metal belts (containing 50 cart. each).

The installation is transported attached to a truck or jeep. It permits also a marching fire. The undercarriage gives a possibility to transport the installation by the truck on the asphalt road with a speed of 70 km/hr and by a jeep with a speed of 40 km/hr.

The trained mounting personnel can set it from field to firing position for 15-20 sec. and from firing to field position for 35-40 sec. The staff consists of 5 persons; commander, gunner, supplier and two persons for loading (right and left).



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SPECIFICATIONS

TYPE	TWIN BARREL A.A MACHINE GUN
Caliber of Barrel (mm):	23
Max. Vertical Distance (m):	1,500
Max. Inclined Distance (m):	2,500
Max. Horizontal Distance (m):	2,000
Tempo of Shooting/with Both Machine Guns (Shots per Min.):	1,600 - 2,000
Firing Rate/with Both Machine Guns (Shots per Min.):	To 400
Horizontal Fire:	360°
Vertical Fire:	From -10° to +90°
Vertical Fire Above the Wheel:	From -30° to +90°
Weight of Cartridge Box with a Cartridge Belt Furnished with Cartridges (kg):	35.5
Weight of Installation in Field Position with Cases and Complete Set of Cartridge Box (kg):	950



MESBAH - AIR DEFENCE SYSTEM

PIC: 000250

DESCRIPTION

The main mission of this **23 mm** air defensive gun is for the protection of strategic, political, economical centers and military fortifications against threats such as; cruise missiles, helicopters, unmanned aircrafts, and air to surface missiles at short ranges and low altitudes.

This low altitude air defensive system includes 8 sets of mechanized twin - barrel 23 mm guns , one set of optical passive fire control , a power supply for making thrust and one set of 3D search-radar system.

This radar system is able to identify air targets up to a distance of 20 km, 20 targets at one time and choose the most dangerous one and directing the data to fire control system.

Fire control of this defensive system at first step tracks down targets at the distance of 10 km then by collecting the entire data about its distance and surrounding conditions, will calculate the coordination of that target and become ready to fire once the target is placed at a distance of 1500 to 2500 meters.



SPECIFICATIONS OF FCU

TYPE	MESBAH
Detection Range (km):	0.2 ~ 20
Max. Speed of Detected Targets (m/s):	300
Max. Angular Speed of Targets (°/s):	90
Motion in Azimuth:	360 °n
Motion in Altitude:	0 ~ 100 °
Angular Precision (milliem):	1
Max. Angular Acceleration (°/s ²):	120

FEATURE OF RADAR

TYPE	MESBAH
Min. RCS (m ²):	0.1
Max. Range (km):	20
Radar Altitude Coating:	45 °
Precision on Azimuth and Altitude:	± 1.5 °





MESBAH 1 - AIR DEFENCE SYSTEM

PIC: 000251

DESCRIPTION

This 8-barrel system is an integration of “chassis & electric system of one 35mm gun” and a number of “4 pieces of twin-barrel 23 mm guns” that is equipped with “one 3D seeker radar”, “one smart fire control unit” and “one set of power supply” for supplying the required energy.

This system utilizing advanced photography sensors and processing the acquired data is able to identify all air targets up to the height of 20 km regardless of being day or night then by conducting complicated calculations, predicts the target's trajectory and shoot towards that target in the end by estimation of its next position.

By connecting to units of fire control (FCU), electro-optic and Sky Guard while having high agility and an acceptable fire volume this has increased the accuracy of the system in firing through reduction of shots diffusion.



SPECIFICATIONS OF FCU

TYPE	MESBAH 1 - AIR DEFENCE SYSTEM
Detection Range (km):	0.2 ~ 20
Max. Speed Detected Targets (m/s):	300
Max. Angular Speed Targets (°/s):	90
Motion in Azimuth:	360 °n
Motion in Altitude:	0 ~ 100 °
Angular Precision:	1 milli
Max. Angular Acceleration (°/s ²):	120

FEATURE OF RADAR

TYPE	MESBAH 1
Min. RCS (m ²):	0.1
Max. Range (km):	20
Radar Altitude Coating:	45 °
Precision on Azimuth and Altitude:	± 1.5 °





SAMAVAT 35 MM AIR DEFENCE SYSTEM

PIC: 000252

DESCRIPTION

This twin barrel defensive system is an automatic version that identifies and tracks down all air targets at low and medium altitudes by use of **Sky Guard** radar.

Then by applying the acquired data, fire control unit calculates shooting angular coordinates and make the system ready to automatically hit down the intended target by salvo fire of 1100 shots per minute.

This system has been designed for protection of strategic, political, economical and military centres against air raids by i.e cruise missiles, helicopters, unmanned aircrafts, fighters and air to surface missiles up to 3000 meters altitude.



SPECIFICATIONS

TYPE	SAMAVAT
Caliber (mm):	35
Number of Barrels:	2
Rate of Fire 2 × 550 (rounds/min.):	1,100
Muzzle Velocity (m/s):	1,175
Barrel Length (90° caliber) (mm):	3,150
GUN - LAYING DATA	
Traverse Arc:	Unlimited
Max. Traverse Speed (rad/s):	2.095
Traverse Acceleration (rad/s ²):	3.142
Elevation Arc:	-5 ° to +92 °
Max. Elevation Speed (rad/s):	1.047
Elevation Acceleration (rad/s ²):	2.095
Max. Range of Fire (m):	11,200
Height of Fire (m):	8,500





60MM MORTAR LAUNCHER STANDARD & LONG RANGE

PIC: 000253

DESCRIPTION

60 mm Standard mortar launcher is a kind of smooth-bore and barrel-loading weapon and has a high elevation angle with a 360° firing field. It is usually operated by two crews, but it can be operated by one crew in emergency. Its sight device is simple and accurate and remains in a steady state during firing. This mortar launcher is able to fire with high accuracy. **Long Range** is used in offensive situations for continuous, close supporting of infantries and in defensive situations for producing concentrated fire within battle field. Simple use, easy repair and maintenance, simple to make it ready to be in combat position, are its advantages.

SPECIFICATIONS

TYPE	STANDARD	LONG RANGE
Caliber (mm):	60	60
Length of Barrel and Breech (mm):	720	1379.5
Max. Range (m):	2550	6000
Min. Range (m):	250	---
Elevation Range (m):	45° ~ 80°	45° ~ 80°
Rate of Fire:	25 in the first minute then 7 continuously	
Weight in Firing Position (kg):	17.5	
Crew:	1	2



STANDARD

LONG RANGE

60MM COMMANDO MORTAR LAUNCHER

PIC: 000254

DESCRIPTION

60 mm Commando mortar launchers are in two types, **Fateh** and **Narollah**. They are light weight, simple and can be classified in portable weapons group. They are able to produce a mass of fire accurately at the shortest time and with high rate of fire on any kinds of targets within its range. These mortar launchers can be used in infantry level, airborne commands, at any kind of battle fields and under all weather conditions. Their sight devices are simple and remain in a steady state during firing. In order to protect the operators' hands from burning while carrying the commando mortar launchers, the barrel is covered with a piece of canvass. To protect the barrel from dust, a cap is used on its top and there is a piece of cord, which connects in one end to breech, for carrying the weapon.

SPECIFICATIONS

TYPE	FATEH	NAROLLAH
Caliber (mm):	60	60
Length of Barrel and Breech (mm):	570.5	570.5
Max. Range (m):	1050	800
Min. Range (m):	100	200
Elevation Range	45° ~ 80°	45° ~ 80°
Travers Range:	Curved & direct fire 360°	Curved & direct fire 360°
Rate of Fire:	20 in the first minute then 7 continuously	20 in the first minute and 7 continuously
Total Weight (kg):	6.25	6.25
Crew:	1	1



FATEH

NAROLLAH



81 MM STANDARD MORTAR LAUNCHER

PIC: 000255

DESCRIPTION

This mortar launcher is a kind of smooth-bore and barrel-loading weapon with a 360° firing field. Due to its high elevation angle, it can be fired from behind obstacles and against targets behind obstacles, also in reverse slopes and finally for supporting own forces. To make it easy for infantry to carry the weapon in difficult terrains, it can be disassembled into small pieces. Its simple structure enables to set up the mortar by three crews. Because of its nearly small dimensions, it can be easily hidden in hole in the ground, which protects the mortar launcher as well as the crews.

SPECIFICATIONS

TYPE	STANDARD MORTAR LAUNCHER
Caliber (mm):	81
Length of Barrel and Breech (mm):	1,530
Max. Range with Mortar Bomb Model M 91 (m):	4,900
Min. Range with Mortar Bomb Model M 91 (m):	200
Elevation Range:	45° ~ 85°
Rate of Fire:	15 in the first minute and 5 continuously
Weight in Firing Position (kg):	53.2
Crew:	3



81 MM SHORT BARREL MORTAR LAUNCHER

PIC: 000256

DESCRIPTION

This mortar launcher is a barrel-loading, smooth-bore weapon which is able to produce a large volume of fire. It is used in offensive situations for continuous, close support for infantries and in defensive situations for producing concentrated fire in the battle field. Rapid displacement, easy use, high accuracy and large volumes of fire are its advantages.

SPECIFICATIONS

TYPE	SHORT BARREL MORTAR LAUNCHER
Caliber (mm):	81
Length of Barrel and Breech (mm):	768.5
Max. Range (m):	2100
Min. Range (m):	100
Elevation Range:	45° ~ 80°
Travers Range:	360°
Total Weight (kg):	37
Crew:	2





120 MM MORTAR LAUNCHER

PIC: 000257

DESCRIPTION

The **120 MM** mortar launcher is a barrel loading type with 360° firing field.

Because of possessing high shooting angle this weapon enables to one fire from behind of obstacles. In regions where roads are impracticable, the mortar can be taken apart and carried by infantry.

Due to its simple structure, it can be erected and operated only by two or three persons.

It is necessary to remind that heating of the barrel is so slow that the rate of firing can not be effected.

Due to its nearly small dimentions, it is possible to dig a trench easily and put the weapon into it, so that the weapon and crew are protected from hurting, specially on defence conditions.



SPECIFICATIONS

TYPE	MORTAR LAUNCHER
Caliber (mm):	120
Weight (kg):	140
Barrel Length (mm):	1700
Rate of Fire (rds/mi):	10 in the first minute and 4 continuously
Max. Range (m):	6600
Min. Range (m):	550
Elevating Range:	45° to 85°
Crew:	4



MORTAR SIGHT

The mortar sight is used for the indirect aiming 120 mm and 81 mm mortars.

STRUCTURE:

The sight consists of three parts:

- 1- Sighting Telescope
- 2- Sight mount with exact elevation and azimuth worm pinions
- 3- Illuminator for night operations





3

73 MM SPG-9 RECOILLESS ANTI TANK GUN

PIC: 000258

DESCRIPTION

The **73 mm SPG9** is a lightweight anti-tank and anti armored gun, normally carried by two soldiers, crewed by four and mounted on a tripod for firing. It can be towed with using a small two-wheel carriage and may be mounted on vehicles. The optical sight is provided for direct fire.

The **SPG-9** fires a fin-stabilised round with either a **HEAT** or **HE** warhead. The propellant charge is wrapped around a boom which extends behind the folded fins and terminates in a perforated disc of bore diameter.

This construction bestows some ballistic advantages and also allows the extension boom to break free from the rest of the projectile but be ejected from the gun muzzle. The projectile consists of the warhead and a tubular rocket motor which is ignited by a delay element shortly after ejection from the muzzle of the gun.

The spare parts are completely compatible with eastern model, and also mountable on vehicle with special rotary stand (optional) with high maneuverability. We can also support Training Aid items.



SPECIFICATIONS

TYPE	SPG-9
Caliber (mm):	73
Anti Tank Rocket Max. Range (m):	1,300
Anti Tank Rocket Effective Range (m):	800
Anti Tank Personnel Rocket Max. Range (m):	4,500
Firing Rate (rockets/minute):	2 to 4
Penetration Depth in Armours (cm):	30 to 40
Elevation:	-3° ~ +7°
Traversing:	-15° ~ +15°
Weight of Weapon with Tripod (kg):	69
Dimensions (cm):	215 x 99 x 80



TRAINING AID SET



106 MM RECOILLESS ANTI-TANK GUN

PIC: 000259

DESCRIPTION

The **106 mm** recoilless rifle is a high efficient anti-tank gun & anti armored weapon with high penetration depth. This rifle is equipped with an aiming rifle, optical sight and is supplied with the usual range of **HEAT** and anti-personnel ammunition.

The standard mounting is the tripod ground mount, but it can also be mounted on any suitably modified light vehicle.

The sub rifle (training aid) and sectional 106 mm gun for training purposes can also be provided.

The spare parts are completely compatible with western model **M40A2**.



ADVANTAGES OF TRAINING BY 106 MM SUB RIFLE

- Lower cost (saving real 106 mm ammunition)
- Higher safety
- Coincidence of firing results as if real 106 mm ammunition is used.
- No wear - out of barrel of 106 mm gun
- Utilization of smaller fire - field
- Time - saving



SPECIFICATIONS

TYPE	106 MM GUN
Calibre (mm):	106
Effective Range (m):	1100
Max. Range (km):	7.5
Allowed Normal Firing Rate (shots/hour):	36
Penetration Depth in Armors (cm):	40 - 50
Elevation on M-79 Tripod:	-7° ~ +27°
Traverse on M-79 Tripod:	270°
Weight of Gun with all Accessories (kg):	216.5
Weight of Projectile (kg):	16.3
Dimensions (cm):	340 x 152 x 112





107 MM SINGLE & DOUBLE BARREL ROCKET LAUNCHER

PIC: 000260

DESCRIPTION

This weapon equipped for battle forces and can launch a **107 mm** rocket to destroy and disturb enemy camping-site, airports, harbours, warehouse, transport lines and so on, to annihilate and suppress enemy effective hidings in light defence workers.

M.o.D of Iran is able to produce these type of weapons in any number of barrels, as customer's request. Sectional cut unit as training aid can be provided as well.

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SPECIFICATIONS

TYPE	SINGLE-BARREL	DOUBLE-BARREL
Calibre (mm):	107	107
Number of Barrel (pcs):	1	2
Length of Barrel (cm):	80	80
Max. Range (m):	8500	8500
Elevation:	0° to 45°	0° to 45°
Traverse:	-11° to +11°	-11° to +11°
Net Weight (kg):	21.5	30
Weight of Each Rocket (kg):	18.9	18.9
Dimensions (cm):	150 x 83 x 81	162 x 126 x 96





107MM MULTI ROCKET LAUNCHER

PIC: 000262

DESCRIPTION

The **107 mm** multiple launcher is manufactured in two standard and mobile versions. In its standard, the two wheeled trailer can be moved by a vehicle, animals or personnel.

All of these launchers deploy unguided rockets.

The launcher has three banks of four **107 mm** barrels and in the trailer type, is mounted on a rubber-tyred split pole type carriage. In the firing position the wheels are normally removed and the launcher is supported by its two trails at the rear which are fitted with spades which are carried on top of the trails when travelling, and two short legs in

front of carriage. It can also be fired with the two rubber tyred road wheels still in position on their stub axles.

The unguided rockets are spin stabilised and the full 12 round load can be salvo-fired in 7 to 8 seconds.

Reloading is performed manually by the crew. The optical sight is fitted on the left side of the launcher.

The standard **107mm** (12 round) launcher assembly is available with a 50 kg weight, 1.4x0.585x0.22m base plate for fitting to light 4 x 4 vehicles or 6 x 6 truck chassis. This weapon

is used in battle fronts to destroy the enemies' positions both continuously and in short intervals. It should be pointed out that this weapon has considerable applications efficiencies in short distances and it is capable to cover a vast area under its fire power.

M.o.D Defence Industries produces this weapon in two versions as:

- **107 mm** rocket launcher with standard carriage
- Self propelled **107 mm** Rocket launcher installable on any kind of jeep.



SPECIFICATIONS

TYPE	MULTI ROCKET LAUNCHER
Calibre (mm):	107
Number of Barrels (pcs):	12
Elevation:	0° ~ 60°
Traverse:	-15° ~ +15°
Max. Range(km):	8.5
Rate of Fire:	12 shots / 8 - 10 sec
Dimensions with Carriage (cm):	280 x 135 x 102
Dimensions with Base Plate (cm):	93 x 140 x 81
Weight of Rifle with Carriage (kg):	475
Weight of Rifle without Carriage (kg):	380
Weight of Rifle with Base Plate (on vehicle) (kg):	430





107 MM NAVAL 11 & 19 BARREL ROCKET LAUNCHER

PIC: 000262

DESCRIPTION

The 11 barrel & 19 barrel naval rocket launchers are installed on top of the cabin of assault or patrol fast gun-boats.

The boat must be a class 11 fast boat with well reinforced cabin framework, suitable to tolerate 800 kg load with force 3. The motor power of gun-boat must be 400 hp or more.

This weapon operate in both electrically and mechanically in different degrees.

The sectional cut of naval launcher as Training Aid is also available.



SPECIFICATIONS

TYPE	11 BARREL	19 BARREL
Calibre (mm):	107	107
Number of Barrels (pcs):	11	19
Max. Range (km):	8.5	8.5
Firing Rate:	11 shots / 12 - 14 sec	19 shots / 20- 24 sec
Elevation:	-5° to 45° (with precision of 3 mil-russian)	-5° to 45° (with precision of 3 mil-russian)
Elevation Speed:	25° / second	18°/second
Traverse:	-90° to +90°	-90° to +90° (in steps of 30°)
Fire Control:	Electronic fire control with joystick	Electronic fire control with joystick
Weight of Weapon (kg):	280	400
Weight of each Rocket (kg):	19	19
Dimensions (cm):	120 x 93 x 65	190 x 110 x 65



122 MM ROCKET LAUNCHER

PIC: 000263

DESCRIPTION

The 122 mm launcher produce in four versions:

» 122 MM SINGLE - BARREL ROCKET LAUNCHER

It is a weapon for launching free low-range rockets. It can be carried by 2 crews. Due to its light weight, it is of a rapid displacement. The firing mechanism is electrical and its elevation and traverse mechanisms are mechanical.

» 122 MM 8 BARREL ROCKET LAUNCHER

Due to its light weight, this launcher can be mounted on light vehicles and steady platforms and can be rapidly displaced.

Elevation and traverse mechanisms are mechanical type; therefore, their vulnerability in comparison with similar systems is at the minimum level. Its trigger system is simple with high safety. Training is very easy and fast to set-up and operate.

» 122MM 16 BARREL NAVAL ROCKET LAUNCHER

It is a weapon for launching free low-range rockets and is mounted on floats and boats. High maneuver and rapid displacement of boats, possibility of firing from cabin and being equipped with firing gate control system, expedite the destruction of enemies' targets. It fires 122 mm rockets. The firing system is electrical. Elevation mechanism is mechanical and electro-hydraulic.

» 122 MM 40 BARREL ROCKET LAUNCHER

This weapon is used in corps level of military and is mounted on trucks. It can be firing from trenches and cabin and ability to drive through water with fording depth of maximum 1m expedite enemy's destruction. This launcher is equipped with hydraulic stability jacks to remove vibrations caused by the firing.



SPECIFICATIONS

TYPE	SINGLE BARREL	8 BARREL	16 BARREL	40 BARREL
Caliber (mm):	122	122	122	122
Length of barrel (mm):	3,000 & 1,900	3,000	1,900	3,000
Max. Range (m):	23,000	21,000	15,000	20,400
Min. Range (m):	2,000	---	12,000	---
Elevating Range:	+5° ~ +50°	0° ~ +52°	-1° ~ +25°	-1° ~ +55°
Traversing Range:	± 12°	At max ± 30°	360°	72° ~ 102°
Weight of Launcher (kg):	90	730	800	with truck 13,150



122 MM & 155 MM HOWITZER

PIC: 000264

DESCRIPTION

» 122 MM HOWITZER D30

This Howitzer is a semi-heavy artillery gun and is deployed at brigade level of army divisions. It is a breech-loading howitzer without lock, with a vertical wedge breech that is loaded by hand. Shell is extracted after each firing and the howitzer is cocked automatically.

Ammunition is of varied and changeable kinds, and its firing system is of a pulling type. Its recoil mechanism (recoil and recuperator) is independent hydropneumatic, and the equilibrator is pneumatic.

This howitzer is on three trails in firing position. It can fire the shell directly, in a curve way and at high elevation angle.



» 155 MM HOWITZER

This Howitzer is a semi-heavy artillery gun with long range. It is used for supporting infantries. It is able to fire 155 mm ammunition. This Cannon is on three trails in firing position. It has a muzzle brake to reduce recoil force.

independent hydro-pneumatic type. Equilibrators are of nitrogen gas type, traverse and elevation mechanisms are mechanical and can be operated by rotating a hand wheel.

Its breech is of a screwing shape, mechanical type with a constant slope. Firing lock is of a continuous pulling and impact type. Recoil mechanism is of a changeable and

Its brake system is pneumatic and operates manually. It should be mounted on trucks with over 10 tons weight. Firing control tools and equipment are carried by 2 crews for both direct and indirect firing.



SPECIFICATIONS

TYPE	122 MM D30	155 MM
Caliber (mm):	122	155
Max. Range (m):	15,300	30,000
Rate of Fire (r/m):	6 - 8	4
Traversing Range:	With elevation from -5° ~18°:360°	23.5° Left and 25° Right
Elevation Range:	-7° ~70°	0° ~ 66°
Number of Rifling:	36	48
Total Weight (kg):	3,440	6,890





3

40 MM MARINE GUN

PIC: 000265

DESCRIPTION

- Anti aircraft & helicopter
- Anti gun-boat & warship
- Anti land missile
- Useable for aerial defense

SPECIFICATIONS

TYPE	MARINE GUN
Calibre (mm):	40
Effective Range - Air Target (m):	4,000
Effective Range - Land Target (m):	6,000
Max. Range (m):	12,000
Firing Rate (shots/min):	300
Elevation:	-10 ~ +84
Traverse:	No limit (360°)
Weight of Weapon (kg):	3,300
Rotating Radius (mm):	3,315



100 MM ANTI AIRCRAFT GUN

PIC: 000266

DESCRIPTION

It is a medium caliber anti-aircraft gun that is controlled by a firing control center to fire against targets flying at height of 36,000 feet (12,000 m). Also it can be used for destructing low-to-medium height, marine and land targets. In each artillery,

four guns are connected with cable and fire mechanically or electrically (semi-automatically). In mechanical firing position, 7 crews are used, while it is reduced to 3 crews in semi-automatic or automatic firing positions.

SPECIFICATIONS

TYPE	100 MM AA GUN
Muzzle Velocity (m/s):	900
Max. Range (m):	21,000
Effective Range (m):	18,000
Max. Range (vertical) (m):	13,000
Effective Range (vertical) (m):	12,000
Length of Barrel (mm):	5,358
Weight in Combat Position (kg):	9,350
Weight in Travel Position (kg):	9,600
Total Length of the Gun (mm):	9,170
Total Width of the Gun (mm):	2,390
Height of the Gun (mm):	2,123
Barrel Life (rds):	Up to firing 800





IFADJR-27 76 MM NAVAL GUN

PIC: 000267

DESCRIPTION

It is a standard marine single-barrel gun, complex, fully automatic (electro hydraulic), controlled from a distance that is suitable for spot objectives and fighting against targets like warships, aircrafts (fighters) and missiles.

It has great reliability and good hit-the-target records. As a main weapon, it can be mounted on small battleships of over 200 tons weight, warships, frigates and destroyers.

This marine gun is controlled by a firing control system from a distance. It can also be controlled by an optical director in emergency.

Its electrical connections and rotation are in harmony with each other and all assemblies except electrical assembly are located inside the gun.

It is loaded by two crews and loading operation from rotary cartridge to the barrel and firing is fully automatic. When it is fully loaded, its rate of fire is 80 rds/min. with no need to reload.

It can be operated even when the sea is turbulent or stormy. Due to high velocity, the shells move in a ballistic-like arc.

The gun can fire 50 shells against target flying at a speed of 300 m/s and 35 shells against target flying at speed of 600m/s.



SPECIFICATIONS

TYPE	FADJR-27
Caliber (mm):	76
Length of Barrel and Breech (mm):	4,700
Rifle Length (mm):	4,012
Chamber Length (mm):	688
Pressure at Muzzle Brake (bar):	540
Max. Pressure (bar):	3,400
Muzzle Velocity (m/s):	925
Max. Range (m):	16,300
Max. Length of Recoil with Gun at 85° (mm):	360
Max. Length of Recoil with Gun at 0° (mm):	330
Recoil Average Force (kg):	7,000
Weight (kg):	7,200
Elevation Range:	-15° ~ 85°





240 MM NAVAL ROCKET LAUNCHER

PIC: 000268

DESCRIPTION

Due to high maneuverability of the boat and high speed of elevation mechanism that operates electro hydraulically the maneuverability of the launcher increases. It can create a rapid fire in two ways, both cyclic and single shot. An automatic and precise targeting is achieved through the launcher connection to a mileage indicator (optional). It is regarded as an efficient marine weapon in naval combats.



A R M A M E N T

SPECIFICATIONS

TYPE	NAVAL ROCKET LAUNCHER
Caliber (mm):	240
Number of Barrel (pcs):	8
Launcher Width (mm):	1,797
Launcher Length (mm):	1,453
Launcher Height (mm):	1,567
Launcher Weight (kg):	576
Consumption Electricity:	24 V DC
Mechanical Elevation Mechanism:	Moveable from inside and outside of the cabin
Hydraulic Elevation Mechanism:	Automatic and Semi-Automatic
Max. Range at 45° (m):	10,112
Min. Range (m):	300
Elevation Range:	0° ~ 38°
Traverse Range:	No traversing



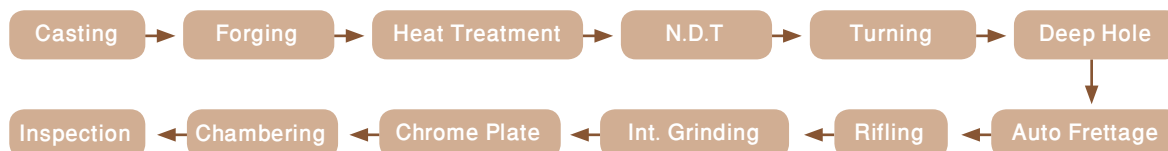


CANNON BARRELS

PIC: 000269

DESCRIPTION

Defence Industries of M.o.D also able to produce different types of medium and heavy cannon barrels for armored vehicles and artileries as following industrial procedure and specifications.



SPECIFICATIONS

ITEM	CALIBRE	WEIGHT (KG)	LENGTH OF BARREL	NUMBER OF RIFLING	RIFLING TWIST	MUZZLE VELOCITY (M/S)	LENGTH OF RIFLING (MM)	LENGTH OF CHAMBER (MM)
1	122	605	3,995	36	Progressive	690	3,405	590
2	155	1,776	6,095	48	1 turn in 20 caliber	826	5,064	996
3	155	1,132	6,045	48	1 turn in 20 caliber	684	5,122	923
4	130	1,780	6,850	40	1 turn in 30 caliber	930	5,860	990
5	130	1,658	6,840	40	1 turn in 30 caliber	930	5,850	990
6	155	1,750	7,046	48	1 turn in 20 caliber	898	5,820	1,180
7	105	325	2,360	36	1 turn in 20 caliber	472	1,980	380
8	105	760	5,347	28	1 turn in 18 caliber	1,174	4,713	633-5
9	125	1,125	6,000	No rifling	No rifling	----	---	840
10	122	537	3,995	36	Progressive	690	---	590



INFANTRY MOBILE UNIT

PIC: 000270

DESCRIPTION

Capability of repairing all types of light weapons including: KL rifle, Pistol (alarming, Browning Breta) all types of RPG, Girenov machine gun, Deraganov rifle, 12.7 mm MG. Increasing the mobility and operational speed of supporting units. Reducing the repair costs.

FEATURES

- Type of vehicle: Mini - Truck
- Permissible load: Maximum 6000 kg
- Room dimensions: 4800 x 2200 x 2000mm
- Type of walls: panel sandwich insulated against coldness, heat and dust. (With the thickness of 70mm)
- Equipped with all types of: Instruments and equipment for the repair of light weapons.
- Equipment and public tools.
- 250A welding transformer.
- Acetylene welding equipment.
- Heating and cooling system.
- Ventilating fan of system.
- Washing up and water tank one hundred litre.
- Special wardrobe tool and equipment.
- Electrical generator.
- Compressed air compressor 80 litre.
- Cumulative stairway.



WIRELESS FIRE CONTROL

PIC: 000271

DESCRIPTION

Wireless fire control device has been designed to control, plan and execute explosive operations from far distance. With this system, it is possible to plan trapping of 400 point of an area with 5-kilometer radius.

FEATURES

- Possibility to identify every receiver by every transmitter.
- Possibility to test and show outputs of all receivers in 20 channels at the same time and fully schematic with pre-defined indexes.
- Display all menus in fully schematic.
- Choice menus by touch panel or push button.
- Possibility to single or group explosions of one or more receiver outputs contemporaneously.
- Ability to plan outputs of one or more receiver for delayed explosion.
- Possibility to ignore delayed settings.
- Possibility to test receiver and transmitter battery conditions and display battery's voltage.
- Possibility to disable receivers for collections.
- Possibility to identify and change system code.



chapter 4

MILITARY VEHICLES





PREVIEW

4



Since 1986, **Armor Industries (AI)** of Iran has started its activity with repair works on Armored Personnel Carriers (**APC**) and **T** series Tanks in service with Armed Forces of I.R.Iran. In 1989 the first **APC** was produced and tested and in 1991 it was delivered to the Armed Forces after successfully passing all tests.

In 1990 the repairing of tanks entered the upgrading stage, and in 1992 the first modernized tank (**type T-72Z**) was tested and presented to the market.

The other activity is, manufacturing **BMP-2** Armored Personnel Carrier and **T-72S** Main Battle Tank. Manufacturing of hull and turret, suspension, mechanical-electrical parts and final assembly is accomplished in the AI facilities with technical assistance and documentations under the license of the original manufacturing country.

Due to experienced employees and skillful specialists, applying the state-of-the art equipment and facilities in manufacturing the armored articles, M.o.D of Iran ready to render services as follow:

- Design and manufacturing of armored vehicles wheeled / tracked.
- Design and manufacturing of special multi purposed vehicles. Overhauling and upgrading of armored vehicles (APC, Tank).
- Manufacturing of different armored protection equipment.
- Rendering training services in armored fields (operating, repair, and maintenance).
- Design and manufacturing of different simulators for armored vehicles.

With respect to obtaining quality & environmental international certifications such as **ISO 9001**, **ISO 9002**, **ISO TF 2002**, **ISO TS 16949** and Military Standards which have key role for entering the worldwide competition.





CHAPTER 4



4

ITEM	TABLE OF CONTENT	PAGE
1	BORAQ ARMORED PERSONNEL CARRIER	118
2	BORAQ VEHICLE FAMILY	119
3	BMP2 ARMORED PERSONNEL CARRIER	121
4	TRACKED RESCUE AND EMERGENCY VEHICLE	122
5	RAKSH - 4X4 WHEELED APC	123
6	RAAD 1 - SELF PROPELLED GUN HOWITZER	124
7	RAAD2 & RAAD2M SELF PROPELLED GUN - HOWITZER	125
8	UPGRADED M-60 MBTS	126
9	MODERNIZED TANK - T72 & T72Z	127
10	EQUIPPING CHIEFTAIN MBTS	128
11	TANK PASSAGE BRIDGE	128
12	RECOVERY TANK	129
13	KAT-72 EXPANDABLE FIRE CONTROL SYSTEM	130
14	3/4 TON (THISTLE) TACTICAL VEHICLE	131
15	1/4 TON (THUNDER) TACTICAL VEHICLE	131
16	1 1/4 TON TACTICAL VEHICLE	132
17	MOBILE WORKSHOP	132
18	ARMOR TRAINING SIMULATOR	133
19	LIGHT MODERN BALLISTICS ARMOR	133
20	TAFTAN 2-MINE CLEANER	134
21	ROLLER TANK MINE CLEANER	135
22	MINE SIFTER	136
23	ARMORED BULLDOZER MINE CLEANER	137



BORAQ ARMORED PERSONNEL CARRIER



PIC: 000101

DESCRIPTION

The **Boraq APC** is one of the tracked amphibious types of APCs and well characterized among its other similar armored vehicles at the same level by proper armor protection, higher mobility, and capability of crossing obstacles and uneven roads at high speed as well as accomplishing heavy maneuvers.

FEATURES

- Provided with micro alloy steel body 5-19 mm in diameter, and ballistic tolerance of 12.7 mm cartridges.
- Incorporating live track with rubber shoes.
- Incorporating communication and navigation equipments (VRC-222 radio set, with 40W power and 40 km range).

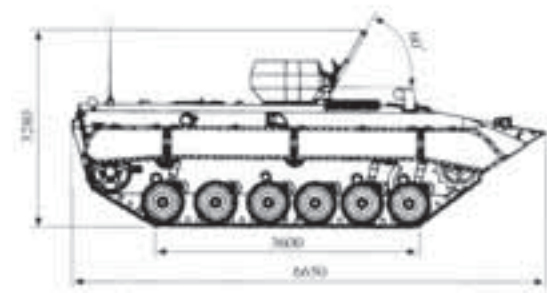


BORAGH VEHICLE FAMILY

Different types of boraq vehicle family includes: Commander, Ambulance, Engineering, Ammunition, Tow missile launcher, Mortar, Gunner and different naval families.

The support vehicle (**Boraq**) with high travelling ability can protect effectively the infantry in tactical and fighting operations.

The vehicle is equipped with 12.7 mm machine gun (**DUSHKA**) with traversing angle of 360°, which can be used directly by a gunner.



SPECIFICATIONS

TYPE	BORAQ APC
Mass (ton):	13
No. of Crews:	12 Men (Driver, Gunner, and 10 Crew members)
Engine:	Deutz, BF8L 513 model, Power 330 HP
Engine Type:	Diesel, 4 - Stroke, 8 - Cylinder with, V - Shaped Arrangement, Air - Cooled with Turbocharger
Power to Weight Ratio (hp/t):	25.4
Speed on High Way(1st Class Road) / While Afloat (km/h):	65/ 5
Capability of Crossing Vertical Obstacle (m):	Max. 0.7
Negotiable Gradient :	Max. 30 deg.
Negotiable Side Slope:	Max. 25 deg.
Main Armament (mm):	12.7 DSK MG
Rate of Fire:	600 Round per minute(RPM) , Velocity of 800 m/s
Elevation/ Traverse Angle :	26 - 78 deg./ 360 deg.
Max. Fire Range (m):	1,600
Cruising Range on Dry Land (fuel endurance) (km):	625



BORAQ VEHICLE FAMILY

» BORAQ MORTAR VEHICLE

The fully tracked **AMV** providing capability for operation and firing of 120-mm mortar in diverse conditions of battle field (soft, sandy, rocky, and swampy grounds) with 360° angle of fire in traverse and ammunition storage of 50 rounds proved itself to be a decisive factor in irregular wars and unequal battles.

PIC: 000102



» BORAQ MISSILE LAUNCHER VEHICLE

The fully tracked amphibious **AIFA** armed with **TOW** missile launcher system serves as a mobile missile site.

In order to confront enemy sites and because of proper armor protection, very high mobility and also capabilities of crossing obstacles and bumps at high speed and exercising heavy maneuvers, it would have a decisive role in the battle fields.

PIC: 000103



» BORAQ AMBULANCE VEHICLE

The **AAV** is designed and manufactured for quick transport and evacuation of injured personnel to behind the front lines which provided with equipments such as air conditioner, respiratory, medicine cabinet and cardiography, orthopedic apparatus as well as four beds for carrying casualties.

PIC: 000104





BORAQ VEHICLE FAMILY



» BORAQ GUN VEHICLE

The **AICV** is capable of accomplishing operations and firing 30-mm rounds within fire angle from 6 to 75 degree in elevation. It serves to confront enemy ground and aerial targets.

The 7.62-mm MG coaxial with 30-mm cannon is located at turret.

PIC: 000105



» BORAQ COMMAND VEHICLE

The **ACV** is designed for supporting of operational units and serves as a mobile command station in battle fields.

Due to equipments and facilities provided in the vehicle and its proper armor protection, mobility and maneuverability as well as combat employment on variable terrains, it plays a decisive role in direction of battle scenario and protection of battle field commanders.

PIC: 000106



» BORAQ AMMUNITION CARRIER

The full track **AACV** is designed for support of armor and infantry units with the capability to carry ammunition required for battle fields through areas difficult to pass and variable terrains and due to its proper speed and high maneuverability, the **AACV** will play a decisive role in quick supply of ammunition to the operational units.

PIC: 000107





BMP2 - ARMORED PERSONNEL CARRIER



4

PIC: 000179

DESCRIPTION

The fully tracked **APC BMP-II** is capable of accomplishing operations and firing **30-mm** rounds.

Its high mobility provides **APC BMP-II** with effective higher firepower and protection of infantries from nuclear weapons.

It serves to confront enemy ground and aerial targets such as tank and combat helicopters.

This vehicle is also armed with a **7.62-mm MG** coaxial with a **30-mm gun** and a stationary- mobile missile launcher for destruction of armor targets.



FEATURES

- Fully tracked amphibious with live padded track (with rubber bush).
- Incorporating communication and navigation equipments (equipped with a 25W power and min. 40 km range R-173 radio set).
- Provided with smoke generating equipment, smoke grenade launcher system, and **NBC** protection system.

SPECIFICATIONS

TYPE	BMP2
Mass (ton):	14 + %2
Engine:	YTD - 20 Model, Power 300(hp) at 2600 RPM
Engine Type:	4 - Stroke, 8 - Cylinder, V - Shaped Diesel Engine, Liquid - Cooled
Gearbox:	Mechanical, Constantly Engaged, with 5 Forward Gears,
Max. Speed on Flat Road/ Fording Speed (km/h):	65/ 7
Negotiation of Cross Trenches (m):	Max. 2.5
Capability of Crossing Vertical Obstacle (m):	Max. 0.7
Negotiable Gradient:	Max. 35 deg.
Negotiable Side Slope:	Max. 25 deg.
Cruising Range (Fuel endurance) on Main Road (km):	550 - 600
No. of Crews:	10 Men (3 +7) (Crews +Combat forces)
Main Armament (mm):	30 - Automatic Gun/ Cannon 2A42 with two(2) Separate Feeding Belts
Effective Range for Ground Targets (m):	2,000 - 4,000 (Depending on Type of Round)



TRACKED RESCUE AND EMERGENCY VEHICLE



4

PIC: 000180

DESCRIPTION

TREV-1 serves to provide relief in time of earthquake, flooding, landslide, snow, storm with driving snow, excavating debris, removing obstructions, and also provides relief and aid for the vehicles involved, transportation and loading of foodstuffs and equipments.

FEATURES (TREV-1)

- Incorporating live padded track with rubber shoes.
- Equipped with dozer for clearing roads and removing obstructions.
- Equipped with life detector, thermal vision device, rescue pneumatic jacks, scissors, hydraulic jack, and gas meter, etc.
- Provided with rescue equipments including winch, rope, pulley etc. and machines for cutting concrete, wood and metal.
- Incorporating communication and navigation equipments (equipped with a 40W power and 40 km range VRC-222 radio set).
- Incorporating a telescopic crane with 3.3 t capacity and operating range of 1.9 m for removing objects and obstructions.



TREV-1

FEATURES (TREV-2)

- Equipped with armored cabin providing the capabilities as follows:
 - Necessary resistance to burst and blast wave.
 - Fitted with bulletproof and anti-explosion glass.
 - Proper insulation for sound and heat.
 - Proper field of view and operator convenience.



TREV-2

SPECIFICATIONS

TYPE	TREV-1	TREV-2
Mass (ton):	10	13
Engine:	Deutz, BF8L 513 model, with 330 HP Power	Deutz, BF8L 513 model, with 330 HP Power
Engine Type:	Diesel, 4 - Stroke, 8 - Cylinder with V - Shaped Arrangement, Air - Cooled with Turbocharger	
Gearbox:	5 Gear Forward, 1 Reverse Gear	5 Gear Forward, 1 Reverse Gear
Max. Speed (km/h):	65	65
Negotiation of Cross Trenches (m):	Max. 2.4	Max. 2.4
Capability of Crossing Vertical Obstacle (m):	Max. 0.7	Max. 0.7
Negotiable Gradient (deg):	Max. 30	Max. 30
Negotiable Side Slope (deg):	Max. 25	Max. 25
Cruising Range (Fuel Endurance) (km):	500	500
Winch Drag Force (kg):	4,082	4,082



RAKSH - 4X4 WHEELED APC

PIC: 000108

DESCRIPTION

Characterized by high speed on road and excellent off-road mobility.

The vehicle can transport 10 men.

The vehicle is fitted with a 12.7 mm Automatic Gun and can be fitted with different cupolas for various types of armaments, Such as 7.62 mm machine gun and water spraying gun for civil defense.

The crew can see and fire through 10 vision blocks and firing ports and can be quickly embarked or deployed through one rear door and two side doors.

OPTION EQUIPMENT

- Ceramic-composite layer
- Protection against 12.7mm up to 14.5 mm
- Smoke discharger
- N.B.C Protection system

ADVANTAGES

- Central inflating system
- Run flat system
- Winch
- Differential lock
- Air condition system

SPECIFICATIONS

TYPE	RAKSH
Crew:	8 + 2
Length (mm):	6,067
Width (mm):	2,405
Max. Height (mm):	3,100
Fully Loaded Weight (kg):	9,000
Max. Load on Front Axel (kg):	4,200
Max. Load on Rear Axel (kg):	5,600
Engine Type:	MAN D0824 LF09
Max. Power:	155HP/2,400 RPM
Torque:	580 NM/ 1,500 RPM
Transmission:	ZF - 62S50(6+1)
Brake System:	Air Pressure, 2 - Way
Max. Speed (km/h):	100





RAAD 1 SELF PROPELLED GUN HOWITZER



PIC: 000182

DESCRIPTION

Raad-1 122 mm self-propelled gun-howitzer is the first **SPGH** designed and manufactured in Defense Industries of I.R.IRAN.

High stability when firing, good maneuverability, firing at different angles, indirect and direct fires, rapid and independent firing, working properly at temperatures -45° to +45°C, rapid change of position, and easy and fast maintenance are among the main characteristics of this system.

Raad-1 is equipped with a computer and a monitor to display and monitor the ballistic data. Enough space has been provided for the crew inside the vehicle and ergonomic conditions have been properly considered.



ADVANTAGES

- Stability during firing
- Various shooting angles
- High mobility
- Easy maintenance

SPECIFICATIONS

TYPE	RAAD1
Crew:	4
Combat Weight (kg):	17,500
Calibre(mm):	122
Gun Length:	39 - cal
Gun Type :	2 A 31
Gun Weight(kg):	1,440
Barrel Life (rounds):	3,000 (with charge 8)
Traverse:	3,600
Gun Elevation/Depression:	-3° to 72°
Fire Control System:	Semi Automatic
Firing Range (m):	15,200
Max. Rate of Direct Fire (rds/m):	4 - 5
Max. Rate of Indirect Fire:	
❑ Loading From the Ground (rds/m):	1.5 - 2
❑ Loading From Ammunition Racks (rds/m):	4 - 5
Muzzle Velocity (complete charge):	686 m/s (HE) 721 m/s (ZBK - 13)



RAAD 2 & RAAD 2M SELF PROPELLED GUN - HOWITZER

PIC: 000109 , 000110

DESCRIPTION

RAAD 2 & RAAD 2M 155mm self-propelled gun-howitzer are armored tracked vehicle designed and manufactured by M.o.D of Iran.

The engine is located in front of the vehicles. There are 5 crews including commander, gunner, driver, and two ammunition loaders.

The hull and the turret are made of all welded special alloy steel and provide enough space for the crew to operate freely.

RAAD-2 & RAAD 2M are equipped with automatic laying system, direct fire telescope, night vision device, NBC system, automatic and manual fire extinguishing equipment, radio communication, digital communication, gunner's control panel and display, commander's control panel and display, GPS system, and air conditioner system.

The engine is multi-fuel with high power to weight ratio.



ADVANTAGES

- Stability during firing
- Various shooting angles
- Quick and independent firing
- Automatic fire control system
- High mobility

SPECIFICATIONS

TYPE	RAAD-2	RAAD-2M
Crew:	5	5
Combat Weight (ton):	36	36
Length (gun forwards) (mm):	9,140	9,140
Width (mm):	3,380	3,380
Height:		
Up to Turret Roof (mm):	2,600	2,600
Up to Machine Gun (mm):	3,467	3,467
Turret Height (mm):	840	840
Ground Clearance (mm):	450	450
Height of Fire Line to 0°(mm):	2,240	2,240
Engine:	V 84 MS	Diesel 5TDF
Max. Engine Power (hp):	840	700
Power-to-Weight Ratio (hp/ton):	23	19.4
Ground Pressure (Average) (kg/cm²):	0.67	0.67
Transmission:	Hydraulic, Sun Gear, Spat-1200, 7 Speeds Forward and One Reverse	
Suspension:	Torsion Bar with Hydraulic Shock Absorber	
Track:	Live - Padded	



UPGRADED M-60 MBTs



PIC: 000184

DESCRIPTION

As **M-60 MBTs** lacks any gun stabilizer and night vision system, there are some changes in its modernized version, including:

EQUIPPING WITH DRIVER'S NIGHT VISION PERISCOPE WITH FOLLOWING SPECIFICATION

- Vision field: 30 degree
- Magnification: 1
- Installation type: at night time and on the driver's hatch
- Time of assembly & disassembly: 30 sec.



EQUIPPING WITH GUN STABILIZER IN ELEVATION & TRAVERSE MAIN COMPONENTS OF M-60 MBT STABILIZER INCLUDE:

- Elevation & traverse electro hydraulic drivers.
- Upgraded hydraulic system.
- Upgraded control panel of gunner.
- Central control unit.
- Gyroscope & 3 phases feed source.

EQUIPPING WITH STABILIZED PANORAMIC COMMANDER SITE.(OPTIONAL)

- Independent of turret in elevation & azimuth.
- Stabilized in elevation & traverse.
- Magnification: 7 times (variable, optional)
- Vision field: 8.5 degree (variable, optional)
- Laser range finder with range of 20 km and precision of 10m.
- Capable of being linked with anti airgun (optional)

FIRE CONTROL SYSTEM

EFCS-3 FIRE CONTROL SYSTEM

- Laser range finder with precision of 5 m.
- Automatic receiving of ballistic & environmental parameters.
- Automatic ballistic and environmental calculation.
- Fully digitally ballistic calculation at less than 0.1 sec.
- Night vision system of Generation II
- Field of view: 7.5 m
- Magnification: 7 times
- Elevation and azimuth stabilized mirror.





MODERNIZED TANK - T72 & T72Z

PIC: 000185

DESCRIPTION

As regards combat employment of advanced and modern equipments, the **T72Z** tank is upgraded version of **T-55 MBT** that provides the same capability of today's tanks worldwide in terms of mobility, fire accuracy and power as well as armor protection.

So that it can compete with various similar upgraded models and today's modern tanks.



FEATURES

- Quick mounting and dismounting of power plant (**MPP**) in shortest possible time due to its integration and modularity.
- Provided with automatic and semi-automatic system of gear shifting.
- Equipped with fire control system **EFCS-3** and laser range finder, model **ND-YAG** with range of 10 km.
- Capability of tracking target (aiming) during firing at night without searchlight.
- Incorporating stabilizer system with capability of tracking and hitting moving and stationary targets.
- Armor protection from single and double stage missiles by installation of explosive reactive armor boxes of Generations 1 & 2.
- Communication and navigation equipments (incorporating VRC-222 radio set and capability to install VRC-122 & R-173).
- Armed with anti aircraft **MG 12.7 mm cal.** and coaxial **MG cal. 7.62 mm.**

SPECIFICATIONS

TYPE	MODERNIZED TANK T-72 & T72Z
Mass (ton):	40
No. of Crews (men):	4
Engine:	Model V46 - 6 with 780 HP
Engine Type:	Diesel, 4 - Stroke, 12 - Cylinder with V-Shaped Arrangement, and Water - Cooled with Supercharger
Power to Weight Ratio (hp/ton):	19.5
Max. Speed (km/h):	65
Negotiation of Vertical Obstacle (m):	Max. 0.85
Negotiable Gradient/ Side Slope (deg):	Max. 30°/ 25°
Main Armament (km) :	105 - mm Gun, Rifled(grooved), model M68 with Fire Range of 14.6



EQUIPPING CHIEFTAIN MBTS



4

PIC: 000186

DESCRIPTION

Chieftain MBT has a primitive fire control system which includes a periscope coaxial with the gun by a mechanical parallelogram.

Target and range finding, measuring the ballistic parameters, ballistic calculations and applying them are all done automatically by the fire control system **EFCS-3** that is substituted for the present fire control system of the tank.

Upgraded Chieftain **MBT** is equipped with driver night vision periscope and because of fire control system has a night vision system, so finally this tank is equipped with a night vision system Generation II.



STABILIZER UPGRADING

Upgrading Chieftain **MBT** stabilizer system in terms of accuracy and speed.

TANK PASSAGE BRIDGE

PIC: 000183

DESCRIPTION

A 24 meter, easily erected bridge is mounted on 81 **SERAT** bridge layer tank, which is applicable for the rapid and reliable passing of heavy and armored vehicles and military equipment over canals and trenches with the maximum width of 23 meters.

The width of the bridge is 3/5 meters and its loading capacity is 60 tons. It is made of aluminium alloy and is assembled on T72 tank as a sliding drawer. The bridge assemblage (extension) and disassemblage are carried out hydraulically.

The maximum time required for the erection of the bridge over canals and trenches is 6 minutes.

The length, width, height and weight of the bridge layer tank (with the bridge being mounted) are 10/6 m, 3/59m, 3/84m and 42 tons, respectively.

The crew number of the tank is 2. The bridge layer tank travels on the earthy roads at the speed of 45 km/h, and on asphalt roads at the speed of 50 km/h.





RECOVERY TANK

PIC: 000187



DESCRIPTION

This tank is an armored assisting tank, being designed and manufactured for the recovery and rescue of armored vehicles and heavy military equipment in the battle field.

The recovery tank is equipped with a powerful **30-ton** winch, which, using its bobbins, has the capability of elevating loads with the approximate weight of **90 tons**.

By means of this winch the tank is capable of pulling out all heavy vehicles, getting stuck in the muddy terrains or being fallen in ditches.

A telescopic crane with the loading capacity of **19 tons**, and the mast turning capability of 360°, has been

mounted on the tank to be applied for loading, repairing and recovering of tanks and military vehicles.

The aforementioned crane simply lifts the engine and turret of T72 tank, and transports them to the desired destination. At the front part of the unit, an excavator blade has been mounted for digging trenches and leveling uneven terrains.

When applying the winch and the crane, this blade will be applicable as the stabilizer arm of the unit.

Other equipment being mounted on the tank are as follows:

Auxiliary winch, welding and cutting equipment. Load carrying package along with a position assigned for mounting and reliable carrying of this tank engine.

This tank carries 2 crews. For the easy and rapid application of the winch and crane, a cable, portable, controlling panel (with the cable length of 15 meters) has been mounted on the tank.

Due to the necessity of the presence of this recovery tank in the war operations in which T72 tank is applied, its maneuverability should be as that of T72 tank, so it should be carried with a cart having the same capabilities as the cart which carries T72 tanks.

In addition, the recovery unit should be equipped with the same equipment mounted on T72 tanks, including the NBC system and night vision device.

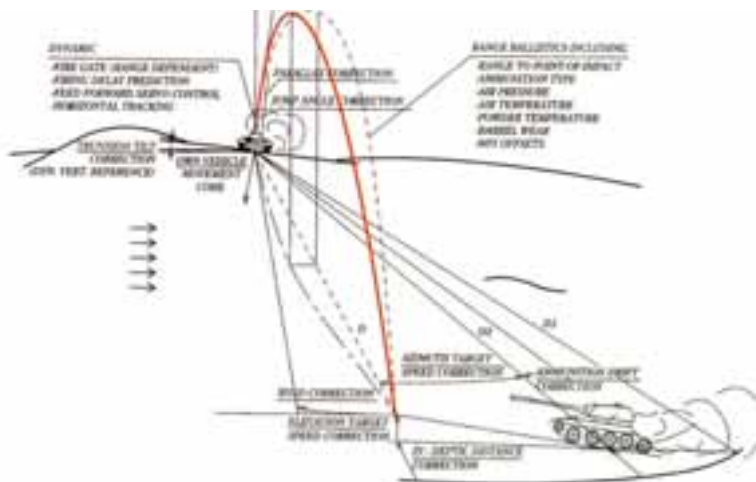


KAT-72 EXPANDABLE FIRE CONTROL SYSTEM

PIC: 000188

DESCRIPTION

- Fast and simple installation, maintenance and repair.
- Original fire control equipment stays in use as a reserve.
- High first round hit probability and short engagement time.
- Day and night operation capability.
- Firing -on-the-move with stabilized gunner's field of view and gun-independenet aiming line.
- A variety of system configurations and optional equipment.
- Optimal performance-to-cost ratio.



SPECIFICATIONS

STABILIZED GUNNER'S DAY/NIGHT LASER SIGHT:

DAY CHANNEL:

Magnification:	10x
Field of View:	6°

Night Channel:

Magnification:	7.5x
Field of View:	7.5°
Image Intensifier:	Second Gen

Stabilized Top Mirror:

AZ Deflection (mils):	± 45
EL Deflection (mils):	-22.5 + 67.5

Laser Range Finder:

Wave Length (um):	1.064
Output Energy (mj):	>30
Number of Targets:	Two Targets
Accuracy (m):	±7.5
Max. Range (m):	9995

Fire Control Computer:

Input Range:	Laser or Manual
Ammunition:	Up to 7 Types



3/4 TON THISTLE TACTICAL VEHICLE

PIC: 000189

DESCRIPTION

It can be used by military operations forces as a platform for weapons and military equipment for deployment in military and peace support operations.

SPECIFICATIONS

TYPE	3/4 TON TACTICAL VEHICLE
Overall Length (cm):	500 ± 20
Overall Width (with Side mirrors)(cm):	200 ± 10
Net Weight (kg):	Max. 2,700
Maximum Towable Load off Road (kg):	750
Traversal Lateral Gradient:	33%
Traversal Longitudinal Gradient (mm):	300
Engine Model:	CYQD32TI
Cylinders Arrangement:	4 in Line
T-turn Radius(m):	7.5
Engine Power:	133 HP / 3,600 RPM
Emission Standard:	EURO II



1/4 TON THUNDER - TACTICAL VEHICLE

PIC: 000190

DESCRIPTION

Double differential, 1/4 ton Thunder tactical vehicles. A four door tactical vehicle with suitable equipment for different requirements of headquarters and operational units of armed forces with special capabilities.

SPECIFICATIONS

TYPE	1/4 TON TACTICAL VEHICLE
Overall Length(cm):	330 ± 20
Width (with spare tyre) (cm):	180 ± 10
Gross Weight (kg):	1,600 ± 50
Net Weight (kg):	Max. 1,600
Maximum Towable Load off Road (kg):	250
Maximum Passable Depth of Water (cm):	60
Longitudinal Gradient:	% 50
Stability in Latitudinal Gradient:	% 30
Engine Model:	Z24 Produced by Mega Motor
Cylinders Arrangement:	4 - in Line
Cylinder Diameter (mm):	89





1 1/4 TON TACTICAL VEHICLE

PIC: 000191

DESCRIPTION

1 1/4 Ton Tactical Vehicle equipped with two speed transfer case and mechanical engagement system, 4WD, with differential lock and driver cab. Also equipped with suspension system including leaf spring and telescopic shock absorber. Hard and soft top, folding wind screen. Having two combat and civil lighting system.

SPECIFICATIONS

TYPE	1 1/4 TON VEHICLE
Number of Cylinders:	4
Engine Type:	OM 314 M.B
Max. Power:	85 HP
Brake System:	Hydraulics with Boosters
Suspension System:	Steel Straps and Telescopic Type
Net Weight (unloaded) (kg):	3,060
Max. Weight (loaded) (kg):	4,310
Max. Speed (km/h)	100
Steering Wheel:	Regular
Clutch:	Single Disk
Power Source:	Single Battery (12 - Volt, 150 - Ampere)



MOBILE WORKSHOP

PIC: 000192

DESCRIPTION

Manufactured to support equipment, mechanically & electrically, in the battle field.

SPECIFICATIONS

Dimension: 6000 × 2450 × 2120 mm
Cabin material: 2 hulls with metal frames, hulls made of steel sheets which are injected with polyurethane foam.
Equipped with working table, tools, foldable 3-ton crane, welding rectifier, acetylene welding, bed and refrigerator.





ARMOR TRAINING SIMULATOR

PIC: 000193

DESCRIPTION

M.o.D of Iran design and manufacture **T-Series Tank Simulators** (driver, gunner, comander) and **APC's Simulators**.

Training simulators are designed for a joint instruction and training their groups of vehicle crew (commander, gunner and driver) in primary level. The 3 dimensional animation is used to simulate different climates, day/ night , and emergency condition.

The crew will operate with all kinds of ammunition. **ATGM** (Anti Tank Guided Missile) in the basic and emergency mode at halt and on the move, stationary and moving tank against the stationary or moving targets as well as flying targets (e.g.helicopters,...) without using real ammunition and consuming service life of combat or training vehicles in condition of A class.



LIGHT MODERN BALLISTICS ARMOR

PIC: 000194

DESCRIPTION

By light and modern ballistics armor, all types of personnel carriers, vehicles and significant centers and establishments resist the following projectiles in accordance with standard stanag 4569:

7.62 × 39 AP (AK47)	14.5 mm AP
30.06 AP M2(NIJ)	20 mm FSP
12.7mm AP (0.50 calibre AP)	23mm APDS and 30mm AP

In Iran , this armor is used to make resistant against the personnel carriers such as **BMP2** and **M113**.

As effects of present weapons on armor equipment in term of energy and calibre increase, it is necessary that personnel carriers get resistant against these threats.

It should be mentioned that M113 and BMP2 resist only up against calibre 7.62 mm and BMP2 and BORAQ resist only up to calibre 12.7 mm ammunition.

The countries that take the advantage of these armors technology include; U.S, France, zionist regime and Iran.





4

TAFTAN 2 - MINE CLEANER

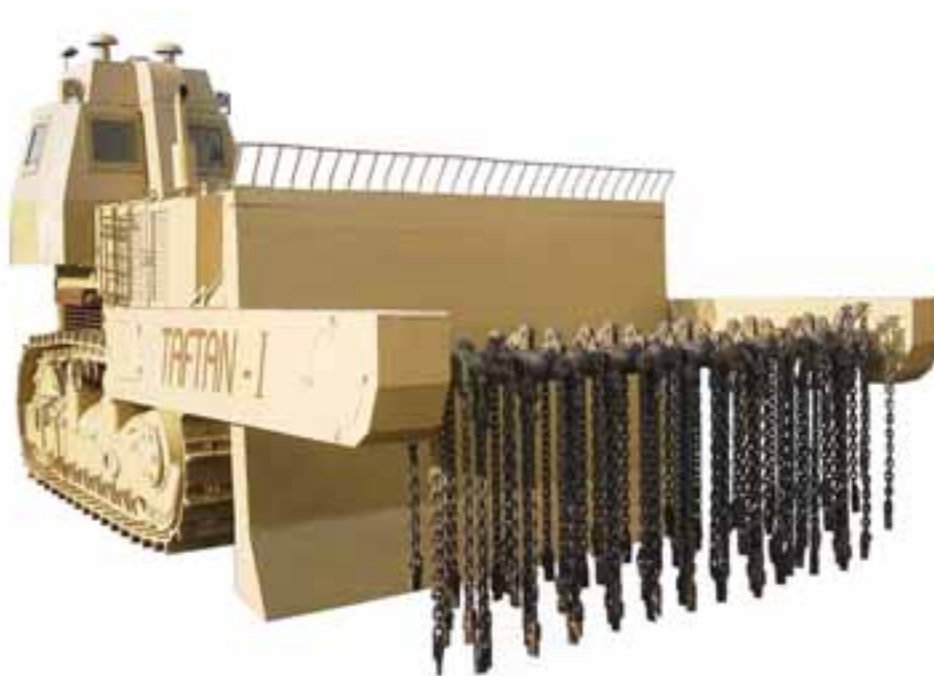
PIC: 000195

DESCRIPTION

The mine clearing vehicle **Taftan 2** fitted with mine clearing flail system is capable to clear both anti-personnel and anti-tank mines and provides the vehicle with a higher maneuverability and mobility due to the equipment proper weight.

FEATURES

- Proper function of flail system on negative and positive slopes.
- Driver proper field of view in four directions.
- Provided with cooler, heater, and air conditioner.
- Possible continuation of flailing operations in one working shift (8 hours).
- Communication and navigation equipments (equipped with a multi-channel radio, gyro radio set of vehicle).
- Provided with hydraulic steering, drive hydraulic joy stick (triple positions) and pneumatic brakes.



SPECIFICATIONS

TYPE	TAFTAN 2
Max. Clearance Power and Mine Detonation (kg):	10 (TNT)
Clearance Depth (cm):	30
Clearance Width (m):	3.04
Flailing Mechanism:	Hammer and Chain
Clearance Speed (km/h):	On surface 2 - 3
Engine Type (HP):	MB - OM 355 - A
Engine Power:	280
Vehicle Protective System:	Armored
Power Transmission System:	Hydrostatic
Cabin:	Armored with Anti Blast Glass
Crews:	2 Men



ROLLER TANK MINE CLEANER

PIC: 000196

DESCRIPTION

The mine clearing roller vehicle is designed and prototyped regarding the requirement for higher reliability and completion of clearance process (using a combination of mechanized equipments) with the capability of detonation of various anti-personnel and anti-tank mines.

FEATURES

- Maintaining the regular arrangement and array of minefields during clearance by MRT and minimum displacement of mines.
- Provided with ergonomic cabin (fitted with cooling and heating, communication, control and navigation systems).
- The equipment can withstand explosion of the most powerful and destructive anti-tank mines and consequent fragments and bursts.
- Capability of operating to enhance clearance assurance coefficient during different clearance processes.
- Communication and navigation equipments (equipped with a multi-channel radio, gyro radio set of vehicle).
- Possible replacement of damaged rollers in less than 15 minutes.



SPECIFICATIONS

TYPE	MINE CLEANER
Type of Vehicle:	T - 55 Tank
Max. Vehicle Speed (km/h):	15
Flailing Mechanism:	Roller
Max. Clearance Power and Mine Detonation:	4.5 kg TNT (Resistant to Anti - Tank Mines)
Min. Clearance Speed (km/h):	On Surface 4
Clearance Lane Width (m):	3.4
Max. Clearance Depth (cm):	30
Roller System Weight (ton):	12
Weight (ton):	45
Cabin:	Armored With Anti Blast Glass
Crews:	2 Men



MINE SIFTER



PIC: 000197

DESCRIPTION

The **Mine Sifter** is designed and prototyped for completion of clearance process (use of combined mechanized equipments) and for collecting residual mines and **UXOs** (after mine clearance operations by mine clearing flail and roller equipments).

FEATURES

- Use of sifter mine sweeper mechanism and automatic discharge of sieved particles (debris, fragments etc.).
- Enhancement of clearance efficiency by other mechanized clearance equipments.
- Equipped with a safe and ergonomic armored cabin for the operator.
- Full protection of hydraulic assemblies of flail mechanism drive from probable blasts.
- Ease of operation similar to road machinery and more similarity of parts with other road construction machines.
- Fitted with depth sensor and monitoring system.



SPECIFICATIONS

TYPE	MINE SIFTER
Type of Vehicle:	Bulldozer D-155- A2
Max. Vehicle Speed (km/h):	11.8
Flailing Mechanism:	Sifter
Max. Clearance Power and Mine Detonation:	4.5 kg TNT (Resistant to Anti - Tank Mines)
Min. Clearance Speed (km/h):	2
Cleared Lane Width (m):	3
Max. Clearance Depth (cm):	35
Sweeper System Weight (ton):	6
Operational Weight (ton):	35
Cabin:	Armored with Anti Blast Glass



ARMORED BULLDOZER MINE CLEANER

PIC: 000198

DESCRIPTION

The armored bulldozer is designed and prototyped for completion of mechanized clearance cycle and upon request of state de-mining center taking advantage of bulldozer D-155-A2 as the mover vehicle fitted with armor body and plowshares mainly used for final mine clearing and plowing of agricultural areas.

FEATURES

- Provided with communication system.
- Provided with heating and cooling system as well as wiper.
- Fitted with armored plowshare.



SPECIFICATIONS

TYPE	ARMORED BULLDOZER
Type of Vehicle:	Bulldozer D-155- A2
Max. Traveling Speed (km/h):	11.8
Max. Power (hp):	314(at 2000 RPM)
Max. Clearance Power and Mine Detonation:	4.5 kg TNT (Resistant to Anti - Tank Mines)
Min. Plowing Depth(cm):	40
Plowing Width(m):	3
Distance Between Plowshares(cm):	10
Max. Gradient and Side Slope:	30°
Operational Weight (ton):	45
Cabin:	Armored with Anti Blast Glass
Crew:	1 Man

The background of the entire page is a photograph of a missile launcher. Four missiles are mounted on the launcher, pointing diagonally upwards towards the top right corner. The launcher itself is a complex mechanical structure with various straps, cables, and mounting points. The entire image has a greenish-grey tint. Two horizontal dotted white lines are positioned above and below the main title.

chapter

5

**ROCKET
AND
MISSILE**



PREVIEW

Rocket and Missile Industries are a leading industrial and military organization which is affiliated to Ministry of Defense and Armed Forces Logistic of I.R of Iran.

More than 25 large factories with over 12,000 expert personnel are involved in manufacturing a vast variety of military products, which partially can be exported. The factories are equipped with modern facilities and long experiences which are led to high quality products according to the optimum international standards.

1- **Products:** different types of rockets and missiles such as: air defense missiles and guns, anti ship missiles, surface to surface rockets and missiles and launchers, gyroscopes and etc.

2- **Technical and engineering services:** precision machining by CNC machines, metal forming ,training & software services, quality control, environmental test, heat treatment & coating, C.M.M measurement equipped metrology lab casting of metals & super alloys by investment casting- manufacturing of non metal parts (composite, rubber, ...) analysis & metallography by equipment (electronic microscopes, ...) .

3- Based on the technical and professional capability of its experts, our industries have achieved the necessary know how in many fields of rocket and missile systems, tests for flights, static and explosives, it constantly promotes the products.

4- Rendering technical assistance to the domestic and external customers for designing and developing missile system are among the activities that are currently being performed.

5- Other services: overhaul, modification and modernization of different types of missiles together with their relevant ground equipment is another task carried out by this organization.

By applying management systems like Project Management, **ISO 140001**, **9001/2008**, **OHSAS 18001**, etc., the industries have set before the eyes the quality and safety measures as their working patterns.





CHAPTER 5



ITEM	TABLE OF CONTENT	PAGE
1	FATH & NAFEZ - 40MM ANTI ARMOR ROCKETS	142
2	SAEGHEH 40 MM ANTI PERSONNEL ROCKET	143
3	AIR TARGET OKM IMITATOR ROCKET	143
4	ZAFAR & GHADIR - ANTI ARMOR ROCKETS	144
5	BASIC TOOPHAN - HEAVY ANTI ARMOR GUIDED MISSILE	145
6	TOOPHAN 2 - HEAVY ANTI ARMOR GUIDED MISSILE	146
7	TOOPHAN3 - HEAVY ANTI GUIDED MISSILE	147
8	TOOPHAN MISSILE - ADVANCED GUIDANCE SET	148
9	TOOPHAN LAUNCHER	148
10	SAEGHEH - LIGHT WEIGHT ANTI ARMOR GUIDED MISSILE	149
11	SAEGHE 2 - LIGHT WEIGHT ANTI ARMOR GUIDED MISSILE	150
12	FADJR 1 - 107MM ROCKET	151
13	ARASH - 122MM ROCKET	152
14	FADJR 2&3 - 240MM ROCKETS	153
15	FADJR 5 - 333MM ROCKET	154
16	FADJR 3 & 5 - BOXY LAUNCHER	155
17	FADJR 3 & 5 - BOXY LOADING MACHINE	156
18	LOADING CRANE	156
19	FALAGH 1 & 2 ROCKETS	157
20	NAZEAT 6-H - SURFACE TO SURFACE ROCKET	158
21	NAZEAT 10-H - SURFACE TO SURFACE ROCKET	158
22	ZELZAL 3 - SURFACE TO SURFACE ROCKET	159
23	ZELZAL 3-B - SURFACE TO SURFACE ROCKET	159
24	FATEH 110 - SURFACE TO SURFACE ROCKET	160
25	ROCKETS & MISSILES COMMAND CABIN	161
26	MOBILE REPAIR UNIT	162
27	METEOROLOGICAL VEHICLE FOR ROCKETS & MISSILES	162
28	MISAGH-1 - PORTABLE SHOULDER LAUNCHER	163
29	SHAHAB THAQEB - AIR DEFENCE MISSILE	164
30	SHALAMCHE - AIR DEFENCE MISSILE	165
31	ZAFAR - ANTI SHIP MISSILE	166
32	NASR - ANTI SHIP MISSILE	167
33	NOOR - ANTI SHIP MISSILE	168
34	NOOR MISSILE - RADAR AND COMMAND VEHICLE	169
35	NOOR MISSILE - LAUNCHING VEHICLE	169



FATH & NAFEZ 40MM ANTI ARMOR ROCKETS

PIC: 000272

DESCRIPTION

It is designed to destroy tanks, self propelled guns and other armored vehicles. It can also be used to destroy enemies' troops in field shelters and other fortifications.



FATH



NAFEZ



SPECIFICATIONS

TYPE	FATH	NAFEZ
Maximum Effective Range (m):	300	200
Minimum Range (m):	50	50
Rate of Fire (r/m):	4	4
Caliber (mm):	40	40
Total Weight (kg):	2.350	2.700
Muzzle Velocity (m/s):	120	112
Maximum Velocity (m/s):	300	270
Length (mm):	920	960
Warhead:	HEAT	HEAT
Penetration (mm):	300	500
Fuse:	Electro Mechanical Impact	Electro Mechanical Impact
System:	Anti Tank Rocket Launcher RPG-7	Anti Tank Rocket Launcher RPG-7



SAEGHEH 40 MM ANTI PERSONNEL ROCKET

PIC: 000273

DESCRIPTION

It is designed for targeting the enemies' troops in open trenches, field shelters and brick-walled fortifications.

SPECIFICATIONS

TYPE	SAEGHEH
Maximum Effective Range (m):	700
Minimum Range (m):	100
Rate of Fire (r/m):	4
Caliber (mm):	40
Total Weight (kg):	1.550
Muzzle Velocity (m/s):	140
Length (mm):	590
Warhead:	HE Fragmentation
Fuse:	Mechanical Impact
System:	Anti Tank Rocket Launcher RPG -7



AIR TARGET IMITATOR OKM ROCKET

PIC: 000274

DESCRIPTION

It is designed to be used as an aerial target for practice firing of IR homing surface to air missile.

SPECIFICATIONS

TYPE	OKM ROCKET
Maximum Range (m):	5000
Maximum Altitude(m):	1200
Maximum Velocity(m/s):	275
Caliber (mm):	132
Total Weight (kg):	45
Length (mm):	1420
Launch Angle:	30 to 40
Flight Duration (s):	30
Number of Flares:	6
Visisbility of Flares(km):	up to 5





ZAFAR & GHADIR ANTI ARMOR ROCKETS

PIC: 000275

DESCRIPTION

It is Intended for destroying tanks including those with explosive reactive armor (ERA), self-propelled guns and other armored vehicles.

It can also be used for suppressing infantry and firing position in brick and reinforced concrete buildings and earth shelters.



GHADIR LAUNCHER



GHADIR



ZAFAR



ZAFAR LAUNCHER

SPECIFICATIONS

TYPE	GHADIR	ZAFAR
Maximum Sighting Range (m):	500	800
Blank Point Range (m):	300	
Minimum Range (m):	50	
Rate of Fire (r/m):	4	2
Caliber (mm):	105	73
Total Weight (kg):	6.700	4.390
Muzzle Velocity (m/s):	230	435
Length (mm):	1100	1100
Warhead:	Tandem HEAT	HEAT
Penetration:	600 mm after ERA	300 mm
Fuse:	Electro Mechanical Impact	Mechanical impact
System:	Anti Tank Rocket Launcher RPG 29	Anti Tank Rocket Launcher SPG-9



BASIC TOOPHAN HEAVY ANTI ARMOR GUIDED MISSILE

PIC: 000301

DESCRIPTION

TOOPHAN is one of the most advanced anti-armore missiles of the world which is capable of destroying stationary and moving targets.

Toophan missile system is semi- automatic using command to line of sight guidance, that is to say , the operator shall simply keep the cross hairs of the sight on the target, confirm tracker lock on and then launch the missile.

During the flight, control commands are sent automatically to the missile by missile guidance set (**M.G.S**).

Meanwhile there is capability for re-lifing and upgrading these missiles.

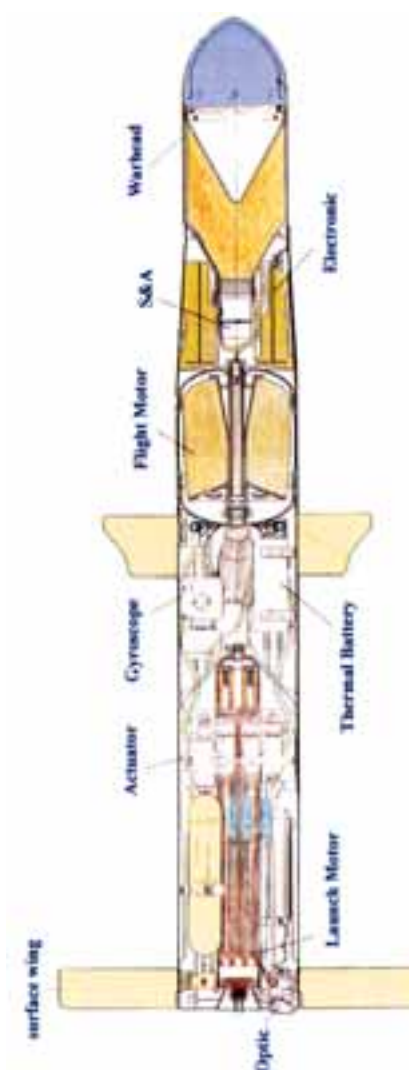


LAUNCHING PADS

- Ground Tripod
- Armored Vehicle
- Helicopter

SPECIFICATIONS

TYPE	BASIC TOOPHAN
Guidance:	SACLOS
Warhead:	Shaped Charge
Warhead Weight (gr):	3,600
Penetration Depth (mm):	550
Missile Weight(gr):	18,500
Missile Length (extended probe) (cm):	116
Missile Diameter (cm):	15
Fin Length (cm):	45
Time of Flight (S):	13
Min. & Max. Range (m):	70 - 3,850
Effective Range in Day (m):	3,500
Effective Range in Night(m):	2,500
Max. Velocity (m/s):	310
Average Velocity (ms):	240
Reliability (%):	95
Rate of Fire (per Min.):	2 - 3
Operation Tempreture (°c):	-40 - +65
Packed Missile Weight (gr):	42,000





TOOPHAN 2 HEAVY ANTI ARMOR GUIDED MISSILE

PIC: 000302



DESCRIPTION

TOOPHAN 2 is an upgraded version of the **basic Toophan**. In order to increase armor penetration depth, a mechanical telescopic probe has been added to the front of the missile.

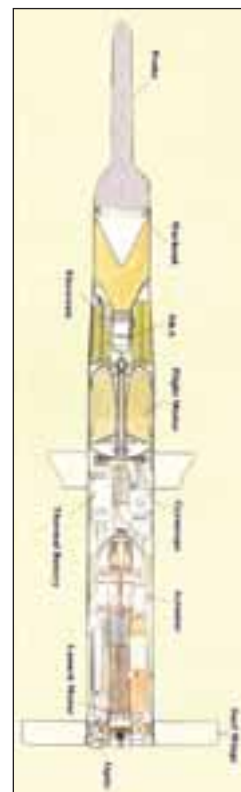
The missile carries two warheads ; precursor and main warhead. The precursor warhead located on the nose of telescopic probe destroys Explosive Reactive armor (**ERA**) and allows the main warhead to penetrate and destroy the target.

LAUNCHING PADS

- Ground Tripod
- armored Vehicle
- Helicopter

SPECIFICATIONS

TYPE	TOOPHAN 2
Guidance:	SACLOS
Main Mission:	Penetrating in Explosive Reactive Armure (ERA)
Warhead:	Tandem Heat
Warhead Weight (gr):	4100
Probe Caliber (mm):	40
Penetration Depth (mm):	760
Missile Weight(gr):	19,001
Missile Length (extended probe) (cm):	145
Missile Length (closed probe) (cm):	116
Missile Diameter (cm):	15
Min. & Max. Range (m):	70 - 3850
Effective Range in Day (m):	3500
Effective Range in Night(m): (Equipped with Thermal Sight)	2500
Max. Velocity (m/s):	310
Average Velocity (ms):	240
Reliability (%):	95
Rate of Fire (per Min.):	2 - 3
Operating Tempreature (°c):	-40 - +65





TOOPHAN 3 HEAVY ANTI ARMOR GUIDED MISSILE

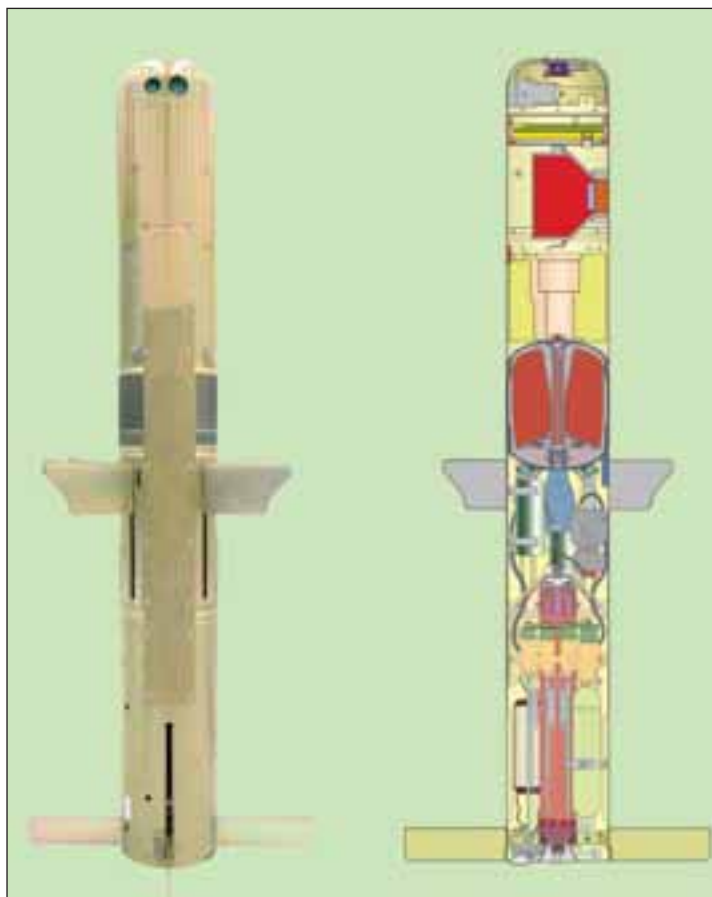
PIC: 000303

DESCRIPTION

TOOPHAN 3 is the new generation of Toophan **ATGM** and is designed to attack the top of the tanks.

Toophan 3 is able to destroy tanks behind any barrier and is equipped with anti-jamming system.

Operation Capability: engagement of magnetic and laser sensors and flying over line of sight.



SPECIFICATIONS

TYPE	TOOPHAN 3
Guidance:	SACLOS
Warhead:	EFP
Penetration Depth:	Min. 80mm (In RHA Armor)
Warhead Weight(kg):	1.7
Warhead Diameter (cm):	12.8
Missile Weight (kg):	19.1
Missile Length (cm):	116
Missile Diameter (cm):	15
Min. & Max. Range(m):	650 - 3,500
Effective Range in Day (m):	3,500
Effective Range in Night (m):	2,500
Max. Velocity (m/s):	310
Average Velocity (m/s):	240
Rate of Fire (r/m):	2 - 3
Operation Temperature (°c):	- 40 + 65



TOOPHAN MISSILE ADVANCED GUIDANCE SET

PIC: 000304

DESCRIPTION

TOOPHAN launcher launches, tracks and guides the missile through two wires.

The major part of the launcher is **Missile Guidance Set (MGS)**.

Regarding series production of Toophan missiles and the need for MGS for the launcher, Iran Defence Industries defined and started the design and manufacture the MGS's with necessary improvement and more capability to the basic set.

MGS demonstrated and showed its capabilities in numerous field tests.



TOOPHAN LAUNCHER

PIC: 000305

SPECIFICATION , TECHNOLOGICAL AND OPERATIONAL CAPABILITIES

- Advanced guidance algorithms.
- Advanced tracking.
- Adaptability for launching various new types of Toophan missile.
- Capability of confronting enemy's Electro-Optic war (Anti-jammer)
- Capability of performing automatic self test.
- The ability to locate the troubles regarding the set parts (to facilitate and hasten service and repairs).
- A digital indicator to perform self-test and bore sight increasing of hit probability in long ranges.
- Increasing of missile operational area from the point of Min. and Max. range.
- Decreasing of calibration costs and periodical repairment.
- Tracking of high speed targets.
- Light weight compared with MGS.





SAEGHE - LIGHT WEIGHT ANTI ARMOR GUIDED MISSILE

PIC: 000306

DESCRIPTION

SAEGHE missile, which is competitive of **DRAGON** and **METIS**, is one of the short-range missiles which has a carrying weight of 14.5kg and can be easily carried by a person.

Guidance of the missile is through wire and Commands to the Line Of Sight (**CLOS**). Due to high accuracy of the tracking system, it is capable To destroy different kinds of moving and stationary targets.

It has more than 90% reliability in maximum range of 1000 meters.

DESTROYING CAPABILITIES

- Tanks
- Armored Vehicles
- Concrete Structures

LAUNCHING PADS

- Ground Bipod
- Armored Vehicles

SPECIFICATIONS

TYPE	SAEGHE
Missile & Launcher:	
carrying weight(gr):	14,500
Length(cm):	115.4
Tracker:	
Guidance:	SACLOS
Weight(gr):	3100
Accuracy (m):	± 4 (at full range)
Missile :	
Warhead:	Shaped Charge
Penetration Depth (mm):	500
Rate of Fire (fire/Min.):	3. 4
Weight(gr):	6100
Length(cm):	74.4
Diameter (cm):	12.7
Muzzle Velocity (m/s):	76
Max. Velocity (m/s):	100
Time of Flight (s):	11.3
Range (m):	65 - 1000
Operational Temperature (°c):	-32 - +62





SAEGHE 2 - LIGHT WEIGHT ANTI ARMOR GUIDED MISSILE

PIC: 000307

DESCRIPTION

SAEGHE 2 is an upgraded version of the **BASIC SAEGHE**. Warhead of this missile is improved in order to destroy tanks with Explosive Reactive Armor (**ERA**).

This missile has an approximate weight of 15.8 kg which can be easily carried by a person. Saeghe2 is fired through tracker and its max. range is 1000 meter in 11.5 seconds.

DESTROYING CAPABILITIES

- Tanks with Reactive Armor
- Armored Vehicles
- Concrete Structures

LAUNCHING PADS

- Ground Bipod
- Armored Vehicles

SPECIFICATIONS

TYPE	SAEGHE 2
Missile & Launcher:	
carrying weight(gr):	15,800
Length(cm):	115.4
Tracker:	
Guidance:	SACLOS
Weight(gr):	3100
Accuracy (m):	± 4 (at full range)
Missile :	
Warhead:	Tandem Heat
Probe Caliber (mm):	40
Penetration Depth (mm):	650
Rate of Fire (fire/Min.):	3 - 4
Weight(gr):	7400
Diameter (cm):	12.7
Max. Velocity (m/s):	92
Time of Flight (s):	11.5
Range (m):	65 - 1000
Operation Temperature (°c):	- 32 + 62
Main Mission:	Penetrating Explosive Reactive Armor (ERA)
Missile Length (closed probe) (cm):	84
Missile Length (extended probe) (cm):	100





FADJR1 - 107MM ROCKET

PIC: 000283

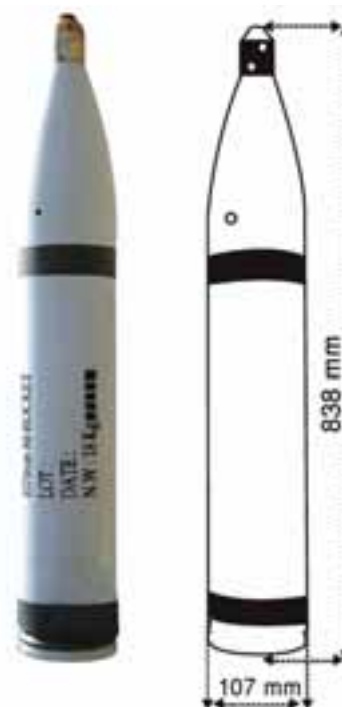
DESCRIPTION

107 mm Fadjr1 rocket has been designed and produced to destroy enemy's offensive force and hidden troops in trenches.

It can also be used to target artillery batteries fires and destroy platoons of tanks, armored vehicles and technical equipments.

This rocket is capable to strike ground equipment, command posts and ordnance depots.

A 12 tube for army and 16 tube for marin forces used as their launchers.



SPECIFICATIONS

TYPE	FADJR 1
Maximum Range in Sea Level (km):	8.3
Fragmentation Radius (m):	100
Maximum Speed (m/s):	375
Average Time of The Motor Operation (s):	0.6
Rocket Length (mm):	838
Rocket Caliber (mm):	107
Rocket Weight (kg):	18
Warhead Weight (kg):	7.9
Quantity Per Box (round):	Two
Type of Warhead:	High Explosive
Type of Propellant:	Double Base
Warehousing (years):	15



ARASH - 122MM ROCKET

PIC: 000284

DESCRIPTION

122 mm rockets are used against enemy man-power in the open and in the field shelters, soft skinned equipment, armored personnel carriers, command posts and other targets.

This rocket can be fired with a launcher that is compatible to **BM-21**.

FEATURES

- The launcher is capable to fire 120 shots per minute.
- Fast maneuverability of the truck.
- The rocket can be equipped with proximity fuze.



SPECIFICATIONS

TYPE	ARASH 1 SHORT RANGE	ARASH 2 STANDARD RANGE	ARASH 3 LONG RANGE
Maximum Range (km):	12	21	40
Minimum Range (km):	1.5	1.6	4
Rate of Fire (r/s):	40 Rounds in 20 seconds	40 Rounds in 20 seconds	40 Rounds in 20 seconds
Caliber (mm):	122	122	122
Total Weight (kg):	44	65	67.5
Maximum Velocity (m/s):	530	710	1100
Length (mm):	1850	2860	2892
Warhead:	HE Fragmentation	HE Fragmentation	HE Fragmentation
Weight of Warhead (kg):	19	19	22
Fuse:	Mechanical Impact	Mechanical Impact	Mechanical Impact



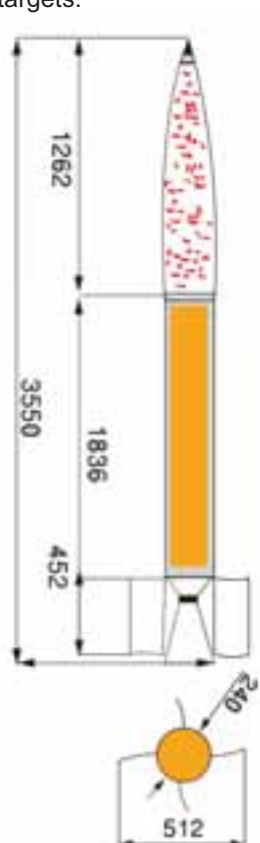
FADJR 2&3 240MM ROCKETS

PIC: 000285

DESCRIPTION

Solid propellant, wrap around fins, multiple launch rockets have been designed and generated for different artillery missions.

240 mm Fadjr 2 & 3 multiple rocket systems are deployed to attack enemy's special force such as command, logistic, economical and political centers, radar stations, communication networks, airfields and other important targets.

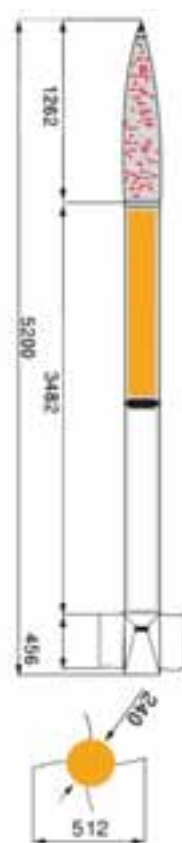


FADJR 3 LAUNCHER



FADJR 2

FADJR 2 LAUNCHER



FADJR 3

SPECIFICATIONS

TYPE	FADJR 2	FADJR 3
Maximum Range in Sea Level (km):	23	43
Fragmentation Radius (m):	300	300
Maximum Speed (m/s):	670	930
Average Time of The Motor Operation (s):	4	4
Rocket Length (mm):	3,550	5,200
Rocket Caliber (mm):	240	240
Rocket Weight (kg):	275	407
Warhead Weight (kg):	85	85
Type of Warhead:	High Explosive	High Explosive
Type of Propellant:	Double Base	Double Base
Warehousing (years):	15	15



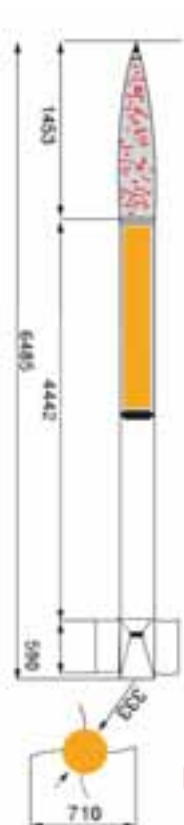
FADJR 5 - 333MM ROCKET

PIC: 000286

DESCRIPTION

Solid propellant, wrap around fins, multiple launch rocket have been designed and generated for different artillery missions.

333 mm Fadjr 5 multiple rocket system is deployed to attack enemy's special force such as command, logistic, economical and political centers, radar stations, communication networks, airfields and the other important targets. Two stages of this rocket extended range upto 180 km.



FADJR 5 ROCKET



FADJR 5 TWO STAGES LAUNCHER



FADJR 5 ROCKET LAUNCHER



FADJR 5 TWO STAGES

SPECIFICATIONS

TYPE	FADJR 5	FADJR 5 (TWO STAGES)
Maximum Range in Sea Level (km):	75	180
Fragmentation Radius (m):	500	500
Maximum Speed (m/s):	1,100	
Average Time of The Motor Operation (s):	5.3	
Rocket Length (mm):	6,485	9,400
Rocket Caliber (mm):	333	333
Rocket Weight (kg):	907	
Warhead Weight (kg):	175	175
Quantity Per Box (metal pallet):	One Round	One Round
Type of Warhead:	High Explosive	High Explosive
Type of Propellant:	Double Base	Double Base
Warehousing (years):	15	15



FADJR 3 & 5 - BOXY LAUNCHER

PIC: 000287

DESCRIPTION

Fadjr 3 and Fadjr 5 mobile launchers are efficient short range launchers, with high mortality radius and accuracy, plays an effective role in providing heavy fires. Having high maneuverability, low seating and launching time, it can help the launcher to deploy fast and be more effective in surprisal enemy's attack.

These multiple launch rocket system are used to create massive and effective fires to support operating forces during combat, destroy economic and military installations and stations, and insecure naval routes in range radius.

Capability of displacement in difficult terrains, performing launching operations at night, ability of firing in single shot or salvo and reloading operations, and rapid deployment has increased its operating power.

Using automation and firing control system, high efficient electromechanical and communicative mechanisms, disuse of hydraulic mechanisms which require particular maintenance, has given more efficiency to the rocket launcher in difficult terrains.

Advanced electronic systems are able to provide a network of different launchers. Fire management plays an important role in guidance of the fire volumes to produce an effect on the launcher within specified time loading operation accompanied with firing tube's box replacement in fastest possible time and simplicity to operate, will increase the weapon loading operation speed during combat operations.



FADJR 3 BOXY LAUNCHER



FADJR 5 BOXY LAUNCHER

SPECIFICATIONS

TYPE	FADJR 3	FADJR 5
Launcher Length in Traveling State (mm):	10,000	10,000
Launcher Width in Traveling State (mm):	2,500	2,500
Launcher Height in Traveling State (mm):	3,340	3,340
Maximum Width of The Launcher in Azimuth State (57°) (mm):	6,135	7,000
Maximum Width of The Launcher in Azimuth State (90°) (mm):	5,450	6,810
Firing Tube Length (mm):	5,360	5,360
Weight Without Rocket (kg):	15,000	15,000
Weight With 12 Rounds Rocket for Fadjr 3 & 4 Rounds for Fadjr 5 (kg):	19,900	19,900
Maximum Range Angle:	57°	53°
Maximum Azimuth Angle (right):	90°	+45°
Maximum Azimuth Angle (left):	100°	- 45 °



FADJR 3 & 5 BOXY LOADING MACHINE

PIC: 000288

DESCRIPTION

333 mm Fadjr 3 & 5 boxy loader has been applied to load twin barreled box of the Fadjr 3 or 5 launchers. A railed chainwheel system is used for loading the barrel box in this mechanism.

Fastest possible loading time and highest accuracy, simple operation and increasing the loading safety, decreasing the rocket body damage and human dangers during manual loading, time saving, lightness and portability are the most important features of the 333 mm Fadjr boxy loader. Since this product can be rapidly assembled and disassembled, if necessary, it is possible to move it to any location.



SPECIFICATIONS

TYPE	FADJR 3 LOADER	FADJR 5 LOADER
Length (mm):	10,400	120,000
Width (mm):	1,100	1,600
Height (mm):	1,300	1,050
Type of Packing:	Waterproof Cover	Waterproof Cover
Quantity of Loadable Rockets in Every Stage (Round):	3	1
Loading Time for Every Rocket (Minute):	2	2
Consumed Electricity:	Uni-phase and Triple Phase	Uni-phase and Triple Phase

LOADING CRANE

PIC: 000276

DESCRIPTION

This crane has been deployed to displace the existent tubes, rockets, boxes that have the capability of installing on the different type of vehicles.

It can be more helpful to the possibility of loading and transferring the rockets and related Fadjr 3 and Fadjr 5 boxes, loading the filled boxes on the carrier vehicles in the fastest possible time.

The crane has 4 rounds hydraulics jacks which make the loading possibility safer and easier.



SPECIFICATIONS

TYPE	LOADING CRANE
Type of Crane:	Joint Folding
Model:	EFEER 155 - 25
Loading Weight in Maximum Distance (kg):	2,200
Loading Weight in Minimum Distance (kg):	4,800



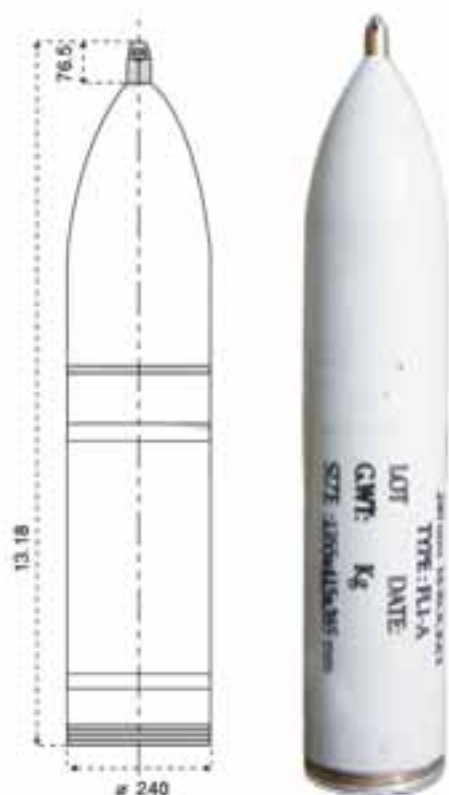


FALAGH 1 & 2 ROCKETS

PIC: 000289

DESCRIPTION

Having a wide spreading explosion rockets, **240 mm Falagh1** and **333mm Falagh 2** have been designed to be deployed in attack and defend positions. Reinforcement of artillery fires and destroying enemy's force and equipment are the other purposes of their design. The rockets can be installed on combat vessels.



FALAGH 1



FALAGH1 LAUNCHER



FALAGH2 LAUNCHER



FALAGH1 ON VESSEL



FALAGH 2



SPECIFICATIONS

TYPE	FALAGH 1	FALAGH 2
Maximum Range in Sea Level(km):	10	10. 8
Maximum Flight Altitude (m):	3,500	3,200
Fragmentation Radius(m):	150	300
Maximum Speed(m/s):	445	376
Average Time of Motor Operation(m/s):	1.3	1.85
Rocket Length(m/s):	1,320	1,820
Rocket Caliber(mm):	240	333
Rocket Weight (kg):	113	256
Warhead Weight (kg):	50	117
Quantity per Wooden Box (can):	One Round	One Round
Type of Warhead:	High Explosive	High Explosive
Type of Propellant:	Double Base	Double Base



NAZEAT 6 - H SURFACE TO SURFACE ROCKET

PIC: 000308

SPECIFICATIONS

TYPE	NAZEAT 6-H ROCKET
Max. Range (km):	100
Min. Range (km):	80
Length (mm):	6,290
Diameter (mm):	356
Initial Weight (kg):	960
Warhead Weight (kg):	130
C.E.P:	<5% Max. Range
Average Action Time (sec):	9
Specific Impulse (sec):	240
Propellant Weight (kg):	420
Type of Propellant:	Solid (HTPB)
Service Life (Years):	7



NAZEAT 10-H SURFACE TO SURFACE ROCKET

PIC: 000309

SPECIFICATIONS

TYPE	NAZEAT 10-H ROCKET
Max. Range (km):	130
Min. Range (km):	110
Length (mm):	8020
Diameter (mm):	457
Initial Weight (kg):	1830
Warhead Weight (kg):	230
C.E.P:	<5% Max. Range
Average Action Time (sec):	10
Specific Impulse (sec):	240
Propellant Weight (kg):	865
Type of Propellant:	Solid (HTPB)
Service Life (Years):	7





5

ZELZAL 3 SURFACE TO SURFACE ROCKET

PIC: 000310

SPECIFICATIONS

TYPE	ZELZAL 3 ROCKET
Max. Range (km):	200
Min. Range (km):	180
Length (mm):	9,600
Diameter (mm):	616
Initial Weight (kg):	3,870
Warhead Weight (kg):	900
C.E.P:	<5% Max.Range
Average Action Time (sec):	20
Specific Impulse (sec):	240
Propellant Weight (kg):	1,980
Type of Propellant:	Solid (HTPB)
Service Life (Years):	7



ZELZAL 3-B SURFACE TO SURFACE ROCKET

PIC: 000311

SPECIFICATIONS

TYPE	ZELZAL 3-B ROCKET
Max. Range (km):	250
Min. Range (km):	235
Length (mm):	9,000
Diameter (mm):	616
Initial Weight (kg):	3,600
Warhead Weight (kg):	600
C.E.P:	<5% Max.Range
Average Action Time (sec):	20
Specific Impulse (sec):	240
Propellant Weight (kg):	1,950
Type of Propellant:	Solid (HTPB)
Service Life (Years):	7



ROCKET & MISSILE



FATEH 110 SURFACE TO SURFACE ROCKET

PIC: 000312

DESCRIPTION

Destroy bases and main military buildings , important centers and airports.



SPECIFICATIONS

TYPE	FATEH-110
Max. Range (km):	250
Min. Range (km):	150
Max. Velocity (km/s):	1.5
Launching Capability:	Mobile and Fixed Launcher
Diameter (mm):	616
Warhead Weight (kg):	450
Packet Missile Weight (kg):	4,900
Length (mm):	8,768
C.E.P:	< 0.3% Max.Range
Type of Propellant:	Solid Composite HTPB Base
Average Action Time (sec):	30
Total Weight (kg):	3,620



ROCKETS & MISSILES COMMAND CABIN

PIC: 000313

DESCRIPTION

Command cabin vehicle is a mobile system responsible to manage fire control of the Fajr 3 and Fajr 5 launchers equipped with automation system. Giving a precise definition of the mission of every launcher plays an appropriate role for weapon efficiency.

The most important responsibility of field management is to determine the required mass fire, relative to the type of targets and scheduled fires between different launchers. Here the key point harmonizing with available weapons in battlefield. The command cabin must obtain geographic coordinate of targets (geographical longitude and latitude and height of sea level) and sends them to the related launcher in any possible manner.

It also sends meteorology information (such as wind direction, wind speed, temperature and humidity) to the launchers to modify firing table. By having the target coordinates with meteorology information and geographical longitude and Latitude related to the launcher position, the launcher enables accurate aim on the target.



The command cabin has been made of fiber glass materials. It has standardized thermal and acoustical isolation.

ACCOMPANYING EQUIPMENT

- Industrial computer with L.C.D monitor, with touch screen capability.
- Meteorology box, having static weather sensor
- GPS
- Diesel generator



SPECIFICATIONS

TYPE	COMMAND CABIN
Room's Exterior Dimensions (m):	4.8 x 2.4 x 2.3
User's Interior Dimensions(m):	3.15 x 2.2 x 1.97
Diesel's Room Interior Dimensions(m):	1.35 x 2.2 x 1.97



MOBILE REPAIR UNIT

PIC: 000314

DESCRIPTION

Mobile repair unit is used to perform all type of ground equipment repairs, which are deployed in launching site in level 2 .

The unit has ability to perform all the related tests, repairs of the ground equipment rocket parts, and conduct every kind of repair at night.

ACCOMPANYING EQUIPMENT

- Semi-isolated cabin
- High ampere fuse (cut - out switch) in battery direction
- Fuse and safety relays
- Electricity panel
- Alarm and flashing light
- Mobiles repairs general equipment
- Mobiles repairs mechanical equipment
- Mobiles repairs electrical equipment
- Mobiles repairs communication equipment



METEOROLOGICAL VEHICLE FOR ROCKETS & MISSILES

PIC: 000315

DESCRIPTION

This cabin consists of equipment (marvin) are used to define wind direction, speed and temperature at the different layer of atmosphere.

And also for predicting the weather situation, detecting the wind speed , receiving the radio signals within 100km distance, receiving the weak radio signals, preparing the emergency power source, printing the received data , hydrogen generated.



CREW ACCOMODATION AND FIRE SAFETY EQUIPMENT

- 1- Refrigerator
- 2- Bed
- 3- Air Conditioning System
- 4- Fire Extinguisher
- 5- Cupboard



MISAGH - 1 PORTABLE SHOULDER LAUNCHER

PIC: 000316

DESCRIPTION

MISAGH-1 missile system is the second generation of short range air defence system which features infrared passive homing with omnidirectional attack capability.

It's a powerful weapon system used to encounter aerial target at straight line of sight.

The system can be deployed by an infantry, parachuts, marine corps, for protection of fixed objectives such as road, bridges, airbases and command posts.

COMPONENTS

Packing case includes:

- One firing unit
- Two tubed missiles
- Four ground battery / coolant units



SPECIFICATIONS

TYPE	MISAGH-1 PORTABLE SHOULDER LAUNCHER
Missile Calibre(mm):	71
Missile Mass (gr):	10,860
In-Operating Mass of System (gr):	16,900
Missile Length(mm):	1,477
Lethal Altitude of the Target(m):	30 - 4,000
Lethal Slant Distance(m):	500 - 5,000
Control System:	Self Spinning and Single Channel Control
Guidance Mode:	Proportional Navigation +TAG
Operational Temperature (°c):	- 40 to + 60





5

SHAHAB THAQEB AIR DEFENCE MISSILE

PIC: 000317

DESCRIPTION

SHAHAB THAQEB weapon system is a short-range , low and very low-altitude air defence missile system.

The **SHAHAB THAQEB** missile has a radio-command guidance system.

The **SHAHAB THAQEB** missile is guided according to the three point method, also called command to the line of sight method (**CLOS**).

The **SHAHAB THAQEB** can be fired by **YZ3**, **YZ5** that are similar to **FM-80** system.



SPECIFICATIONS

TYPE	SHAHAB THAQEB
Length (m):	2.9
Diameter (m):	0.154
Weight (kg):	84
Guidance:	Radio Command to Line of Sight
Missile Overload (g):	25
Velocity (m/s):	740
Guidance Mode:	Radio Command
Guidance Law:	CLOS
Reaction Time (s):	5 - 6
RCS (m ²):	1
Fire Power (Ready Missile):	4
Target Tracking:	Autonomous
Number of Targets Engaged:	2
Number of Missile Firing at one Time:	2
Target Velocity (m/s):	440





SHALAMCHE AIR DEFENCE MISSILE

PIC: 000318

DESCRIPTION

The **SHALAMCHE** is used to defence battle fields, military areas, economical centers, strategic installations and identification of targets at low altitudes at a distance of **70 km** such as aircrafts, helicopters.

The **SHALAMCHE** missile can be fired by **MERSAD** air defence system which is almost similar to **HAWK**.



SPECIFICATIONS

TYPE	SHALAMCHE
Mission:	Surface to Air, Medium Range, Guided Missile
Guidance Method:	Semi-Active Radar Homing (SARH)
Guidance Method:	PN (Proportional Navigation)
Max. Range (km):	40
Max. Speed (mach):	2.7
Weight (kg):	636
Lenght (m):	5
Diameter (cm):	35.5
Rocket Motor:	Solid Propellant, Single Chamber, Dual Thrust
Warhead:	HE- Fragment- 74 Kg



5

ZAFAR - ANTI SHIP MISSILE

PIC: 000319

DESCRIPTION

ZAFAR missile with Maximum range of 25 Km are launched from boat, truck and helicopter to destroy medium and small warships.



SPECIFICATIONS

TYPE	ZAFAR
Minimum Effective Range (km):	8
Maximum Effective Range (km):	25
Speed (Mach):	0.8
Launching Angle in Pitch Direction:	12°
Offset Launching Angle in Yaw:	± 10°
Launching condition:	
Temperature (°C):	-25 to +55
Wind Speed (m/s):	≤ 15
Humidity:	< %93 ± 3 (IN 40°C)
Sea State:	4 Sea Force
Guidance System:	Fire & Forget
Control:	Self Control & Radar Seeker
Boat Speed (Knot):	30
Warhead (kg):	30 - SAP
Fuse:	Contact-Delay
Weight (kg):	120
Length (mm):	2685
Diameter (mm):	180
Wing Span (mm):	586
Propulsion:	Dual Chamber Solid Engine with Booster and Sustainer



NASR - ANTI SHIP MISSILE

PIC: 000320

DESCRIPTION

NASR missile with Maximum range of 35 Km are launched from warships, truck, hovercraft, helicopter and submarine to destroy medium and small warships.



SPECIFICATIONS

TYPE	NASR
Minimum Effective Range (km):	8
Maximum Effective Range (km):	35
Speed (Mach):	<0.8
Offset Launching Angle in Yaw:	$\pm 30^{\circ}$
Launching Method:	Single and Salvo
Launching condition:	
Temperature (°C):	-25° to +55°
Wind Speed (m/s):	≤ 15
Humidity:	< %93 ± 3 At 40° C
Sea State:	Force 3
Control:	Self Control & 8 mm Homing Radar Seeker
Warhead (Kg):	130 - SAP
Weight (Kg):	360
Length (mm):	3284
Diameter (mm):	280
Propulsion:	Dual Chamber Solid Engine with Booster and Sustainer



5

NOOR - ANTI SHIP MISSILE

PIC: 000321

DESCRIPTION

NOOR missile with maximum range of 120 km can be launched from beach and ship to destroy medium and small warships.



SPECIFICATIONS

TYPE	NOOR
Minimum Effective Range (km):	10
Maximum Effective Range (km):	120
Speed (Mach):	0.8 - 0.9
Launching Sector:	± 30°
Launching condition:	
Temperature (°C):	-25 to +50
Wind Speed (m/s):	≤ 15
Humidity:	< %98 IN 250° C
Sea State:	≤ 5 Sea Force
Length (mm):	6383
Diameter (mm):	360
Wing Span (mm):	1220
Weight (kg):	715
Kind of Control:	Self Control & Homing Phase
Propulsion:	Consist of two Parts, Booster & Sustainer
Warhead (kg):	155 - SAP- HE
Fuse:	Contact - Delay



NOOR MISSILE RADAR AND COMMAND VEHICLE

PIC: 000322

DESCRIPTION

Searching sea area and detecting enemies vessels, display of tactical situation and missile status Tracking & Targets simultaneously and assigning 4 targets to missile launching vehicles, controlling the fire of missile on four launching vehicles.



SPECIFICATIONS

TYPE	RADAR & COMMAND VEHICLE
Adopting frequency agility system and ability to counter active jamming and interference sea clutter (ECCM)	
Radar (tws):	2031
Max. Range (km):	210
Effective Range (km):	130
Min. Detecting Range (km):	5
False Alarm Probability:	10^{-6}
Detection Probability (%):	90
Azimuth:	360°
Elevation:	3°

NOOR MISSILE - LAUNCHING VEHICLE

PIC: 000323

DESCRIPTION

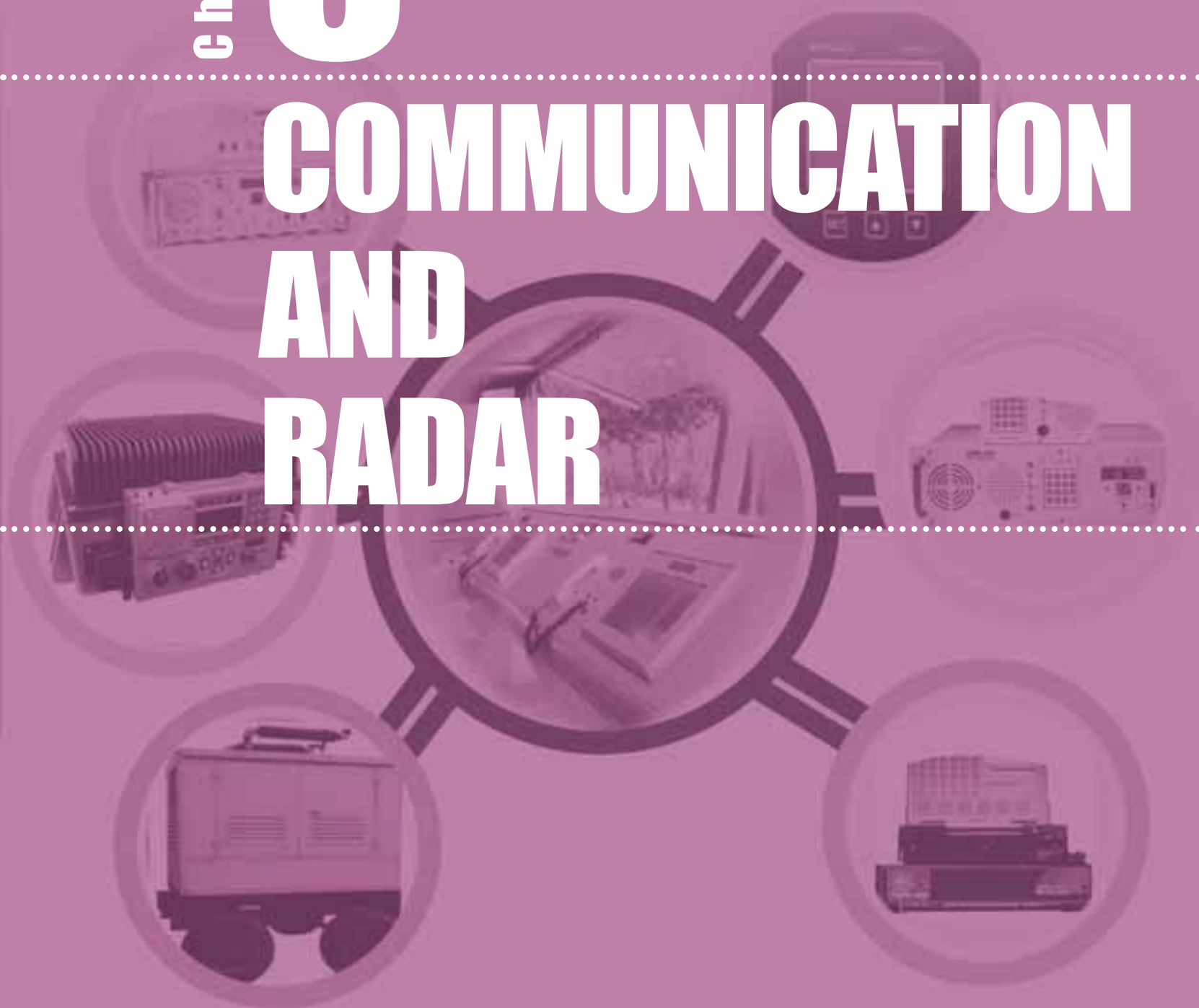
Missile storage traveling with missiles on roads or cross-country roads. Supplying sufficient power to the launching device leveling automatically and manually if necessary. Providing initial attitude angle for the missile launching and carrying out missile pre launch check. Launching missile under the control of command vehicle or independently.

SPECIFICATIONS

TYPE	MISSILE LAUNCHING VEHICLE
Dimension of the Vehicle in Combat State (m):	9.456 x 2.490 x 3.429
Dimension of the Vehicle in Normal State (m):	9.676 x 2.490 x 4.20
Number of Missile:	3
Weight of Truck (kg):	22,000
Max. Speed Without Missile (km/h):	85
Speed With Missile (on roads) (km/h):	50 - 80
Speed With Missile (on soil roads) (km/h):	20 - 40
Accuracy of Auto Leveling (Min.):	<15

Chapter 6

COMMUNICATION AND RADAR





PREVIEW

One of the major and mega industries which is affiliated to M.o.D of Iran is Iran Electronic Industry (**I.E.I.**). This industry was established in 1972 and presently is the major producer of electronic systems and products in Iran.

With eight subsidiaries, over one hundred different kinds of electronic products and 3200 experienced personnel, including 1600 qualified and highly trained engineers in different disciplines, Electronics Industry is considered as the largest electronics company in Iran.

In military field, IEI's activities cover a diversified range including optics, Electro-optic & Laser, Communication, Telecommunication Security, Underwater Communication, Electronic Warfare, Radar, Radar Tube, Manufacturing & Refurbishment and Missile Launchers.

In military communication, Modern Tactical Communication Systems in HF, VHF and UHF and als Electronic Field Telephone and Switchboard are manufactured by Iran Electronic Industries. In optics and Electro-optics this industry is manufacturer of Night Vision Systems (NVS), Laser Range Finders and Day Sight optics like Binocular & Periscope.

In the space field IEI Designs, develops, produces, tests and utilizes various types of research, remote sensing, and communication satellites in addition to various ground stations (Image Receiving, TT&C, Flight Control Center, User terminals) and develops the level of research in the field of space science and technologies in cooperation with the other related entities.

This Industry is currently enjoying a modern manufacturing capability such as SMT and fully automated assembly lines and well equipped & Professionally managed Calibration & Environmental Laboratory.

Iran Electronic Industries is an **ISO-9000** company and its products meets International military standards.

In addition to many versatile products, Electronic Industry has developed a high potential of Research and Development (R&D) which is the technological backbone of the company.





CHAPTER 6



ITEM	TABLE OF CONTENT	PAGE
1	BATTLEFIELD AUTOMATIC TELEPHONE	174
2	MULTI BAND RADIO COMMUNICATION	175
3	(RTX-58) VHF TRANSCEIVER	176
4	(PRC-58) VHF TRANSCEIVER	177
5	HF GROUND RADIO COMMUNICATION SYSTEM	178
6	(TRC - 128) RADIO RELAY	179
7	(VRC - 58) VHF TRANSCEIVER	180
8	(VRC - 110/R) HF TRANSCEIVER	181
9	(IM - 800) PROFESSIONAL MOBIL RADIO	182
10	(TP - 780) TRUNKED TWO-WAY RADIO	183
11	(IR - 800) CONVENTIONAL REPEATER	183
12	VHF AIRBORNE GROUND RADIO COMMUNICATION SYSTEM	184
13	(GBR - 1000) UHF AIRBORNE GROUND RADIO COMMUNICATION	185
14	(ARC-160) UHF AIRBORNE RADIO COMMUNICATION SYSTEM	186
15	ARC-1000 UHF AIRBORNE RADIO COMMUNICATION SYSTEM	187
16	IFF IDENTIFICATION FRIEND OR FOE	188
17	(RES-2111) RADIO ENCRYPTION SYSTEM	189
18	(PCS-8030) PHONE CIPHERING	190
19	(FCS-5030) FACSIMILE CIPHERING	191
20	(NED-4182) BULK CIPHERNG SYSYTEM	192
21	(NED-2223) VIRTUAL PRIVATE LAN	193
22	(NED-123X) FIREWALL	194
23	(NED - 2121) NETWORK SECURITY GATEWAY	195
24	SECURE MOBILE PHONE	196
25	SECURE COMPUTER (PES - 3121)	197
26	SECURE NOTEBOOK (PES - 7631)	198
27	GSM, DCS & 3G JAMMER	199
28	MILITARY SATELLITE JAMMER	199
29	VIP BOMBING JAMMER	199
30	(MATC-401) MOBILE AIR TOWER	200
31	TACTICAL INTEGRATED COMMUNICATION	200
32	SURVEILLANCE & CONTROL NETWOERK	201
33	CRISIS MANAGEMENT NETWORK	201
34	TACTICAL MULTI LAYERS SHELTER	202
35	SPECIAL REPAIR MOBILE UNIT	202
36	SEMI LONG VHF AIR SURVEILLANCE RADAR	203
37	AIR SURVEILLANCE SHORT RANGE RADAR	204
38	(TM-ASR-1) TACTICAL MICROWAVE AIR SURVEILLANCE RADAR	205
39	(PAR-X30) PRECISION APPROACH RADAR	206
40	SHORT RANGE GROUND SURVEILLANCE RADAR	207
41	(MVR-88) MARINE VESSEL RADAR	208
42	(TRM - 8C - 3000) TACTICAL RADIO MONITORING	209
43	(HRM-330) HF RADIO MONITORING	210
44	(SDF-230) WIDEBAND DIRECTION FINDER	211
45	(HDF-230) HAND HELD DIRECTION FINDER	212
46	ELINT	213
47	CUSTOMIZED EW COMMAND AND CONTROL SYSTEM SOLUTION	214
48	(UWP27-1) UNDERWATER PINGER	215



BATTLEFIELD AUTOMATIC TELEPHONE

PIC: 000701



DESCRIPTION

BAT-730 is the latest generation of field telephones. It is a fully automatic system and does not require any switchboard or operator.

Thirty (30) BAT units (subscribers) can be interconnected by a pair of 5km battlefield wires to establish communication with the desired subscribers through direct dialing (7 simultaneous calls).

FEATURES

- Communication among subscribers via battlefield wire (only 2 wires)
- Conference capability for subscribers (up to 10)
- Extension capability up to 30 subscribers with 7 simultaneous calls
- Auto tester for line/battery check
- Whisper mode capability
- Call alarm via LED buzzer
- Light weight / small / simple operation
- Easy connection / disconnection to battlefield lines
- Without any influence on communications
- Fast Selective call by direct dialing
- Doesn't need to switch operator
- Great reduction in using amount of wires
- 12 Emergency entrance of commander through calls

SPECIFICATIONS

TYPE	BAT-730
Transmission Line :	Twin-Wire Field Cable
Comm. Channel :	Battlefield Wire (one pair)
NO. of Channels:	7Full Duplex, Simultaneously
Modulation:	NBFM-FDM with Crystal Synthesizer
Frequency Range (khz):	150 - 300
Supply Voltage :	6V, 4C Size Cells (Internal) , 11 To 30VDC (External)
Battery Life:	About 100 Days (for 3 Minutes Operating per Hour)
Audio & Ring Level Adjustment :	By G-step Selector
Network Capacity:	By G-step Selector
Timer:	Call 180S (after on), Ring 60S
Dimension (mm):	149 x 253 x 91
Weight (kg):	2
Range of Operating Temperature (°c):	-30 to +65



MULTI BAND RADIO COMMUNICATION

PIC: 000702

DESCRIPTION

MBRC-500 is a high technology Manpack transceiver that operates in multi band (20-520MHz). This radio establishes communication in a part of **HF & UHF** band and whole **VHF** band in short & long distance.

Foundation of design & construction of MBRC-500 is “**software defined radio**” in which DSP processor fulfill different kinds of base band processing such as voice compaction, analog and digital modulation, coding, encryption, ...

This system has a capability of transmitting and receiving data & SMS and can be connected to GPS, also equipped to RTC.

Output power of this system is selectable in HIGH, and LOW mode. Since designing of this system is base of software therefore it is feasible for updating and adding software options.



MBRC- 500

SPECIFICATIONS

TYPE	MBRC-500
Frequency Range(MHz):	20 ~ 520
Frequency Increasing Step (KHz):	12.5
Frequency Accuracy (ppm):	± 2
Modulation Type :	DQPSK, AM, FM
Voice Transmission:	FM=5 KHz , AM=60%
Data Transmission:	4.8 Kbps, DQPSK
Short Message Sending Facility :	Keypad, Preprogrammed (Memory Selection), Computer Chat
GPS Connection Facility :	Geographical Location , GMT Time , Altitude
RS - 232 Connection Port:	Computer Controlled Panel, SMS Programming, Data Transferring
Programmable:	External (JTAG or Serial)
Operating Temperature Range:	-20 ~ 65°C With LCD , -40 ~ 65°C With VFD
Input Supply:	DC: 18 ~ 30V
Output Supply:	DC: 21V / 200MA , DC: 8V / 1.5 Amp
Dimension (mm):	90 x 25 x 100
Transmitter spesification:	
Out put Power:	4W , 100 m W
Harmonic Emission:	-40 dB for 20 ~ 40 MHz , -50 dB for 40 ~ 520 MHz
Carrier Attenuation (db):	-40
IM/D Attenuation (db):	-30
Current Consumption:	24V/3Amps in High Power
Modulation Deviation:	FM: ± 5 KHz
Spurious Signal Rejection :	-50 dB
Input Voice Amplitude:	10 mV p-p Max.
Reciever Spesification:	
Sensitivity:	10dBSinad $\rightarrow$ SSB: -107 dBm , AM:-103dBm,FM: -107 dBm



(RTX-58) VHF TRANSCEIVER

PIC: 000703

DESCRIPTION

The **RTX-58** handheld radio is designed for establishing communication between infantry and armored units and is a suitable substitution for the **PRC-77**.

The **RTX-58** is an extremely compact and lightweight VHF transceiver for a short range voice and data communication. Smaller and lighter than its predecessors, it covers the 30- 88MHz frequency range with synthesizer controller channels at 12.5KHz spacing.

This provides compatibility either majority of FM VHF

transceivers currently in service world wide. It supports the 150Hz tone and noise operated squelch facility.

The **RTX-58** provides programmed pre-selection of 100 channels. The radio can be programmed to operate in either simplex or two frequency simplex (half duplex) mode.

The transceiver has been designed to operate with external digital voice encryption unit.

SPECIFICATIONS

TYPE	RTX-58
Frequency Range (mhz):	30 - 88
NO. of Channels (khz):	4640 / 2320
Channel Spacing (khz):	12.5 - 25
NO. of Pre-set Channels:	100
Modulation Type :	FM Narrow - band
Supply Voltage:	7.4 VDC (nominal)
Data Transmission:	QPSK
Transmitter	
Output Power:	H: 2W , L: 100mW
Frequency Stability :	5 PPM
Squelch Tone Frequency:	150
Harmonic Emission :	Better than 48 dB
Maximum PTT Duration :	3 minutes
Microphone Input Level:	At least 20mW (600 Ω load)
Battery Life Time :	More than 10 hours (1 - 1 - 8 ratio)
Current Consumption:	HP: 1.4A, LP:450mA
Receiver	
Sensitivity :	-115 dBm with 10 dB SINAD
Image Rejection :	Better Than 65 db
IF Rejection :	Better Than 65 db
Adjacent Channel Rejection :	Better Than 70 db
Frequency Response :	± 6 dBr (300 HZ ~ 3400 HZ)
Distortion Factor :	Less Than 5%
Squelch Sensitivity Selection :	-120 dBm UP TO - 100dBm (no ise)
Current Consumption :	150 mA in call waiting mode, 300 mA in receiving voice maximum level
Squelch Type :	Noise + Carrier , Tone



RTX- 58



(PRC-58) VHF TRANSCEIVER

PIC: 000704

DESCRIPTION

The **PRC-58** portable radio is designed for establishing communication between infantry and armored units, and is a suitable substitution for the **PRC-77**.

Besides compact size and lightweight, the facilities are improved to hold interoperability. Smaller and lighter than its predecessors, it covers the 30-88MHz frequency range with synthesizer controller channels at 12.5KHz spacing.

This provides compatibility either majority of FM VHF transceivers currently in service world wide. The range of this radio depends on variety of factors like output power, antenna length, minimum level of received signal,

environmental limitation, noise, etc. Considering this factor, range of the system is 5 Km.

The **PRC-58** provides programmed pre-selection of 100 channels.

Output power can be controlled by the keyboard, also SMS, selective call and internal / external ciphering unit are some other features of the system.

By adding a 5 watt booster for amplifying output power from 100mW up to 5 watt and a 12V power supply, the handheld transceiver will be changed into manpack transceiver with the same capabilities but 5 watt power output.

SPECIFICATIONS

TYPE	PRC-58
Frequency Range(MHz):	30 - 88
NO. of Channels:	4640 / 2320
Channel Spacing (KHz):	12.5 - 25
NO. of Pre-set Channels:	100
Modulation Type :	FM Narrow - Band
Supply Voltage:	12 VDC (Nominal)
Data Transmission:	QPSK
Transmitter	
Output Power:	H: 5W, L: 100mW
Frequency Stability :	5PPM
Squelch Tone Frequency (Hz):	150
Harmonic Emission :	Better than 56 dB
Current Consumption	HP: 2A , LP:1.5A
Maximum PTT Duration :	3 Minutes in TX Mode
Microphone Input Level:	At least 20mW (600 Ω load)
Battery Life Time :	More than 12 Hours (8 - 1 - 1 Ratio)
Receiver:	
Sensitivity :	-115 dBm with 10dB SINAD
Image Rejection :	Better Than 65 db
IF Rejection :	Better Than 65 db
Adjacent Channel Rejection :	Better Than 70 db
Frequency Response :	± 6 dB (300HZ ~ 3400HZ)
Distortion Factor :	Less Than 5%
Squelch Sensitivity Selection :	-120 dBm Up To - 100dBm
Squelch Type :	Noise + Carrier , Tone



PRC-58



HF GROUND RADIO COMMUNICATION SYSTEM

PIC: 000705

DESCRIPTION

GRC - 110 - 400W is a high power, full band **HF** communication system to meet military, long range information exchange requirements.

This system is designed to provide comm. between fixed command centers and mobile tactical units within battle areas, where immunity against interference & jamming is required. This system can also be used as key element for strategic intercontinental links for secure communications.

GRC-110-400W has selectable output power 100,200,400 W and has high receiver sensitivity to establish excellent communication quality.

FEATURES

- AM, SSB, CW modes
- 284,000 Channels at 100 Hz Spacing
- 100 Preset Channels
- High dynamic range receiver
- High sensitivity
- Selectable output power
- Audio amplifier system built in

SPECIFICATIONS

TYPE	GRC-110-400W
Frequency Range(MHz):	1.6 - 30
Channel Spacing (Hz):	100
Channels (KHz):	284,000
Preset Channels :	100
Modulation :	AM, SSB, SSB-CW
Emission:	Simplex
Frequency Stability :	± 1 PPM (-10°C ~ 45°C)
Supply Voltage:	220 VAC
Speaker:	0.2 W / 8 Ω
Transmitter	
Output Power:	HP:400W, MP:200W, LP:100W
Harmonic Emission :	Better than - 40 db
Spurious Emission :	Better than - 30 db
Carrier Suppression :	Better than - 40 db (SSB,CW)
Inter Modulation Distortion :	Better than -20 db
Current Consumption :	RX:1 A (MAX), TX: 40 A FOR 400 W
Receiver	
Sensitivity :	
SSB:	1 μ V PD for SINAD = 12 dB
CW:	1 μ V PD for SINAD = 12 dB
AM(30%Modulation):	4 μ V PD for SINAD = 15 dB
Selectivity :	SSB: 6 DB @ 2 KHZ, 40DB @ 5 KHZ
Current Consumption :	400mA MAX.
Out put Impedance :	50 Ω
If Rejection :	70dB
Image Rejection :	70dB
Harmonic Suppression :	30dB
AGC :	6dB for 100dB Variation above 1 μ VPD



GRC-110 400 W



TRC - 128 RADIO RELAY

PIC: 000706

DESCRIPTION

The digital radio relay **TRC-128** is a full duplex communication system which sends and receives voice, data and image with the maximum capacity of 8Mb/s through the frequency range of **225 to 400 MHz**.

This system could be applied as point to point, in a network or as a repeater.



FEATURES

- Voice, data and image transmission
- Controlling output power
- Relay facility
- Networking
- Internal monitoring and test
- Remote monitoring and test
- Automatic duplexer setting
- Easy and quick installation
- Protected against eavesdropping and jamming
- Changing data transmission rate
- Plain and ciphered operating mode
- Changing output power from 1 to 20 W
- O.W supervisor channel with the rate of 64Kbps
- Displaying received signal strength
- Displaying output power
- Displaying internal temperature
- Signal reception and connection indicator
- Subjected to **MIL-STD-810**



TRC-128 RADIO RELAY



VRC - 58 VHF TRANSCEIVER

PIC: 000707



VRC-58

DESCRIPTION

The **VRC-58** is an extremely reliable vehicular **VHF** transceiver for long range voice and data communication. The radio is designed in VHF band (narrow band FM) for using in vehicular/ base station.

This system has removable panel which can be separated from the main transceiver and operates all functions via a pair wire up to **4Km**.

This system has the variety of capabilities such as high / low power selection, manual / automatic channel scanning, SMS sending, ...

The **VRC-58** provides 100 pre selection programmed channels. This system has compatibility with majority of FM VHF transceivers currently in service world wide, according to construction of FM modulation, noise reduction in this method is better than the others.

It supports the 150Hz tone and noise operated squelch facility.

The radio can be programmed to operate in either simplex or two frequency simplex (half duplex) mode. The transceiver can operate with internal digital voice encryption unit.

FEATURES

- 30 - 880 MHz synthesizer controlled
- FM narrow band
- Remote panel up to 4Km
- 100 preset channels
- Connectable to external ciphering system, modem and data terminal
- Simplex and half duplex
- Handset / headset / hands free audio output
- External encryption key data entry via gun / PC
- Tone / Noise squelch operation
- Power output selection
- LCD facility
- User ID and password facility
- Emergency key erase facility
- Simple man-machine interface (MMI)
- Audible/visual warning facility
- External programming facility
- Retransmission facility
- Internal SMS (sell call, group call, broadcast)
- Scanning facility
- Channel space selection facilities



(VRC - 110/R) HF TRANSCEIVER

PIC: 000708

DESCRIPTION

The **VRC-110/R** is a vehicular **HF** transceiver with selectable power out put (25.50,100 W) with **SSB,SSB-CW** and **AME** modulation in the frequency range of 1.6~29.9999MHz.

The radio is equipped with external automatic antenna matching unit which can operate with rod, dipole and wide band antennas to establish communication in Mores, audio and data mode for long range distance approximately 1000 ~ 2000Km, between vehicular and base stations.

The **VRC-110/R** uses advanced hardware & software to control function of the radio up to **3Km**. So in case that commander wants to be far from propagation location, he will be in safe place if deploys the radio remotely.



VRC-110/R

SPECIFICATIONS

TYPE	VRC-110/R
Range Frequency (MHz) :	1.6 - 29.9999
Channel Spacing (Hz):	100
No. of Frequency Channel :	284,000
NO. Preset Channel (CH):	100
Frequency Offset (Hz):	10
Type of Modulation :	AME,CW(SSB) , SSB(LSB,USB)
Power Supply :	24~28 VDC
Audio Band Width (HZ):	500 ~ 2500
Input Audio Level for Modem :	350mVrms
Antenna :	Vehicular Antenna (AT - 230) Wide Band Antenna. Folded Dipole Antenna
Antenna Matching :	Less than 2 Second (Automatically with the First PTT)
SWR :	2.5
RF Power for Matching :	1~3W (1.5W Nominal)
Transmitter:	
Output Power :	HP: 100 W (±1db) MP: 50 W (±1db) LP: 25 W (±1db)
Current Consumption in TX Mode:	HP: 15A, MP: 13A, LP: 10A
Frequency Stability :	±1PPM (-10°C ~ +50°C) ±2PPM (-40°C ~ +65°C)

FEATURES

- Removeable panel control up to 3KM
- 100 preset channels
- Display for showing frequency, power,...
- SWR showing
- Facility of connecting to computer
- Built in internal modem and facility to connecting to external modem
- Built in internal encryption and facility of external encryption connection
- Audio and squelch processor for improving quality
- Audio volume control
- Protection against reversed polarity
- Protection against mismatch or antenna short connection
- Automatic antenna tuning
- Built in internal speaker
- Equipped with clarifier
- Equipped with internal voltage regulator
- Portable facility



IM - 800 PROFESSIONAL MOBILE RADIO

PIC: 000709



IM-800

SPECIFICATIONS

TYPE	IM-800
Frequency Range (MHz):	136 - 174 , 350 - 400 , 400 - 470, 440 - 490 , 450 - 512 , 480 - 526, 450 - 520
Channel Capacity :	512
Zone Capacity :	256 (Max.)
Channel Spacing (KHz):	25/12.5
Operating Voltage :	13.6V DC $\pm$ 15%
Current Drain:	
Standby:	0.5A
Receive:	<2.0A
Transmit :	<5A(5W), <8A(25W), <12A(50W),
Frequency Stability :	$\pm$ 2PPM
Operating Temperature (°C):	-30 ~ +60
Antenna Impedance :	50 Ω
Dimension (mm) :	175 X 48 X 164
Weight (kg):	1.7
Transmitter	
RF Power Output :	5W / 25W / 50W ; 5W / 25W / 45W
Modulation :	16K0F3E/11K0F3E
Spurious and Harmonics :	-36 dBm < 1GHz -30 dBm > 1GHz
FM Noise (Wide / Narrow) :	48 / 42 db
Audio Distortion :	$\leq$ 3%
Receiver	
Sensitivity (Wide / Narrow) :	0.28 / 0.35 μ V
Selectivity (Wide / Narrow) :	75 / 65 db
Intermodulation (Wide / Narrow) :	70 / 65 db
Spurious Response Rejection (db):	80 - 85
S/N (Wide / Narrow) :	48/42 db
Internal Audio Power Output : (Rated):	5W (Less than 5% Distortion), 3W (Less than 3% Distortion)
External Audio Power Output : (Rated):	13W (Less than 5% Distortion), 12W (Less than 3% Distortion)

FEATURES

- Wide-band Coverage
- Powerful Audio
- Remote Mount Control Head
- CTCSS, CDCSS
- DTMF Encode
- 5-Tone Model: 5-Tone + HDC 2400
- 2-Tone Model: 2-Tone + HDC 1200
- Built-in Scrambler
- Auxiliary Port
- Automatic Number ID
- Short Message
- Dot Matrix Alphanumeric LCD
- 8 Programmable Backlight Keys
- Built-in Modem
- Front Panel Programming
- Emergency Siren
- PC Programming
- Multi-language Capability
- Stun/Revive
- Over-heating Protection
- Ignition Sense, Timed Power Off
- Wired Clone



(IP - 780) TRUNKED TWO-WAY RADIO

PIC: 000710

FEATURES

- 256 Conventional Channels
- DTMF Encode / PTT ID
- Priority Channel Scan
- Revert Channel
- Nuisance Channel Delete
- Busy Channel Lockout
- 3 Programable Function Buttons
- Time – Out Timer
- Emergency



IP-780

SPECIFICATIONS

TYPE	IP-780
Frequency Range (MHz):	VHF: 136 - 174 , UHF: 350 - 390 , 400 - 440, 440 - 470 (trunked), 400 - 470 (conventional)
Channel Capacity :	256
Channel Spacing (KHz):	12.5 - 25
Operating Voltage:	7.4 V
Battery :	1700 mAh Li - ion Battery 1800 mAh Ni - MH Battery (optional) 2100 mAhLi - ion Battery (optional)
Battery Life (1700mAh Li-ion Battery) :	About 13 Hours
Frequency Stability (ppm):	± 2.5
Operating Temperature (°c):	-25 ~ +60
Antenna Impedance (Ω):	50
Dimensions (HxWxD)	
(with battery, without antenna) :	126 x 55 x 34mm
Weight (with antenna & battery) :	330.5g

(IR - 800) CONVENTIONAL REPEATER

PIC: 000711

FEATURES

- Powerful, Varied Speed and Thermal-Controlled Fan
- Integrated Front Panel with Repeater Controller
- CTCSS/CDCSS
- Repeater Hang Time
- Real-time Scan
- Adjustable Power Output
- Time-out Timer
- Floating Charge
- Bi-directional & Uni-directional Operation
- Optional Duplexer



IR-800

SPECIFICATIONS

TYPE	IR-800
Frequency Range (MHz):	VHF: 136 - 174 UHF: 350 - 400 , 400 - 470 MHz 450 - 512
Channel Capacity :	512
Channel Spacing (KHz):	12.5 - 25
Operating Voltage:	13.8V DC ± 20% 88 - 264V AC 50 / 60Hz
Duty Cycle:	100% Continuous (5 / 10 / 25 W) 50% Intermittent (45 / 50 W)
Current Drain: Standby: Receive: Transmit :	<1.8A <2.5A <13A
Frequency Stability (ppm):	±2 @ -30 ~ +60° C
Operating Temperature (°c):	-30 ~ +60
Antenna Impedance (Ω):	50
Dimensions (mm):	310 x 190 x 362
Weight (kg):	11
LED Indicators:	Battery On, Tx , Rx , RPT , OPT , Setup



VHF AIRBORNE GROUND RADIO COMMUNICATION SYSTEM

PIC: 000712

DESCRIPTION

GBR-160 is **VHF** transceiver for using in air traffic control towers, radar center or coastal sites.

Due to its wide frequency range and **AM/FM** modulation, many application in **VHF** band such as civil and military **ATC**, commercial **VHF** communications and maritime / navy communications is supported.

By using a remote control unit, all functions and voice communication can be established up to 500 meters distance.

This system can be used in air surveillance and traffic control towers (military and commercial aircrafts), command shelters and vessels to establish ground to air communication.

ARC-160 operates in 108-174 MHz frequency range, from 108-118MHz operates as a receiver for navigation purposes and transmission in not permitted, from 118- 137 MHz with AM modulation and 25 or 8.33 KHz channel spacing the radio is used for civil air traffic control (ATC)communication.

From 137 to 156 MHz with AM& FM modulation and from 156 to 174 MHz with FM modulation maritime communication is provided.

FEATURES

- AM, FM , VHF airband and marine base radio
- Suitable for using in watchtower and airborne control center
- Guard receiver with AM modulation in 121.5 MHz
- Usable in desktop and rack mount mode
- Output power selectivity in low and high mode
- Retransmission mode
- ADF mode facility
- Subjected to military standard
- High receiver sensitivity
- Remote control facility
- High frequency accuracy
- Squelch ON/OFF indicator
- Frequency & test process display via LCD
- Mains supply and battery facilities
- Retransmission mode



GBR-160

SPECIFICATIONS

TYPE	GBR-160
Frequency Range (MHz):	108 – 174 (108 – 118 Only in Receiving)
Modulation Type :	AM - A3E , FM - F3E
Channel Spacing (KHz):	8.33 / 25
Input Power Supply :	110 / 220 VAC -50 ~ 60 Hz, 24V DC
Preset Memory :	99
Control Interface:	RS - 232 & RS-485
Frequency Stability :	5 PPM
Guard Receiver :	121.5MHz - AM
Receiver	
Sensitivity :	AM: -103 dBm (10dB SINAD) FM: -110 dBm (12dB SINAD)
Selectivity :	25KHz 6dB > 15KHz , 50dB < ± 25KHz 8.33KHz 6dB > 7.5 KHz , 40dB < ±10KHz
Audio Output :	2W ON Speaker + Headphone Output
Line Audio Output :	0dBm / 600Ω Balanced
IF & Image Rejection :	70 dB
RF Input Protection :	UP to 1 WATT (+30dBm)
Transmitter	
Output Power :	AM:40W CARRIER(± 1dB) FM:60W ± 1dB
Low Power Mode Selectable:	10 / 15W (AM / FM)
Modulation Depth :	80% MIN(AM), ± 5KHz(FM)
Harmonics Rejection :	-60 dBC
Spurious Signals :	-70 dBC
Modulation Distortion :	MAX. 5%



IGBR - 10001 UHF AIRBORNE GROUND RADIO COMMUNICATION

PIC: 000713

DESCRIPTION

GBR-1000 is a **UHF** transceiver applicable in **ATC** towers, radar center and so on for ground to air communications. This radio can be used and controlled either locally or via a remote control unit.

In addition to conventional AM mode, frequency hopping and other spread spectrum techniques are used in conjunction with built in encryption ensure a high level of security for communication in military applications.

An independent guard receiver at 243 MHz allows simultaneous reception of emergency signals and main receiver channel signal .

This system works in two major operational modes: **AM & ECCM**, in **AM** mode system operates as a conventional UHF-AM radio.

In **ECCM** (Electronic Counter Counter Measure) mode, using a CVSD codec, built - in digital encryption, combination of spread spectrum techniques such as frequency hopping, system provides as secure communication link against jamming & interception.

In **ADF** mode in conjunction with ADF antenna and related indication, system is for homing purpose.



GBR-1000

SPECIFICATIONS

TYPE	GBR-1000
Freq. Range (MHz):	225-400
Freq Spacing (KHz):	25
Preset Channel NO. (CH):	99
Power Supply :	110 / 220 VAC - 50 ~ 60 Hz 24VDC
Voice Interface :	AM: Analog Voice, ECCM:16KB / S CVSD
Data and Control :	RS 232 , RS 422
Transmitter:	
Out Put Power :	AM - 20W , ECCM - 100W Peak
Self Protection (Reduced Power):	VSWR & Temperature
Harmonic Radiation :	-60 dbc
Spurious Radiation :	-70 dbc
Modulation :	AM:A3E , ECCM: FSK
Frequency Stability (ppm):	5
Local Audio Input :	-35 ~ +10 dBm
Local Input Impedance (Ω):	150 ~ Unbalanced
Line Audio Input :	0 dbm @ 600 Ω Balanced
Modulation Distortion :	%5
Receiver	
Sensitivity :	AM: -103 dBm - (10DB SINAD) ECCM: -90 dBm(BER<0.0001)%5
Selectivity AM :	6db > 20 KHz , 50db < ±25 KHz
Audio Output :	2W (Speaker) 10MW (Headphone)
Line Audio Output :	Adjustable up to +10dBm
Line Impedance (Ω):	600 Balanced

FEATURES

- UHF AM airborne transceiver
- Applicable in surveillance towers and airborne control centers
- Advanced ECCM capability
- Internal digital encryption
- Remote control facility
- Guard receiver 243 MHz
- Military standards conformance
- Modern software and hardware technologies



(ARC-160) UHF AIRBORNE RADIO COMMUNICATION SYSTEM

PIC: 000714

DESCRIPTION

This **VHF** airborne radio is designed to establish communication among military aircrafts, helicopters and ground control stations/ units and civil applications.

ARC-160 can apply two types of standard modulation such as : Amplitude (**AM**) and frequency (**FM**).

This system has rugged mechanical construction in such a way that can be easily installed on military platforms such as aircraft, helicopter, and ships.

An independent guard receiver on 121.5 MHz allows to receive emergency message automatically.

Standard version of this radio is designed for cockpit installation and remote control version is also available.



ARC-160

SPECIFICATIONS

TYPE	ARC-160
Frequency Range (mhz):	108 – 174 (108 – 118 Only in Receiving)
Modulation Type :	AM - A3E , FM - F3E
Channel Spacing (Khz):	8.33 - 25
Input Power Supply :	28 VDC, Subjected to MIL - STD -7 04
Preset Memory :	99
Data & Control Interface:	RS - 232 & RS - 485
Frequency Stability (ppm):	5
Guard Receiver :	121.5 MHz - AM
Transmitter:	
Output Power :	AM:16W Carrier (± 1 dS) FM:18W (± 1 dS)
Power Select:	In10 Step
Modulation Depth :	80% min(AM), ± 5 KHz(FM)
Harmonics Rejection (dBc):	-60
Spurious Signals (dBc):	-70
Modulation Distortion :	Max. 5%
Audio Input (dbm):	-35 ~ +10
Line Audio Input :	0dbm @ 600 Ω Balanced
Inout impedance :	150 Ω ~ Unbalanced
Receiver:	
Sensitivity :	AM: -103 dBm (10db SINAD) FM: -110 dBm (12db SINAD)
Selectivity :	25KHz 6db > 15KHz , 50db < ± 25 KHz 8.33KHz 6db > 7.5 KHz , 40db < ± 10 KHz
Audio Output :	5 - 150 m W Adjustable (150 / 600 OHM)



ARC-1000 UHF AIRBORNE RADIO COMMUNICATION SYSTEM

PIC: 000715

DESCRIPTION

ARC-1000 is an advanced **UHF ECCM** Radio Communication System, which is specially designed to establish secure anti-jamming communication among air-to-air as well as air-to-ground stations.

This system can be installed on fighters, transporter planes, and helicopters.

This system works in two major operational modes: **AM& ECCM**, in **AM** mode system operates as a conventional **UHF-AM** radio.

In **ECCM** (Electronic Counter Counter Measure) mode, using a **CVSD** codec, built - in digital encryption, combination of spread spectrum techniques such as frequency hopping, system provides a secure communication link against jamming & interception. In **ADF** mode in conjunction with ADF antenna and related indicator, system is for homing purpose. This system uses Advanced Hybrid ECCM techniques (spread spectrum) to provide a high level of security and resistance against electronic warfare.



ARC-1000

SPECIFICATIONS

TYPE	ARC-1000
Freq. Range (MHz):	225 - 400
Frq Spacing (KHz):	25
Preset Channel NO. (CH):	99
Power Supply :	28vdc . 5A (Mil - STD - 704)
Voice Interface :	AM: Analog Voice , ECCM:16KB / S CVSD
Data and Control :	RS 232 , RS 485
Transmitter:	
Out Put Power :	AM - 20W , ECCM - 100W Peak
Self Protection (Reduced Power):	VSWR & Temperature
Harmonic Radiation (dBc):	-60
Spurious Radiation (dBc):	-70
Modulation :	AM:A3E , ECCM:FSK
Modulation Distortion :	%5
Frequency Stability (ppm):	5
Local Audio Input :	-35 ~ +10dBm
Local Input Impedance :	150Ω ~ Unbalanced
Line Audio Input :	0 dBm @ 600 Ω Balanced
Receiver:	
Sensitivity :	AM: -103 dBm - (10dB SINAD) ECCM: -90 dBm - (BER < 0.0001) %5
Selectivity AM :	6db > 20KHz, 50db < ±25KHz
Audio Output :	5 - 150 mWAdjustable (150/600W)
Line Audio Output :	0dBm
Line Impedance :	600 Ω balanced
Audio Distortion :	%5 (-90 ~ 0dBm)
ADF Output :	4 Vp - p @ 20 kΩ
ADF Response :	50 Hz ~ 10 KHz
RF Input Protection :	up to 1W (+30dBm)



IFF - IDENTIFICATION FRIEND OR FOE

PIC: 000716

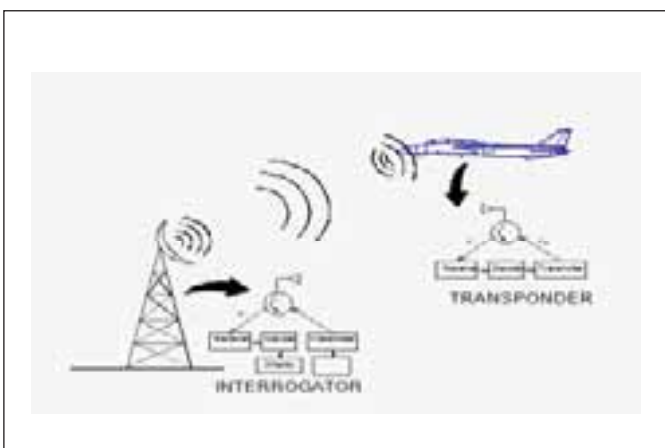
DESCRIPTION

In peacetime or during conflict, it is critical to maintain a situational awareness that distinguishes friendly forces from hostiles or unknowns. Identification Friend or Foe (**IFF**) is an identification system designed for command and control. It is a system that enables friendly forces to recognize friendly platforms among potential targets in the battle space. This information provides critical lead time to counter an attack or prevent fratricide between friendly and/or coalition forces. The system was initially intended to distinguish between enemy and friend but has evolved such that the term "**IFF**" commonly refers to all modes of operation, including civil and foreign aircraft use.



SYSTEM COMPONENTS

IFF solution consists of ground interrogator, ground extractor, airborne transponder, airborne interrogator and antennas. The ground interrogator transmits interrogation pulses to the airborne transponder which is installed in an aircraft or a helicopter. The transponder transmits reply pulses to the ground interrogator and extractor to be analyzed in order to determine and display the target identification. The airborne transponder is utilized as Air Traffic Control transponder (mode A and C) in civilian aircrafts or helicopters. Airborne interrogator challenges airborne targets and process target response to determine whether the target is a friend, foe or unknown. The whole system is fully compatible with **STANG-4193** Standards.



SPECIFICATIONS

TYPE	GROUND INTERROGATOR	AIRBORNE TRANSPONDER	AIRBORNE INTERROGATOR
	SIMS-GI23	SIMS-AT101	SIMS-AI76
Transmitter Frequency (MHz):	1030	1090	1030
Receiver Frequency (MHz):	1090	1030	1090
Receiver Sensitivity (dBm):	- 84	- 76	- 80
Input Power:	90 - 230 VAC 40 - 400 Hz	20 - 32 VDC	115 VAC (400 Hz) 20 - 33 VDC
Output Power (dBw):	33 ± 1	27 ± 3	29 ± 1
Operating Temperature(°C):	-20 to +65	-40 to +71	-40 to +71
Humidity:	0 to 95%	0 to 95%	0 to 95%
Altitude:	Up to 20,000 ft	Up to 70,000 ft	Up to 70,000 ft
Built-in Test:	√	√	√
RSLS:	√	-	√
ISLS:	√	√	√



RES-2111 - RADIO ENCRYPTION

PIC: 000718

DESCRIPTION

VHF / UHF Radio Encryption System (RES - 2111), is a digital cipher that has a very powerful cryptographic algorithm and signal processor for high security communication.

RES-2111 plug to Motorola professional (**GP338**) and other commercial radio.

Rate of transformation of RES-2111 is 2400 bps and voice delay is less than 200 ms.

ADVANTAGES

- High level of security for radio communication
- Plug-in installation in GP338 Motorola radio
- Solder installation into other VHF / UHF Radio
- 2400 bps voice transformation rate
- Plain and code override in receiver
- Erasing keys and algorithms in emergency condition
- Operating current in Stand by

CRYPTOGRAPHY

Encryption Algorithm:	Straem CIPHER
Programming:	Via PC



RES - 2111

SPECIFICATIONS

TYPE	RES-2111
Mode (mA) :	20
Operating Current (mA):	110
Operating Temperature (°C):	-20 ,+60
Storage temperature (°C):	-30 ,+65
Humidity (%):	Up to 90



PCS-8030 - PHONE CIPHERING

PIC: 000719



PCS-8030

DESCRIPTION

Phone ciphering system, **PCS-8030**, is designed for making security in telephone calls. It uses **CELP 4800** as a vocoder. A very powerful cryptographic algorithm is used for high security communications.

PCS-8030 can operate on all kinds of telephone Analog lines and Panasonic Centres and connected between the phone and line (**PSTN**). **PCS-8030** is easy to install and switches of a touch from plain to encrypted mode.

FEATURES

- Usable with variety of phone
- Voice alarm for plain telephone calls
- High level of security
- Voice signal compression using CELP technique
- Emergency clear function
- Easy Installation and operation
- LCD display
- Key distribution by KG-8030

SPECIFICATIONS

TYPE	PCS-8030
Analyze and Synthesize Algorithm :	CELP
Bit Rate (bps):	4800
Communication Standard :	V.32 LCD DISPLAY
Cryptography	
Encryption Algorithm :	Stream Cipher
Communication Key Length (bits):	128
Message key length (bits):	32
Number Communication keys (bit):	16 x 128
Period :	$>10^{250}$
Key Management	
Key Distribution System:	KG-8030
Mechanical & Environment	
Dimensions (mm) :	305 x 220 x 60
Weight (kg):	1.8
Operation Limit (°c):	0 ~ 50
Storage Limit (°c):	-10 ~ 60



FCS-5030 FACSIMILE CIPHERING

PIC: 000720

FCS-5030



DESCRIPTION

The **FCS-5030** facsimile ciphering system is designed to offer a very high level of security during transmission and reception of fax documents.

By connecting this unit to a fax machine documents can be transmitted in either encrypted or plain modes. This system allows for automatic detection of coded or plain mode during reception.

A high level of security is provided by using a long period key stream and very complex ciphering algorithm to prevent unauthorized access to documents during transmission or reception.

This unit offers a key variety of settings, up to sixteen of which can be stored by the user.

Each key setting is consisting of 32 digits (128 bits), which can be distributed via keypad or keygun.

Communication security can easily be maintained by FCS-5030 due to easy installation, operation and maintenance.

FEATURES

- High level of security
- Compatible with Group 3 (G3) fax machines
- Key distribution via keypad and keygun
- Emergency clear function
- Automatic detection of cipher / plain modes during reception
- Easy communication with fax machines without encryptor in plain mode
- LCD display

SPECIFICATIONS

TYPE	FCS-5030
Cryptography:	
Encryption Algorithm :	Stream Cipher
Communication Key Length (bits):	128
Message Key Length (bits):	32
Period :	$>10^{100}$
Key Management:	
Communication Key Variety :	3.4×10^{38}
Synchronous Key Variety :	4.3×10^9
Key Generator Period:	$> 10^{100}$
Mechanical & Environmental:	
Dimensions (mm) :	305 x 220 x 60
Weight (kg):	1.72



NED-4182 BULK CIPHERING SYSTEM

PIC: 000721



NED-4182

DESCRIPTION

The **NED-4182 ciphering system** guarantees secure data transmission in G.703 and G.704 standards. This system supports the ATM, PCM and Frame Relay protocols. The NED-4182 cipher/decipher data in transmitting or receiving with rate of 2Mbits/s.

In the special model which could be installed in rack, totally 8 cards could be installed in the system, so 8x2M bit/s data could be transferred.

This system includes hardware and software locks for increasing security. Low current consumption, capability of connecting to 220VAC or 48VDC, independence from cooling fan, special alarm in plain mode, etc are some of the features in the system.

FEATURES

- Reporting And Programming Via PC
- Emergency Clear
- Using New Ciphering Algorithm According To User Demand
- Independent Code For Each Communication In E1 (PCM) Mode, For Increasing Security
- Upgrading Via Software
- Applying In Various Switches With PCM, Frame Relay, ATM, etc Standards
- Automatic Detection Of The Ciphering System In The Other Side and Automatic Switching TO Cipher Mode

SPECIFICATIONS

TYPE	NED-4182
System Type :	Ciphering System for E1 2Mbit Links
Ciphering Method :	Digital
Input/output Impedance (Ohm):	75 and 120
Ciphering Algorithm Specification	
Type of Algorithm:	Stream or Block Ciphering
Non-linear and non Recursive Resistance Against Correlation Attacks	
Code Key	
Communication key Length (bit):	256
Number of Key Sets :	1024 Sets of 256 bit
Number of key Banks :	One 32KB or two 16KB
Supply	
Input Voltage :	220VAC / 48VDC
Power Consumption (w):	Less than 5
Environmental	
Operating Temperature Range (°C):	0 ~ +55
Storage Temperature Range (°C):	-10 ~ +70



NED-2223 VIRTUAL PRIVATE LAN



PIC: 000722



NED-2223

DESCRIPTION

NED-2223 is a network security product which uses encryption and tunneling mechanisms in data link layer to authenticate the computers or devices in the network and protect their communications.

NED-2223 can be used to establish secure connections between communication parties in the network.

So it makes possible to create a Virtual Secure **LAN** over the existing physical network. Each computer in the network can be either a member of the physical

(insecure) **LAN** or the virtual secure **LAN**.

NED-2223 is presented in two form factors: Host and bridge. **NED-2223** in host mode is installed as special software on computers in the network to provide secure connection with other **NED-2223** systems.

NED-2223 in bridge mode on the other hand is used as a hardware appliance to secure the connection between the subnet under protection with other **NED-2223** agents in the network.

FEATURES

- Authentication of computers or devices in the network
- Confidentiality and integrity of the exchanging information
- Replay attack protection
- Limited traffic flow confidentiality
- Second layer Virtual Private Network
- Securing communications between computers and/or devices in the network
- Iran native algorithm (or any ordered by customer)
- Mutual and two factors authentication based on DES-1551 hardware security module
- Secure automatic key management based on key file method
- Strong protection of software and its sensitive information using DES-1551
- Plug & Play installation and totally transparent from network applications
- Invisible and undetectable by the end users (Bridge mode)
- User friendly, simple to configure and use



NED-123X FIREWALL

PIC: 000723



NED-123X

DESCRIPTION

NED-123x firewall is a plug and play device and a really transparent (wire level) frame inspector for small and medium level enterprises.

Its installation is as easy as an ordinary Ethernet switch. After installation, all **IP** packets with wrong checksums will be dropped. All **TCP** connections will be checked.

From the checksum to the state transition and sequence numbers of **TCP** sessions.

If configured, the logs about all fetched **URLs**, Email senders/recipients, **FTP** users and commands, besides hundreds of statistical information will be sent to the log collector system.

Logs of **NED-123x** are sent by a special purpose protocol which will not be saturated by SPAM and/or junk Sync floods.

NED-123x is a 4-ports, layers 2-7 switch with many advanced features including.

FEATURES

- Serial port console with full setup capabilities.
- Traffic shaping features.
- Full layer three, packet filtering with automatic IP checksum control.
- Tight TCP stateful inspection.
- Application layer protocol monitoring and violation control for Telnet, SMTP, FTP, and HTTP (e.g: invalid HTTP requests cause TCP connection termination).
- URL filtering.
- Regular expression matching with space for 100 regular expressions to match with HTTP requests.
- SMTP session filtering based on entries defined in databases of username, domain name, user@domain.
- LAN user authentication.
- Java GUI based setup program for Windows.
- Ultra fast log protocol.
- NAT/PAT support.
- IP/MAC database as source and/or destination in rules.



NED-2121 NETWORK SECURITY GATEWAY



PIC: 000724



DESCRIPTION

Internet, as a public and widespread network, is rapidly progressing in the world. This is where its technical properties and non-private nature of Internet reduce the chance of using it as a secure network for some activities, including commercial and sensitive data interchange.

NED-2121, as a security gateway, provides services in order to protect the supervised network as well as making secure internet work connections. Taking advantages of **IPSec** and **IKE** standard protocols, **NED-2121** is able to establish secure tunnels among private networks connected to the Internet or other public networks to securely interchange information packets (VPN Service).

Moreover, **NED-2121** provides intrusion detection and also firewall services. It controls the network traffic according to its firewall security policy and detects hacking attack signatures as well as any unusual modification in the gateway system configuration.

NED-2121 security gateway is deployed transparently in the network. It can be securely controlled from any host in the network using a web based management console.

FEATURES

- Supporting IPSec protocols (AH, ESP)
- Using strong encryption algorithms including 3DES, AES and proprietary algorithms
- Manual and automatic key management based on the IKE protocol
- Interoperability with Windows 2000 and XP as tunneling endpoints
- Supporting firewall
- Equipped with host and network-based intrusion detection systems (HIDS and NIDS)
- Management, alert classification, generating reports and drawing various kinds of statistical diagrams
- Secure management and configuration of the system using a web-based management console from any node of the network



SECURE MOBILE PHONE

PIC: 000725



DESCRIPTION

Mobile phone, as a new communication tool, plays an essential role in the human life and its penetration is getting ahead of the ordinary telephone in some parts of the world.

Most of the countries use GSM standard for mobilephone systems. In order to protect conversations, **GSM** standard encrypts digital data transmitted across the air channel. Despite this fact, a weak encryption algorithm has been offered for the non-European countries.

Moreover, eavesdropping of the conversations is still possible in other parts of the mobile phone network. So the mobile phone conversations are subject to serious threats. **DES-1331** is a new product for securing mobile phone conversations. It consists of a pocket PC phone and a secure phone software. Users can make secure conversations with **DES-1331** without any limitations.

DES-1331 secure phone software is executable under **Windows CE** operating system. It transmits compressed and encrypted speech signals to the destination by means of data service of the mobile phone. In the destination, speech signals are decrypted and reconstructed for hearing.

FEATURES

- Full duplex and real time secure conversation
- Powerful encryption algorithm
- Synthesized speech with toll quality
- Selectable conversation mode (secure/plain) by caller
- Automatic recognition of conversation mode (secure/plain) in the called part
- Phone book and incoming outgoing and missed calls recording
- Caller ID facility
- Compatible with DES-1152 cryptophone software
- Executable in Windows CE (Pocket PC Phone edition) operating system



SECURE COMPUTER IPES-3121I

PIC: 000726



PES-3121

DESCRIPTION

Today, data is the main asset of every organization and company.

Therefore protecting data and guaranteeing its security and safety is essential and vital for companies.

In today's competitive markets, data security is highly being endangered by rival companies to be stolen, cheated or destroyed through developed techniques.

For this reason, Secure computer **PES 3121** is designed to bring security and safety for data while supervising users to confront any threats.

Secure computer **PES 3121** owns facilities based on which the level of security and safety increased for counteracting various threats.

FEATURES

- User's Authentication before start-up based on E-card and password
- Hardware Protected through diagnostic sensors
- Network Ciphering System to protect links and data exchanged
- Encryption and Decryption files and folders
- Wiped files and folders
- Creating secure virtual partitions on system
- Encryption and Decryption CD/DVD (IDE) writer
- Managing level of users' access by E-card
- Custom algorithm supported
- Desktop or Rack Mount Case
- Special software for secure Control of Operating System
- Electromagnetic radiations controlled



SECURE NOTEBOOK (PES-7631)

PIC: 000727



PES-7631

DESCRIPTION

Today, data is the main asset for organization and company.

Noting that using portable computers or notebooks are becoming more popular everyday and security threats are also increasing accordingly because of many threats, protecting data and guaranteeing its security is essential for every company.

Secure notebook **PES 7631** eliminates some **I/O** ports like Blue tooth, **Wlan**, **GPS**, **3G** which eavesdropping would be possible through their ports.

This system is providing confidentiality and authentication through a comprehensive method based on which ciphering files, virtual drives and **VPN** client are being applied with hardware lock.

FEATURES

- Supplying confidentiality and Authentication in Notebook
- Compatible with Windows 2000 - 2003 - XP - Vista
- Removing some I/O ports for prevent sniffing (Bluetooth, Wlan, Modem, GPS, 3G)
- Supporting McAfee anti virus
- Utilizing RS232 serial port
- Secure login to operating system
- Military standard support (**MILSTD - 810F**, **MIL-STD 466**, **MIL-STD 266**)



GSM, DCS & 3G JAMMER

PIC: 000728

DESCRIPTION

Mobile phone technology plays a sensitive role in the lives of the human beings. because of the different advantages it can not be separated from our lives. but the disadvantages of this technology are undeniable.

Using the system at military sites will increases the danger of spying or to be tracked by spies which is considered as an important threat.

So, jamming on 3 bands of the mobile phones simultaneously or separately should be done. Each jammer provides jamming on 3 bands in a specific radius according to the output power.

The system can be controlled remotely.



MILITARY SATELLITE JAMMER

PIC: 000729

DESCRIPTION

The **GPS/GLONASS** jamming system is designed for effective jamming on the enemy's receivers, protecting the targets against cruise missiles and guiding bombs.

This system could jam on UAVs and other ground, marine and airborne communication systems which are equipped with GPS/GLONASS receiver systems.



GPS/GLONASS JAMMER

VIP BOMBING JAMMER

PIC: 000730

DESCRIPTION

The **VBJ-100** jamming system is designed for three frequency bands: 100 to 500MHz (50W output power), GSM-900 (30W output power) and GSM-1800 (20W output power).

These three sub-systems operate separately with the total output power of 100W.

Each frequency band includes an independent synthesizer, power amplifier and antenna. A common power module supplies the power of the system.

The subsystems could be turned ON/OFF separately.



VBJ-100



[MATC-401] MOBILE AIR TOWER

PIC: 000731

DESCRIPTION

The **Moveable Air Traffic Control Tower** is a complete set of equipment which are necessary for a control tower in an airport; and are integrated in a moveable tactical structure.

The **MATC-401** is designed for guiding the aircrafts and helicopters while taking-off or landing. Moveability of this control tower is useful for the condition that quick and temporary establishing of an airport is desired.



GBR-1000

DIGITAL DISPLAY

VRC-11-150

GBR-160

GENERATOR

VOICE RECORDER

TACTICAL INTEGRATED COMMUNICATION

PIC: 000732

DESCRIPTION

This mobile communication site is designed for asymmetric war to increase the maneuver power of the entities and also saving the operators/soldiers health.

The **T-ICS** system is able to establish various types of radio communications (VHF, UHF, HF), PSTN, PBX and PHONE PATCH, as well as audio/data transmission via field wire or optical fiber and CAT5 (IP base Ethernet) between different sites i.e. aircrafts, helicopters, command and control posts, LOS, troposcatter, radar and missile.





SURVEILLANCE & CONTROL NETWORK

PIC: 000733

FEATURES

- Collecting the surveillance information and transferring the gathered information to repeater station via a transceiver, then to CP (command post) point and from CP point to center through fiber optic.
- Each network contains 64 surveillance and control location.
- Each location is equipped with a data terminal and a manpack transceiver.
- Maximum distance between observer and manpack radio is 100m.
- Facility for data/voice reception and transmission is provided.

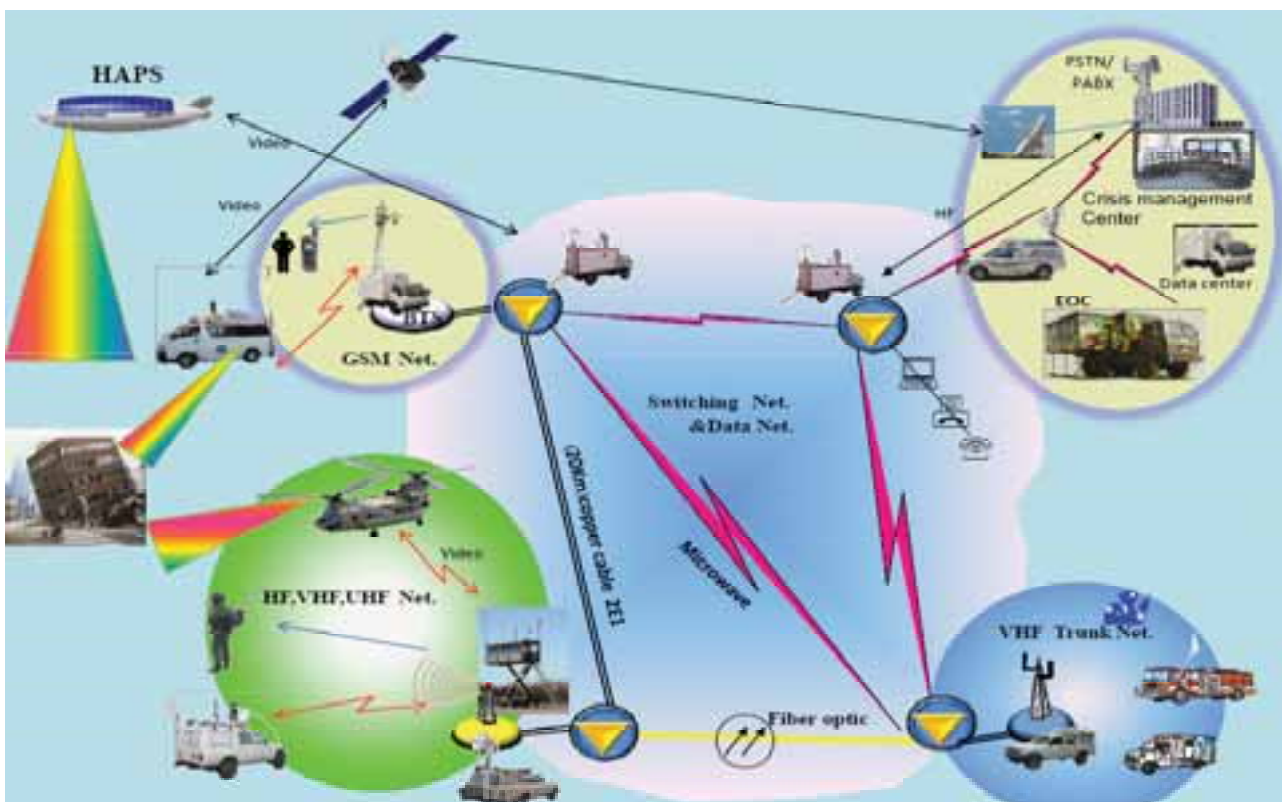


CRISIS MANAGEMENT NETWORK

PIC: 000734

DESCRIPTION

All necessary communication could be integrated as communication network in both civil & military bands to cover in crisis area.





TACTICAL MULTI LAYERS SHELTER

PIC: 000735

DESCRIPTION

Modern tactical operations require high degree of mobility and flexibility with high speed data communications between command posts.

S-4000 is a tactical communication shelter. A group of these shelters in a wide area can establish tactical communications network between main command posts, sensors and weapons.

S-4000 is one of the major communications system of an army force at echelons corps and division and could established a big flexible network with lower levels. (S-2000 for division & S-1000 for brigade level and CV-1000 for battalion level)



S-4000
TACTICAL SHELTER

SPECIAL REPAIR MOBILE UNIT

PIC: 000175

DESCRIPTION

Repair of electrical and mechanical equipment, small aerial vehicles and class four vehicles equipments. Repair of weapons and military equipments. The capability of making communications through HF, VHF UHF and Airband. The capability of presenting self – sufficient, basis meteorological information. The capability to help people suffered by flood, earthquake and fire, air, automobile accident and The capability of demolishing concrete, consolidated walls, metal and wooden walls and bodies. Receiving and sending images through GPS, GPRS and internet.



FEATURES

- Rear towing system for carrying caravan.
- Rimouk (and its relevant accessories).
- Electrical, mechanical and electronical and weapon repair equipment.
- Fire proof clothing and equipment
- System of detecting drowned live people up to the depth of six metre.
- Pneumatic jacks with measurement ton till 24 ton.
- Two way electric and gasoline water pumps.
- Generator 5Kw starter electric generator for supplying electricity under emergency conditions.
- Shock and water proof military computer applicable below 10°C temperature (amid with bluetooth printer).



SEMI LONG VHF AIR SURVEILLANCE RADAR

PIC: 000736



SL-ASR-1

DESCRIPTION

Mobile Solid State Digital **VHF** Band is a medium range radar designed to detect air targets within the radar's coverage, determine their range, azimuth, range rate and height and transmitting radar information to air defence information network.

SL-ASR-I is a new generation radar which combines advanced radar developments, modern digital technologies and hardware with constructive technological solutions.

The radar can operate in self-contained mode and as a component to automation **C3I** system as well.

SL-ASR-I is a highly mobile, fully solid state and medium altitude surveillance radar developed based on the operation demands by radar forces. This radar is vehicle carried; uses full coherent pulse compression and **MTD** detection techniques.

The radar is characterized by good performances such as high mobility, high automation,... etc.

FEATURES

- Early detection of air targets and determination of their coordinates including range, azimuth , radial velocity and height.
- Capability of **IFF** system conjunction as secondary radar.
- **ECCM** capabilities against wide variety of jamming and interference.
- Target tracking using **TWS** technique.
- Detection of targets with small RCS & low altitude.
- Supplying peripheral radars with target extracted data.

SPECIFICATIONS

TYPE	SL-ASR-1
Transmitter	
Model:	Fully Solid State
Detection Range (km):	>300
Azimuth Resolution (°):	6 - 8
Antenna Type:	Yagi
Output Power (peak power) (KW):	12
Duty Cycle (%):	10
Coherent Receiver	
No of Receiving Channels:	2
Receiver Sensivity (dBm):	-115
Noise Figure of the Receiver (db):	< 5
Dynamic Range (db):	80
STC Requirements	
RF STC: RF attenuation (dbm):	≥ 18



AIR SURVEILLANCE SHORT RANGE RADAR

PIC: 000738



ASRX-A

DESCRIPTION

ASRX-A is an air surveillance **X-band** pulse radar used to detect air targets with **low RCS** and **low altitude**. The system is a 2-dimensional radar displaying azimuth and range of targets. It detects targets with 1 m^2 **RCS** at 18 Km and 0.05 m^2 **RCS** targets at 10 Km.

APPLICATIONS

- May be used in fire control unit (**FCU**) by transmitting target data to this system.
- Able to detect small targets such as **UAV**, **RPV**, **cruise missile**, etc.
- Using **TWS** technique, it can track targets and be linked with optical tracker.

SPECIFICATIONS

TYPE	AIR SURVEILLANCE SHORT RANGE RADAR
Frequency (mhz):	8600 - 9500
Number of Frequencies:	5
Transmitter Peak Power (kw):	25
Pulse Length (μs):	0.3 1.4
Antenna Rotation in Azimuth (°):	360
Antenna Rotation Velocity (rpm):	12, 24, 60
Resolution:	75m with the Pulse of 1.4 μs
detectable speeds (m/s):	15 to 1000
Detection Accuracy in Range:	310 m with the Pulse of 1.4 μs
Minimum Range:	75 m with the Pulse of 0.3 μs 310 m with the Pulse of 1.4 μs
Maximum Range (km):	20.2
TWS:	Capability of Simultaneous Tracking of 8 Targets
Antenna Beam Length in Azimuth (°):	1.7
Antenna Gain (db):	28
Antenna Pattern:	Extended Cossec ²
Power Supply:	3Phase - 400Hz - 110V



[TM-ASR-1] TACTICAL MICROWAVE AIR SURVEILLANCE RADAR

PIC: 000739

DESCRIPTION

TM-ASR-1 radar is an magnetron pulse surveillance radar with maximum power of 800kw. Since this radar is able to detect the targets with maximum range of up to 150 km, it is used as primary warning radar.

This radar is also used to detect the flying targets and is able to measure the distance and angle (R,O) of the target and display the "X" and "Y" coordinates of the target on the monitor.

FEATURES

- 2D medium range air surveillance radar.
- Detection of low "RCS" targets such as PRV, Cruise missiles,...
- Deliver the target information to command & control center.
- Searching at 360° and review speed of 3,6,12 RPM.
- Displaying the clutter map and amplifying the critical areas.
- Mobility , fast set-up and break-down.
(maximum time for setup is about 30 minuts)



TM-ASR-1

SPECIFICATIONS

TYPE	TM-ASR-1
Operating Frequency:	S - Band
Peak Power (kw):	800
Average Power (w):	800
Coverage 2 Modes:	Normal Mode 360° Range = 150 Km Altitude = 14 Km
Resolution:	a) Range 300 m b)Azimuth 1.3°
Detection Coverage:	a) R max = 150Km pd = 80% RPM = 6 rcs = 10 m ² b) R max = 120 Km pd = 80% RPM = 12 rcs = 10 m ²
Antenna:	a) Type = Parabolic b) gain = 30 dB
Beam Width:	a) Horizontal = 4° b) Vertical = 40°
Target Processing Capacity:	≥ 100 Target
Display:	2 LCD Displays (with zoom display 30/ 60/ 120/ 150)
Track While Scan (TWS):	20 Target
BITE:	Included
ECCM:	Selectable/freq. 10 Point
Pulse width (µs):	1 , 0.8 , 0.4
Pulse Repetition (hz):	866 , 1250 , 1666
Frequency band width (mhz):	240
Transfer Data Information:	To Command Control
Local Network:	Included
Portable:	Track
Power:	60 kw 3 φ 220v , 50 Hz
Power Consumption:	30 kw 3 φ
Secondary Radar (IFF):	Included



PAR-X30 PRECISION APPROACH RADAR

PIC: 000740

DESCRIPTION

Precision Approach Radar (**PAR**) is a type of radar guidance system designed to provide lateral and vertical guidance for an aircraft pilot during landing, until the landing threshold is reached.

After the aircraft reaches the Decision Height (**DH**) or Decision Altitude (**DA**), guidance is advisory only.

This system is used in air traffic control and precision guidance of aircrafts and helps pilots keep the aircraft on

course and glide path during final approach.

PAR-X30 is one of the most important sensors in Air Traffic Control Systems and plays a very important role in providing the flight safety, especially in landing operation, for both civil and military aircrafts. This system is a very high precision, 3D system working in X-band frequency and specifically designed to help pilots in landing fighters even on the shipboards.

FEATURES

- Increasing the flight safety during the landing especially at night, rainy and dusty weather.
- Checking and controlling the aircraft function and guiding the aircraft in three dimensions (range, azimuth, altitude) on the airport glide path from the landing point to the distance of up to 30 km.
- Communicating with other airport radars such as PSR and SSR.
- Checking and measuring flight altitude in order to improve the landing operation.
- Being full function controlled and monitored in air traffic control tower through optical or radio links.



PAR-X30

SPECIFICATIONS

TYPE		PAR-X30
Operating Frequency:		X Band (9.0~9.2 ghz) in 10 Channel
Antenna:	Azimuth	Beam Width: 0.85*3.5degree Max. Gain:38.5 dB (in linear Polarization)
	Elevation	Beam Width: 0.65*3degree Max. Gain:39.5 dB (in linear Polarization)
Transmitter:	Type	Solid State
	Power	50Watt (peak), 7.5 Watt (Average)
	Pulse width	Short 500ns ong 40µs in LFM or NLFM Modulation
Receiver:	Type	Full Coherent Solid State
	Gain	Max. 75 dB (Controlled by STC)
	Noise Figure	3.5 dB
	Band width	4 MHz
Environmental Conditions:	Temperature	Indoor : -10°C ~ + 45°C Outdoor: -30°C ~ + 60°C
	Humidity	90% @ 40°C
	Wind speed	90km/h (Operation) 120km/h (Nonoperation)



SHORT RANGE GROUND SURVEILLANCE RADAR

PIC: 000741

DESCRIPTION

GSR - 110 S is an advanced new generation of short range tactical radar which is portable and can be easily carried by soldiers. A light weight of less than 25kg, solid state & low power have made it one of the most effective and efficient battlefield surveillance radars available in the military market.

It is also capable of fulfilling tactical surveillance mission efficiency as fixed - site surveillance radar.

GSR - 110 S is used for detection, localization, recognition and tracking of any moving target on the ground such as pedestrian, vehicle or low flying helicopter during day or at night under decayed climatic conditions such as fog, smoke, snow and etc.

FEATURES

- Excellent ECCM capability.
- Low probability of intercept (LPI).
- Pulse duppler filter processing.
- Hight mobility and fast set-up.
- Modularized design.
- CFAR processing.
- Audible detection alarm.
- Solid state transmitter.
- Micro strip array antenna.
- Manual & automatic target detection & tracking.
- Digital wave form generation and pulse compression.
- Rugged computer as a radar terminal.
- Full function remot control (FFRC).
- Automatic sectorial surveillance with sector axis and width selectable.



GSR-110S

SPECIFICATIONS

TYPE	GSR-110S
Detection Range (km)	
Pedestrian (km):	2.8
Light Vehicle (km):	5
Heavy Vehicle:	6.4
Resolution	
In Range (M):	40
In Azimuth (deg.):	4.5
Location Accuracy	
In Range (m):	10
In Azimuth (deg.):	±1.25
Azimuth Coverage (deg.):	0 - 360
Detectable Target Velocity (km/h):	3 - 100
Power Consumption (w):	Less than 30
Operating Temperature	
Operator Console (°C):	-25 to +55
Outdoor Equipment (°C):	-30 to +60
Storage Temperature (°C):	-30 to +60



[MVR-88] MARINE VESSEL RADAR

PIC: 000737

DESCRIPTION

Radars are employed by almost all Maritime and Coastal surveillance organizations, with powerful signal processing ability.

Modern radar systems are capable of extracting targets and determining their velocity from the surrounding background. Processing the radar video with modern processing techniques improves the radar detection performance.

Processor unit performs signal processing, tracking and display under the operator's control and generates high resolution data suitable for use, all in real time.

This radar is presented in "X" and "S" band versions, with technical specification on the back page.



MVR-88

FEATURES

- Developed to enhance the radar performance, user-friendly operation and visibility.
- Utilizing software-based digital signal processing with high performance computer-based technologies.
- Full range tracking, data extractor (Networking) and high angular accuracy.

SPECIFICATIONS

TYPE	X-BAND			S-BAND	
	A1	A2	A3	A1	A2
Length (ft):	4	6.5	8	10	12
Beamwidth (H)(°):	1.9	1.23	0.95	2.3	1.8
Beamwidth (W)(°):	20	20	20	25	25
Sidelobe Within (db):	-24	-28	-28	-24	-24
Sidelobe Outside (db):	-30	-32	-32	-30	-30
Polarisation:	Horizontal				
Rotation (rpm):	24 or 42				
RF Transceiver					
Frequency:	[(9410 MHz (x-band), 3050 (s-band))] ±30 MHz				
Output power (KW):	25 (x-band) , 30 (s-band)				
If Frequency (mhz):	60				
Noise Figure (db):	6 (typical)				
Duplexer:	Ferrite Circulator with Diode Limiter				
Range Accuracy (%):	Within 1				
Bearing discrimination (°):	0.95				
Bearing accuracy (°):	< 1				



TRM - 8C - 3000 TACTICAL RADIO MONITORING

PIC: 000742

DESCRIPTION

TRM-8C-3000 is an Eight Channel Radio Monitoring System working at 0.5 to 3000 MHz frequency band. It detects all conventional analog and digital modulation. This system contains one receiver for searching and seven hand off receivers for monitoring. By IF spectrum processing method, it has the maximum scan rate of 10GHz/s. All 7 receivers can monitor 1000 channels. Taking advantage of using two separate racks enables the operators to manage radio monitoring and channel searching independently. This feature extends the system application specially during tactical operations.

FEATURES

- 8 Channel simultaneous interception.
- Fast search.
- Interception of all conventional analog and digital modulations.
- High quality antenna (omni & direction) for better sensitivity.
- 8 x 1000 monitoring channels.
- High quality antenna distribution system for easy operation.
- Remote control operation and network based interception.
- 8 channel digital recording & play back system.
- Wide band IF (BW=10MHz) recording.
- Tactical & stationary application.



TRM - 8C - 3000

SPECIFICATIONS

TYPE	TRM-8C-3000
Frequency Range:	HF Band: 500KHz - 30MHz ,V/UHF Band:30MHz - 3GHz
Sensitivity:	-110 dBm for FM , 12 dB SINAD (AT 1000 MHz)
Modulation Type:	Analog: WFM - FM - AM - CW - USB - LSB Digital: Adjustable
IF Band Width:	FSK - MFSK - GFSK - DPSK - BPSK - QPSK
Scanning Rate:	Monitoring Receiver: 100 Ch/s Searching Receiver: 10GHz/s for BW<10 MHz 2GHz/s FOR BW>10 MHz
Memory Capacity:	Minimum 80 GB Hard Disk & 1GB flash
Squelch:	Adjustable
Simultaneously Monitoring Channel:	8 Channel Maximum
Directional Antenna (MHz):	30 - 1300
Omni Directional Antenna:	10 KHz - 3 GHz
Panorama Band Width (MHz):	V/UHF>80 , HF>5
Power Supply (VAC):	220/110
operating Temperature (°C):	0 - 50 In Door
Operating Humidity:	85% at T=40° C/NON - Condensing
Dimension (mm):	650 X 600 X 800
Weight (kg):	Searching Rack 75 , Monitoring Rack 75
Racks:	2 Pcs (19" X 14u)



HRM 330-HF RADIO MONITORING

PIC: 000743

DESCRIPTION

HRM-330 is designed for automatic and manual detection and analysis of Low Probability Interception (**LPI**) signals (Burst, hoppers), wide and narrow band, signal search and surveillance, interception, monitoring, recording and analysis of radio communications in the frequency range of 300 KHZ to 30 MHZ.

Signal processing functionally includes detection of conventional and short time signals, tuner scan functionally, interception of voice signals, demodulation & decoding for all analog and digital narrowband signals and at least two wideband signals with definite protocol.



HRM-330

FEATURES

- Wide band signal interception, recording and decoding (0.5, 1, 5 MHZ) to a maximum of two simultaneous separate spread spectrum signals.
- Simultaneous interception and record of 8 fix frequency signals in narrow band width.
- Remote control operation and network based intercepting.
- Capability of self test.
- VOR capability.
- Tactical & stationary applications.

SPECIFICATIONS

TYPE	HRM-330
Frequency Range:	300 KHZ - 30 MHZ
Phase Noise:	≤ -90 dBc/hz (1KHZ offset)
3 rd Order Intercept Point:	≥ 15 dBm (Pre Amplifier off, 1.5MHz to 30 MHz) ≥ 15 dBm (Pre Amplifier off, 0.2 MHz to 1.5 MHz) ≥ 15 dBm (Pre Amplifier on, 1.5 MHz to 20 MHz) ≥ 15 dBm (Pre Amplifier on, 20 MHz to 30 MHz) ≥ 15 dBm (Pre Amplifier on, 0.5 MHz to 1.5 MHz)
2 nd Order Intercept Point:	With Sub Octave Filters of Pre Selector ≥ 75 dBm (Pre Amplifier off) ≥ 75 dBm (Pre Amplifier on) With Broad Band Filters of Pre Selection ≥ 300 KHZ to 1.5 MHZ or 300 KHZ 30 MHZ ≥ 50 dBm (Pre Amplifier off) ≥ 30 dB (Pre Amplifier on)
Image Frequency Rejection:	≥ 60 dB
IF Rejection:	≥ 60 dB
Noise Figure:	2 DB (Without Pre Amplifier) , 8 dB (With Pre Amplifier)
Demodulation Mode:	AM, FM, USB, LSB, CW, ISB



SDF-230 WIDEBAND DIRECTION FINDER

PIC: 000744

DESCRIPTION

SDF-230 with its extremely high scan speed is the latest generation of scanning direction finders. It can find the direction of frequency/time hopping, burst and fixed frequency signals.

Having a wide frequency range from **20 MHz to 3000 MHz**. The system uses Correlation/interferometer method, 15 direction finding algorithm. Antenna system is designed for application in stationary or semi-mobile **VHF/UHF DF** systems.

Using of large **DF** antennas with nine elements, the system can reach highly accurate bearings.

In this system four **DF** modes are available: scan mode (fast **DF** scanning), fixed frequency mode (**HM**), search mode and wideband mode (**WFFM**).



SDF-230

FEATURES

- Enhanced DF accuracy and sensitivity by using wide aperture DF antenna.
- Extremely high frequency resolution.
- Remot control via virtually any data link.
- GPS support.
- Direction finding of narrow band digital-analog signals with QAM, ASK, FSK, AM, FM, CW, PULSE, SSB, and IQ modulations.
- Direction finding of 2 wideband signals with 2000 H/S in the 20 MHz band width simultaneously.
- Networking capability for Position finding.
- Self test capability.
- Tactical & stationary application.

SPECIFICATIONS

TYPE	SDF-230
Frequency Range (mhz):	20 to 3000
DF Method:	Correlative Interferometer
DF Capability:	AFF, DFF, FH/TH, BURST
Hop Rate (hp/s):	2000
Frequency Span in Wideband Mode (mhz):	20
System DF Accuracy (°rms):	3 (with 12m Antenna Height)
Display Resolution (°):	0.1 or 1 (Selectable)
Dynamic Range:	≥ 70
Image Frequency Resolution:	≥ 90
IF Rejection:	≥ 90
Signal Duration (µs):	≥ 400
Scan speed (full band)(sec):	3
Noise Figure	
F < 2 GHz (db):	≤12
f = 2 GHz to 3 GHz (db):	≤15
Modes of Demodulation:	CW, AM, FM, SSB, FSK
Signal - to - Noise Ratio	
F < 2 GHz	AM, Lpin = -107 dbm
f = 2 GHz to 3 GHz	AM, Lpin = -104 dbm
B = 6 KHz, m = 0.5, F _{MOD} = 1 KHz	SINAD 10db
F < 2 GHz	FM, Lpin = -107 dbm
f = 2 GHz to 3 GHz	FM, Lpin = -104 dbm



HDF-230 HAND HELD DIRECTION FINDER

PIC: 000745

DESCRIPTION

Hand held Direction Finder Systems are used for mobile **DF** applications where radio emitters are not accessible by vehicle and the operator has to dismount and continue the mission on foot.

HDF-230 is a light weight portable radio direction finder designed to detect the radio emitters by tracing radio emissions and proceeding toward the direction of the arriving signal.

By selecting the appropriate hand -held antenna wide frequency range from 20 MHZ to 3GHZ can be covered.



HDF-230

FEATURES

- Small, lightweight, easy to use, suitable for rapid deployment.
- Coverage of frequency range , 20 MHZ to 3 GHZ , depending on selected antenna.
- Capability of displaying signal direction & strength at day or night on a pocket pc monitor.
- Operation can last for a duration of seven hour or mor using a charged battery.
- Hight stability receiver with phased locked loop and synthesized oscillator.
- Versatile functions such as spot search, scan and priority reception.

SPECIFICATIONS

TYPE	HDF-230
DF Type:	Amplitude Comparison
Frequency Range:	Ant - 20 - 500 20 - 200 MHZ Ant - 200 - 500 200 - 500 MHZ Ant - 500 - 3000 500 - 3000 MHZ
selectivity SSB, CW:	More than 1.8 KHZ/ -6 dB
AM, FM:	More than 12 KHZ/ -6 dB More than 30 KHZ/ -60 dB
WFM:	More than 150 KHZ/ -6 dB
Polarization:	Vertical
Bearing Display:	Numerical and Bar graph
Accuracy:	+/- 15 Degrees
Receiver:	Frequency includes 150 KHZ to 3304.999 MHZ
Reception Manual:	(Digital up/ Down/ Dial) Spot, Search and Scan
Memory:	1250 (1000 Regular, 50 Scan edge and 200 Auto Memory Write)
Power:	Power consumption: DC 5V 700 mA or Less
Battery:	3.6 VDC, 9 AH
Car Battery:	Through Cigarette Lighter
Charger:	Power supply from AC 100 - 230 VDC
Weight:	Main unit Less than 3 kg Including Battery
Antenna:	Less than 2 kg



ELINT

PIC: 000746

DESCRIPTION

Electronic Intelligence (**ELINT**) is a mean of observing the signals transmitted by radar systems to obtain information about their capabilities. It is the remote sensing of remote sensors.

Through **HINT**, it is possible to obtain valuable information while remaining far from the radar itself.

ELINT is most useful in hostile situations. The distinctive advantage of **ELINT** is that it provides on time information about threatening systems such as radars that guide aircraft or missiles to targets.

FEATURES

ELINT is an advanced passive receiver for the radar signals which can be used in truck; naval and ground based applications. The system uses **IFM** (instantaneous frequency measurement) receiver, channelized receiver and super heterodyne receiver technology covering a frequency range of 0.4-18 GHZ also Omni directional antennas and directional antennas for covering 3600 azimuths and 600 elevations.



The system measures the signal parameters including angle of arrival (**AOA**), pulse width (**P.W**), pulse repetition frequency (**P.R.F**), RF frequency, radar antenna scan rate, pulse amplitude, radar antenna scan type and radar antenna polarization.

This system also may be used as a sensor for Position Finding (PF) network by using 2 or 3 ELINTs.

SPECIFICATIONS

TYPE	EL 418	EL 118	EL 818
Frequency Range:	0.4 - 18 GHZ	1 - 18 GHZ	8 - 18 GHZ
Frequency Accuracy:	1 MHZ		
Type of Receiver:	IFM, SUP, Channelized	IFM, SUP	
Receiver Sensitivity:	-80 dBm		-85 dBm
Dynamic Range:	70 dB		75 dB
Pulse with Range:	100 nsec to 300 µsec		
Pulse with Accuracy:	5% ± 20nsec		
PRF Range:	100 HZ to 300 KHZ		
PRF Accuracy:	5%	5%	2.5% lower than 50 khz 5% upper than 50 khz
DF Accuracy:	Better than 5° RMS		
Azimuth Coverage:	30° Instantaneous 360° Selectable	360°	120° Instantaneous 360° Selectable
Elevation Coverage:	-5° to +20°	-30° to +30°	-5° to +20° (10° tilt)
Data Base:	Yes		1000 Radars
Working Continuity:	24 Hours		
Power Supply:	220 VAC (50 Hz)		
Built in Test:	Yes		
GPS:	Yes		
MTBF:	2000 Hours	1500 Hours	
Type of Intercept able Signal:	Pulse, CW, Chrip - Pulse Duppler - PRF staggered - PRF jitter - Agile - Pulse compression		
Type of Antenna:	Directional & Omni Directional		
Platform:	Ship Born/ Land Base	Land Base	



CUSTOMIZED EW COMMAND AND CONTROL SYSTEM SOLUTION

PIC: 000747

DESCRIPTION

This system is an advanced **EW** system designed to intercept, monitor, analyze, record and implement direction finding and position finding of narrow band/ wide band signal from **1 MHz to 3 GHz** to acquire strategic/ tactical knowledge and information of enemy's activities.

In addition to monitoring and direction finding, this system has also the ability to jam communication signals in the frequency range of **20 to 500 MHz**.

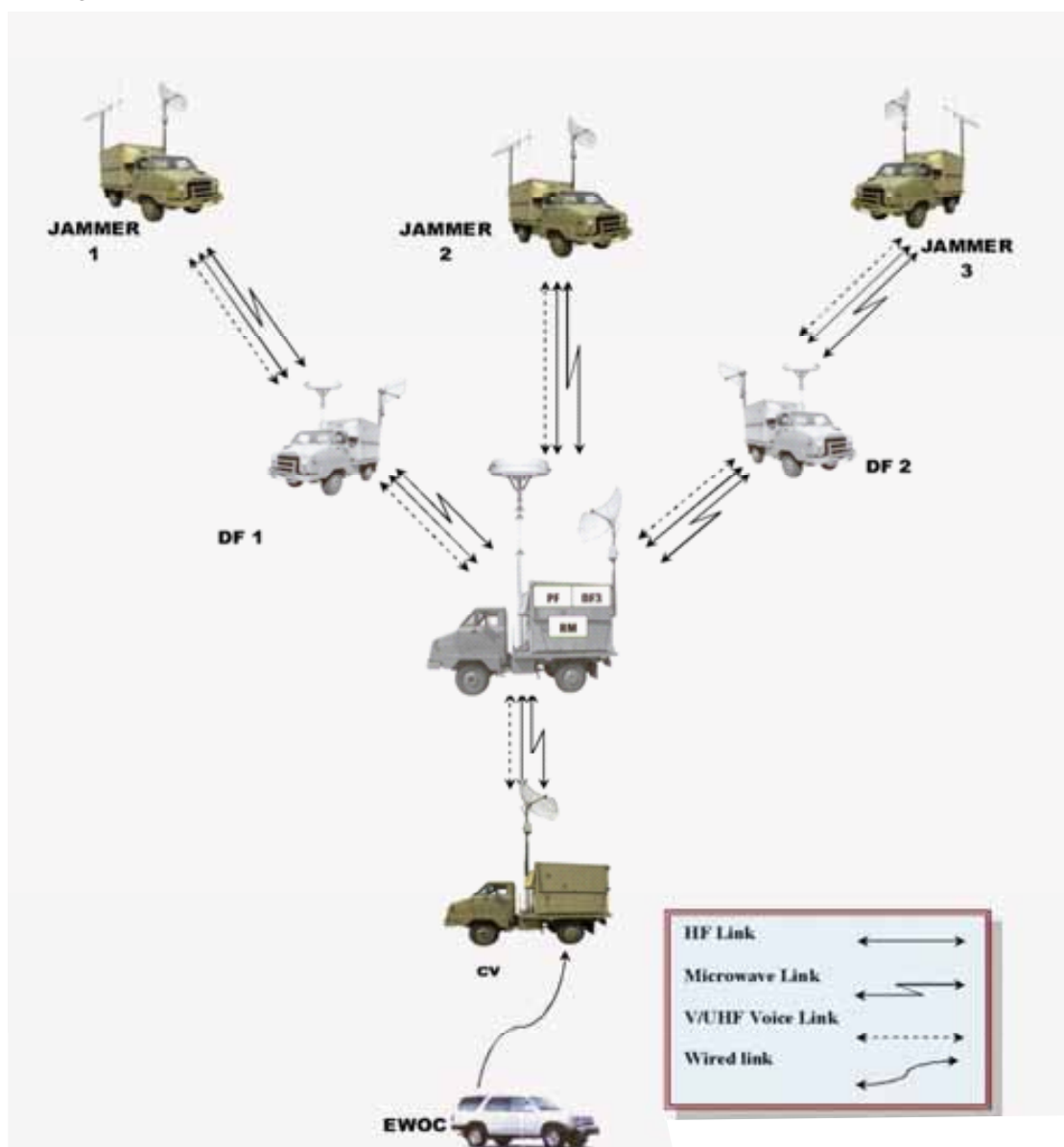
The whole cell is equipped with an Electronic Warfare Operation Center (**EWOC**) which controls and manages Radio Monitoring Systems, three direction finder stations (**DF's**) and three jammer command centers. Each **DF** finds the position of targets in their assigned area and each jammer control center controls its related jammer vehicles through microwave links.

The system provides remote control operating and GIS capabilities.

Several types of antennas are used depending on the operation of each subsystem such as dipole and whip passive antennas in jamming system, omni directional and 8 active crossed loop antennas in radio monitoring and direction finding systems.

An important advantage of this system is a network which is used for controlling and managing total system by **EWOC**.

Communication between jammers command center (**JCC**) and jammer vehicles are through microwave links and the network media for other equipments can be fiber optic or the existing communication backbone (**ISDN** lines) which should be prepared by the customer.





IUWP27-1 UNDERWATER PINGER

PIC: 000748

DESCRIPTION

Pingers are guiding devices used to provide an acoustic energy source that can be heard with a hydrophone.

Used as an acoustic beacon, pingers commonly produce and send out the regular pulses in specific rate in water by a transducer. **IUWP27-1** as an underwater pinger is a small, rugged and reliable acoustic signaling device used for marking equipment and underwater sites.

IUWP27 -1 offers flexible features that allow you to tailor it to your requirement.

Factory selectable acoustic power out puts, frequencies, and time delays give the ability to order **IUWP27-1** in many configurations. Using an off-the-shelf 12 volt battery, adds the advantage of cost effective maintenance.

IUWP27 -1, continuously sends out an acoustic signal for up to 10 days.



SPECIFICATIONS

TYPE	IUWP27-1
Frequency (khz):	27
Output Power (watts):	10
Pulse Length (m/sec):	5
Pulse Period:	1 Pulse per 3 Sec
Operation Voltage (vdc):	12
Battery Type:	Lithium
Battery Life (days):	Up to 10
Operation Depth (m):	Up to 1830

chapter

7

OPTICS





Iran Optics Industries (IOI) was established in 1987 with the aim of setting up vigorous and modern optics industries. The employment of expert engineers and state-of-the-art equipment has made it one of the most capable industries in the country.

IOI has extensive experience in designing and manufacturing optical equipment according to **MIL standards**. Comprehensive training facility has enabled us to offer in depth technical support both in Iran and in abroad. In order to produce reliable optical systems we have experienced how to make superior optical components and calibrated instrument.

IOI has developed the technology of manufacturing Opto-mechanical and Electro-optical elements and devices such as:

Electro-Optical products:

Electro-Optical Fire Control Systems, Aerial Camera, Electro-Optical Surveillance Systems, Marine and Helicopter Digital Display Systems

Opto-Mechanical products:

Hand Held Binoculars, Observatory Binocular, Compass, Aiming Circle, Sighting Scope, Submarine Periscope, Periscopes (Eastern and Western series), Glasses and Optical parts.

Simulators:

Laser Target Simulators, Firearm Training Simulators (**FTS**) and Dome with Anti aircraft Gun Simulators.

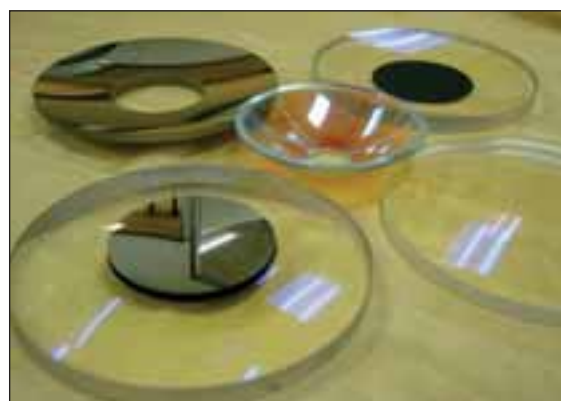
Glass:

Aircraft Windshields, Aerospace Transparency System, Security and Safety Glass, Proof Glass and Cars.

Also we have been able to get both of national and international standards certificates such as **ISO, EFQM, ISO 9001/2000, ISO/IEC17025, Kharazmi National Award** for innovation and some of customer's satisfaction certificates.

To provide our customers with high quality products, we have high-tech standards labs and facilities.

We can also provide our customers with engineering and designing consultations.





CHAPTER 7



ITEM	TABLE OF CONTENT	PAGE
1	HAND HELD BINOCULARS	220
2	OBSERVATORY BINOCULARS	221
3	FADAK 8 - LASER RANGE FINDER	222
4	GHADIR 4 - RIFLE SCOPE	222
5	ESR 20-H LASER RANGE FINDER	223
6	KAVOSH 2 - ZOOM LENSES	224
7	NVS-400 - NIGHT VISION SIGHT	225
8	ANG-86 - AVIATOR NIGHT VISION GOGGLE	226
9	MNG-87 MONOCULAR NIGHT VISION GOGGLE	227
10	NDG-87 - NIGHT DIVER GOGGLE	228
11	THERMAL CAMERAS	229
12	SADAD 103 - MONITORING SYSTEM	230
13	ELECTRO OPTICAL SURVEILLANCE SYSTEM	231
14	OGHAB 11	232
15	EOVP 4 - AERIAL CAMERA	233



HAND HELD BINOCULARS

PIC: 000749

DESCRIPTION

We produce various types of prismatic hand held and observatory binoculars.

Binoculars are designed for observation of stationary and moving near and far objects and targets at daytime conditions.

The reticle which is inserted in right telescope focal plane, enables the observer to estimate the distance from observer to the specified viewing objects or targets.



FEATURES

- Fully Coated Optics
- Easy Center Focus
- Equipped With Graduation Reticule
- Completely Water Proof
- Fully Covered by Shock-Resistance Rubber Armor
- Nitrogen Gas Filled



SPECIFICATIONS

TYPE	ROOF PRISM 10 X 40	PORO PRISM 7 X 42	PORO PRISM 7 X 42 WITH COMPASS	PORO PRISM 15 X 56
Model:	BDS - 101	BDS - 102	BDS - 111	BDS - 109
Magnification (x):	10	7	7	15
Objective Diameter (mm):	40	42	42	56
Real Field of View (degree):	6.5	7.3	7.3	4.5
Exit Pupil Diameter (mm):	4	6	6	3.7
Relative Brightness:	16	36	36	13.7
Inter Pupillary (mm):	58 ~ 72	58 ~ 72	58 ~ 72	58 ~ 72
Diopeter Adjustment (diopeter):	-5 ~ +5	±4	-6 ~ +6	-6 ~ +6
Weight (g):	850	1,800	1,000	1,302
Size (mm):	165 x 135 x 56	165 x 190	165 x 190	220 x 195



OBSERVATORY BINOCULARS



PIC: 000750



FEATURES

- Supplied With Complete Holding Arm and Sturdy Tripod
- Fully Coated Optics
- Airtight Mechanism/Protection Against Water, Dust, Rust Moisture, etc.
- Nitrogen Gas Filled
- Included Ocular Focusing

SPECIFICATIONS

TYPE	20 X 80	20 X 120	30 X 120	40 X 150
Model:	BDS - 201	BDS - 202	BDS - 203	BDS - 401
Objective Diameter (mm):	80	120	120	150
Magnification (x):	20	20	30	40
Real Field of View (degree):	3	3	2.2	1.5
Exit Pupil Diameter (mm):	4	6	6	3.75
Relative Brightness (mm):	16	36	38.5	14.06
Range of diopter adjustment (diopter):	- 4 ~ +2	- 4 ~ +2	- 4 ~ +2	- 4 ~ +2
Range if Inter Pupillary Distance Adjustment (mm):	58 ~ 72	58 ~ 72	58 ~ 72	58 ~ 72
Azimuth (mil):	360	6,400	6,400	6,400
Elevation (mil):	50	1,000	1,000	1,000
Depression (mil):	20	500	500	500
Weight (kg):	8	14	17	32.6



FADAK 8-LASER RANGE FINDER

PIC: 000751

DESCRIPTION

- 15 km range finding capability (In 20 km atmospheric visibility range) in adverse weather and battlefield obscurants
- High reliability and low maintenance
- Equipped with 12X optical sight and CCD camera
- Finding defects by operator capability
- Detecting and saving 3 targets simultaneously
- Range finding in 1-5Hz operation with highest measurement accuracy
- Optical & Electronical test without laser pulse Fire
- Installation on electro optic, surveillance & seeking systems
- Compact, lightweight and easy operation



FADAK 8

SPECIFICATIONS

TYPE	FADAK 8
Laser Type:	Nd: YAG
Wave Length (NM):	1064
Laser Output (mj):	Up to 40
Laser Classification:	Class 4
Repetition Rate (Hz):	1 - 5
Pulse Width (ns):	12 ± 5
Range Finding:	500m to 15km at 20km visibility
Target Dimensions (m²):	1.5×1.5
Accuracy (m):	± 5
Target Discrimination (m):	30
Divergence (mRad):	1
Operating Voltage (V DC):	20 - 30
Operating Current (A):	10
Interface:	RS 422
Operating temperature (°C):	-30 to +55
Storage temperature (°C):	-40 to +70
Cooling:	Air-Cooled
Dimensions (mm):	250 × 220 × 105
Weight (Kg):	< 7

GHADIR 4 - RIFLE SCOPE

PIC: 000763

SPECIFICATIONS

TYPE	GHADIR 4
Magnification:	3X-12X(Zoom)
Entrance Pupil Diameter (mm):	50
Exit Pupil Diameter (mm):	16.5(3x) to 4.1(12x)
Eye Relief (mm):	77 - 84
Range of Diopter Adjustment:	-3 to + 3 diopteries
Field of view in 1000 meter:	31.5 m(12x) to 101.3 m(3x)
Field of view:	1.8(12x) to 5.8(3x) degree
Elevation Adjustment Control (mil):	22
Lateral Adjustment Control (mil):	12 ± 6
Parallax Adjustment:	50 to ∞ (50 - 100 - 200 - 300 - 500-800 - 1000 - ∞)
Total Length (mm):	Approx 350



GHADIR 4



ESR 20-H LASER RANGE FINDER



PIC: 000752

DESCRIPTION

ESR20-H is a professional long range binocular eye-safe laser range finder for measuring distance up to 20 Km. This hand-held model is easy to use and withstand the harshest environmental conditions. It uses single-pulse, erbium-glass laser technology for range finding and operates at an eye-safe wave length of 1.54μm.

The system is equipped with an external communication connector, enabling remote operation and output of electronic ranging information for further processing and display. It is mainly used as a hand-held unit, but can also be mounted on a tripod.



ESR 20-H

FEATURES

- Low battery warning
- Easy and safe operation
- One-click button operation
- High reliability, low maintenance
- Built-in 1/4-20 tripod adapter mount
- Combination with a night vision device possible
- Ruggedized, meets severe environmental conditions
- Capable of being used as 6x monocular camera for monitoring

SPECIFICATIONS

TYPE	ESR 20-H
Transmitter:	
Laser Type:	Erbium Glass
Wavelength (nanometers):	1540
Pulse Repetition (ppm):	12
Range Display:	
Measuring Range:	80 m to 20 Km
Range Accuracy (m):	± 5
Data Display:	2 × 5 - Digit, LED, Brightness Adjustable
Viewing Optics:	
Field of View (mrad):	0.7
Magnification:	6X
Visibility (Km):	20
Electrical Interface:	
Power Supply (Volts):	9 to 33
Data Interface:	RS - 422 & RS - 232
Environmental Conditions:	
Operating Temperature (°C):	-30 to +60
Storage Temperature (°C):	-40 to +70
Mechanical Interface:	
Dimensions (mm):	197 × 175 × 94
Weight (excluding battery) (Kg):	1.7



KAVOSH 2 - ZOOM LENSES

PIC: 000753

DESCRIPTION

Zoom lenses are designed to present a range of continuous focal planes which are apply with interchangeable **CCD sets** for various applications such as imaging, aiming and targeting of objects and targets.



KAVOSH 2

SPECIFICATIONS

TYPE	KAVOSH 2
Useable CCD Format:	1/2 Inch Format CCD Cameras
Focal Length (mm):	10 - 500
With 2x (Internal Extender) (mm):	(120 - 1000)
Zoom Ratio (x):	50
Maximum Numeric Entrance (mm):	F4.0 (10 - 350)_ F5.6 (500)
(With 2x internal extender) :	(F8.0 (20 - 700mm)_ F1 1.2 (1000mm)
Numeric Entrance Adjusting Limit:	F 4.0 - 125 (F 360 option)
(With 2x internal extender):	(F 8.0 - 250)
Image Size:	6.4x 4.8mm Ø 8.0
Flang Length:	17.526mm (C mount)
Minimum Imaging Distance (m):	4
(hvx)Sight Field Angle:	35.5° X27.0° (10mm) 0.7° X 0.6° (500mm)
(With 2x Internal Extender):	18.2° X13.7° (20mm) 0.4° X 0.3° (1000mm)
Operating: Diaphragm	Motorized (MI/ AI - MI Changeable type) Auto (AI/ AI - MI Changeable type)
Operating: Focus	Motorized
Operating: Zoom	Motorized
Size (mm):	300 × 124 × 106
Weight (kg):	3.2



7

NVS-400 NIGHT VISION SIGHT

PIC: 000755

DESCRIPTION

NVS-400 is a modern multi-purpose night vision weapon scope built to fit the most demanding military specifications. This night vision sight is designed to be mounted on individual –served weapons and used as an aiming system at darkness and night time. It will benefit military snipers and law enforcement marksmen. One of the essential advantages of this device is light weight and easy maintenance.

FEATURES

- Long observation range
- Mountable on various types of individual served weapons.
- Adjustable brightness reticule
- Various weapon mounts are available
- Shockproof, built for use with heavy recoil weapons
- Lightweight reinforced plastic body
- Powerful infrared illuminator (optional)



SPECIFICATIONS

TYPE	NVS-400
Resolution (mrad):	0.36
Magnification :	4X
Field of View :	8°
Diopter Adjustment :	- 4 to +6
Range of Focus :	10 M to ∞
Power Supply :	One 3V Battery
Dimensions (mm):	320 × 100 × 100
Weight (kg):	1.35
Image Intensifier Tube	
Type :	Direct
Size (mm):	18
Generation :	2+
Operating Range (°C):	-30 to +50
Storage Range (°C):	- 40 to +60



ANG-86 AVIATOR NIGHT VISION GOGGLE



PIC: 000756

DESCRIPTION

ANG-86 is an aviator's night vision imaging system that consist of binocular assembly, herein referred to as the binocular, a power (battery) pack , a carrying case, and ancillary equipment including a binocular holder (helmet mount), consisting of a mounting block ,which allows the binocular to be aligned and adjusted.

The binocular consists of two monoculars and a holder . Each monocular consists of a body (monocular housing),an objective lens assembly , an image intensifier assembly.

The holder provides each monocular to move sideways (interpupillary distances adjustments).

The power pack includes two battery compartments, each accepting two 1.5Vdc AA-size batteries in series.

The carrying /storage case provides protection for the binocular, power pack and ancillary equipment against conditions associated with carrying these items in the field environment and also damages associated with transportation , handling and storage. It shall be used for commercial practices.



ANG - 86

SPECIFICATIONS

TYPE	ANG - 86
Magnification (X):	1
Field of View :	40°
Resolution (cy/mr) :	0.76
System Gain (fl/fl):	2000
Vertical Adjustment (mm):	16
Fore and Aft Adjustment (mm):	16
Tilt Adjustment (mm):	8°
Weight (g):	550
Objective Focal Length (mm):	27
T-Number:	1.35
F-Number:	1.23
Focusing Range:	28 cm to Infinity
Eyepiece Focal Length (mm):	27
Diopter Adjustment (mm):	+2 to -6 Diopter
Interpupillary Adjustment (mm):	55 to 71
Eye Relief (mm):	25
Operating Voltage (Vdc):	2.7 to 3
Battery Type:	4× AA Batteries
Operation Temperature Range (°C):	-32 to +52
Storage Temperature Range (°C):	-35 to +71
Humidity (%):	95
Image Intensifier Tube:	18 mm, Inverter
Generation:	2nd or 3rd



MNG-87 MONOCULAR NIGHT VISION GOGGLE



PIC: 000757

DESCRIPTION

MNG-87 night vision goggle is a battle-proven technology designed for the most demanding night time applications and rigors of combat.

It is used in any special operations by elite armed forces and enables the user to perform a variety of tasks at night such as walking, driving weapon firing, patrol, map reading and administering medical aid.

It can be used on soldiers' head helmet. **MNG-87** is designed in a variety of ground based night operations.



MNG-87

FEATURES

- Light weight.
- Ergonomic design.
- Rugged construction.
- Easy operation and maintenance.
- 18mm super generation image intensifier tube to offer crystal clear super bright viewing in visible and near infrared.
- Capability of continuous passive operation over a 15-hour period without the need for battery replacement.
- Modular construction.

APPLICATION

- It can be used as hand held, head helmet mounted and weapon mounted operation.
- Nocturnal observation, target location, perimeter surveillance and reconnaissance for tank commanders, forward area observers, boarder patrols/customs and special forces.
- Command and control.
- Rapid night time artillery fire control.
- Terrain navigation.
- Selection of fighting positions.
- Search and rescue.
- Law enforcement and drug trafficking control.



NDG-87 - NIGHT DIVER GOGGLE



PIC: 000758

DESCRIPTION

Night Diver Goggle is a lightweight and high performance underwater night vision goggle permitting observation both above and below water up to a depth of 30 meters. The assembly incorporates an IR-emitting light source that provides assistance in close-up inspection of objects or checking underwater instruments. Night Diver Goggle can also be used for above water surface reconnaissance as well as underwater searching.



NDG-87

FEATURES

- Light Weight.
- Ergonomic Design.
- Rugged Construction.
- Easy Operation and Maintenance.
- 18mm second generation image intensifier tube to offer crystal clear super bright viewing in visible and near infrared.
- Capability of continuous passive operation over a 15-hour period without the need for battery replacement.
- Modular Construction.

SPECIFICATIONS

TYPE	NDG - 87
Magnification (X):	1
Field of View (in air):	40°
Resolution Power (cy / mr):	0.68
System Gain (fl / fl):	2000
Weight (kg):	<1
Objective Lens Data	
Focal Length (mm):	26
T Number:	1.30
F Number:	1.20
Focusing Range:	25 cm to Infinity
Eyepiece Lens Data	
Effective Focal Length (mm):	26
Diopter Adjustment (diopter):	- 6 to + 2
Interpupillary Adjustment (mm):	55 to 71
Power Source Type	
Operating Voltage (dvc):	2.7 to 3
Battery Type:	2 x AA
Ambient Characteristics	
Temperature Range (operation) (°c):	-51 to +45
Temperature Range (storage) (°c):	-51 to +71
Maximum Humidity (%):	94
Operational Depth (meters):	30



THERMAL CAMERAS



PIC: 000759

DESCRIPTION

» TBIR-240 THERMAL CAMERA

TBIR-240 is a medium weight thermal camera equipped with FPA cooled detector which may be used for surveillance purposes and detection of targets at day and night. It may be used on vehicles or airplanes.



TBIR-240

» TBIR-800 LONG RANGE THERMAL CAMERA

TBIR-800 long range thermal camera with high sensitivity, low noise and equipped with image processing system and automatic tracing is used for surveillance and detection of targets at day/night even in obscure weather conditions.

It may be installed on vehicles, vessels, airplanes and utilized for tracking military targets.

PIC: 000760



TBIR-800

SPECIFICATIONS

TYPE	TBIR-240	TBIR-800
Image Performance		
Detector Material:	HgCdTe	HgCdTe (MCT)
Pixel:	320 x 256	320 x 256
Pixel Pitch (u m):	30	30
Spectrum Range (u m):	3.7 - 4.8	3.7 - 4.8
Cooled Method:	Stirling cooler	Stirling Cooler
Cooling Time (minutes):	6 to 7	6 to 7
NETD:	<15 mK @ 25° C	<15 mK @ 25° C
Lens:	240/60 MM / F# / 2	800/ 200/ 70 F#/ 4
FOV:	2.3° x 1.8° (NFOV), 9.2° x 7.3° (WFOV)	0.69° x 0.55° (NFOV) 2.76° x 2.2° (WFOV), 7.8° x 6.3° (WFOV)
Power Consumption		
Power for Cooling (w):	<30	<30
Interface		
Remote Control:	RS 422 & RS 232	RS 422 (Keyboard), RS 232 (Tracker)
Environmental		
Operating Temperature (°C):	-40 ~ +60	-40 ~ +60
Storage Temperature (°C):	-40 ~ +60	-40 ~ +60
Physical Characteristics		
Size (cm):	41 x 15 x 17	90 x 25 x 28
Weight (kg):	9.5	32



SADAD 103 MONITORING SYSTEM



PIC: 000761

DESCRIPTION

The **SADAD 103** is an ideal system for long range surveillance, air and sea related operation.

The system is designed in a modular construction pattern for conversion and adapting system performance to mission parameters.

A variety of sensors and control equipments have been integrated, including high speed payload, digital infrared and visible cameras with variable focal length lenses and a laser range finder.



FEATURES

- Easy integration
- Easy installation
- Advanced image processing algorithm
- Checkpoint surveillance
- Multi sensor control via networking
- Modular design to accommodate to appropriate sensors
- Range measurement-optional

APPLICATIONS

- Border and coastal surveillance
- Critical infrastructure protection
- Airport security and surveillance
- Capability to monitor for suspicious activities



SADAD 103

IR CAMERA SPECIFICATIONS

Detector:	HgCdTe Cooled FPA , 3 – 5 um, 320*256 Pixels
Fields of View:	9.2 * 7.3 (Wide), 2.3° × 1.8° (Narrow) deg
Brightness/Gain Adjustment:	Automatic, Manual
Non-Uniformity Correction (NUC):	1 Point ,2 Point, External, Internal
NETD:	≤13mK @ 30°C
Storage Temperature (°C):	-45 to +65
Operation Temperature (°C):	-40 to +50
Start-up Time (min):	≤7
Power Consumption (W):	14
Water Resistance:	IP67
Communication Port:	Standard RS422 / 232



ELECTRO OPTICAL SURVEILLANCE SYSTEM



PIC: 000762

DESCRIPTION

Electro-Optical Surveillance Systems (EOSS series) are military automated and computer controlled long range electro optical systems designed for day and night border surveillance and monitoring in all whether condition.

New electro-optical imaging and tracking system includes: director, thermal camera and TV camera, monitor and control unit.

A variety of models are available for civil and military applications, could be installed on mobile and fixed bases.

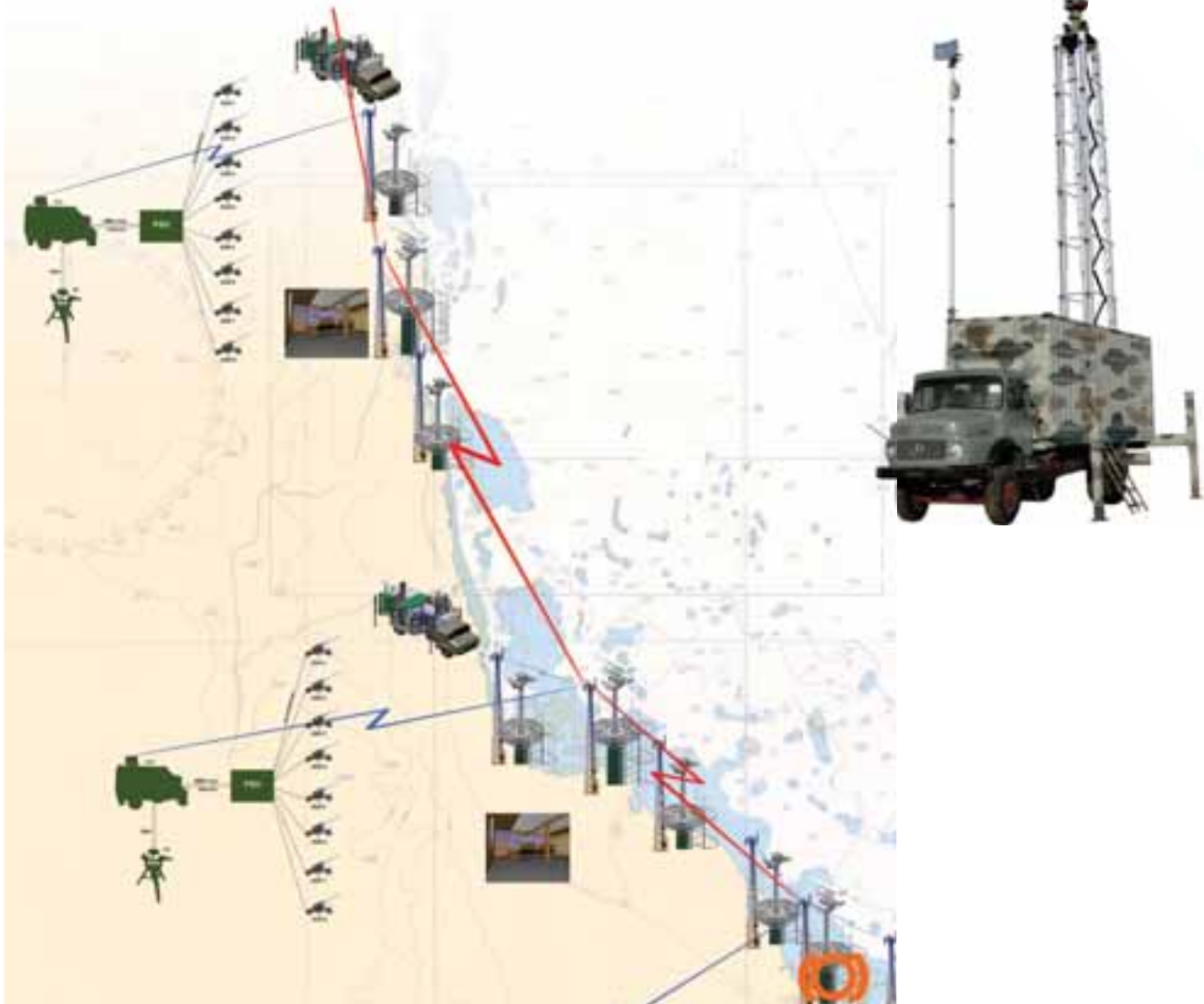
MAIN FEATURES:

High resolution TV camera equipped with motorized zoom lens having variable focal lens, High performance cooled thermal imager, Remotely controlable dual axis Pan / Tilt platform which is mounted on stable mast/tripod.

All the functions of Electro-Optical Surveillance Systems are controlled from a compact control unit.



EOSS





OGHAB 11



PIC: 000754

DESCRIPTION

Aerial video and photography cameras have a wide range of defense and military applications, used on aerial vehicles including UAVs, helicopters and airborne tactical, enhance day/night surveillance capability.



OGHAB 11

SPECIFICATIONS

TYPE	OGHAB 11
TV Camera:	
Detector Type,Band,Sensitivity:	Color 1/4", SUPER HAD CCD
Field of View- Optical (deg):	1.6 – 42.2
Optical Zoom Ratio:	1 : 26
Digital Zoom Ratio:	12 X (312 X With Optical Zoom)
Focus, Iris, Shutter, Exposure:	Manual, Auto
Gain:	Auto / Manual (-3 to 28dB, 16 steps)
Video out:	PAL
Tracker:	
Output Video:	PAL
Resolution:	720*576
Included Algorithm:	Correlation, Edge Based Centroid
Protocol:	RS232
Platform:	
Power:	50w (peak), 10w(continuous) 24V (15-32V)
LOS Stabilization (mrad):	1
Field Of Regard:	N × 360° AZ, -85 to +10 EL
Slewing Rate:	60 deg/sec max
Weight (Kg):	1
Dimension (mm):	D = 120, L = 150





EOVP 4 - AERIAL CAMERA



PIC: 000717

DESCRIPTION

EOVP 4 is a new electro-optical multi-sensor gyro stabilized payload for aircrafts, helicopters, UAVs and naval vessels providing reconnaissance and surveillance missions over land and over sea.

The **EOVP 4** is equipped with infrared and TV cameras also a laser rangefinder (Optional). The system is designed in a modular construction for conversion and adapting system performance to mission parameters.

The system provides azimuth and elevation positioning to the EO sensors allocated inside, in order to provide video in the visible and infrared spectrum, automatic tracking of fixed and mobile targets and also target range measurement.

FEATURES

- Day and night surveillance
- Modular payload
- All weather, long range identification
- Automatic target tracking capability
- Stabilized two axes - Azimuth and Elevation
- Joystick controllable-Optional



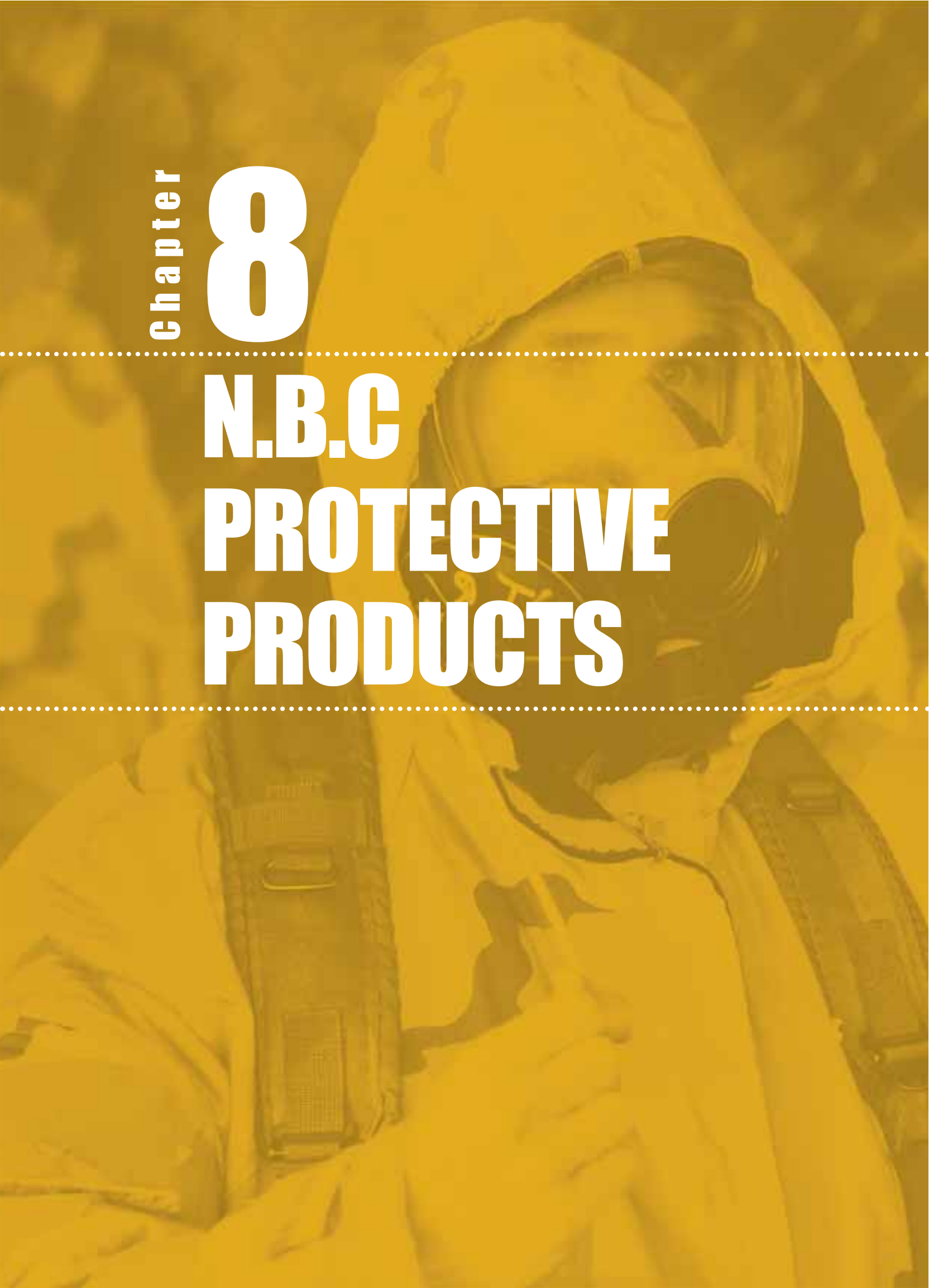
EOVP 4

APPLICATIONS

- Search and rescue
- Border and coastal surveillance
- Protection of ground troops
- Police operations

SPECIFICATIONS

TYPE	EOVP 4
IR Camera	
Detector:	Cooled FPA, 3.7 - 4.8 μm , 320 * 256 pixels
Fields of View:	9.2 * 7.3 (Wide), 4.6 * 3.7 (Narrow) deg
AGC:	Automatic, Manual
Non-Uniformity Correction (UNC):	1 point, 2 point, External, Internal
Frame Frequency (Hz):	50
NETD:	$\leq 13 \text{ mK @ } 30^{\circ}\text{C}$
Start-up Time (min):	≤ 7
TV Camera	
	SONY FCB-980SP
Detector type, Band, Sensivity:	Color 1/4", SUPER HAD CCD
Field of View - Optical (deg):	1.6 - 42.2
Optical Zoom Ratio:	1 : 26
Digital Zoom Ratio:	12 X (312 X with optical zoom)
Focus, Iris, Shutter, Exposure Gain:	Manual, Auto
Gain:	Auto / Manual (-3 to 28dB, 16 steps)
Tracker	
Output Video:	PAL
Resolution:	720 * 576
Included Algoritm:	Correlation Edge Based, Centroid
Protocol:	RS232
Power (v):	12 - 32



chapter

8

N.B.C PROTECTIVE PRODUCTS



PREVIEW

As a result of years of constant chemical bombardments during the Iraqi imposed war, Iran is definitively the biggest victim of chemical attacks **since World War I**. Hence, extensive research and development in the field of NBC defense products were carried out to minimize the effects of such attacks.

The unfortunate first hand experience with chemical agents and its victims provided a suitable platform to manufacture a complete array of **NBC defense products** ranging from sophisticated electronic detectors, permeable NBC suits, practical and affordable masks and canisters, medium size decontamination containers and vehicles to self aid kits containing most common chemical antidotes and skin decontamination powders.

In keeping with world class producers, our research laboratories are actively engaged in developing lighter, faster acting, more precise and affordable NBC supplies. Our manufacturers strictly adhere to international military standards and have obtained the necessary standards from well-known institutions.

Now, with more than 25 years of quality production reputation, our **NBC products** have reached the export stage and are already becoming the institutional NBC ware of a number of countries.

With respect to achievement of quality & environmental international certifications such as **ISO 9001, ISO 9002, ISO-QS134852003** and **Military Standard** which have key role for entering the worldwide competition, we are proud to have put up the company's image, reliability and respect for committing simultaneously the mutual obligations in companion with customers in internal & external markets.

These products and services include:

1- Sales of technical know-how of several high technology products based on the qualified standards and international rules & regulations.

2- Design, installation and maintenance of a number of production lines aimed at producing industrial products such as: blood bags, microbial hoods and providing technical manpower to design and execute clean rooms of all Sterility classes.

3- Presents various levels of training courses in the area of NBC protective products and defensive equipment and materials.





CHAPTER 8

ITEM	TABLE OF CONTENT	PAGE
	DETECTION DEVICES	
1	IRANCAM - WAR CHEMICAL FACTOR DETECTOR	238
2	DIGITAL DOSIMETER (302 NDRAD)	239
3	DETECTOR KITE (NDK-1)	239
	PROTECTION EQUIPMENT	
4	NBC PROTECTIVE SUIT	240
5	MASK (IM2)	241
6	PROTECTIVE HOOD OF MASK	241
7	RESPIRATORY SYSTEM	242
8	IF2 FILTER	242
9	NBC PROTECTIVE GLOVES	243
10	NBC BOOT COVER	243
	DECONTAMINATION	
11	DRAKSH 7	243
12	DSM 2500, 3500 & 5000 - NBC DECONTAMINATION SYSTEMS	244
13	D3 CAPSULE	245
14	OBIDOXIM CHLORIDE	245
15	MRC-200P DECONTAMINATION	246
16	MRC-500N RINSER SOLUTION	246
	TREATMENT	
17	N.B.C SELF AID KIT	247
18	ATROMIL	247
19	OXOMIL	247
20	NITROMIL	247
21	PROMIL	248
22	DECOMIL	248
23	DECODERM	248
24	MOBILE LABORATORY SYSTEM	249
25	MOBILE MEDICAL SERVICE SYSTEM	249



8

IRANCAM WAR CHEMICAL FACTORS DETECTOR

PIC: 000151

N.B.C PROTECTIVE PRODUCTS



SPECIFICATIONS

TYPE	IRANCAM
Detecting Method:	IMS
Function:	Automatic
Identifiable Factors:	Nerves, Vesiculation, Blood and Drowning
The Least Extent of Recognition:	About ppb
Detecting Time (s):	Less than 5 seconds
Dimensions (cm):	40 × 14 × 19
Weight (kg):	5.7
Outer Supply:	220 - 240 V / 12-16 VDC
Battery:	Rechargeable Li-ion
Work Temperature Range (°C):	-20 to +50
Ionization Source:	63 Ni – 15 mci



8

DIGITAL DOSIMETER (302 NDRAD)

PIC: 000152

FEATURES

- Detecting X and gamma rays with higher energy than 60 keV
- Measuring field intensity
- Saving and displaying total received dose
- Saving and displaying function time
- Equipped with outer detector, installed on one meter antenna, water and dust proof keyboard
- Capability of regulating warning levels

SPECIFICATIONS

TYPE	302 NDRAD
Maximum Responding Field (mSv/h):	999
Main Machinery Dimensions (cm):	16 × 9 × 3.5
Supply:	Chargeable Battery 7.2 V
Continuous Function Duration (hours):	20



N.B.C PROTECTIVE PRODUCTS

DETECTOR KITE (NDK-1)

PIC: 000178

SPECIFICATIONS

TYPE	NDK-1
Detecting Method:	Colorimetry
Function:	Non – Automatic
Identifiable Factors:	GA, GB, HD, AC, CK, CG
The Least Extent of Recognition (mg/lit):	HD : 0.003 CG : 0.004 AC, CK : 0.005 GA, GB : 0.0001
Detecting Time (seconds):	Less than 30 seconds
Dimensions (cm):	17 × 13 × 11
Weight (kg):	1
Work Temperature Range (°C):	-20 to 55
Determination Concentration Levels:	Three hazardous levels, very hazardous, fatal





NBC PROTECTIVE SUIT

PIC: 000153

DESCRIPTION

The **NBC protective suit** provides whole body full protection against chemical, biological and nuclear dusts affecting the skin in the form of vapor, gas and aerosol, the suit is permeable and allows body heat and perspiration vapor transmitted out of garment while prevents the entrance of toxic agents by enhanced filter.

The NBC protective suit consists of two main layers.

The outer shell is oil and water repellent durable **poly/cotton** or **PA/cotton** blend fabric.

The filter layer is activated carbon impregnated **PU foam** or activated **carbon fabric**.



FEATURES

- Comfort
- Easy movement
- Machine washable
- Flame retardant (optional)
- Anti IR camouflage (optional)
- Different sizes including S,M,L,XL & XXL
- Up to 10 years guarantee in vacuum packed
- According to Iranian Defense Standard **IDS 034**
- Different model including 2-pieces jacket-trousers suit with/without open closure zipper & coverall
- Different colors & camouflage patterns





MASK (IM2)

PIC: 000154

DESCRIPTION

The protector mask is a sure device for full protection of respiratory system, eyes and face against the NBC military elements. This mask is equipped in different systems to be comfortable for the user.

The design of this mask is in the method that not only the user does not have any kind of difficulty in breathing but also he/she can breathe very easily.



SPECIFICATIONS

TYPE	IM2
The Time of Putting the Mask (s):	Less than 10
The Ability of a Part of the Mask in Passing the Liquids (liquid opening part of the mask) (cc/min):	150
The Ability of a Part of the Mask in Passing the Air (air opening part of the mask):	2/2 inch 2 lit air/min
Pressure Comedown of Inhalation Valve:	4 mm H ₂ O
Pressure Comedown of the Exhalation Valve:	7 mm H ₂ O
The Weight of the Mask (gr):	495

PROTECTIVE HOOD OF MASK

PIC: 000155

DESCRIPTION

Protective hood, keeping head and neck away from aerosols, chemical vapors, biologic particles and nuclear dust, is utilized as a complementary for IM2 mask.

Protective hood can improve mask efficiency, meanwhile it makes more security area and minimizes the contaminations from mask surface in contact with aforesaid agents, consequently the decontamination procedure would become easier.





8

RESPIRATORY SYSTEM

PIC: 000156

DESCRIPTION

The respiratory system is designed and used for the protection of the human respiratory system against poisonous gases and when the oxygen in the air is less than general (**less than 17%**).

This system has great applications in fire fighting and petrochemical industries as well.



SPECIFICATIONS

ACTIVITY TIME IN TERMS OF ACTIVITY TYPE			THE AIR VOLUME IN CYLINDER	CHARGE PRESSURE	TYPE OF CYLINDERS	TYPE OF THE RESPIRATORY SYSTEMS
100 LITERS	40 LITERS	20 LITERS				
12 MIN	30 MIN	60 MIN	1200 Liters	200	6 Liters	Single Cylinder
18 MIN	45 MIN	90 MIN	1800 Liters	200	9 Liters	Single Cylinder
12 MIN	30 MIN	60 MIN	1200 Liters	300	4 Liters	Single Cylinder
18 MIN	45 MIN	90 MIN	1800 Liters	300	6 Liters	Single Cylinder
16 MIN	40 MIN	80 MIN	1600 Liters	200	2*4 Liters	Two Cylinder

IF2 FILTER

PIC: 000157

DESCRIPTION

The filter, holder the model **IF2**, is a complex filter and designed in accordance with all military standards.

This filter with the **mask IM2** organizes a full respiratory protection system against chemical, biological elements and radioactive floating dust.



SPECIFICATIONS

TYPE	IF2 FILTER
Weight (gr):	220 ± %5
Dimensions (mm):	110 × 60
Pressure Comedown:	Max. 18 mm H2O
Paper Filter:	% 99/99
Crust Color:	Black – Olive Green
This Filter is Useless in Less than 18% Oxygen	





NBC PROTECTIVE GLOVES

PIC: 000158

DESCRIPTION

These gloves, protecting hands against toxic warfare agents, are part of personal gear. Gloves would be made of polymerized latex (**Chloroprene**) and could resist remarkably against breakthrough of aforementioned agents.

Pair of cotton gloves, avoiding hands perspiration and skin irritation.

This prepared improve the efficiency of protective gloves. Cotton gloves should be worn before wearing protective ones.

Weightless, optimal safety factor, incredible flexibility and compatibility with user body are protective gloves advantages.



N.B.C PROTECTIVE PRODUCTS

NBC BOOT COVER

PIC: 000159

DESCRIPTION

Boot cover, keeping feet away from **NBC** agents. It would be made of elastic polymeric materials which have a good abrasive resistance.

This cover is used as a complementary of military boot and should be worn quickly after boot wearing. The protective cover, casting as an integrated form without crack or split, is made of **PVC** and **NBR** elastomer compounds.



DRAKSH 7

PIC: 000160

DESCRIPTION

» TACTICAL DECONTAMINATION SYSTEM

Advanced system of protective operations against NBC attack, capable of being equipped with meteorological supplies, various detectors, operation signals, portable decontamination ability, warning items and equipment, board and warning cordon, 6kgs. Decontamination cylinders, device for taking sample, instrument for determining the radius of polluted areas, installing the place gun and wireless. An engine coupled with floral pump provides flow and necessary pressure for spraying high range of decontamination solution like calcium hypochlorite. The system with suitable dimension has been mounted on chassis and the function of decontaminating during the system's movement when the engine of vehicle is turned off is also possible.





DSM 2500, 3500 & 5000 NBC DECONTAMINATION SYSTEMS

PIC: 000161

DESCRIPTION

High pressure decontaminating systems model 2500 , 3500 and 5000 designed & built based on the latest standards capability of design & installation on different semi – heavy vehicles and differences model have only the different capacities.

DSM – 3500 is one of them, equipped with fire – fighting system feeding pump & water and blasting equipment, Capable of being installed with (detectors & warning equipment) suitable for rapid decontaminating of people, items, equipment & machinery, places & installation, cloths, individual equipment, fuel tanks & tankers.



SPECIFICATIONS

TYPE	DSM 3500
Dimension of the Equipment Cabin (m):	4.2 × 2.30 × 1.8
Total Dimension of the Vehicle (m):	5.4 × 2.30 × 3.10
Weight of Equipments and Cabin (tone):	6
The Operation Pressure (bar):	10 (Max)
Flow of High Pressure System (lit/hour):	2000
Monitoring System for Decontaminating of the Place With High Flow:	25 - 30 in the Spraying Range
The Temperature of Hot Water (°C):	80 (max)
Dried Steam Output:	30 bar-270 Lit/Hour 150° C, 20 bar-300 Lit/Hour 210°C
The Mixing Ratio of the Chemical	Up to 8% Powder or Liquid
Decontaminating Material:	12.5 Percent Powder
Fuel:	8-12 Diesel / Lit / Hour
Engine (tow units):	3000 rpm/4 Cycle/2 Cylinder/17 Horse Power/220 V/500 W
The Possibility of Electrical Output:	220 V/500 W / AC-12V/5W/DC
Water Tank (lit):	3500 (Galvanized)



8

D3 CAPSULE

PIC: 000162

DESCRIPTION

D3 NBC decontamination device is intended to be used for decontamination of military vehicles, related personals or weapons.

DS2 liquid is used as the decontamination solution for this device.



SPECIFICATIONS

TYPE	D3 CAPSULE
Capacity (lit):	1.26
Weight (filled) (kg):	3
Effective Spray Range (m):	1.8-2.4
Duration of One Charge (sec):	30

N.B.C PROTECTIVE PRODUCTS

OBIDOXIM CHLORIDE

PIC: 000165

DESCRIPTION

OBIDOXIM CHLORIDE is a combination of two strong oxims that is used as a revival of Colin Stras in a cure of initial poisoning, patients of nerve chemical components, and patients of Phosphorus Organo.





MRC-200P DECONTAMINATION

PIC: 000163

DESCRIPTION

MRC-200P decontamination powder is used to clean all chemical and biological agents and destroys them.

In order to remove pollution from military instruments and equipment surfaces, aircrafts and vessels' surface, buildings and forces disposition, this decontamination powder is used (Do not use for skin cleaning).

CONSUMPTION

Each package should be mixed with 140-180 liter of water by DSM System or non-mechanized manner and spread on the contaminated surfaces.



MRC-500N RINSER SOLUTION

PIC: 000164

DESCRIPTION

MRC-500N Rinser Solution is used to remove radioactive materials pollution.

CONSUMPTION

Washing surfaces with rinser solution without dissolving in water or other solvents that shall be spread directly on the contaminated surface by spreader machines like DMS 2000 for at least 30 minutes, then wash the surfaces with plenty of water.



APPLICATION TERMS

Desirable conditions for removal of nuclear pollution by the rinser is temperature of 20°-50° C and PH=4.



N.B.C SELF AID KIT

PIC: 000166

DESCRIPTION

This compact **Self Aid Kit** contains all needed medical antidotes and prophylactic creams to protect soldiers in chemical attack situations.

Each kit contains 2 Atromil and 10 Oxomil autoinjector, 4 Nitromil ampulse, 1 Promil cream, 1 Decoderm and Decomil Powder.



ATROMIL

PIC: 000167

DESCRIPTION

An autoinjector that provides fast and reliable delivery of 2mg of atropinesulfate in cases of nerve agent attacks. Atromil formulation conforms with the specifications of **U.S.P.N.F XXLL** and **B.P. 1998**. Its mechanism of action bears several advantages compared to some of the conventional autoinjectors. Two atomic autoinjectors are available in 1 Self Aid Kits. Atromil also comes in bulk boxes of 100 units. Blue colored Atromil devices are available for training purposes.



OXOMIL

PIC: 000168

DESCRIPTION

Oxomils are known antidotes of nerve agents. Their rapid injection along with atropine sulfate, in chemical attacks, will prevent undesirable symptoms and help to neutralize their side effects to a large degree.

Obidoxime chloride has been synthesized and formulated in 220mg/0.8ml autoinjectors under the trademark of Oxomil. The blue color Oxomil autoinjectors do not contain the injection cartridge and are designed for training purposes.



NITROMIL

PIC: 000169

DESCRIPTION

Iso – Amyl Nitrite ampoules are used to combat blood poisonings with cyanide compounds. Four 0.3 ml ampoules wrapped in woven absorbent coverings are placed in Self Aid Kit. Also, boxes of 250 ampoules are packed separately. When it is needed, the ampoules should be broken between fingers and the volatile contents inhaled. Since the fabric pouch retains Nitromil contents for at least 5-10 minutes, repeated ampoules of Iso-Amyl Nitrite may be utilized within fifteen minutes of each other if poison blood gases or their side effects persist.





8

PROMIL

PIC: 000170

DESCRIPTION

Promil is an **NBC** protection cream designed to protect the skin from blister agents and at the same time, to serve as a barrier to both blister and nerve agents.

Promil also contains a chemical agent neutralizing compound that will interact with chemical agents on the molecular level and detoxify them.

Promil should be applied to unprotected areas of the skin such as the wrists, ankles and neck, in a thick and visible layer. The cream must be wiped off immediately after exiting the contaminated area.

Promil is available in 15g aluminium and 30g polyethylene tubes.



DECOMIL

PIC: 000171

DESCRIPTION

Decomil powder bag is included in Self Aid Kits to decontaminated personal belongings such as clothing and weapons.

Decomil also acts as a detector and turns red in the presence of chemical agents.

Decomil powder comes in a porous bag and is packed in an air and light proof cover to guarantee the stability of its active ingredients.



DECODERM

PIC: 000172

DESCRIPTION

Skin decontaminated powder (**Decoderm**) is filled in mitt shaped double layer woven covering.

The mitts are placed in a water and light proof plastic packaging to ameliorate their effective shelf life.

Decoderm comes with a small towel to first absorb liquid droplets of chemical agents and then placing the fingers inside the mitt, patting motion on suspected areas of the skin should continue for at least 30 seconds.





MOBILE LABORATORY SYSTEM

PIC: 000173

DESCRIPTION

This system is a **strike vehicle**, which is equipped by laboratory equipments to identify and diagnosis nuclear, biological and chemical factors in crisis and when laboratory centers accessibility is not possible.

FEATURES

- Carrying capability of the laboratory by tugger.
- Sampling and diagnosis of NBC factors in site
- Servicing civilian section
- Being a dual-Purposed laboratory during war and peace
- Increasing defending capability of armed forces against NBC attacks
- Low costs in comparison with its foreign counterpart



N.B.C PROTECTIVE PRODUCTS

MOBILE MEDICAL SERVICE SYSTEM

PIC: 000174

DESCRIPTION

This device is transferable and is capable of carrying the injured and medical equipments.

- Having illumination and environmental protecting tent
- Having illumination system and electrical tableaux
- Having conducting equipments
- Having brake system
- Transferable with appropriate thill trackers
- Having holding jacks
- Very light
- Operative in different climate conditions
- Equipped with suspension system



Chapter 9

AVIATION





AVIATION INDUSTRIES ORGANIZATION (A.I.O)

AIO is a sub-division of Ministry of Defence and Armed Forces support of the Islamic Republic of Iran.

AIO is not only the major and the top organization in Aviation Industries, the policy center and decision making in the field of aviation, but also is the strategic, and the midterm planner for buying and selling both aircraft rotary and fixed wings, and aerial engines as well.

It is the coordinator to manufacture, overhaul, repair, and support of various types of airplanes, helicopters, and their related engines mainly conducted by its subsidiaries.

One of its main tasks are supplying the spare parts for the aircraft in addition to obtaining the different kinds of technologies required for manufacturing various types of parts, tools and assemblies for the Islamic Republic of Iran Armed Forces and other civilian or governmental sections.

Research Center of Aviation Industries Organization with more than 20 years of experience in research & exploration in different aircraft design, manufacture & tests turns to an acquirer organization which has provided considerable services during its activity history for aviation industry of the country, relying on its specialized & liable human resources & its laboratorial infra-structures In different field of designing and testing.

These important missions and different tasks are conducted and achieved through its 6 affiliated industries which are:

- Iran Aircraft Manufacturing Industries (**HESA**)
- Iran Aircraft Industries. (**I.A.C.I**)
- Fajr Aviation & Composite Industries (**FACI**)
- Iran Helicopter Support & Renewal Company (**IHSRC**)
- Qods Aviation Industries (**Q.A.I**)
- Research Center





CHAPTER 9



ITEM	TABLE OF CONTENT	PAGE
1	HESA IRAN AIRCRAFT MANUFACTURING INDUSTRIES	254
2	IRAN-140 AIRCRAFT	255
3	FAJR AVIATION & COMPOSITE INDUSTRIES	257
4	FAJR 3 COMPOSITE AIRCRAFT	258
5	FAJR 20 - ULTRA LIGHT AIRCRAFT	259
6	FAEZ AIRCRAFT	259
7	IACI - AIRCRAFT INDUSTRIES	260
8	IHSRC - IRAN HELICOPTER SUPPORT & RENEWAL COMPANY	263
9	SHAHED 278 HELICOPTER	266
10	QODS AVIATION INDUSTRIES	267
11	SAEGHE 2 - AERIAL TARGET SYSTEM	268
12	MOHAJER 2 - UAV SURVEILLANCE SYSTEM	269
13	ABABIL 2 - UAV SYSTEM	270
14	ABABIL 3 - UAV SYSTEM	270
15	PARACHUTE	271
16	AIRCRAFT WINDSHIELD	272
17	AIRCRAFT TIRES	273
18	JATO MK-34	274
19	FLIGHT SIMULATOR	275



HESA - IRAN AIRCRAFT MANUFACTURING INDUSTRIES

9

DESCRIPTION

Iran Aircraft Manufacturing Industries “HESA” of Isfahan, located in a vast land from which about 250,000 square meters are allocated for design & engineering centers , manufacturing & assembly workshops, laboratories and hangars . Having experienced personnel highly qualified technician, experts, well educated engineers; holding **B.S, M.S, PHD.** degrees, has granted this company an outstanding rank among the aircraft and part manufacturers and designers.

In addition to various and magnificent activities that was mentioned in the field of design, engineering (A/C or R.PV) and manufacturing, reverse engineering, part fabrications, the main activity is production of **IRAN140**, a 52 seat regional airplane, which is made through a joint venture project with Ukraine.

HESA, founded in 1964,now is responsible for the designing and manufacturing of the first Iranian made commercial plane. In an international tender , in 1978 it was handed over to an American company (Bell- Textron), and after the Islamic Revolution of 1979, the company was only % 11 completed.

No partial measure was taken to complete the plant up until 1985. When construction work got off the ground again in 1986, the plant manufactured its first **UAV (ABABIL)**. Construction of the company came to an end in 1995 and the plant was fully ready for manufacturing **IRAN-140** civil aircraft.





IRAN-140 AIRCRAFT

9

PIC: 000914

DESCRIPTION

The regional 52-seat **IRAN-140** (AN-140) high wing aircraft is one of the latest aviation technology products at the present time.

Low operating cost, low fuel consumption, with the fatigue life of 50,000 flights of 1 hour, 25 years calendar life, are some features of this multipurpose aircraft.

This aircraft is equipped with 2 turboprop engines at 2500 horse power each.

Taking advantage of the latest utility technology, it is designed according to **FAR-25**, **JAR-25** and **AP-25** which

have made this aircraft a suitable one for domestic and international carriers.

Luxury VIP aircraft, spacious interior and baggage compartment, full **IFR** operating, internal and external low noise level, independence from ground equipments, capable of landing in airports with the least landing facilities.

However, the aircraft initially was intended for passenger, but could be designed for cargo, releasing parachutes or military items, radar patrol and guidance.



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52 PASSENGER LAYOUT





IRAN-140 AIRCRAFT

9

Iran Aviation Industries also manufacture engine **TV3-117VMA-SBM1** as **IRAN-140** power plant with following specifications. This engine is capable to be used as an industrial power plant.

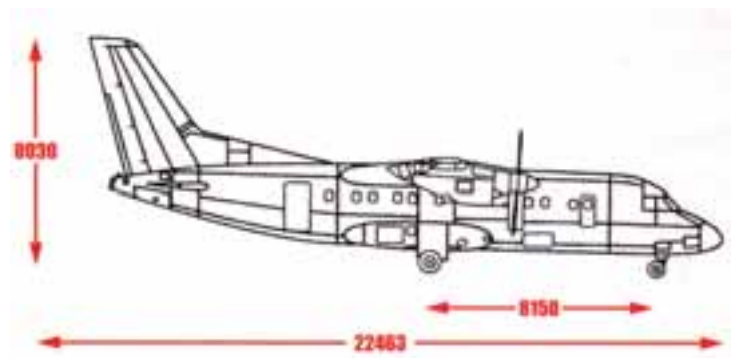


SPECIFICATIONS

TYPE	IRAN-140
Max. Take Off (kg):	21,500
Max. Landing (kg):	21,000
Empty Weight (kg):	19,200
Max. Pay Load (kg):	6,000
Max. Fuel Tanks (kg):	4,370

IRAN 140 VERSATILE PERFORMANCE

- High payload cargo transport
- Comfortable 52-seat regional airliner
- High endurance flying sensor platform
- Spacious VIP aircraft
- Versatile military workhorse
- 2100 km range with full pax and reserves
- Operation out of unprepared airfields possible
- Among the highest speed in pressurized class
- Superior fuel performance
- Excellent maneuverability



Electronic Warfare



Maritime Patrol



Staff Transport



Radar Patrol & Guidance



Troop & logistics Transport



Crew training



DESCRIPTION

FACI is established to design and develop the composite technology in Iran Aviation Industries, and is now the most important center in composite activities, which has several certifications due to Joint Aviation Regulations from International Civil Aviation Organization (ICAO).

These industries are located in an area of 30,000 square meters, consisting of design offices, manufacturing hangers, workshops and administration offices. A group of experienced engineers and technicians are involved in design, manufacturing, quality control and testing.

ENGINEERING CAPABILITIES:

1.DESIGN : **FACI** has taken design organization approval (D.O.A) according to **JAR.21** to design light aircraft. Design departments provide engineering qualification in design and testing of any composite products.

High technology including **CAD.CAM** techniques are employed to explore design alternatives and supply

optimum production configuration.

2.MANUFACTURING: **FACI** has taken production certification (P.C) according to **JAR.21** for producing **F.3** aircraft. **FACI's** manufacturing capabilities feature: the latest technologies and techniques with established reputations in the composite industries. **FACI** ideal position to manufacture any type of composite products, no matter what size the production will be.

Repair of composite structures can range from easy cosmetic repairs through almost complete reconstruction of damaged vital components, and also manufacturing the composite masters, molds and fixtures are done on **FACI**.

3.QUALITY ASSURANCE: Quality department of **FACI** with a good number of qualified engineers and quality control experts guarantee the quality of **FACI** products. In this field standard department of **FACI** with a highly qualified engineers are familiar with latest aviation regulations such as **ISO, JAR** and **FAR**.





FAJR 3 COMPOSITE AIRCRAFT

9

PIC: 000902

DESCRIPTION

FAJR 3 can be used for the following purposes: pilot training, light transport, roads traffic & boarder controls, coastal patrol, VIP transfer, postal purpose and aerial photography.

Low stalling speed, good flying qualities, good handling characteristics and maneuverable and stable, insensibility to side winds during take off and landing, double stick and foot operated flying controls, double braking system and an additional land operated parking brakes, easy and low maintenance and repair, are some of the specification of F3 aircraft.



SPECIFICATIONS

TYPE	FAJR 3
Length (m);	8.1
Height (m):	3.1
Cabin Width (m):	1.3
Wing Area (m ²):	14
MAC (m):	1.4
Wing Span (m):	10.5
Empty Weight (kg):	1,100
Max.Weight (kg):	1,580
Fuel Capacity (liter):	280
Max.Cruise Speed (knot):	165
Cruise Speed at %75 Power (kts):	148
Absolute Ceiling (ft):	19,000
Rate of Climb (ft/min):	920 - 1030
Range (km):	1,300
Endurance (hrs):	5.5
Stall Speed (knot):	56



FAJR 20 - ULTRA LIGHT AIRCRAFT

9

PIC: 000908

SPECIFICATIONS

TYPE	FAJR 20
Length (m):	7.1
Height (m):	2.3
Cockpit Height (m):	1.2
Cockpit Width (m):	1.1
Wing Span (m):	8.6
Wing Area (m2):	11.6
Empty Weight (kg):	350
Max. Weight (kg):	650
Fuel Capacity (lit):	130
Max. Useful Load (kg):	300
Max. Speed (kn/h):	259
Cruising Speed at 75% Power (km/h):	231.5
Service Ceiling (m):	3657
Rate of Climb (ft/min.):	800
Take off Distance (m):	183
Landing Distance (m):	305



FAEZ AIRCRAFT

PIC: 000921

SPECIFICATIONS

TYPE	FAEZ
Wing Span (m):	4.9
Wing Chord (m):	0.63
Cabin Width (m):	0.58
Cabin Height (m):	0.49
Overall Length (m):	3.9
Height (m):	1.13
Max. Width (m):	0.61
Wheel Track (m):	1.1
Wheel Base (m):	1.15
Wing Area (m2):	3.1
Empty Weight (kg):	78
Max. Fuel Weight (kg):	35
Max. Take-off Weight (kg):	190
Stall Speed (km/h):	72
Max. Cruise Speed (km/h):	180
Max. Level Speed (km/h):	260





PIC: 000901

DESCRIPTION

Iran AirCRAFT Industries (**IACI**) as one of the largest industry which is affiliated to Defence Ministry of Iran was established in 1970, to overhaul and perform technical modification (**T.C.T.O**) in addition to the promotion of capabilities for manufacturing aircraft and power plants in its coming years. **IACI** is reputable for gaining valuable technical experience in the field of aviation and science technologies, which is concluded with an unique industrial position both in Iran and abroad. **IACI** with nearly 30 years of excellent experience in aviation industry, conducting more than fifty million work hours in overhauling of various level of civil and military flying objects.



POSITION, FACILITIES AND EQUIPMENT

- 3.5millions total sq. meters in the aircraft sector, 34 thousand square meters shop area for manufacturing turbine engines and 2500 specialized personnel in power plant overhaul sectors and assembly lines.
- The overhaul of more than four types of aircraft: fighter, passenger, tanker, cargo and also conducting heavy and exclusive service bulletins.
- The capability to repair, overhaul, modify, and to perform **T.C.T.O** of gas turbines and turbine engines according to the latest standards.
- Conducting all types of repairing according to the latest technical bulletins considering the highest international standards.
- Employing the most exceptional engine test cells in the middle east, being able to test jet engines with ratings up to 100,000 pounds thrust, in two separate part: shaft cell and jet cell.
- Part manufacturing with making use of data process as well as modern facilities in the form of ordering in large scales.
- Metal spraying with the use of modern systems (plasma, thermospray, dimonjet), modern welding, electro beam welding (**E/B/M**), vacuum brazing (**V/B**), and diffusion coating.
- Excellent past records and valuable experience in repairing and modification of damaged aircraft, even in unfacilitated conditions.
- Precision instrumentation, metallurgical and chemical laboratories for analyzing various structures under static and dynamic loads.
- Structural strengths analysis as a metal fatigue point of view, aerodynamic fatigue and fluid mechanics.
- Non-destructive test: penetrative-magnetic, focco, beam and ultrasonic.
- Electronic and mechanical precision instrument calibration.
- Corrosion removal and complete strip of coatings as well as plating and repainting all types of aircraft.





DISTINCTIVE PRIORITIES

- Fully equipped and well appointed plants for the overhaul of various aircraft and aviation industrial power plants.
- Modern libraries with up-to-date software.
- **IACI** is the only possessor of **ISO 9002** certificate in aviation industries in Iran.
- Sufficient technical knowledge concerning twenty million existing aviation technical documents on aircrafts and power plants.
- Manufacturing of more than 20 thousand types of parts concerning modern technologies.
- Running a main calibration center possessing the right to issue necessary certificates.
- Optimistic usage of possessed facilities by co-operative and subsidiary industries of Ministry of Defence for accomplishing new projects.
- Precision testing in an engine test collection (Unique test cell).
- Providing required guarantees in appropriate packages according to the technical manuals.
- Performing the after delivery services of various power plants such as: turbojet, turbofan, turboprop, turboshaft and industrial.. etc.





HISTORY OF ACTIVITIES OF AIRCRAFT REPAIRS & MANUFACTURING INDUSTRY

ARMI a well known center in **IACI** is established for the purpose of carrying out repair, heavy maintenance & overhaul of aircraft such as:

C130, B707, F14, F4, F27, F28. B747, AIRBUS, ILYUSHIN, ANTONOV, TOPOLOV , CESSNA, P3F, JET STAR, TURBOCOMMANDER, YAK,... are amongst aircrafts which are repaired and applying time compliance technical orders (**TCTO**) to military and some of the civil airplanes available in **IRAN** with anticipation of being able to manufacture aircraft components.





IHSRC - IRAN HELICOPTER SUPPORT & RENEWAL COMPANY

9

PIC: 000912

DESCRIPTION

In 1969, following the purchase of considerable number of helicopters from Italian company "Agusta" initial step to establish a helicopter support and maintenance unit, under the name of "Iran Helicopter Company Joint Stock" was taken.

By purchasing a substantial number of helicopters from American "Bell Textron" in 1973, extensive planning to support new fleet was made and 3 main sections namely: (NMP), (HDMC) and (NICP) were defined for it. Accomplishing this "Iran Helicopter support and renewal company" (IHSRC) became the largest helicopter maintenance center in the Middle East. Following the victorious Islamic Revolution in 1979, severance of ties with the USA and the imposed war inflicted to the

new regime, loyal experts of "IHSRC" with the aim of supporting Islamic combatant in the battle fronts were able to reach self-sufficiency in maintenance and renewal of Iran helicopters fleet.

In the wake of these achievements, the responsibility to repair and support of 10 different models of military and civil helicopters, totally to 16 types of helicopters belonging to domestic companies, such as, "Iran Red Crescent Organization", "Iran Telecommunication Organization", "Tavanir" and National Iranian Oil Company (NIOC), was given to "IHSRC".

In 2002 "IHSRC" obtained necessary licences issued by "Civil Aviation Organization" to perform basic repairs and overhaul of civil helicopter models of 205, 212, 214 and 412.





DESCRIPTION

FUSELAGE REPAIRS :

Based on the repair schedules and for renewal and repair activities helicopters are placed in the proper jigs & fixtures.



TRANSMISSION WORKSHOP :

All power transmission systems of helicopters such as Clutches, Main 42 and 90 degree Gearboxes are overhauled and tested at this workshop by sophisticated test equipments and are prepared for installation on helicopters.



BLADE REPAIRS :

Defective blades are paint stripped and dispatched to the blade shop for necessary repairs, painting and static balancing.



FUEL CELL :

Producing all types of polymeric and composite aircraft fuel cell by research and development in Iran.



DESCRIPTION

JIG AND FIXTURE DESIGN AND MANUFACTURING :

Design and manufacturing of jig & fixtures are part of main activity to build or rebuild cabin and tail booms for different structures. Experienced designers can assist in these cases to make subassembly and final assembly fixture for different needs.



FINAL ASSEMBLY :

After removal of cabin from fixture and transferring to the final assembly area, all parts and repaired components are reinstalled on helicopter according to the repair procedures and necessary adjustments and calibrations are carefully implemented to ensure the conformity of the performed repairs with aviation standards.



DESIGN AND MANUFACTURING ENGINEERING SECTION:

To assess the possibility of manufacturing process for metal and nonmetallic parts, preparation of production process, determination of required technology, selection of standard materials, preparation of workshop drawings and finally assembly plan for newly fabricated parts is part of expert engineers tasks in which by computer aides are responsible for such activities.





SHAHED 278 HELICOPTER

PIC: 000904

DESCRIPTION

Iran Aircraft Manufacturing Company (**HESA**) also capable to produce and repair variety of aviation products and components.

Shahed 278 is a single engine , two bladed sea-saw main rotor helicopter designed for military and civil operations. During manufacturing, safety is the main concept.

SHAHED 278 APPLICATIONS

- Transportation
- Training
- Emergency medical service Police patrol
- Traffic observation
- Aerial photography

FEATURES

- Composite airframe
- High efficiency tail rotor blades
- New design of horizontal stabilizer
- Self sealing fuel tanks

Shahed 278 is a mature design having accumulated a vast experience in all environments. In particular 278 is:

- Used for several missions in hot climate
- Suitable for air rescue



SPECIFICATIONS

TYPE	SHAHED 278
Standard Configuration Weight (kg):	682
FAA Normal Gross Weight (kg):	1451
Useful Load (kg):	769
Max. Speed at Sea Level (km/h):	240
Max. Speed at 1524 m(km/h):	230
Service Ceiling (m):	6400
Fuel Capacity (liters):	287
Max. Rate of Climb at Sea Level (m/s):	12.7
Rang (km):	340



DESCRIPTION

Qods Aviation Industries (QAI) as one of the subsidiary industries of Iran aviation organization, produce a wide range of Unmanned Air Vehicles. (U.A.V) that are used for surveillance and reconnaissance missions and target acquisition. Having commenced the design and manufacture of UAVs in 1985 QAI now finds itself with a wide portfolio of success fulfilling a number of requirements.

In addition to the UAVs, this industry produce several types of parachutes such as: personnel parachutes, cargo parachutes, recovery parachutes, and para gliders.





SAEGHE 2 AERIAL TARGET SYSTEM

9

PIC: 000913

DESCRIPTION

SAEGHE 2 is a medium to high performance remotely controlled missile target designed for surface-to-air armament systems training and evaluation. The **UAV** airframe is constructed mainly of composites.

This system can be equipped with three **IR** flares, and a radar corner reflector to enhance **IR** radiation and radar image (**RCS**).

GUIDANCE AND CONTROL

Full microprocessor-based autopilot control, programmable with or without control. Two alternative ground control systems: in-sight flight unit for visual control without telemetry data, and **GPS** - based programmable navigation system which allows target to be controlled, tracked and viewed on monitor during both in sight and out-of-sight missions.

GROUND CONTROL STATION

The ground control station (**GCS**) provides all of the functionality necessary for mission planning and in flight monitoring of the **UAV**. The **GCS** is designed as a mobile or portable.

POWER PLANT: One 25 hp Two-cylinder, two-stroke engine, two-blade propeller.

Launch: by booster rocket from a short-length ramp

Recovery: parachute recovery system, belly skid landing on land (Emergency)



SPECIFICATIONS

TYPE	SAEGHE2
Air Frame:	Composite
Wing Span (m):	2.60
Length (m):	2.81
Max. Take-Off Weight (kg):	60
Operating Range (km):	50
Endurance (min):	45
Operating Ceiling (ft):	11,000
Max. Speed (km/h):	250



MOHAJER 2 UAV SURVEILLANCE SYSTEM



9

PIC: 000903

DESCRIPTION

MOHAJER II was designed as a reconnaissance and surveillance **UAV** system. This **UAV** can be based on shipboard or deployed on land to carry out different missions such as: real time video imaging reconnaissance, aerial picturing and battle areas surveillance for covering various targets on the surface.

GUIDANCE AND CONTROL

MOHAJER II can be operated either under remote control from the **GCS** or automatically, controlled in the latter case by its GPS-based onboard automatic navigation system. Flight path is governed by a digital autopilot set to either automatic (preprogrammed-99 waypoints) mode or to semi-automatic mode for **GPS** operator.

Launch: Pneumatic Launcher

Recovery: Skid landing or Parachute recovery system

MAIN TASKS AND FUNCTIONS OF THE UAV :

- Aerial reconnaissance and battle field surveillance
- Border patrol
- Aerial photography
- Target positioning, artillery shooting adjustment
- Communication relay
- Traffic control

SPECIFICATIONS

TYPE	MOHAJER 2
Air Frame:	Composite
Wing Span (m):	3.80
Length (m):	2.91
Max. Take-Off Weight (kg):	85
Operating Range (km):	50
Endurance (min):	90
Operating ceiling (ft):	11,000
Max. Speed (km/h):	200





ABABIL 2 - UAV SYSTEM

PIC: 000918

DESCRIPTION

With simple commands, the target area will be in your hand in real time. **Ababil 2**, is an UAV highly automated and simple to use, full pre-programmed, in flight programmable and/or pilot controlling from the ground station for:

- Observation and target location
- Reconnaissance and surveillance
- Communication relay
- Police and civil safety
- **Launching:** by jato
- **Recovery:** by parachute or skids



SPECIFICATIONS

TYPE	ABABIL 2
Operational Radius (km):	100
Endurance (min):	75-120
Flight Altitude (ft):	11000
Max. Speed (km/h):	250



ABABIL 3 - UAV SYSTEM

PIC: 000907

DESCRIPTION

Real-time situation awareness is a critical matter for decision-makers in every operation, both in civil and military conditions. The key of success depends on timely delivery of information.

Ababil 3 is an UAV with a low radar-cross section used for:

- Surveillance and Reconnaissance
- Communication relay
- Electronic war

CAPABILITIES:

- Real time transmission of the acquired data
- Digitized map with aircraft position and main flight data
- Full pre-programmed, in flight programming and/or semi automatic programming
- Photographing according to the program
- Reconnaissance and surveillance
- **Take off:** by wheel or rocket-assistant
- **Recovery:** by parachute or runway





PARACHUTE

9

PIC: 000916

DESCRIPTION

Iran Aviation Industries also design & manufacture various types of sky diving and UAV's recovery parachutes, paraglider canopies, lightening parachutes & also different types of harnesses by using the latest experience, modern technologies and high quality materials.

Following is the specifications of some current products:



SPECIFICATIONS

TYPE	OFOGH	HOMA 2000-270	HOMA 2000-280	EMDAD III	FAKHTEH 270	FAKHTEH 280
Shipping Weight (lb):	7.2	12.2	12.7	7.2	9	9.9
Pack Volume (in3):	416	790	830	416	584	662
Area (m²):	26.9	250	260	26.9	24.3	27.5
Operating Limited in Ordered (lb):	154 - 264	140 - 180	150 - 200	154 - 264	165 - 257	187 - 291

ASSEMAN II Harness / Container System (Javelin)

Featuring back mounted main & reserve canopy containers. The **ASSEMAN II** equipped with the 3-ring release system. Other standard features include main and reserve pilot chute which are not limited to the strength of the stitches and the junction are stronger than the webbing itself.



SPECIFICATIONS

TYPE	SIZE	MAIN CANOPY	RESERVE CANOPY	CONTAINER
ASSEMAN II	M	HOMA / FAKHTEH	EMDAD III	J - 8

SPECIFICATIONS

ODGE 77 (SYMPHONY)

TYPE	S	M	L	XL
Number of Cells:	22	24	26	28
Wing Span (m):	9.4	10.0	10.8	11.6
Area (m²):	24.6	27.2	29.7	32.3
Aspect Ratio:	3.1	3.3	3.6	3.8
Speed (mph):	13 - 25	13 - 25	14 - 26	14 - 26
Take-off Weight (lbs):	121 - 165	132 - 187	143 - 220	187 - 264
Motorized T/O Weight (lbs):	154 - 209	165 - 220	176 - 253	209 - 297
Glider Ratio L/D:	6.0	6.0	6.0	6.0
Minimum Sink (f/s):	4.26	14.2	14.2	4.2
Main Lines/Risers:	4	4	4	4



AIRCRAFT WINDSHIELD

9

PIC: 000767

DESCRIPTION

- Cockpit windows must withstand various stains and constraints
- Electric heating windshield
- Ultra thin wire grids technology
- Windows are equipped with built-in temperature sensors
- Metal frames
- Excellent optical quality
- Bullet-proof
- Anti-explosion



SPECIFICATIONS

TYPE	MATERIAL	STRUCTURE	MIL-STD
P3 F Windshield Center Panel:	Glazing Float Glass	Glass or PC	MIL-STD-25871
P3 F Pilot & CO-Pilot:	Glazing Float Glass	Fully Glass	MIL-STD-25871
F-4 Center Windshield:	Glazing Float Glass	Fully Glass	P.S.23501
F14 Center Windshield Bullet Proof:	Glazing Float Glass	Fully Glass	MIL-G-5485D
B-707 Sky Light Window:	Glazing Float Glass	Fully Glass	SAE-AMS-G-25871
F-27 Side Windshield:	Glazing Float Glass	Glass or PC	SAE-AMS-G-25871
F-27 Co-Pilot Windshield:	Glazing Float Glass	Glass or PC	SAE-AMS-G-25871

MORE FEATURE

BIC PART NO:	PART NAME	MATERIAL	STRUCTURE	MIL-STD
911104200	Center Windshield C130	Glazing Float Glass	Non-Organic Glass	SAE-AMS-G-25871(1999)
911104200	Panel Assy Front Windshield LHC130	Glazing Float Glass	Non-Organic Glass	SAE-AMS-G-25871(1999)
911104300	Panel Assy Front Windshield LHC130	Glazing Float Glass	Non-Organic Glass	SAE-AMS-G-25871(1999)
911104300	Panel Assy Front Windshield LHC130	Glazing Float Glass	Non-Organic Glass	SAE-AMS-G-25871(1999)
911104150	Panel Assy Clear Vision Windshield LHC130	Glazing Float Glass	Non-Organic Glass	SAE-AMS-G-25871(1999)
911104150	Panel Assy Clear Vision Windshield LHC130	Glazing Float Glass	Non-Organic Glass	SAE-AMS-G-25871(1999)
911104100	Panel Assy Clear Vision Windshield RH C130	Glazing Float Glass	Non-Organic Glass	SAE-AMS-G-25871(1999)



AIRCRAFT TIRES

9



PIC: 000181

DESCRIPTION

Aircraft Tire industries is another subsidiary of Defence Ministry of Iran.

Tire industry is one of the most advanced & modernized complex of its kind.

Various types of aircraft tire such as Boeing, Airbus, military aircraft (**C130N**), fighters (**F4, F5 & F14**), eastern aircraft (**Illusion, Migs, Sokho**) and heavy vehicle tires are produced in this industry as per international standards.



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SPECIFICATIONS

AIRPLANE TYPE	TIRE TYPE	SPEED RATING (KM/H)	MAX. LOAD (KN)	TT OR TL	EQUIVALENT LAYER	TIRE SIZE
F5E.F	Main	401	55.6	TL	18	24 x 8-13
F14	Main	350	138.8	TL	28	37 x 11.5-16
F4M	Main	389	115.6	TL	26	30 x 11.5-14.5
F5A.B	Main	315	44.5	TL	16	22 x 8.5-11
C130	Nose	241	56.9	TL	12	12.5-16
C130	Main	322	206.4	TL	26	20-20
F4N	Nose	442	27.5	TL	14	18 x 5.5
CHINOOK	Main	111	25	TL	10	8.5-10
F5E.F	Nose	413	22.2	TL	12	18 x 6.5-8
F14	Nose	350	53.3	TL	20	22 x 6.6-10
F7	Main	330	34	TL	8N	660 x 200-11
F7	Nose	290	15	TL	6N	500 x 180G-1
IL76	Main	330	118.7	TT	14N	1300 x 480
IL76	Nose	310	66.71	TL	10N	1100 x 330
SU24	Nose	320	23.5	TT	8N	660 x 200
SU24	Main	400	89.18	TL	14N	950 x 300
MIG29	Nose	340	25.5	TT	8N	570 x 140
MIG29	Main	330	83.35	TL	12N	840 x 290



JATO MK-34

PIC: 001133

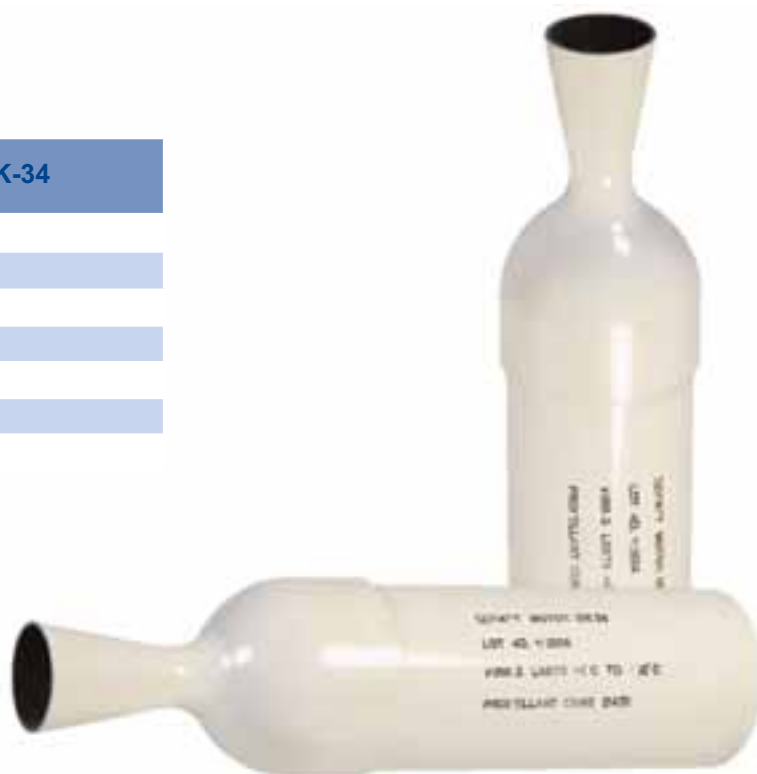
DESCRIPTION

This product is a **Jet-fuel Assisted Take Off (JATO)** device to provide additional thrust to heavy overloaded airplanes to assist them for a take off from short or damaged runways, high-altitude airfields, or to give rapid acceleration for a quick take off.

It can be used for a wide range of applications such as providing additional energy to air launched targets like **RPVs** to test their performance or other characteristics.

SPECIFICATIONS

TYPE	JATO MK-34
Total Weight (kg):	11.7
Length (mm):	544
Diameter (mm):	136.7
Propellant:	Solid
Nominal Thrust (lb):	2690
Thrust Duration (sec):	0.7
Nominal Impulse (lb.sec):	1870





FLIGHT SIMULATOR

9

PIC: 000765

DESCRIPTION

Simulation is known as one of the most strongest, reliable and economic tools, specially for military, industrial and education purposes, it is because of considerable reduction of costs, preventing the mass casualties and various unforeseeable accidents.

ADVANTAGES

- As a new educational method
- Profound and more efficiency
- Salient reduction of hazards
- Considerable reduction of costs
- Saving the item of education
- Possibility of education in an emergency and unfavorable conditions.
- Possibility of repeat, review and evaluation
-

CAPABILITIES

- Consultancy and provision of aid educational total solution systems.
- Managing the market of local simulator systems to prevent any parallel activities.
- Design, production and optimization for all kinds of air, ground and sea(surface and under surface) simulators.
- Design and implementation all kinds of War Game simulators.
- Design and implementation all kinds of educational multimedia simulators upon computer (CBT)

SOME PROJECTS

- Complete flight simulator of **Helicopter Bell-206**
- Complete flight simulator of **Airplane AN-74**
- Complete flight simulator of **Helicopter Mil-171**
- Optimization and development of (second cabin) educational simulator for **Airplane F-14**
- Complete flight simulator of **Airplane B-707**
- Car driving simulator.
- Complete of undersurface **Ghadir Submarine simulator**
- Optimization and upgrade the image system of **Airplane B-707**
- Optimization and upgrade the image system of **Airplane F-27**
- Mechanized **Armorial shot** fields simulator systems
- Ground War Game systems



chapter 10

MARINE





The Marine Industries Organization (**M.I.O**) as one of the major subsidiary industries of Ministry of Defence, began its activity in 1986, aiming of producing a variety of marine vessels with their bodies made of fiberglass, steel and aluminium, in the two categories of military and non-military vessels.

Considering the 3,000 km of marine borderline around Iran and the good market for building and producing various types of marine vessels and export markets, this group has taken action for creating specialized factories in the various parts of Iran in order to have full-fledged presence in the market.

While employing effective , specialized manpower and benefiting from the most appropriate methods of design and production , and with more than 27 years of experience , marine industries organization is presently leading the market for building and producing a variety of marine vessels.

According to the programs, MIO has focused on the following:

- To design , build and produce a variety of military vessels (surface , surface effect and subsurface types)
- To design, build and produce a variety of non - military vessels: (General Cargo, Multi - Purpose Landing Craft, Tug...)
- To provide designing and engineering services
- To provide purchased engineering services
- To provide and manage projects
- To produce without factory
- To implement research projects in the field of marine industries.
- To provide repair and maintenance services for a variety of vessels
- To perform joint venture projects.

With respect to existing facilities of the organization, it can be mentioned that the shipbuilding facilities in different cities such as Bandar Abbas , Bandar Khorram shahr and Bandar Anzali , are very large and well-equipped with workshops and warehouses and building berths computerized ship lift system cranes of 1600 tons lifting capacity , as well as excellent repair and maintenance facilities supported by floating docks and dry docks , where the ships (up to 5000 DWT) can be accommodated.





CHAPTER 10



ITEM	TABLE OF CONTENT	PAGE
1	MISSILE LAUNCHER BOAT (JM18 & JM16)	280
2	MIDGET SUBMARINE	281
3	MOWJ FRIGATE	282
4	SINA-FAST MISSILE BOAT	283
5	ZOLFAGHAR-FAST PATROL CRAFT	284
6	17 M-R & 13 M-R - MULTI-PURPOSE PATROL BOAT	285
7	10M - R & 12M - MULTI-PURPOSE PATROL BOAT	286
8	HIGH SPEED PATROL BOAT 10 M-S	287
9	13M & 16M - MULTI-PURPOSE PATROL BOAT	288
10	12M - ROCKET LAUNCHER PATROL BOAT	289
11	14M - MULTI- PURPOSE PATROL BOAT	290
12	SAR 20 & 20 -2 SEARCH & RESCUE VESSELS	291
13	SUPPLY & CREW BOAT	292
14	TUG BOATS	293
15	LANDING CRAFT	294
16	FLOATING DOCKS	295
17	OCEANOGRAPHIC VESSEL	296
18	FLAT & TANKER BARGES	296
19	HOVERCRAFT	297
20	RECOMPRESSION CHAMBER	297



MISSILE LAUNCHER BOAT (JM18 & JM16)



10

PIC: 000501

DESCRIPTION

JM 18 is a high speed & patrol boat with 3x1200 HP engines and two surface to surface missiles with fire control system and it has good performance, sea-keeping and maneuvering capabilities for sea-states 2-3. The propulsion system of boat is surface piercing propellers.

JM 16 is a high speed & patrol boat with 2x1300 HP engines and two surface to surface missiles with fire control system.

It has good performance, sea-keeping and maneuvering capabilities for sea-states 2-3. The propulsion system of boat is surface piercing propellers.

Above mentioned boats have capabilities for installation of various light marine weapons such as 12.7 mm gun, 23 mm gun; surface to surface torpedo, 107 mm rocket and 122 mm rocket depend on customer requirements and general arrangement.



JM - 18

SPECIFICATIONS

TYPE	JM-18	JM-16
Length (m):	21.15	16.3
Breadth (m):	5.77	3.75
Height (m):	2.1	1.93
Draft (m):	0.87	0.67
Light Weight (tons):	28.16	13.75
Engine Power (hp):	3 x 1200	2 x 1300
Speed (knots):	53	54
Endurance (n.m):	450	320
Crew:	6	3
Weaponry System:	Missile	Missile



JM - 16



MIDGET SUBMARINE

PIC: 000511

DESCRIPTION

The **Midget Submarine IS-120**, is specially designed to get good ability to meet a naval battle in a shallow sea in conformity with demands of the 21st. century modern wars.

The midget submarine is designed not only to attack enemy's surface warships and submarines but also to block up enemy's transportation waterway.



SPECIFICATIONS

TYPE	MIDGET
Length Overall (m):	29.00
Length Pressure Tight Hull (m):	19.20
Beam Overall (m):	2.75
Diameter of Pressure Tight Hull (m):	2.75
Height From Keel to Deck (m):	3.20
From Keel to Bridge (m):	4.80
Height of Metacenter Surfaced (m):	0.3
Height of Metacenter Dived (m):	0.21 - 0.25
Draft [Normal] (m):	2.5
Displacement at Surface (tons):	115
Weight at Dived (tons):	123



MOWJ FRIGATE



PIC: 000509

DESCRIPTION

Mowj is an Iranian class frigate. It is highly capable for mounting various type of armament (such as missile control, torpedoes launcher, 76mm and 40mm artillery...), it is also capable of long distance sailing for military & naval operations.

This class combines anti-submarine assets with other systems of weapons capable of dealing with surface and air threats as well. The ship is designed for 150 crew.



SPECIFICATIONS

TYPE	MOWJ FRIGATE
Length (m):	94.46
Breadth (m):	11.1
Draft (m):	3.2
Weight (tons):	1500
Engine Power (hp):	4 x 4250 / 4 Units
Speed (knots):	28
Diesel Generator (kw):	4 x 500
Weaponry System:	Artillery / Missile / Torpedo





SINA - FAST MISSILE BOAT

PIC: 000522

DESCRIPTION

This fast patrol craft is capable for mounting various types of armament which has fast operating speed & reaction for reconnaissance, attack, destroy the target and quick return. This craft is equipped with heavy artillery.



SPECIFICATIONS

TYPE	SINA (F.M.B)
Length (m):	47
Breadth (m):	7.04
Draft (m):	1.92
Weight (tons):	275
Engine Power (hp):	4 x 3500 / 4 Units
Speed (knots):	35
Diesel Generator (kw):	3x140
Weaponry System :	Artillery / Missile



ZOLFAGHAR - FAST PATROL CRAFT

10

PIC: 000503

DESCRIPTION

This vessel with catamaran hull form is designed and manufactured with 2 surface to surface missiles which controlled by **FCS** (Fire Control System) and tracer radar. Power for this vessel is provided by 2x1150 HP engines.

The vessel has surface drive propulsion system and can reach to maximum speed of 55 knots in sea states 2-3. This vessel is also equipped with advanced navigation and communication system and safety appliances such as life boat according to the rules of **IMO**.



SPECIFICATIONS

TYPE	ROCKET LAUNCHER FAST PATROL CRAFT	MISSILE LAUNCHER FAST PATROL CRAFT
Length (m):	13.65	13.75
Breadth (m):	4.8	4.8
Height (m):	1.9	1.9
Draft (m):	0.7	0.75
Light Weight (tons):	17	17
Engine Power (hp):	2 x 1150	2 x 1150
Speed (knots):	55	51
Endurance (n.m):	300	250
Crew:	5	5
Weaponry System:	Rocket Launcher / Machine Gun	Missile / Machine Gun



17 M-R & 13 M-R MULTIPURPOSE PATROL BOAT

PIC: 000517

DESCRIPTION

17M-R is a vessel made by fiberglass with 2x1200 HP engines and propulsion system.

The vessel has good performance & reliability that can reach maximum speed of 65 knots.



13M-R is a fast patrol boat with high performance & special characteristics.

This boat has good ability for maneuvering in calm water and wave. It is powered by 2 x 660 HP engines with the surface propulsion system.



SPECIFICATIONS

TYPE	17M-R	13M-R
Length (m):	16.50	12.9
Breadth (m):	3.70	2.64
Height (m):	2.50	2.70
Draft (m):	0.80	0.80
Light Weight (tons):	13.95	8.137
Engine Power (hp):	2 x 1200	2 x 660
Speed (knots):	65	60
Endurance (n.m):	400	400
Crew:	5	3
Weaponry System:	Rocket-Artillery Single Barrel Naval Gun Naval Mine	Rocket-Artillery Single Barrel Naval Gun Naval Mine



10M - R & 12M MULTI PURPOSE PATROL BOAT

10

PIC: 000505

DESCRIPTION

10M-R is a fast patrol boat with good maneuvering in calm sea and wavy condition.

It can reach to maximum speed of 55 knots. It has 2x350 HP outboard engines.

The structure and hull are made of fiberglass. This vessel can operate in tropical weather.



12M is a fast patrol boat with capability of maneuvering in different sea states with 2x350 HP outboard or 2x420 HP inboard engines.

It can reach to maximum speed of 60 knots.



SPECIFICATIONS

TYPE	10 M - R	12 M
Length (m):	10	11.85
Breadth (m):	2.70	2.85
Height (m):	2.15	2.25
Draft (m):	0.60	0.80
Light Weight (tons):	3.22	5.302
Engine Power (hp):	2 × 350	2 × 420
Speed (knots):	55	60
Endurance (n.m):	300	400
Crew:	3	3
Weaponry System:	Rocket-Artillery Single Barrel Naval Gun	Rocket-Artillery Single Barrel Naval Gun Naval Mine



HIGH SPEED PATROL BOAT 10 M-S

PIC: 000523

DESCRIPTION

10M-S is a fast patrol boat with good maneuvering in calm sea and wavy condition. It can reach to maximum speed of 46 knots. It has 2 x 300 HP outboard engines. The structure and hull are made of fiberglass. This vessel can operate in tropical weather.

This boat has capabilities for installation of various light marine weapons such as 12.7 mm gun, 107 mm rocket depend on customer requirements and general arrangement.



SPECIFICATIONS

TYPE	10 M-S
Overall Length (m):	10.45
Overall Breadth (m):	2.9
Draft (m):	0.7
Depth to Deck (m):	1.2
Range (mile):	200
Max. Speed (knot):	46
Crew:	3
Weapon (mm):	12.7
Main Engine (hp):	2 x 300
Propulsion:	Outboard
Material:	Composite



13M & 16M MULTI PURPOSE PATROL BOAT

10

PIC: 000514

DESCRIPTION

13M is a fast patrol boat with 2x 660 HP engines and surface propulsion system. It can reach to maximum speed of 53 knots.



16M is a fast patrol boat with V-shape hull which can reach to 68 knots. The hull and super structure are made of composite light materials.

This boat was made based on international standards and safety rules of IMO. This boat has 2x 1200 HP engines and surface propulsion system.



SPECIFICATIONS

TYPE	13 M	16 M
Length (m):	12.90	16.43
Breadth (m):	2.64	2.85
Height (m):	2.70	3.50
Draft (m):	0.82	0.84
Light Weight (tons):	6	15.33
Engine Power (hp):	2 × 660	2 × 1200
Speed (knots):	53	68
Endurance (n.m):	400	400
Crew:	3	5
Weaponry System:	Rocket-Artillery Single Barrel Naval Gun	Rocket-Artillery Single Barrel Naval Gun Naval Mine



10

12M ROCKET LAUNCHER PATROL BOAT

PIC: 000518

DESCRIPTION

This vessel is fast attack boat which is made of marine aluminum and it has catamaran hull form.

The boat can reach maximum speed of 50 knots in calm sea and it has good manufacturing and sea-keeping in different sea states condition. The vessel has surface propulsion system with 2 x 660 HP engines according to the rules of maritime societies.



SPECIFICATIONS

TYPE	12M PATROL BOAT
Length (m):	11.90
Breadth (m):	3.1
Height (m):	1.5
Draft (m):	0.65
Weight (tons):	8.5
Engine Power (hp):	2 × 660
Speed (knots):	50
Endurance (n.m):	250
Crew:	2
Weaponry System:	Rocket

M A R I N E



14M MULTI PURPOSE PATROL BOAT



10

PIC: 000504

DESCRIPTION

This vessel has catamaran hull form which is designed and manufactured from marine grade aluminum. It is also has capability for maneuvering in sea states 2-3 and it has 2x1200 engines and surface propulsion system.

It is equipped with rockets and advanced navigation and communication system. It can reach maximum speed of 55 knots.



SPECIFICATIONS

TYPE	14M PATROL BOAT
Length (m):	13.75
Breadth (m):	4.9
Height (m):	2.25
Draft (m):	1
Light Weight (tons):	18.5
Engine Power (hp):	2 × 1200
Speed (knots):	55
Endurance (n.m):	300
Crew:	5
Weaponry System:	Rocket-Artillery Single Barrel Naval Gun, Naval Mine
Capacity (passengers):	25



10

SAR 20 & 20 -2 SEARCH & RESCUE VESSELS

PIC: 000502

DESCRIPTION

The vessels are designed and manufactured for Search and Rescue operations. They have a Mono Hull series with high manoeuvre and navigation performance.

The hull is made of marine aluminium grade 5083 and it has strong structure and high resistance against the corrosion.

One of the special characteristics of these vessels are Self Righting and the propulsion systems. Maximum speed of both vessels are 35 knots.

The vessels are designed according to the rules of the GL institution of Germany. They are equipped with central ventilation system, fuel tanks with a capacity of 3600 liters, fresh water 700 liters, internal and external fire fighting system advanced communication system, navigation equipment, and health and welfare facilities.

Cruise speed of vessels are 20 knots in calm sea and 45°C weather temperature, 35°C sea temperature and relative humidity of 95% during 24 hours a day.

Considering the special duty of this vessel, a hook with a pulling capacity of 4 tons and crane with a capacity of 1.2 ton are designed and installed on the vessel.

These vessels also have the capability to install weapons and carry troops.



SAR 20 - 2



SPECIFICATIONS

TYPE	SAR 20&20 -2
Length (m) :	20
Breadth (m):	4.70
Draft (m):	0.95
Seakeeping – Sea State:	3 - 4
Speed (Kn):	35
Endurance (N.M):	500
Displacement (Full Load) (ton):	25
Main Engine (HP):	2×820
Propulsion System:	Water Jet



SUPPLY & CREW BOAT

PIC: 000506

DESCRIPTION

CREW BOAT

The vessel constructed as a medium speed, sea kindly, crew and cargo boat for operation in ambient air temperature of max 50°C and 95% RH and seawater max temperature of 32°C.

The vessel is operatable at a sea state of 5. The bridge is designed to allow all round vision. Entry to the passenger decks is via rear doors. Aft control station is located on the top deck aft of the main bridge station within the air conditioned bridge area. A suitable counter equipped with refrigerator for serving snacks which is arranged in the passenger area.

This ship can be used as supporter boat on the islands for transporting of troops, equipment, machinery and also installation of heavy and light weapons and rocket launcher.

SUPPLY BOAT

This vessel is a twin supply crew boat and is designed for harbour coastal and unrestricted tropical service operating. The vessel is designed to carry 90 passengers, 65 tons of fuel and oil cargo, 50 tons fresh water cargo, 8m³ deep freeze cargo, 8m³ refrigerated cargo, 12m³ provisional store cargo and be able to carry 50 tons equally spreaded cargo or 30 tons truck on aft deck.

This vessel is designed to withstand the sea state of 6. Range of operation on continuous cruising speed is 1000 nautical miles. The hull and superstructure of the vessel are all welded construction. The hull is made of marine steel, grade "A" shipbuilding certificate. The superstructure is made of aluminium alloy, grade "AG4MC".

The ship can be equipped with rocket launcher or used for transporting 90 soldiers, food, drinking water or fuel.



SPECIFICATIONS

TYPE	CREW BOAT	SUPPLY BOAT
Length Overall (m):	33.50	47
Beam(m):	7.93	
Draft (Mean):	1.50 Meters Approx	2.86
Height(Without mast) (M):	6.20	
Classification:	BV-Special Service-Crew Boat	LR society
Fuel (Litters):	32000	65000
Passengers:	68	90
Water (Litters):	22000	5000
Speed (Cruising) (KN):	20	21
Speed (Max) (KN):	24	
Endurance (N.M):	450	1000
Crew:	6	15



TUG BOATS

PIC: 000520

DESCRIPTION

TUG BOAT 1200 HP

This vessel is a modern seagoing twin screw tug/work boat for coastal waters and features a compact deckhouse on the fore deck with a top deck placed wheel house and a large aft deck with a specific load of 2 ton/m².

The single hull and the superstructure are all welded construction and are made of marine steel materials, grade "A" shipbuilding, certified by **Lloyds Register** of shipping.

All materials and equipment are marine type and appropriate for this kind of vessel. The deck house accommodate officers and crew for a total of 8 persons. The vessel is propelled by two marine diesel engines, each driving a fixed pitch propeller in a fixed nozzle. These engines have a closed cooling water system and air started.

TUG BOAT 2400 HP

This vessel is equipped with suitable pumps, air conditioning and ventilation, air compressors, modern navigation system, hydraulic system, electrical system, ...

on the aft deck a hydraulic crane and a hydraulic towing winch are fitted. This vessel is manufactured under supervision and according to the rules of LR classification society.

TUG BOAT 4400 HP

This vessel is an azimuth stern drive tug and a highly maneuverable high performance tug of modern design. The propulsion consists of two main engines, air started and box cooled, with two steerable rudder propeller unit with fixed pitch propellers in nozzles which is mounted in the aft ship. The vessel is designed for towing on the hook, for pulling and pushing operations and can be used for harbour, coastal berthing operation and offshore towing, including fire fighting.



SPECIFICATIONS

TYPE	1200 HP	2400 HP	4400 HP
Length O.A. (m):	25	25	30.82
Breadth (m):	7.25	9	10.2
Depth at Side (Half Length) (m):	3.75		4.8
Lightweight (ton):	195		380
Displacement (Full Load) (ton):	275		583
Draft (aft) (m):	2.65	4	4.55
Speed (Kn):	10.3	11	12.8
Bollard Pull (ton):	14.7	30	48.6
Main Engine (HP):	APP. 2 x 600	APP. 2 x 1200	APP. 2 x 2200
Auxiliary (KVA):	APP. 2 x 60	APP. 2 x 90	APP. 2 x 97



LANDING CRAFT



PIC: 000515

DESCRIPTION

LANDING CRAFTS 2500 & 3300 DWT

This vessel is a multi carrier ship and has ship shape body in front and a flat bottom construction in amidship capable for loading and unloading of deck cargoes.

The superstructure of the vessel is located at aft deck and is designed to accommodate officers and crew for a total of 11 persons.

A large deck area is provided in fore of superstructure for loading of packed cargoes, containers and trucks and suitable to carry loads up to approx 4 ton/m² on the deck area, 18 long trucks (trailers) .

Fuel oil tanks of the vessel have a double bottom and double skin construction. The propulsion consists of two sets of main engine, gear box, shaft and fixed pitch propeller and two rudder steering units.

We are able to produce landing craft from 150 up to 5000 DWT as customer's requirement.

The ships can be used for handling all kind of light and heavy military equipment between the islands.



SPECIFICATIONS

TYPE	2500 DWT	3300 DWT
Length O.A. (m):	78	80
Breadth (m):	17	17.5
Depth at Pontoon (m):	4	4.75
Lightweight (ton):	985	1,188
Net Weight (ton):	2,285	3,162
Displacement (Full Load) (ton):	3,270	4,350
Draft Light Weight (mm):	900	1,100
Draft Loaded (mm):	2,691	3,238
Speed (Kn):	10	10
Range (N.M):	800	800
Fuel Oil (ton):	70	94
Fresh Water (ton):	50	72
Cargo Oli (ton):	-	2,650
Main Engine (HP):	2 x 1280	2 x 1280
Main Gen. Set (KVA):	2 x 90	2 x 90



FLOATING DOCKS

10

PIC: 000508

DESCRIPTION

800 TON L.C. FLOATING DOCKS

The dock is designed for docking of ships up to 800 tons lightweight at a sea water density of 1025 kg/m^3 by means of three transverse bulk heads and one longitudinal bulk heads, the dock is divided into 8 water tight compartments.

The dock's hull cross section should be U-shaped and has a flat bottom construction and tow platforms at fore and aft the hull and superstructure of the dock are of an all welded construction and are made of marine steel, grade "A" shipbuilding, certified by LLoyd's register of shipping.

4000 TON L.C. FLOATING DOCKS

The dock is designed for docking of ships up to 4000 tons lightweight at a sea water density of 1025 kg/m^3 . there will be 700 tons ballast water available in the ballast tanks for the compensation of trim and heel and for reduction of the longitudinal deflection. Discharge of compensation ballast water can increase the lifting capacity up to about 4700 tons under certain circumstances.

The metacentric height will be 2.5m when docking a ship weighting 4000 tons and having a center of gravity 9m above the keel blocks when the bottom of the ship is at water level for weighting 4000 tons approximately.

All kinds of our floating docks can be used for repairing of different type of military vessels and also use to prevent them from sinking in the sea.

SPECIFICATIONS

TYPE	800 TON	2000 TON	4000 TON
Length O.A. (m):	62	110	130
Length Pontoon (m):	50.60	95	117.81
External Breadth (m):	20.00	22	28
Internal Breadth (m):	15.00	17	21
Depth Total (m):	8.00	8	11
Depth Pontoon (m):	2.00	2.2	3.15
Draft Ballast (m):	6.50	8	-
Draft Loaded (m):	1.55	1.9	2.75
Draft (light weight) (m):	0.75	0.8 m	1.15
Lifting Capacity (ton):	800	2,000	4,000
Light weight (ton):	680	1,750	3,300





OCEANOGRAPHIC VESSEL

PIC: 000516

DESCRIPTION

In order to prepare military and civil sea map, an equipped and accured oceanography vessel is necessary.

Scientific Equipments:

- Side Scan Sonar System
- Multibeam
- Scientific Echo Sounders
- Echo Sounders For Biological Use
- Bottom Chrip System
- Gravimeter System
- Multiparametric System
- High Definition Seismic System
- Nets For Scientific Use
- Data Processing System
- Drilling System, Corer, Grabs Etc.
- Acoustic Positioning System
- Data Transmission
- Magnetometer System
- ADCP For Deep And Shallow Water
- ROV For Deep And Shallow Water



FLAT & TANKER BARGES

PIC: 000510

DESCRIPTION

350 UP TO 3500 DWT

This vessel is designed to carry cargoes on the deck and has a flat bottom construction. On the end of fore deck there is a mechanical operated ramp for loading and unloading of deck cargoes.

A large deck area is provided for loading of packed cargoes, containers and trucks and suitable to carry loads up to approx 3.5 ton/m². A one meter height bulwark is installed around the deck.



SPECIFICATIONS

TYPE	FLAT & TANKER BARGES
Length O.A. (m):	65
Breadth (m):	19.30
Depth (m):	4.5
Draft Loaded (m):	3.45
Draft Light Weight (cm):	70
Lightweight (ton):	720
Displacement at Full Load: (ton):	3,500



10

HOVERCRAFT

PIC: 000513

DESCRIPTION

The Uness 22 is a fully amphibious hovercraft which is capable of transporting up to 16 service men or equal weight of freight. The main structure is out of aluminum and the power is provided by single air cooled diesel engine.

The craft is capable of operating in wind speed up to 7m/s and over sea condition with waves up to 1m high (sea states 1).

Due to highly responsive characteristics of the flexible skirt system, the quality of ride and performance over rough water are good. It can reach the speed of 40 knots.



SPECIFICATIONS

TYPE	UNESS 6	UNESS 22
Length (m):	5.6	12
Beam (m):	2.6	5
Height (m):	1.7	3.2
Hover Height (m):	0.3	0.75
Capacity (person):	6	22
All-up Weight (kg):	700	7,200
Velocity (knot):	40	40
Hull Structure:	Composite	Aluminum Alloy

RECOMPRESSION CHAMBER

PIC: 000507

DESCRIPTION

SADAF recompression chamber designed to carry diver under pressure to six-man.

Pressurize and decompression on driver, planned instructive diving for diver and treating diseases cause by deep diving are some of the application of this system.



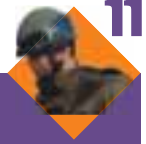
SPECIFICATIONS

TYPE	SADAF
Inspection Authority:	Germanischer Lloyd (GL)
Construction Material:	Aluminium Alloy
Inside Diameter Max (mm):	1,200
Overall Volume (Lit):	1,700
Overall Length (mm):	2,720
Viewport ID (mm):	150
Medical Lock DIA. (mm):	165
Gross Weight (Kg):	550

Chapter 11

MISCELLANEOUS





PREVIEW

The Iranian Ministry of Defence is not directly concerned with the production of civilian goods or manufacturing items such as house-hold commodities, cars, fishing boats ordinary telephones, recorders and computers etc..., however, in peace time, and a prospect of future maybe redundant workforce bothers the authorities. Though current demand to place orders for non-military goods increase,... all the reasons are there to provide enough signals to senior directors to push the same workforce back into factories, this time for producing civilian items.

Among the enourmous military parts and equipment, M.o.D manufactures a good number of popular products mainly suitable for police force or national defence sector that some of them are shown on this section as miscellaneous.

Variety of equipment which are shown in this part of the directory are produced and used also by police. Equipment and devices to keep law and order to prevent crimes and control essential service routes.

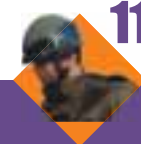
Armed forces also use many of these items to help police and civil authorities to maintain peace in the streets and order in all public places.

Batons, handcuffs, helmets, hunting guns, mines, anti-bullet vests, electronics, portable army tools, camouflages and simulators are to be named as some of the other subjects inserted in this section.





CHAPTER 11



11

ITEM	TABLE OF CONTENT	PAGE
1	ANTI BULLET/FRAGMENT VEST	302
2	ANTI-BULLET VEST (POLICE FORCES)	303
3	ANTI-BULLET BRIEFCASE	303
4	ANTI-BULLET BLANKET	303
5	ANTI-BULLET PORTABLE FORT	304
6	ANTI-BULLET COVER	304
7	COMPOSITE HELMET	304
8	ANTI-RIOT CLOTH AND EQUIPMENT	305
9	ANTI-RIOT SHIELD	306
10	BATON & TONFA	306
11	ANTI-RIOT HELMET	306
12	HANDCUFFS	307
13	FETTER & THUMB CUFFS	307
14	CAR BARRIER	307
15	ANTI RIOT SHOTGUN	308
16	RIFLE BAG	308
17	BOOTS	309
18	PERSONAL DEFENSE SPRAY	309
19	INSECT SPRAY	309
20	ANTI MINE GARMENT	310
21	SELF NEUTRALIZED MINE (YM-I-S)	311
22	SMOKE MINE (YM-I-T)	311
23	BOUNDING MINES	312
24	CLAYMORE MINES	313
25	ANTI TANK & VEHICLE MINES	314
26	ANTI HELICOPTER MINE	315
27	MOBILE BAKERY	315
28	FIELD KITCHEN SYSTEM	316
29	MOBILE KITCHEN SYSTEM	316
30	WATER PURIFICATION SYSTEM	317
31	FIELD ICE MAKING UNIT	317
32	3×1 EXPANDABLE SHELTER	318
33	MOBILE GENERATOR	318
34	MOVING HOSPITAL	319
35	PNEUMATIC BRIDGE	319
36	BATTLE DRESS UNIFORM	320
37	GROUP TENT	320
38	AIRCRAFT SUIT	321
39	TANK CREW SUIT	321
40	INDIVIDUAL FOOD RATION	322
41	LIFE PILL	322
42	ADVANCED FIRST AID BAG	323
43	FOLDING STRETCHER	323
44	BOTTLE OF WATER	324
45	NMK 100 (MARKER)	324
46	BULLET PROOF GLASS	325
47	MULTI-SPECTRAL CAMOUFLAGE SCREEN	326
48	NAKHJIR 1 – HUNTING GUN	327
49	NAKHJIR 2 – HUNTING GUN	327
50	NAKHJIR 3 – HUNTING GUN	328
51	SHAHIN – HUNTING GUN	328
52	MINI FLARE & TRIP FLARE (M49A1)	329
53	FIRING DEVICES	329
54	GEOMATIC SERVICES	330
55	ROCKET FLARE (MK5)	332
56	FLARE TARGET (MK2)	332
57	STS2 - FIREARM SHOOTING SIMULATOR	333



ANTI BULLET/FRAGMENT VEST

PIC: 000277

DESCRIPTION

Anti Bullet vests are designed and made in different types for different missions. They are light, comfortable, flexible, ergonomic and can stand bullets under **NIJ 0101.03 standard** and fragment under stanag **2920 standard**.

The weight of vests depends on the specific threat and the needed body protection level and they varies from 1.3 to 5.5 kg.
The marine type, with life saving vest, is 6.5 kg.



N.I.J BULLET PROOF PROTECTION LEVELS (N.I.J 0101.03 STANDARD)

N.I.J LEVELS	AMMUNITIONS	WEIGHT (G)	VELOCITY (M/S)	TRAUMA (MM)
I	.38 Special RN, Lead .22 LRHV, Lead	10.2 (158 gr) 2.6 (40 gr)	259 - 0 / +15 320 - 0 / +15	≤44 ≤44
II-A	.357 Magnum JSP 9mm FMJ	10.2 (158 gr) 8.0 (124 gr)	381 - 0 / +15 332 - 0 / +15	≤44 ≤44
II	.357 Magnum JSP 9mm FMJ	10.2 (158 gr) 8.0 (124 gr)	425 - 0 / +15 358 - 0 / +15	≤44 ≤44
III-A	.44 Magnum SWC, Lead 9 mm FMJ	15.55 (240 gr) 8.0 (124 gr)	426 - 0 / +15 426 - 0 / +15	≤44 ≤44
III	7.62 mm FMJ	9.7 (150 gr)	838 - 0 / +15	≤44
IV	.30 - 06 AP	10.8 (166 gr)	868 - 0 / +15	≤44

DIFFERENT TYPES OF ANTI-BULLET VESTS, SPECIAL FOR MILITARY FORCES

These vests are used against different fragments according to standard stanag **2920** and against different types of bullet class **I, IIA, II, IIIA & IV** according to standard **NIJ 0101.03** and also can prevent the knife penetration.

POLYMERIC ANTI BULLET PANEL (TYPE III)

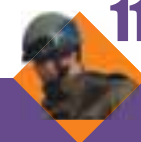
In order to increase the resistance in vest and provide protection against type III bullets (which have greater) velocity and hardness than type IIIA and II) polymeric panels were built. These panels are used along with the vest to stand against bullets of type **III**, and comply with **NIJ 0101.03** standard.

The panel is made of polyethylene which in certain conditions gets hardened and can resist bullets of type III.

COMPOSITE CERAMIC PANEL (TYPE IV)

This panel is used along with anti bullet vest against bullets of type **IV-AP** and complies with **NIJ 0101.03** The hardness of composite ceramic panels, due to their structure, is four times harder than steel. It shatters the steel core of a bullet and stops it.





11

ANTI-BULLET VEST (POLICE FORCES)

PIC: 000278

DIFFERENT TYPES OF ANTI-BULLET VESTS, SPECIAL FOR POLICE FORCES

These vests are used against different types of fragments according to standard stanag 2920 and against different types of bullets class **I,IIA, II, IIIA** according to standard **NIJ 0101.03** and also can prevent the knife penetration. Weight of the above mentioned vests are varied from 1.3 to 2.8 kg.

FEATURES

Changeability in design, color and shape according to the customer's needs for camouflage in different environments, flexible enough for necessary movements and possibility of equipping additional waterproof outer layer.



ANTI-BULLET BRIEFCASE

PIC: 000279

DESCRIPTION

Ballistics efficiency: protection against fragments and bullets class **IIIA** in accordance with standard stanag 2920 V50>550 m/s and **NIJ 0101.30**. Anti bullet briefcase is designed to protect personalities as a shield.

SPECIFICATIONS

TYPE	ANTI – BULLET BRIEF CASE
Open State Dimension (mm):	$(925 \pm 20) \times (450 \pm 20)$
Closed State Dimension (mm):	$(450 \pm 20) \times (350 \pm 20)$
Fiber Ballistics:	Polyethylene
Outer Coating:	Leader 100%
Weight (gr):	$3,900 \pm 400$



ANTI-BULLET BLANKET

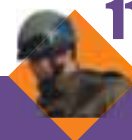
PIC: 001101

DESCRIPTION

The blanket can be used in rooms, inside cars and provides protection against **IIIA** bullets. The outer layer of this blanket is made of water proof fibers and inner layer is made of anti ballistic polyethylene fibers.

The blanket is available in two sizes, small 100×80 cm and large 200×160 cm. Other sizes are available on request.





ANTI-BULLET PORTABLE FORT

PIC: 000291

DESCRIPTION

Anti Bullet Forts are for protection of troops against all kinds of bullets and fragment of type **IIIA**, **III**, **IV**.

Due to its flexibility and versatility, it is possible to use the fort in different vehicles, fences, hedges and bushes and provides protection for the soldier in kneeled or standing position.

This fort is small in size and weights 9 kg only and in 30 seconds is ready to be used.



ANTI-BULLET COVER

PIC: 001103

DESCRIPTION

The cover works like a shield and provides protection for VIPs.

The cover complies with NIJ 01.01.03 standards and protects the person against type **IIIA** bullets.

The outer layer is made of leather and inner layer is made of anti ballistic polyethylene fibers.



COMPOSITE HELMET

PIC: 000280

DESCRIPTION

The Helmets are made of aramid and/or polyethylene fabrics so they can stop bullets and fragments. They are used in military operations and by police forces against potential threats. The body of helmet resists standard fragments under **NATO** standard of **STANAG 2920**, between **550-680** m/sec of velocity. Different weight of helmets in respect of their applications are given below:

SPECIFICATIONS

WEIGHT OF MEDIUM SIZE HELEMET	BALLISTICS VELOCITY
1300 GR	550 m/s
1350 GR	610 m/s
1400 GR	620 m/s
1550 GR	680 m/s





ANTI-RIOT CLOTH AND EQUIPMENT

PIC: 001102

DESCRIPTION

This type of clothing with light weight is used against impacts of weapons. Some of the specifications of these clothing are:

Light weight, high flexibility, ease on wearing and movement, high level of protection and could be used for a long period of time.

The structure of this clothing is in such a way that with the use of anti-impact panels and elastic layers and providing the gap between the impact point and the body, clothes prevents the impact force to the body.

COMPONENTS OF THE CLOTHES:

Is made of polypropylene, the hard panels of its inner part is made of polycarbonate, shock absorber foams, anti-wear material for coating the outer larger of clothes, and sweat absorbing material for inner part of cloth.

COVERAGE LEVEL OF THE CLOTHES:

For arm, elbow, shoulder, chest, abdomen, back, backbone, groin, thigh, knee and shin. Because of its multi-piece structure, it could provide its user suitable movement. It could be offered in different sizes.

A Package of all necessary items as antiriot shield, battons, sprays and ... packed as **Anti Riot Bag** for quick access and transfer.



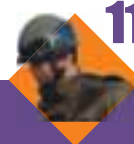
ANTI RIOT BAG

SHIELD

BATTONS

ANTI RIOT CLOTH

HELMET



11

ANTI-RIOT SHIELD

PIC: 000281

DESCRIPTION

Material: Poly carbonate

- High impact strength
- Excellent light transmission
- Dimensional stability
- Fire proof

DIMENSIONS

TYPE	LENGTH (MM)	WIDTH (MM)	THICKNESS (MM)
NO.1	900	500	3
NO.2	1000	500	4



BATON & TONFA

PIC: 000290

SPECIFICATIONS

TYPE	BATON
Material:	Poly Ethylene
Length (cm):	60
Weight (g):	650

SPECIFICATIONS

TYPE	TONFA
Short Length (cm):	30
Long Length (cm):	50
Weight (g):	860



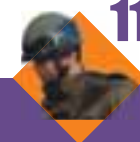
ANTI-RIOT HELMET

PIC: 000282

SPECIFICATIONS

TYPE	ANTI RIOT HELMET
Out Shell:	Plastic A.B.S
Face Protector:	Poly Carbonate
Sheet (mm):	4
Weight (g):	1,150
Color:	Green, White, Blue





11

HANDCUFFS

PIC: 000292

SPECIFICATIONS

TYPE	SM-103	SM-105
Material:	Nickel Plated Steel	Stainless Steel
Type of Connection:	Chain	Arm
Lock:	20 Teeth	21 Teeth
Weight (g):	230	350
Dimensions (mm):	75 × 95 × 2.2 × 230	80 × 100 × 2.2 × 195



SM - 103



SM - 104 (Strip Handcuff)



SM - 105

FETTER & THUMBCUFFS

PIC: 001110

SPECIFICATIONS

TYPE	SM-202	SM-301
Material:	Nickel Plated Steel	Nickel Plated Steel
Lock Position:	22 Teeth	17 Teeth
Weight (g):	1200	113
Dimensions (mm):	95 × 115 × 3 × 560	24 × 44 × 1.8 × 103

SM - 202



SM - 301



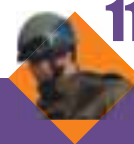
CAR BARRIER

PIC: 000293

DESCRIPTION

This kind of car barrier is so safe due to using removeable nail. Very quick using, resistance against different type of vehicle and long time life time is specific features of this barrier. It can be used by both police and military forces.





ANTI RIOT SHOTGUN

PIC: 001109

DESCRIPTION

Police anti-riot shotgun is a pump-action which can shot various ammo with length of 3 inches. For instance, tear gas ammo or penetrating one which are used by this rifle. Initially this shotgun was used in U.S & EU for sporting & hunting but nowadays police forces use them for anti-riot operations.



SPECIFICATIONS

TYPE	SHOTGUN
Gauge:	12
Butt Stock:	Walnut
Magazine Capacity (round):	5
Rifle Length (mm):	980
Length of Barrel (mm):	470
Weight (gr):	3,100



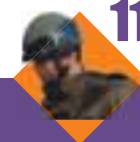
RIFLE BAG

PIC: 000300

DESCRIPTION

Rifle bag helps easy carrying and prevent from any hurt. It can be used by police forces and particularly by hunting and sports persons. Different design and making with different material depends on customer's request.





BOOTS

PIC: 001120

DESCRIPTION

Boots have the main role in both police and military forces. Using of boots in vast missions causes to produce different types of them for different position.

Leather boot, fabric boot and making boots with cordora material are three types of boots.



PERSONAL DEFENSE SPRAY

PIC: 000294

DESCRIPTION

These kind of sprays are used all over the world to control the riots, personal defense, and in a case of danger against invaders.

It is also used as a factor in military training and manoeuvre. The first effect of this product is tearing, instigating eyes and breathing, and scald skin in sensitive parts of the face and body, and at last cause temporary disability.



INSECT SPRAY

PIC: 001106

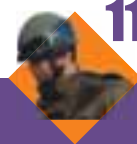
DESCRIPTION

This product made as an spray to use easily and has DI ETILEN TOLO AMID effective material.

That is used to protect open parts of the body from insect biting such as: fly, tick, flea, mosquito,

Once use of this product protect the skin of human body for 4 hours from biting.





ANTI MINE GARMENT

PIC: 000296

DESCRIPTION

Light weight demining clothes is designed to protect against explosive anti-personnel mines which are capable of creating over pressure and fragmentation.

Its flexibility, lightness and high level of protection cause more convenient usage during long and hard times of demining operations.

Compared to its similar clothes ,this clothe without needing to a cooling system ,makes demining operations more possible in hot and humid climates.

BALLISTICS EFFICIENCY

Protection of body against fragment, over pressure, strike and heat of mine blast.

WEIGHT OF DIFFERENT PARTS OF DEMINING ENSEMBLE

- Pants and accessories: 3.856 kg
- Sleeves for protection of hands and arms: 1.770 kg
- Back protection part: 0.896 kg
- Vest and panel for frontal protection; 6.550 kg

VISORED HELMET

A visored helmet is used to neutralize mines. It is a combination of a military helmet and a transparent shield which stands against sharpnel and explosion shock wave for protection of head and face. A visor with 1.9 kg (4.2 bp) weight protects neck, face and forehead and can be attached to the most of the military helmets.

ANTI MINE POLYMERIC & CERAMIC BOOTS

With polymeric or ceramic soles which their physical properties change during mine explosion process leads to reduction in energy transmission.

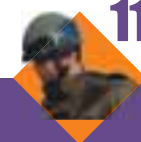
ANTI MINE WEDGED-SOLE

With a construction composed of an angled sole for maximum outwardly diversion of blast waves.

SPIDER BOOTS

The boots maintain a distance between foot and the center of explosion to make energy reduce and divert.





11

SMOKE MINE (YM-I-S)

PIC: 000200

DESCRIPTION

YM-I-S Mine is intended to be used in maneuvers as well as for training purposes. It has a pressure type fuze and when it is activated, the smoke generated for 30 to 40 seconds. The smoke can be in different colors as **red, green, orange and blue.**

SPECIFICATIONS

TYPE	YM-I-S
Case Material :	Plastic (P.C)
Weight (gr) :	160
Diameter (mm) :	90
Height (mm):	45
Weight of Charge (gr):	35
Smoking Time (s):	30 - 40
Operating Force (kg):	7-15
Quantity Per Box:	30
Weight of Box(kg):	11.6



SELF NEUTRALIZED MINE (YM-I-T)

PIC: 001118

DESCRIPTION

YM-I-T Mine is provided with an electronic device which is programmable in factory and is fitted with a safety and arming switch.

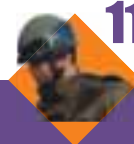
This mine is activated under a pressure of 7-15 kg and at the end of its active life, it will self destruct.

SPECIFICATIONS

TYPE	YM-I-T
Case Material :	Plastic (P.C)
Weight (gr) :	240
Diameter (mm) :	90
Height (mm):	55
Weight of Main Charge:	30 gr C4
Neutralizing Time:	6 months (1 - 4621 hr.)
Operating Force (kg):	7 - 15
Quantity per Box:	30
Weight of Box (kg):	15
Type of Battery:	Lithium (3V)
Delay Time (min.):	10



MISCELLANEOUS



BOUNDING MINES

PIC: 000295

DESCRIPTION

» BOUNDING-MINE (YM-IV)

YM-IV is a bounding mine with a cylindrical cast-iron case and removable fuze mounted on the top. The removable fuze is activated by pressure load of about 7-15 kg and traction load of about 3-8 kg. When the fuze is activated, a detonator ignites the propelling charge causing the body of the mine to be projected about 50 cm into the air and explode.



YM-IV

» FLARE BOUNDING MINE (YM-IV-F)

YM-IV is a flare bounding mine with a cylindrical plastic case. This mine is fitted with a removable trip wire fuze which is activated by pressure load of about 7-15 kg and traction load of about 3-8 kg. When the fuze is activated, a detonator ignites the propelling charge causing the body of the mine to be projected about 100 m. The flare charge fires and comes down slowly by parachute in 20-30 seconds. The flare mine illuminates about 10,000 m² at night.



YM-IV-F

» T-FLARE BOUNDING MINE (YM-IV-F-T)

This is a flare bounding mine with an electronic fuze which by activating, a detonator ignites the propelling charge and mine body is projected into the air about 100 m. The flare charge burns and cases down slowly by parachute in 20-30 seconds. This mine illuminates about 10,000 m² at night.



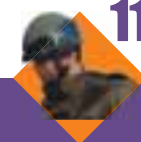
YM-IV-F-T



SPECIFICATIONS

TYPE	YM-IV	YM-IV-F	YM-IV-F-T
Case Material :	Cast iron	Plastic + cast iron	Plastic + cast iron
Weight (kg) :	301	105	1.5
Diameter (mm) :	92	92	92
Total Height (mm):	220	220	225
Weight of Main Charge (gr):	550 TNT	250	250
Jumping Height (m) :		80 -100	80 - 100
Fuze Type :	Pressure & tension	Pressure & tension	Electronic
Delay Time (8min) :			3
Flare Duration (s) :		20 - 30	20 - 30
Quantity per Box (pcs) :	4	4	4
Weight of Box (Kg):	18	12.5	12





CLAYMORE MINES

PIC: 001119

The Claymore mine fires steel balls, out to about 100 meters within a 60° arc in front of the device. It is used primarily in ambushes and as anti - infiltration device against enemy infantry .

It is also of some use against unarmored vehicles. This mine is activated by either detonator wire or electronic timer fuze or remote control. These mines produce in two types which have different specifications.



M18A1

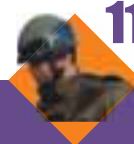


M18A2

SPECIFICATIONS

TYPE	M18 A1	M18 A2
Case Material :	Plastic (ABS)	Plastic (ABS)
Total Weight (kg) :	1.450	5.7
Size (mm) :	220 X 125 X 55	300 X 190 X 100
Weight of Main Charge (gr):	550 C4	2700 COM.B
Fragment :	510 (6 mm steel balls)	2300 (6 mm steel balls)
Effective Radius (m) :	30	50





ANTI TANK & VEHICLE MINES

PIC: 000297

DESCRIPTION

» LIGHT ANTI TANK MINES (YM-II)

YM-II is a light anti tank mine that can be scattered from helicopters or laid mechanically from a vehicle or manually in the ground to a depth of 10 cm. The small size of mine allows easy camouflage on the ground. The mine is provided with a double anti shock device operating mechanically and pneumatically. This mine is resistant to the explosions occurred by other mines close to it.



YM-II

» LIGHT ANTI TANK MINES (YM-II-T)

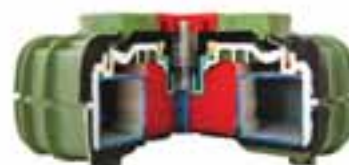
YM-II-T mine is provided with an electronic device which is programmable in factory by a setter and is fitted with a safety and arming switch. This mine is activated under a pressure of 150-300 kg and at the end of its active life it will self destruct.



YM-II-T

» ANTI VEHICLE SMOKE MINE (YM-II-S)

YM-II-S anti vehicle smoking mine is physically similar to the YM-II and can be scattered from helicopters and laid mechanically from a vehicle or manually. It is fitted with detonator and smoking charge. The detonator is stored separately and inserted into the mine before laying.



YM-II-S

» ANTI TANK MINE (YM-III)

YM-III anti tank is a minimum metal blast mine. This mine is provided with an anti-shock device. The pressure plate of the mine is a slightly domed and requires a pressure of between 450-900 kg to move downward.

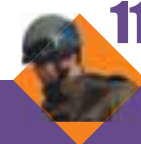


YM-III

SPECIFICATIONS

TYPE	YM-II	YM-II-T	YM-II-S	YM-III
Case Material :	Plastic (ABS)	Plastic (ABS)	Plastic (ABS)	Plastic (ABS)
Weight (kg) :	3.4	3.2	1.6	6.9
Diameter (mm) :	232	232	232	270
Height (mm):	90	90	90	110
Weight of Charge (kg):	2 Comp . B	1.6 Comp . B	350 Smoke	5.7 Comp . B
Operating Force (kg) :	150 - 300	150 - 300	100 - 180	450 - 900
Effective Radius (m) :	30	30	-	100
Quantity per Box (pcs) :	5	5	5	4
Weight of Box (kg):	24	23	17	37





11

ANTI HELICOPTER MINE

PIC: 001104

DESCRIPTION

The anti-helicopter mine is designed to defeat targets by cast iron fragmentation at ranges of about 50 m. The mine can be used to defend military and civil installations against attacks of helicopters and to defend sea side area, where helicopter-landing is possible. **J-AHM** can also be used as bounding mine against personnel & vehicles. In this case the mine is laid with laying angle of 35°-45° in a distance of 250 to 300 meters from target.

SPECIFICATIONS

TYPE	J-AHM
Case Material :	Cast iron
Weight (kg) :	6.2
Diameter (mm) :	112
Height (mm):	177
Weight of Charge:	1.2 kg T.N.T
Effect:	Fragmentation
Jumping Height (m):	150 - 180
Remote Control Distance (km):	5
Effective Radius (m):	50

J - AHM



MOBILE BAKERY

PIC: 000298

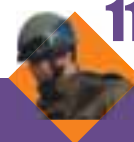
DESCRIPTION

The mobile bakery as tracker with capability of baking different kinds of bread is used in crisis and maneuvers with following features.

FEATURES

- Capability of baking bread
- Baking bread with indirect fire
- Having 6 rollers widening with inventor
- Decrease of bread waste
- Equipped with heating and cooling systems (split)
- Having paste mixer and bread preparation system
- Consumption of oven fuel, every hour 5 liter
- Container sizes : 2.7 x 2.5 x 13.5 m
- Thickness of wall insulator : 4cm
- Capacity of fuel reservoirs : 700 and 1300 liter
- Capacity of water reservoirs : 1500 liter
- Having water heater and diesel generator to supply electricity





FIELD KITCHEN SYSTEM

PIC: 001117

DESCRIPTION

This machine is portable and capable of cooking food for 200 individuals and uses fuels such as petroleum / gas oil, and liquid gas

APPLICATION

- In crisis for establishment and usage in settlement camps
- Applicable in large communities such as maneuvers, recreation and pilgrimage camps , etc.

FEATURES

- This system is capable of carrying with appropriate thill trackers
- Cooking in every climatic conditions
- Cooking in one place and distribution of hot food in another region
- Suspension system by joint rubbers (compressed polymer)
- Capability of modules separation and independent baking



MOBILE KITCHEN SYSTEM

PIC: 000299

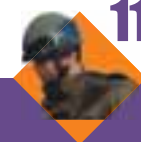
DESCRIPTION

The mobile kitchen system can be used to prepare food for upto 600 people .This kitchen has the capability of storing food which includes below zero refrigeration facility with a capacity of 200 kilograms meat, chicken, fish, butter, etc.

It is also equipped with a pantry with the capacity of storing 2.5 tons of dry foods such as rice, grains and canned food.

This kitchen is equipped with a jumbo streak with a jumbo streak with the capacity of 24 tons, 13.5 meters length and 2.6 meters wide. The distance of the floor from the ground is 130 cm and is equipped with two German 12 ton axes. Two double plied tanks with a fuel capacity of 1500 liters and a water storage tank with the capacity of 2000 liters and an 11 kilogram liquid gas capsule box with the capacity of 12 capsules are installed underneath the floors. In addition a sewage system has been integrated on the floor of the trailer .





WATER PURIFICATION SYSTEM

PIC: 001108

DESCRIPTION

The water purification system (**WPS**) is a self dependant treatment unit which can provide treated or desalinated treated water from brackish or sea water for battle fields. The desalination technology is based on Reverse Osmosis (RO) technic with use of high quality membranes. The quality and flow rate of input and output water is monitored and controlled from input and output control panel.

The unit is mounted on mobile chassis for displacing from area to area. The required power can be supplied from built - in diesel generator or from power line.

SPECIFICATIONS

TYPE	WPS
Total Dissolved Solid in Raw Water (mg/lit):	45,000 - 48,000
Total Dissolved Solid in Product Water (mg/lit):	300 - 500
Treatment Capacity (lit/hr):	5,000
Desalination Capacity (lit/hr):	1,000
Raw Water Temperature (°C):	15 - 45



FIELD ICE MAKING UNIT

PIC: 001127

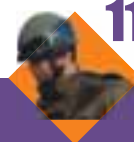
FEATURES

- Production capacity in day and night: 500kg
- Capacity of water reservoir: 600 liter
- Capacity of ice reservoir: 500 kg
- Quality of interior wall sheet: Niger sheet
- Quality of exterior wall sheet : Steel
- Equipped with tent and its light
- Equipped with generator

This unit is used in crisis by establishing in camps and also in maneuvers for producing cubic ice.

This device is mobile and portable with the capacity of producing 500 kg ice every 24 hours.





3×1 EXPANDABLE SHELTER

PIC: 001105

DESCRIPTION

Field hospital operating room, radiography, laboratory, etc. Field clinic, Commander post, Controlling room, Monitoring room, Meeting room, Prayer room and Situation room are some of usages.

FEATURES

- Rapid loading/unloading
- Easy carriage with all types of trailers (due to its standard dimensions (20 foot container).
- The capability of assembling all types of required equipments related to shelter's application type.
- Rapid assemblage/disassemblage of the shelter by two people manually, no need for any special equipments.
- Providing a flexible space with a 37 sq meters space.
- Increasing the mobility and operational speed of warring and supporting units.



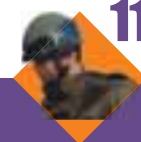
MOBILE GENERATOR

PIC: 001126

SPECIFICATIONS

TYPE	MOBILE GENERATOR
Length (mm):	3,010
Width (mm):	1,850
Height (mm):	2,450
Gross Weight (kg):	2,060
Suspension System:	Pick – up
Brake System:	Mechanical – Manual
Motor:	Disel Engine
Qty of Cylinder:	4
Starter:	12 V
Generator:	Stamford
Model:	BCI 184E1
Output Power (kw):	10
Output Voltages:	110V, 60HZ, 1,800 rpm 220V, 50HZ, 1,500 rpm



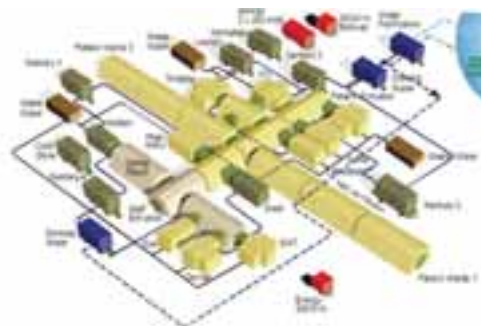


MOVING HOSPITAL

PIC: 000176

DESCRIPTION

Moving hospital is a collection that produces health services holding the capability of moving and shifting from a place to another. Moving hospitals present the condition of a real hospital with necessary equipments and potentialities in the manner of medical and remedial teams that can present all necessary medical and health services at minimum time and distance at the result of events and exigency.



AGENT IN THE FORM OF COMBINATION IS

- Flexible; consists of the veils with metallic and pneumatic frames
- Inflexible: consist of 20 feet standard & 20 feet expandable containers.

FEATURES

- Capability of easy and fast erection (installation).
- Capability of transmission by means of the road transportation , ships , train or plane.
- Supplying all medical , establishments and energy establishments.
- Possibility to stand up in different ecological conditions.
- Possibility of activity in polluted zones.
- Flexibility in arranging in respect to zone's position.
- Capability of extension and expanding of the hospital and increase of the beds.



PNEUMATIC BRIDGE

PIC: 000177

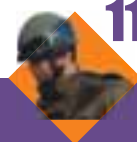
DESCRIPTION

The 40 meters bridge is a kind of quick fixing bridge in air class that has lower volume and lighter weight in comparison with other quick fixing bridges. This kind of bridge and all of its fixing equipments and cluster piles are portable with a van. This bridge is fixable and controllable with the help of only 8 people in 15 minutes on the surface of the stagnant water. This period increases with the speed of 1.5 meters per second to 40 minutes for the flowing water. If the bridge is decided to be closed up , the same time is needed.



SPECIFICATIONS

TYPE	PNEUMATIC BRIDGE
Length (m):	40
Width (m):	1.4
Width Efficiency (cm):	80
The Suspension Capacity of the Bridge (kg):	80 people (every person 100kg)
Type:	Reinforced PVC
The Weight of the Bridge (kg):	About 200
The Dimensions in Packaging Form (cm):	120 × 120 × 140
Blower System:	Subterfuge fan 1200 m3/and the pressure 4 - 6 kpa
Fan Force:	A 8 - horse power petrol engine
The Weight of the Blower Package (kg):	110
Bridge Controller System:	Steel spark plug



BATTLE DRESS UNIFORM

PIC: 001111

DESCRIPTION

The battle dress uniform is the first protective mean of a warrior. Nowadays, thanks to using new innovations, soldiers garment is an effective self defense weapon. Camouflage against night vision devices, flame protection, antimicrobial finishing, etc. are key elements of survival. The uniforms are made of different fabrics such as 100% cotton, poly/cotton & PA/cotton.

FEATURES

- Comfort
- Soft hand feel
- Extra durable
- Ergonomic design
- Sizes: S,M,L,XL & XXL
- Different colors & camouflage pattern
- Good color fastness against washing, rubbing, perspiration & light
- Anti IR camouflage (optional)
- Anti microbial finishing (optional)
- Flame retardant (optional)
- Soil/oil/water repellant (optional)



GROUP TENT

PIC: 001125

DESCRIPTION

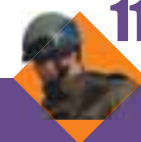
The group tent with beautiful design is an effective mean for settling in war and peace.

Apart from war, it provides a comfortable place for people suffer from phisical and psychological stresses in critical conditions.

FEATURES

- High resistance against wind
- Rapid set up
- Water proof strong duck fabric
- Robust metal skeleton
- Using CAD-CAM technologies in production
- Water proof floor covering
- Ability of installing lamp inside tent
- Easy transport
- Low volume of package
- Using engineering software's in skeleton design
- By connecting the modular type tents together, bigger space is achieved





11

AIRCREW SUIT

PIC: 001130

DESCRIPTION

The **Aircrew Coverall** suits are available from **Aramide** fabrics such as **Nomex** or flame-retardant non-aromatic fabrics.

The coverall is a one-piece unlined garment with a slide fastener front closure and side pass through, a bi-sewing back hook and pile fastener tape adjustments for the waist and sleeves.

FEATURES

- Comfort
- Knife rope
- Flame proof
- Beautiful design
- Easy movement
- Antistatic properties
- Machine washable
- Sewed with aramide thread
- 9 pockets with zipper closure
- 10 main sizes each includes 3 drapes
- According to Iranian defense standard **IDS 441**
- Durable metal zippers with inflammable aramide tape



TANK CREW SUIT

PIC: 001114

DESCRIPTION

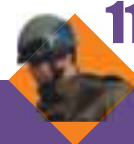
This coverall is made of flame retardant fabric, based on **Aramide** or flame retardant finished fabrics, such as **PES /CO, FR-viscose/PES .etc.**

FEATURES

- Comfort
- Flame proof
- Easy movement
- Beautiful design
- Antistatic properties
- Machine washable
- Sewed with aramide thread
- 9 pockets with zipper closure
- 10 main sizes each includes 3 drapes
- According to Iranian defense standard **IDS 441**
- Life strap for taking out the infected crew from tank
- Durable metal zippers with inflammable aramide tape



MISCELLANEOUS



INDIVIDUAL FOOD RATION

PIC: 001124

DESCRIPTION

Individual food rations, in military and civilian types and individual daily packages , snack and single meals.

FEATURES

- long maintainability of the product for 2 years
- Compactness and flexibility of the products that result in transit and storing easiness
- Maintenance in different geographical and climate conditions and no requirement to springhouse
- Ease of application by consumers

DIFFERENT MENUS OF THE PRODUCTS

- Main packages as daily and snack basis
- Main meals include: breakfast, lunch, dinner
- Supplements include: different snacks usable all day long
- Packages of supplies include: necessary instruments and facilities such as spoon, tissue paper, toothpick and so
- Self heater bags, flame needless



LIFE PILL

PIC: 001131

DESCRIPTION

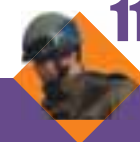
A new phenomenon in the production of food rations. The best nutrition with the least volume and weight able to supply enough nutrition for one person for 15 days stored in room temperature. A complete meal in the form of a pill for long lasting.

In emergency situations , during crises , when there is no access to food for any reason, during unexpected incidents and especially for military men and soldiers, pilots, athletes and mountaineers.

Daily consumption ; if used for survival and vital needs of the body, 12 pills a day is recommended (in case of access to any other source of nutrition this number will decrease).

Packaging; in plastic bottles with screw caps impenetrable to light and oxygen with labels containing information on the compositions , storage , consumption doses and usage instruction etc.





11

ADVANCED FIRST AID BAG

PIC: 001112

DESCRIPTION

Applying advanced first aid equipment, utilizing modern technologies that could protect and save life critical and special conditions of battlefields, which requires abilities to deal with conditions prevailing in these facilities.

For this purpose, advanced first aid bag is provided, having necessary tools and facilities to stop bleeding, sterilizing, quick dressing of deferent cuts, cleavage and other wounds.



FEATURES

- Block Bleeding powder for blocking vast bleeding
- Nano silver sterilizer spray for complete destroying of pathogenic microbes
- Super absorption gluey stomach pad, to absorb secretions and dressing stomach wounds
- Compressed block bleeding gas, to block and dress the wound
- Blocking tampon for nose bleeding
- Gluey crystal bandage to protect the wound against microbial factors and air intake in chest wounds
- Stitch special paste, gas, vaseline gas, scissors, paste, abs lung, emergency drugs and guidance booklet.

FOLDING STRETCHER

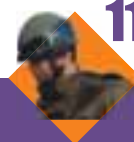
PIC: 001136

FEATURES

- For carrying the injured from damaged sites and critical areas
- Lightweight (about 6.5 kg)
- Transportability by individual as a bag or backpack
- Folding and not bulky
- Well occupied the automobile trunk
- Applicability in military and civil areas
- Applicability in therapeutic centers, ambulance and so
- Color variety and capability of camouflage
- Load tolerance up to 150 kg



MISCELLANEOUS



BOTTLE OF WATER

PIC: 001123

DESCRIPTION

This product consists of a bent bag and other related accessories. The bottle (bag) is used for storing and conveying water for one person.

Some of the specifications of this product are: making use easy, small size in the form of packing, low weight and easy for transportation, having proportion with physics of the body at the time of carrying, resistance against weather conditions, resistance against UV rays and the most serious specification that the components of this product are so hygienic (health).



This product possesses the ability of storing water without any change in its quality for a period of 48 hours. You can use this product in military and non-military services or even in polluted environmental conditions.

SPECIFICATIONS

TYPE	WATER BOTTLE
Water Volume (Liters):	2
Type of the Bag:	PVC
Dimensions of the Bag (cm):	25.5 × 22.5
Weight of Churn in Empty Mode (gr):	450
Weight of Churn in full of Water Mode (gr):	2,400

NMK 100 (MARKER)

PIC: 001128

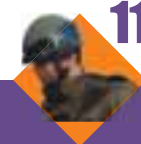
DESCRIPTION

It is a colorful powder, by scattering on the surface can cover a very large surface and making the area distinguished.

The most important application on this product all over the world is used for surviving and helping operation in marine events.

Also it is used in icy and snowy places. There are some more usages for this product in similar places too.





BULLET PROOF GLASS

PIC: 000766

DESCRIPTION

The bullet proof glass provides high level security glazing with a defined resistance against the firing of specified weapons and ammunition from handguns to armor piercing rifle. The bullet proof glass is a composite of asymmetrical glass clad polycarbonate sheet.

This glazing is specially designed to absorb both the impact and the enormous energy of a projectile and to contain it within the glass panel.



THIS PRODUCT HAS A VAST ARRAY OF POTENTIAL APPLICATIONS INCLUDING:

- Provide protection to government , diplomats and politicians.
- Government buildings, museum, banks, VIP buildings, airports, petrol filling stations, cash desks and personnel protection screens in banks.
- Emergency and public safety vehicles.
- Ambulances and fire engines.
- Civil defense vehicles.
- Military installations.
- Cash-in transit vehicles.

FEATURES

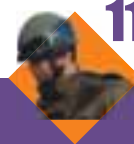
- No-spall
- Meet different levels of **EN 1063** from **BR1** to **BR7** and other standards
- Maximum transparent and minimum distortion
- Different size, shape and color
- Moisture protective edge for long service life in extreme weathers
- Flat and bent from according to customer request



SPECIFICATIONS

CLASSIFICATION, REQUIREMENTS FOR 'NO SPALL' BULLETPROOF GLASS

CATEGORIES OF RESISTANCE	TYPE AND CALIBER OF ARMS USED	APPLICATION	THICKNESS (MM)	WEIGHT (KG/M2)
BR1	Rifle 0,22 LR	Public administration buildings, mansions	14	27
BR2	Gun 9 mm Luger		17	34
BR3	Gun 0,357 Magnum	Telecommunication and computer operators	19	39
BR4	Gun 0,44 Magnum		27	39
BR5	Rifle 5,56	Buildings open to high risk of terrorist attacks or robbery, cashiers' compartments	34	72
BR6	Rifle 7,62	Military facilities, penitentiaries	42	88
BR7	Rifle 7,62	Military facilities and other high risk facilities.	79	173



MULTI-SPECTRAL CAMOUFLAGE SCREEN

PIC: 001113

DESCRIPTION



Multi spectral camouflage screen
(visible, near infrared, thermal and dual-purpose radar stationary and mobile).

- To Camouflage targets in the range of visible wavelength of 400-700 nanometers.
- To Camouflage targets in the range of near infrared wavelength of 700-1200 nanometers.
- To Camouflage targets in the range of Thermal infrared wavelength of 3-5 and 8-14 micrometers.
- To Camouflage targets in the range of radar X-band (8-12 GHZ)

MULTI SPECTRUM CAMOUFLAGE CLOTH

In a modern combat scene, it is necessary to observe without being observed.

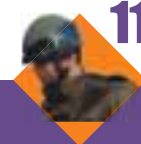
Also, it is essential to approach enemy lines before being recognized. Ordinarily, soldiers who wear thermal cloth and compatible with their back ground (emit less heat) will have more survivability.



MOBILE CAMOUFLAGE FOR 23 MM CANNON

- 360° rotation during operation
- Ability to reduce being detected and recognized by visible, night vision, thermal cameras and radar system in the X-band (8-12 GHZ)
- Useable during operations
- Capable of distorting the image





NAKHJIR 1 – HUNTING GUN

PIC: 001122

DESCRIPTION

“NAKHJIR 1” is a break action single barrel shotgun to shoot birds. The butt stock is made of walnut wood. Barrel manufacture technology belongs to the **Germany**. Its material is of highest quality. Barrel is full choke, so its accuracy and range is unique. Inner barrel is of hard chrome-coated to prolong its life & avoid abrasion. It can be supplied in such a way that once it is loaded, moves into safety mode. Engraving & wood carving & coating is custom made and optional.

SPECIFICATIONS

TYPE	NAKHJIR 1
Calibre (mm):	12
Overall Length (mm):	1250
Height (mm):	183
Width (mm):	40
Length of Barrel (mm):	810
Weight (kg):	3
Length of choke (mm):	65



NAKHJIR 2 – HUNTING GUN

PIC: 001128

DESCRIPTION

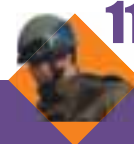
“NAKHJIR 2” is a side by side break-action double barrel shotgun to shoot birds. The butt stock is made of walnut wood. The barrels can also be semi-choke beside full choke barrel to give it different ranges. Barrel manufacture technology belongs to the **Germany**.

Its material is of highest quality. Barrel is full choke, so its precision and range is unique. Inner barrel is hard chrome-coated to prolong its life & avoid abrasion. It can be supplied in a way that once it is loaded, safety mode is automatically applied. Engraving & wood carving & coating is custom built and optional.

SPECIFICATIONS

TYPE	NAKHJIR 2
Calibre (mm):	12
Overall Length (mm):	1136
Height (mm):	190
Width (mm):	58
Length of Barrel (mm):	700
Weight (kg):	3.640
Length of Choke (mm):	65





NAKHJIR 3 – HUNTING GUN

PIC: 001116

DESCRIPTION

“NAKHJIR 3” has a beautiful design & fully ergonomic for hunting. Manufacturing technology of barrel is of **German** origin & barrel is made of stainless steel with 4 right hand grooves which is presented with 270 caliber capable of changes in to **30-06, 308 & 8.57** calibers. The butt stock is made of walnut wood.



SPECIFICATIONS

TYPE	NAKHJIR 3
Gauge (mm):	Win 270 = 6.68
Weight (kg):	3.6
Overall Length (mm):	1,120
Rifle Height (mm):	182
Final Range (mm):	3,000
Effective Range with Optical Site (m):	700
Magazine (mm):	5
Effective Range Without Optical Site (m):	300



SHAHIN – HUNTING GUN

PIC: 001121

DESCRIPTION

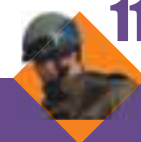
“SHAHIN” is an over-under break-action double barrel shotgun to shoot birds. The butt stock is made of walnut wood along with anti-recoil butt plate. The barrels can be semi-choke besides being full choke to give it different ranges. Barrel manufactured technology is of high precision from **Germany**, its material is of highest quality. Barrel is full choke so its precision and range is high. Inner barrel is hard chrome-coated to prolong its life & avoid abrasion. The barrel can come out in moveable choke. This rifle is equipped with automatic safety & ejector mechanism & can be supplied as a sporting gun for skeet & trap contests with 3 minor changes.



SPECIFICATIONS

TYPE	SHAHIN
Calibre:	12
Overall Length (mm):	1,140
Length of Barrel (mm):	710
Weight (kg):	3.3





11

MINI FLARE & TRIP FLARE (M49A1)

PIC: 001129

DESCRIPTION

To give warning of infiltrating troops by illuminating the field of the advancing enemy, trip flare **M49A1** is recommended.

To use as a distress signaling devise for downed airmen or ground troops or as personal survival, the miniflare is provided.

SPECIFICATIONS

TYPE	MINI FLARE	TRIP FLARE M49A1
Length (mm):	57	151
Diameter (mm):	12.8	40
Weight (g):	14.8	370
Trip Wire Length (m):	-	15
Burning Time (sec):	5 - 7	60
Light Intensity (cd):	2,500 (Green and White) 3,000 (Red)	-
Luminosity (cd):	-	80,000
Illumination Color:	Red, Green and White	Yellow & White
Altitude (m):	75	-
Fuze Delay Time (sec):	-	1.2 - 2



FIRING DEVICES

PIC: 001134

DESCRIPTION

The Firing Devices have been designed to initiate a train of fire or detonation charges, booby traps or mines.

They are of two general types as tubular and box type and are packed as separate items.

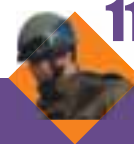
The tubular Firing Devices are **M1**, **M1A1** and **M3**, and the box type is **M5**.



SPECIFICATIONS

TYPE	M1	M1A1	M3	M5
	PULL	PRESSURE	PULL-RELEASE	PRESSURE-RELEASE
Spring Release (lbs):	3 - 5	20	6 - 10	5
Weight (g):	60	100	70	54
Quantity Per Box (pcs):	140	100	140	120
Box Weight (kg):	18	15	19	13
Box Dimensions (mm):	550 × 340 × 180			

MISCELLANEOUS



DESCRIPTION

National Geographical Organization (NGO) was founded in 1921 and started its main activities in the field of geomatic services. NGO successfully has executed both national and international important projects all over the world and at the present time covers a wide variety of services as follows:

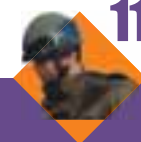
- 1- Producing coverage topography maps in all scales.
- 2- Producing urban large scale maps (1/500, 1/1000, 1/2000).
- 3- Producing Operational Navigation Chart (ONC) and Tactical Pilotage Chart (TPC).
- 4- Producing digital hydrographic charts which are known as Electronic Navigation Charts (ENC).
- 5- Aerial photography with large format digital camera.
- 6- Direct receiving of satellite imagery and image processing.
- 7- Producing Thematic maps.
- 8- Technical services related to Border area and border protocols.

According to the high level of technical capabilities and long duration of successful expertise, NGO presently, is ready to give other countries, following services which they are conformity with both national and international standards:

A- Training services in the field of geomatic sciences and in three levels; General, Advanced and on job Training as follows:

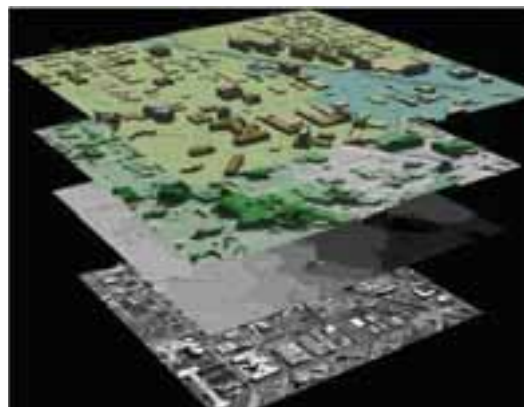
- 1- Geodesy and terrestrial surveying.
- 2- Photogrammetry.
- 3- Remote sensing.
- 4- Hydrography.
- 5- GIS.





B-Technology Transfer by carrying out of following technical services.

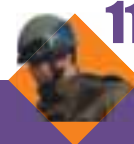
- 1- Designing and implementing digital 3D map production lines via surveying , photogrammetry and Remote Sensing in all scales, based on Geo Database.
- 2- Designing and implementing Aerial Digital Photography and Laser Scanning procedure with man and unman aerial vehicles and Airborne Sensors for photogrammetry purposes and other applications .
- 3- Establishing Fixed and mobile direct receiving Ground station of Satellite imagery and Remote Sensing applications .
- 4- Establishing hydrographic center for producing paper charts, Electronic Navigation Charts (ENC) and other related services like nautical parameters analysis.
- 5- GIS application .
- 6- Design and Implementation of Geospatial Database .
- 7- Design and Implementation of Geoportal and Web Mapping.
- 8- Design and Implementation new Geodetic Networks and related adjustment and analysis.



C- Presenting and Publishing Geomatic products and services as follows:

- 1- Topography maps.
- 2- Operational Navigation Chart (ONC) and Tactical Pilotage Chart (TPC).
- 3- Paper and Electronic Navigation Charts (ENC).
- 4- Aerial and Satellite images and Data.
- 5- Thematic maps.
- 6- Geographical Atlases.
- 7- Lithography services.





ROCKET FLARE (MK5)

PIC: 001132

DESCRIPTION

This product is provided to illuminate the target at night and identify the own or alien forces from various distance and to use as a movable target in maneuvers or training operations.

The rocket is launched from a launcher mounted on the warship and can be adjusted in different angles from 5 to 45 degrees in elevation and from **0** to **360** degrees in azimuth, due to its rotary launcher.

SPECIFICATIONS

TYPE	MK5 - 2" FLARE ROCKET
Length (inch):	51 (130cm)
Weight (lb):	13 (6 kg)
Caliber (inch):	2 (5.08 cm)
Illuminating Material (gr):	435
Illuminating Intensity (cd):	Min. 200.000
Burning Time (sec):	Min. 70
Type of Fuel: (mm):	Solid - Double
Weight of Fuel: (mm):	1200
Fuze:	Electrical
Delay Time (sec):	3.5 ± 0.5
Ignition:	Electrical
Max. Range (yard):	5,000
Rote of Descent (m/s):	Max. 6
Armament:	MK5 Launcher



FLARE TARGET (MK2)

PIC: 001135

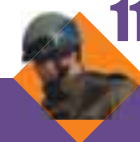
DESCRIPTION

The Flare Target (**MK2**) is a flare device to be used on RPV to make it visible at night and considered as air target for training purposes. It is fired by the 24V battery by remote control.

SPECIFICATIONS

TYPE	MK2
Weight (kg):	1.250
Diameter (mm):	44
Length (mm):	430
Burning Time (sec):	60
Packing:	6 in a Woodem Box
Intensity of Light (cd):	250.000
Ignition:	Electrical





STS-2 FIREARM SHOOTING SIMULATOR

PIC: 000764

DESCRIPTION

Firearm shooting simulator designed and produced for different types of rifles.

By **STS-2** personnel start to use and get familiar with different firearms like **AK-47**, **Dragnov** and **Pistol** and increase their skills on speed, aiming and accuracy. On the other hand, instructors can exam shooters by different training and find out best shooters.

STS-2 has designed and made for train five shooters of firearms simultaneously. **STS-2** software has based on target range and Battlefield simulation.

TARGET RANGE SIMULATOR PROPERTIES

- 3D fix and mobile target simulation
- Definable shooter and weapon
- Changeable target shapes
- Movement ability for targets in each scenario
- Friend and foe target definition
- Ballistic based bullet movement (real shooting conditions like wind, ...)
- Printable result report
- Graphical hit points display
- Hand movement diagram report
- Definable hostage target
- Negative score for shooting friend target
- Trainable five person simultaneously

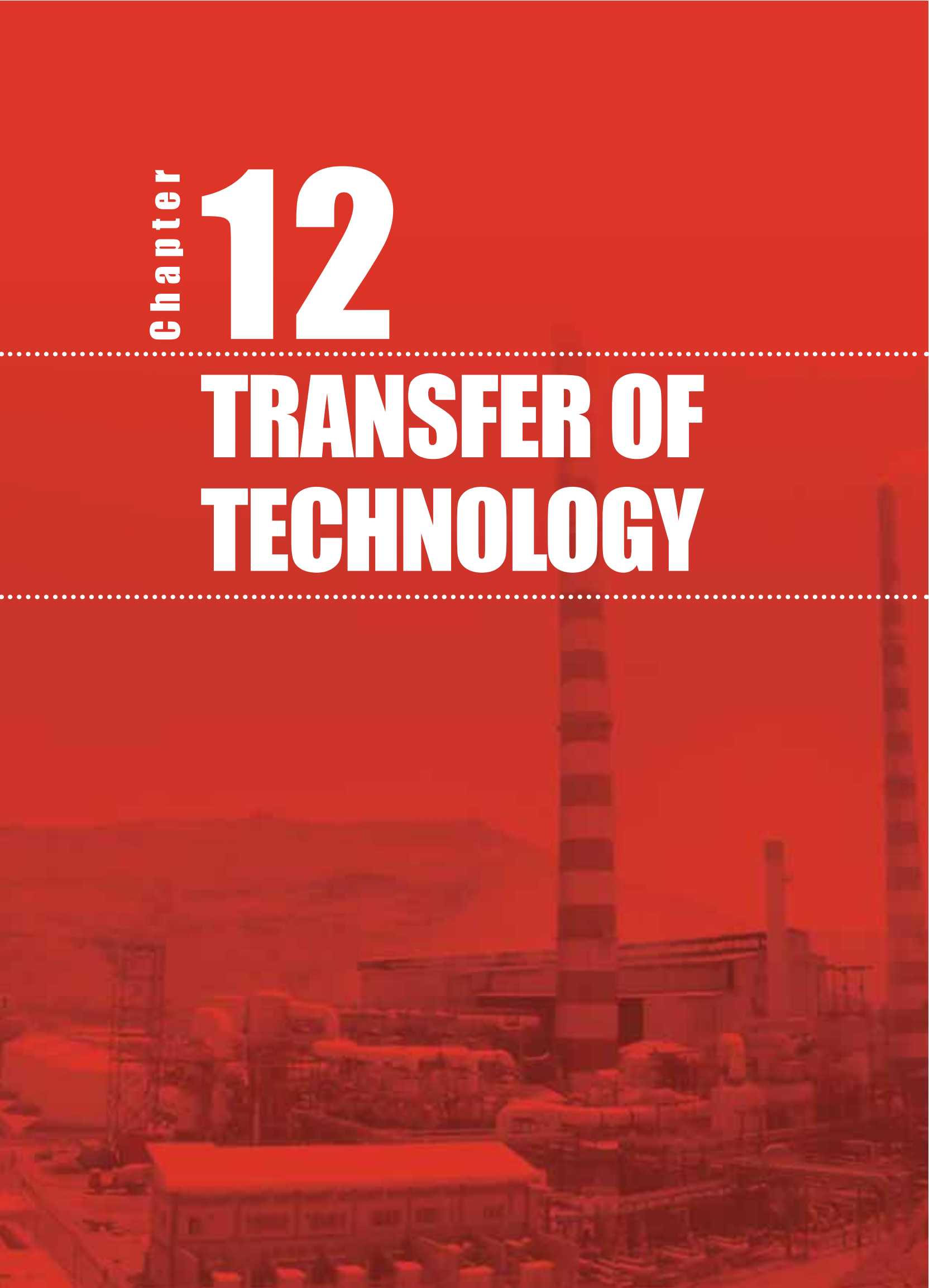
BATTLEFIELD SIMULATOR PROPERTIES

- Definable different 3D scenario by instructor
- Ability to use predefined scenarios
- Shoot back simulation
- Definable fire line with shooters and their weapon up to five person
- Realistic 3D models: Soldier, Tank, Plane, Chopper
- Definable scenario run time (Day/Night)
- Definable different terrain (City, Jungle, Desert,)
- Definable weather condition (rain, snow,)
- Shooters enrollments
- Mobile fire line
- Report workbook and scores
- Display target distance to fire line
- Train five person simultaneously



chapter 12

TRANSFER OF TECHNOLOGY





By employment of experienced personnel having the expertise in various engineering skills as well as using all types of machineries required to execute the customer's required projects, and upon elimination of prevailing UN sanctions against Islamic Republic of Iran, defense industries under M.o.D. are well prepared to render high services in technical engineering procurement, construction, installation, commissioning, training, start-up and operation of large military and civil projects in the form of EPC contracts.

Concerning military projects, the long experience of operating different production lines with wide range of products and applying active interaction with famous technology owners as well as research and optimization of production plants, all have enabled the Ministry to possess the key of the technology.

This capability presently covers turn-key transfer of technology for the production of over thousands items and materials including various types of defensive and offensive equipments, such as; tanks, ground forces vehicles, APC's, guns, anti-air defense systems, missiles, marine equipments, aircrafts and helicopters, communication systems, optics, satellites, infantry equipments, chemicals, explosives and raw-materials and more.

Executive method of related industries to transfer technology is based on project management standards with consideration of the latest improvement made on customer contracted production lines.





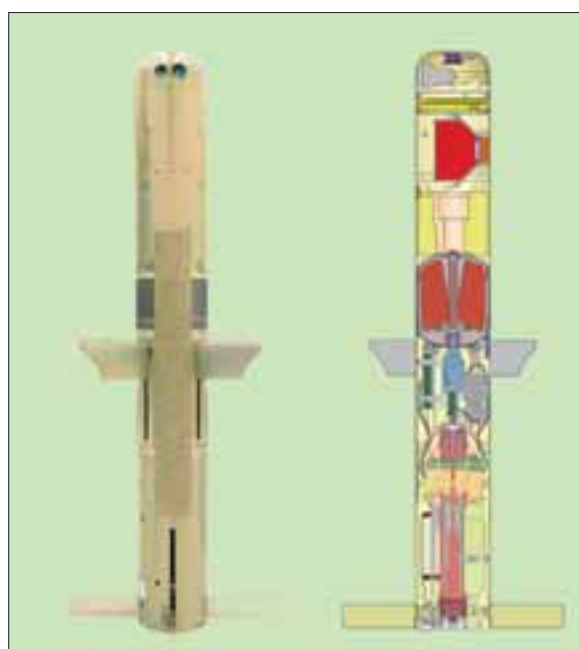
Complying with engineering principles, we pay serious attention to quality promotion, safety and automation of production lines to optimize total product cost by applying the latest improvements on production technology .

A Turn Key factory, the scope may be as follows:

- Know how and engineering design
- Procurement and industrial construction
- Building construction
- Installation and industrial execution
- Training
- Technical assistances
- Start up, trial run, performance test and line deliver

If the customer possesses enough facilities and capability in building construction, installation and commissioning, then the providing industry has just an obligation to supervise the performance of such activities .

Upon customer's request, a technical commercial proposal of the production line would be submitted accordingly . This proposal consists of three separate section including technical sections, commercial section and time schedule . These are so adjusted that the customer can easily realize what services at what cost and when would be provided .





TRANSFER OF TECHNOLOGY

Definition of deliverable services in the field of technology contracts are based on standard models so that all characteristics of a suitable technology are met.

In the engineering phase, the Document Register of the project will be prepared initially and delivered to the customer so that his comments on the types and quality of the design documents can be considered .

Designing is down in two phases; Basic and Detailed. The manuals for commissioning, utilization and quality control are also provided in the engineering phase.

In the procurement phase , the sources are selected from an approved Vendor List, their documents are examined carefully, authenticated and all manufacturing stages would go under supervision and monitoring.

In addition to theoretical training , the practical one is similar to the actual factory so that the customers'





TRANSFER OF TECHNOLOGY

experts are well prepared for operating the real production line .

Provisional and final commissioning is determined by concerned supervising department and after the final performance test, the production line is delivered to the customer .

The warranty period services will be defined and given as per the contracts' conditions after commissioning .

The main strategy in technical engineering contract is to keep credibility and good reputation through enhancement of the customer's satisfaction.



Alphabetical Index

0.38 S&W	5	203 mm How - HE	36,62
0.40 S&W	4	240 mm Rocket	153
0.45 AUto	4	240 mm Naval Rocket Launcher	111
5.56 X 45 mm	64	333 mm Rocket	154
7.62 X 39 mm	5,64	3/4 Ton Tactical Vehicle	131
7.62 X 51 mm	5,52,64	1/4 Ton Tactical Vehicle	131
7.62 X 54 mm	5,65	1- 1/4 Ton Tactical Vehicle	132
9 X18 Makarov	4	3X1 Expandable Shelter	318
9 X19 Luger	4,53	10 M- Multi Purpose Patrol Boat	286
12 mm Anti Riot	7	12 M- Multi Purpose Patrol Boat	286
12.7 X 108 mm	6,54	12 M- Rocket Launcher Multi Purpose Patrol Boat	289
12.7 mm Sniper Rifle	90	13 M - Multi Purpose Patrol Boat	288
20 X 102 mm Ammo.	8	14M - Multi Purpose Patrol Boat	290
20 X 128 mm Ammo.	8	800 & 2000 Tons - Floating Docks	295
23 X 152 mm Ammo.	9		
23 mm Single Barrel A.A Gun	94	A	
23 mm Twin Barrel A.A Gun	95	Ababil 2 - UAV System	270
30 X 165 mm Anti Aircraft	11	Ababil 3 - UAV System	270
30 mm Naval Ammo.	10	Advanced First Aid Bag	323
30 mm Aircraft Ammo.	12	Aerial Target System	268
35 mm Ahti Aircraft Ammo.	13	Air Defence Missile	164 - 165
38 mm Multi Purpose Gun	91	Air Surveillance Radar	203 - 204
40 mm HE & Training Grenade	37	Air Target OKM Imitator Rocket	143
40 mm Naval Ammo.	14	Aircraft Ammunition	12,8
40 mm L60 Seven Perforated	55	Aircraft Manufacturing Industries	254
40 mm Grenade Launcher	92	Aircraft Tires	273
40 mm Automatic Grenade Launcher	92	Aircraft Windshield	272
40 mm Marine Gun	109	Aircrew Suit	321
40 mm Anti Armor Rockets	142	AK - 47 Assault Rifle	85
40 mm Anti Personnel Rockets	143	Ammunition Fuzes	41,42
60 mm Mortar Bomb/ Launcher	15,99	Anit Ship Missile	166 - 168
60/81/120 mm Split Increment Charge	68	Aniti Bullet / Fragment Vest	302
73 mm SPG9	102	Anti - Riot Cloth and Equipment	305
76 mm Naval Gun	109	Anti Aircraft Ammunition	8-11,13
76.62 mm Naval Ammo.	55,21	Anti Bullet Blanket	303
81 mm Mortar Bomb / Launcher	16,17,100	Anti Bullet Briefcase	303
100 mm Anti Aircraft Ammo.	21,67	Anti Bullet Cover	304
105 mm Anti Tank Ammo.	22,70	Anti Bullet Portable Fort	304
105 mm How- HE & III.	27,28,56	Anti Bullet Vest (Police Force)	303
106 mm . HE & HEAT Ammo.	23	Anti Helicopter Mine	315
107 mm Rocket Launcher	104- 106	Anti Mine Garment	310
107 mm Rocket	151	Anti Personnel Rocket	143
120 mm Mortar Bomb / launcher	18,19,20,101	Anti Riot Cartridges	7
120 mm Tank Ammo.	24	Anti Riot Helmet	306
122 mm Tank Ammo.	24	Anti Riot Shield	306
122 mm Howitzer Gun	108	Anti Riot Shotgun	308
122 mm How Ammo.	29,30,31,57	Anti Tank Ammunition	26
122 mm Rocket Launcher	107	Anti Tank and Vehicle Mines	314
122 mm Rocket	152	Anti Tank Rocket (RPG7)	142
125 mm HE & HEAT	25,58	Arash - 122 mm Rocket	152
125 mm APFSDS - T	26	ARC - 1000 UHF Airborne Communication	187
130 mm HE & BB	32,60,66	ARC - 160 UHF Airborne Communication	186
130 mm Smoke & III.	33	Armor Training Simulator	133
152 mm Reduced Charge	59	Armored Bulldozer Mine Cleaner	137
155 mm How - Ammo.	34,35,61	Armored Personnel Carrier (APC)	118
155 mm Base Bleed	71	Atomil	247
155 mm Howitzer Gun	108	Aviator Night Vision Goggle	226

B

Ballistite	67
Basic Toophan - Heavy Anti Armor	145
Baton & Tonfa	306
Battle Dress Uniform	320
Battlefield Automatic Telephone	174
Binoculars	220 - 221
Black Powder	76
BMP2 APC	121
Boots	309
Boraq APC	118
Bottle of Water	324
Bounding Mine	312
Boxy Launcher	155
Bridge	128 , 319
Bullet Proof Glass	325

C

Cannon Barrels	112
Car barrier	307
Chemical Factor Detector	238
Chieftain MBTS	128
Claymore Mines	313
Command Cabin	161
Composition A	72
Composition B	73
Composition C4	73
Crisis Management Network	201
Customized EW Command & Control System	214

D

D3 Capsule	245
Decoderm	248
Decomil	248
Decontamination Vehicle	243
Detector Kite	239
Detonators	44,45
Digital Dosimeter	239
Double Base Propellant	63
Drakhsh 7	243
DSM 2500 , 3500 & 5000	244
Dushka 12.7 mm Machine Gun	90
Dynamite	77

E

EITF Fuze	43
Electro Optical Surveillance System	231
Elint	213
Encryption System	189
EOVP4 - Aerial Camera	233
Expandable Shelter	318

F

F61 Fuze	43
Fadak 8 - Laser Range Finder	222
Fadjr - 27 Naval Gun	110
Fadjr 1 - 107 mm Rockets	151

Fadjr 2 & 3 - 240 mm Rockets	153
Fadjr 3 & 5 - Boxy Launcher	155
Fadjr 3 & 5 - Loading Machine	156
Fadjr 5 - 333 mm Rocket	154
FAEZ Aircraft	259
Fajr 20 - Ultra Light Aircraft	259
Fajr 3 - Composite Aircraft	258
Fajr Aviation Industries	257
Falag 1 &2 Rockets	157
Fast Boat	283 - 284
Fateh 110 - Surface to Surface Rocker	160
Fath - 40 mm Anti Armor Rockets	142
FCS - 5030 Facsimile Ciphering	191
Fetter &Thumbcuffs	307
Field Ice Making Unit	317
Field Kitchen System	316
Filter	242
Firearm Shooting Simulator	333
Firewall	194
Firing Devices	329
Flare Target	332
Flat & Tanker Barges	296
Flight Simulator	275
Floating Docks	295
Folding Stretcher	323
Folding Stretcher	323
Frigate	282

G

G3 Rifle	86
GBR - 1000 UHF Airborne Communication	185
Geomatic Services	330
Ghadir - Anti Anti Armor Rockets	144
Ghadir 4 - Rifle Scope	222
Grenade	38,39,46,47
Grenade Launcher	92
Group Tent	320
GSM , Dcs & 3G Jammer	199

H

Hand Grenade	38
Hand Help Binoculars	220
Hand Help Direction Finder	212
Handcuffs	307
Helmet	304 , 306
HESA - Aircraft Industries	254
Hexogen (RDX)	75
HF Ground Radio Communication	178
HF Radio Monitoring	210
HF Transceiver	181
High Explosive	72
High Speed Patrol Boat	287
Hovercraft	297
Hunting Guns	327 - 328

I

IF2 Filter	242
------------	-----

IFF Identification Friend or Foe	188
IHSRC - IRAN Helicopter Support & Renewal Company	263
IM - 800 Professional Mobile Radio	182
Individual Food Ration	322
Infantry Mobile Unit	113
Insect Spray	309
IR - 800 Conventional Repeater	183
IRAN - 140 Aircraft	255
Iran Aircraft Industries	260
IRANCAM - War Chemical Factor Detector	238
Irritant & Colored Smoke Crenade	47
Tracked Rescue Vehicle	122

J

Jammer	199
JATO	274

K

KAT 72 Fire Control System	130
Kavosh 2 - Zoom Lenses	224

L

Landing Craft	294
Laser Range Finder	222 - 223
Launching Vehicle	169
Law Enforcement Cartridges	4
Life Pill	322
Light Ballestic Armor	133
LP 26.5 & LP 26.5 WT	91

M

M60 - MBTS	126
Machine Gun	89,90
Marine Vessel Radar	208
Marker	324
Mask (IM2)	241
Mesbah - Air Defence Gun	96
Mesbah 1 - Air Defence Gun	97
Meteorological Vehicle	162
MGA3 Machine Gun	87
Midget Submarine	281
Military Cartridges	5
Military Satellite Jammer	199
Mine Cleaner	134 - 137
Mine Sifter	136
Mines	311 - 315
Mini Flare & Trip Flare	329
Misagh 1 - Portable Shoulder Launcher	163
Missile Launcher Boat	280
Mobile Air Tower	200
Mobile Bakery	315
Mobile Generator	118
Mobile Kitchen System	315
Mobile Laboratory System	249
Mobile Medical Service System	249
Mobile Radio	182
Mobile Repair Unit	162
Mobile Workshop	132

Modern Ballestic Armor	133
Modernized Tank T72 & T72Z	127
Mohajer 2 - UAV Surveillance System	269
Monitoring System	230
Monocular Night Vision Goggle	227
Mortar Bomb	12 - 20
Moving Hospital	319
Mowj Frigate	282
MRC - 200 P Decontamination	246
MRC - 500N Rinser Solution	246
Multi Band Radio Communication	175
Multi Purpose Patrol Boat	286 , 290
Multi Spectural Camouflage Screen	326

N

N.B.C Boot Cover	243
N.B.C Protective Gloves	243
N.B.C Protective Suit	240
N.B.C Self Aid Kit	247
Nafez - 40 mm Anti Armor Rockets	142
Nakhjir & 2	327
Nasr - Anti Ship Missile	167
Naval ammunition	14
Nazeat 10 - H Surface to Surface Rocket	158
Nazeat 6 - H Surface to Surface Rocket	158
NED - 123X Firewall	194
NED - 2121 Network Security Gateway	195
NED - 4182 Bulk Cyphering System	192
NED- 2223 Virtual Private Lan	193
Night Driver Goggle	228
Night Vision Sight	225
Nitromil	247
Noor - Anti Ship Missile	168
Noor Missile - Launching Vehicle	169
Noor Missile - Radar & Command Vehicle	169
Zelzal 3 - Surface to Surface Rocket	159

O

Obidoxim Chloride	245
Observatory Binoculars	221
Oceanographic Vessel	296
Oghab 11 - Aerial Camera	232
OKM Imitator Rocket	143
Oxomil	247

P

Parachute	271
Patrol Boat	284 - 290
PCS - 8030 Phone Cyphering	190
Personnel Defence Spray	309
Petrn	75
PKM Machine Gun	89
Plain & Shock Tube Detonators	45
Pneumatic Bridge	319
PRC - 58 VHF Transceiver	177
Precision Approach Radar	206
Primers	44
Promil	248

Protective Hood of Mask	241
Protective Suit	240

Q

QODS Aviation Industries	267
--------------------------	-----

R

RAAD1 - Self Propelled Gun	124
RAAD2 - Self Propelled Gun	125
RAAD2M - Self Propelled Gun	125
Radar & Command Vehicle	169
Radio Monitoring	209 - 210
Rakhsh - 4X4 Wheel APC	123
Recompression Chamber	297
Recovery Tank	129
Repair Unit	162
Repeater	183
RES - 2111 Radio Encryption System	189
Respiratory System	242
Rifle Bag	308
Rifle Grenade	39
Rifle Scope	222
Rocket & Missile Command Cabin	161
Rocket Flare	330
Rocket Launcher Patrol Boat	289
Roller Tank Mine Cleaner	135
RPG7 Launcher	93
RPG7 Training Aid	93
RTX - 58 - VHF Transceiver	176

S

Sadad 103 - Monitoring System	230
Saeghe - 40 mm Rocket	143
Saeghe 2 - Aerial Target System	268
Saeghe2 - Light Anti Armor Missile	150
SAMAVAT - 35 mm Gun	98
SAR 20 - Search & Rescue Boat	291
Search & Rescue Boat	291
Security Computer	197
Security Mobile Phone	196
Security Note Book	198
Self Neutralized Mine	311
Semi Long VHF Radar	203
Shahab Thaqeb Missile	164
Shahed 278 Helicopter	266
Shahin - Huntig	328
Shalamche Missile	165
Short Range Surveillance Radar	207
Shotgun	308
Shotgun & Sport Ammo	7
Shoulder Launcher	163
Similator	333 , 133 ,275
Sina - Fast Missile Boat	283
Single Base Propellant	52
Smoke Mine	311
Smoke Pot	46
Start Ignition Charge RPG7	69
STS 2 - Firearm Shooting Simulator	333
Suits	321

Supply & Crew Boat	292
Surface to Surface Rocket	158 - 160
Surveillance & Control Network	201
SVD Sniper Rifle	88

T

T.N.T	74
T9 &T9S Automatic Gun	84
T9K - Light Submocene gun	83
Tactical Integrated Communication	200
Tactical Multi Layers Shelter	202
Tactical Microwave Radar	205
Tactical Radio Monitoring	209
Taftan 2 - Mine Cleaner	134
Tank Ammunition	22 ,24
Tank Crew Suit	321
Tank Passage Bridge	128
Thermal Cameras	229
Tonfa	306
Tophan 2 - Heavy Anti Armor	146
Tophan 3 - Heavy Anti Armor	147
Toofan Launcher	148
TP - 780 Trunked Two - Way Radio	183
Training Grenade	37
Transfer of Technology	336 - 339
TRC - 128 Radio Relay	179
Triple Base Propellant	70
Tug Boats	293

U

UAV Surveillance System	269
UHF Airborn Communication	185 - 187
Under Water Pinger	215

V

Vessel	291,296
VHF - Airborne Communication	184
VHF Transceiver	176 - 177
VIP Bombing Jammer	199
VRC - 110R HF Transceiver	181
VRC - 58 - VHF Transceiver	180

W

Water Purification System	317
Wideband Direction Finder	211
Wireless Fire Control	113
UHF Airborn Communication	185 - 187
Under Water Pinger	215
UAV Surveillance System	269
UHF Airborn Communication	185 - 187
Under Water Pinger	215

Z

Zafar - Anti Armor Rockets	144
Zafar - Anti Ship Missill	166
Zelzal 3 - Surface to Surface Missile	159
Zoaf Pistol	82
Zolfaghar - Fast Patrol Craft	284

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.